

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T09:30:06

Source: data/json/noise_experiments/v2/noise8i_nocoplayer_memprimary_game/claude-sonnet-4.5/game/noisy8_bidirectional_informed_memprimary_nocoplayer/noise8i_nocoplayer_memprimary_game_003_neutral_rep03.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy8_bidirectional_informed_memprimary_nocoplayer
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 3
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 0
coplayer_history_window: 0
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 3
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard",
  "Agent_3": "standard",
  "Agent_4": "standard",
  "Agent_5": "standard",
  "Agent_6": "standard",
  "Agent_7": "standard",
  "Agent_8": "standard"
}
dyadic_pairing_schedule: {
  "num_turns": 10,
  "role_targets": {
    "Agent_1": 5,
    "Agent_2": 5,
    "Agent_3": 5,
    "Agent_4": 5,
    "Agent_5": 5,
    "Agent_6": 5,
    "Agent_7": 5,
    "Agent_8": 5
  },
  "strategy": "balanced_random_roles_v1"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths,

follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_3; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_4; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_5; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_6; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_7; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_8; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

=====

ROUND 1

=====

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Sender, Agent_4=Receiver, Agent_5=Receiver, Agent_6=Receiver, Agent_7=Sender, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_6=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_8=Sender, Agent_5=Receiver
dyad_4: Agent_7=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$3; received \$9; Agent_6 returned \$3; payoffs Agent_2=\$5, Agent_6=\$6;
noise/visible: sent visible=\$2.25, received visible=\$6.75, returned visible=\$2.81
Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_3 sent \$4; received \$12; Agent_4 returned \$8; payoffs Agent_3=\$9, Agent_4=\$4;
noise/visible: sent visible=\$4.78, received visible=\$14.34, returned visible=\$8.69
Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 12.0, "min": 0, "raw": 8.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_8 sent \$3; received \$9; Agent_5 returned \$3; payoffs Agent_8=\$5, Agent_5=\$6;
noise/visible: sent visible=\$2.3, received visible=\$6.9, returned visible=\$3.37
Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent \$3; received \$9; Agent_1 returned \$5.5; payoffs Agent_7=\$7.5, Agent_1=\$3.5;
noise/visible: sent visible=\$3.77, received visible=\$11.31, returned visible=\$5.51
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 9.0, "min": 0, "raw": 5.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 3.5,
"Agent_2": 5.0,
"Agent_3": 9.0,
"Agent_4": 4.0,
"Agent_5": 6.0,
"Agent_6": 6.0,
"Agent_7": 7.5,
"Agent_8": 5.0

}

Visible balances after round: {

"Agent_1": 5.810000000000005,
"Agent_2": 4.810000000000005,
"Agent_3": 9.69,
"Agent_4": 6.34,
"Agent_5": 3.899999999999995,
"Agent_6": 3.75,
"Agent_7": 7.51,
"Agent_8": 5.37

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:14:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_6",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_6"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the **SENDER**:

- You receive \$5 each round
- You decide how much to **SEND** to the **RECEIVER** (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to **RETURN**
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the **RECEIVER**:

- Sender receives \$5
- Sender decides how much to **SEND** to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to **RETURN**
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to establish a cooperative approach while being reasonably cautious since I have no history with my co-player.

The multiplication factor of 3x creates an opportunity for mutual benefit. If I send a moderate amount, we can both potentially end up better off than the initial \$5.

I'll send **\$3**, which gives my co-player \$9 to work with. This shows trust and willingness to cooperate. If they return approximately \$3-4, we'll both be in a good position, and it could establish a positive pattern for future rounds.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit
2. The receiver then decides how much to return
3. There's communication noise mentioned - amounts may appear different than actually sent
4. This is a multi-agent setting with 8 agents total, pairs rotate
5. This is the first round, so I have no history with any player yet

Since this is the first round and I have no information about other players' behavior, I need to consider:

- Sending \$0 guarantees I keep \$5, but no potential for mutual gain
- Sending everything (\$5) maximizes the total pot (\$15 for receiver) but is risky
- A moderate amount balances risk and potential reward

In trust games, common strategies include:

- Sending about half as a signal of conditional cooperation
- The tripling effect means even partial cooperation can benefit both parties

Let me consider sending \$3:

- I send \$3, receiver gets \$9
- If receiver returns \$3, I end up with \$5 (same as starting)
- If receiver returns \$4-5, I'm better off
- If receiver returns less than \$3, I'm worse off
- This shows trust while keeping some insurance

Alternatively, sending \$2.5:

- I send \$2.5, receiver gets \$7.5
- If receiver returns \$2.5, I end up with \$5
- If receiver returns \$3+, I'm better off
- Less risk than \$3

Or sending \$2:

- I send \$2, receiver gets \$6
- If receiver returns \$2, I end up with \$5
- Lower risk, but also lower potential reward

Given this is the first round and I want to establish a cooperative pattern, I'll send \$3. This shows trust and willingness to cooperate while not being completely vulnerable.

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_1

Timestamp: 2026-08-12T09:14:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
```

```

"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_6"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.25, that is 45.0% of their total endowment

nt. You received \$6.75. How much do you return to the sender? (0-6.75)

Assistant output:

Looking at this situation:

- The sender showed moderate trust by sending \$2.25 (45% of their endowment)
- This was multiplied to \$6.75 for me
- The sender currently has \$2.75 remaining

I want to reward their trust while maintaining a fair distribution. If I return \$3.00:

- The sender ends up with \$5.75 (a gain of \$0.75)
- I keep \$3.75 (a gain of \$3.75)
- Both of us are better off than if no money had been sent

This seems like a reasonable balance that rewards their moderate trust and encourages future cooperation.

```
{"return": 3.00}
```

Reasoning:

Let me think through this situation carefully.

I'm the RECEIVER in this round of a trust game.

The sender:

- Started with \$5
- Sent me \$2.25 (which is 45% of their endowment)
- This was multiplied by 3x, so I received \$6.75

Now I need to decide how much to return.

Key considerations:

1. The sender showed moderate trust by sending 45% of their endowment
2. They could have sent nothing, or they could have sent everything
3. If I return nothing, I keep \$6.75 and they keep \$2.75 (\$5 - \$2.25)
4. If I return some amount, we both benefit from the cooperative exchange

Let me think about fair or reasonable return amounts:

- If I return \$3.00: The sender ends up with $\$2.75 + \$3.00 = \$5.75$ (gain of \$0.75)
- I keep: $\$6.75 - \$3.00 = \$3.75$
- Total: $\$5.75 + \$3.75 = \$9.50$

- If I return \$2.25: The sender ends up with $\$2.75 + \$2.25 = \$5.00$ (break even)
- I keep: $\$6.75 - \$2.25 = \$4.50$
- Total: $\$5.00 + \$4.50 = \$9.50$

- If I return \$3.50: The sender ends up with $\$2.75 + \$3.50 = \$6.25$ (gain of \$1.25)
- I keep: $\$6.75 - \$3.50 = \$3.25$
- Total: $\$6.25 + \$3.25 = \$9.50$

Common approaches in trust games:

I could mirror their proportion-returning about 45% of what I received, which would be roughly \$3.04. Alternatively, I could split the cooperative gains equally: the \$4.50 surplus created through the exchange divided in half means returning \$2.25 to give them an even split of the benefit. Or I could aim for a return that ensures they profit from trusting me, somewhere between \$2.25 and \$3.50....

Usage: {

```
"input_tokens": 529,  
"output_tokens": 1174,  
"reasoning_tokens": 0
```

}

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T09:15:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_2",  
"move_index": 2,  
"opponent_id": "Agent_4",
```

```

"pairing": {
  "agents": [
    "Agent_3",
    "Agent_4"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round with no prior history, I want to signal cooperation and trust while

e being somewhat cautious.

In this trust game, sending money creates value (3x multiplier), but requires trusting that my co-player will reciprocate. Since:

- We'll likely play multiple rounds (possibly with different partners)
- Building a cooperative reputation could benefit me when I'm the receiver
- The socially optimal strategy involves sending substantial amounts to maximize total value

I'll send a significant amount to demonstrate trust, but keep a small buffer given the uncertainty.

```
{"send": 4}
```

Reasoning:

This is the first round of a trust game where I am the SENDER. Let me think through the strategic considerations:

1. **Game Structure:**
 - I have \$5
 - I can send \$0-\$5 to the receiver
 - Whatever I send gets multiplied by 3x
 - The receiver then decides how much to return
 - My final payoff: \$5 - sent + returned
 - Receiver's payoff: 3xsent - returned
2. **Strategic Considerations:**
 - If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
 - If I send \$5: Receiver gets \$15, and could return anywhere from \$0-\$15
 - If they return \$0: I get \$0 total (lost \$5)
 - If they return \$5: I get \$5 total (break even)
 - If they return \$7.50: I get \$7.50 (gain \$2.50)
 - If they return \$10: I get \$10 (gain \$5)
3. **Trust Game Logic:**
 - This is a classic trust game
 - Sending money shows trust
 - The socially optimal outcome is for me to send the maximum (\$5), which creates \$15 total, and then split fairly
 - A fair split might be: I sent \$5, created \$15, so maybe 50/50 of the gain (\$15-\$5=\$10 gain, so \$5 each) meaning they return \$10 and keep \$5
 - Or they could view it as: I took the risk, so 50/50 of total created value: they return \$7.50, we each get \$7.50
4. **First Round Considerations:**
 - This is round 1, so there's no history to build on
 - However, there will be multiple rounds with potentially different partners
 - Establishing a cooperative reputation early could pay dividends later, especially since I'll alternate between sender and receiver roles
 - The communication noise mentioned means amounts might shift slightly, which could complicate trust signals
 - A moderate to high send amount (\$3-5) demonstrates willingness to cooperate while managing risk, potentially encouraging reciprocal behavior in future rounds

I'm leaning toward sending \$4 as my opening move. This signals meaningful trust-it triples to \$12 for the receiver-while keeping \$1 as a buffer. Even a modest return of \$4-5 puts me ahead, and it establishes the cooperative tone I'd want if positions reverse. Classic trust game research suggests starting with substantial trust and adjusting based on responses works better than cautious opening moves, so I'll go with that approach....

```
Usage: {  
  "input_tokens": 500,  
  "output_tokens": 1038,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T09:15:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_2",
```

```

"move_index": 3,
"opponent_id": "Agent_3",
"pairing": {
  "agents": [
    "Agent_3",
    "Agent_4"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_3": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$4.78, that is 95.6% of their total endowment

nt. You received \$14.34. How much do you return to the sender? (0-14.34)

Assistant output:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.78 (95.6% of their endowment)
- They kept only \$0.22 for themselves
- I received \$14.34 after the 3x multiplier
- A fair return should reward their trust

If I return \$8.00:

- Sender ends up with: $\$0.22 + \$8.00 = \$8.22$ (well above their initial \$5)
- I keep: $\$14.34 - \$8.00 = \$6.34$ (also above the initial \$5)
- Both of us benefit from the cooperation

This seems like a fair reciprocal response to their high level of trust.

```
{'return': 8.00}
```

Reasoning:

Let me understand this situation:

1. The sender had \$5 endowment
2. They sent me \$4.78 (which is 95.6% of their \$5)
3. This was multiplied by 3x, so I received $\$4.78 \times 3 = \14.34

Now I need to decide how much to return.

Key considerations:

- The sender showed a lot of trust by sending 95.6% of their endowment
- They kept only $\$5 - \$4.78 = \$0.22$ for themselves
- If I return nothing, the sender ends up with just \$0.22
- If I return generously, both of us can benefit from this cooperation

What's fair/reciprocal?

- The sender trusted me significantly
- A common fairness norm would be to split the gains
- Total created value: \$14.34
- Original sender investment feel: \$4.78
- One approach: Return enough so we both end up roughly equal
- Another approach: Return enough so sender gets significantly more than their original \$5 (rewarding trust)

Calculations:

- If I return \$7.17 (about half of what I received):
 - Sender gets: $\$0.22 + \$7.17 = \$7.39$
 - I get: $\$14.34 - \$7.17 = \$7.17$
- If I return \$8:
 - Sender gets: $\$0.22 + \$8 = \$8.22$
 - I get: $\$14.34 - \$8 = \$6.34$
- If I return \$9:
 - Sender gets: $\$0.22 + \$9 = \$9.22$
 - I get: $\$14.34 - \$9 = \$5.34$

Given that the sender showed exceptional trust (sending 95.6%), I should reward this trust generously. A return that gives them significantly more than their original endowment while still keeping a reasonable amount seems appropriate.

Returning \$8 strikes a good balance-it rewards their exceptional trust while ensuring both of us benefit from the cooperation. They'd end up with \$8.22 versus their original \$5, and I'd keep \$6.34. But given how much they risked, maybe I should push toward \$8.50 to give them even more of the gains we created together....

Usage: {

```
"input_tokens": 529,  
"output_tokens": 866,  
"reasoning_tokens": 0
```

}

GAME - Agent_8 - Sender - dyad_3

Timestamp: 2026-08-12T09:15:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_5"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_5": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this as Round 1 with no prior information about my co-player, I want to demonstrate trust while being reasonably cautious.

If I send \$3:

- My co-player receives \$9 (3×3)
- This shows meaningful cooperation
- If they return a fair portion (say $1/3$ to $1/2$), we both benefit
- My maximum loss is limited to \$3

I'll go with a moderately trusting approach:

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I have \$5 to start with
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. I keep: \$5 - sent + returned
5. The receiver keeps: received - returned

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There may be communication noise
- I don't know who my co-player is

Strategic thinking:

- If I send \$0, I guarantee keeping \$5, but there's no potential for gain
- If I send everything (\$5), the receiver gets \$15, and if they're cooperative, they might return a fair amount
- A moderate amount (like \$2-3) shows some trust but limits risk

Since this is round 1 and I have no information about my co-player:

- I want to signal cooperation and trust
- But I also want to be somewhat cautious
- Sending around \$3 seems like a reasonable middle ground that:
 - Shows good faith (60% of my endowment)
 - Gives the receiver \$9 to work with
 - If they return fairly (say $1/3$ to $1/2$ of what they receive), I could end up with \$5-6.5
 - Limits my maximum loss to \$3

Let me send \$3 as a cooperative but measured opening move.

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T09:16:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_5"
    ]
  }
}
```

```

],
"investor": "Agent_8",
"roles": {
  "Agent_5": "trustee",
  "Agent_8": "investor"
},
"trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.3, that is 46.0% of their total endowment. You received \$6.899999999999995. How much do you return to the sender? (0-6.899999999999995)

Assistant output:

Looking at this situation:

- The sender showed trust by sending \$2.3 (46% of their endowment)

- I received \$6.9 after the 3x multiplier
- The sender currently has \$2.7 remaining

I want to:

1. Reward the sender's trust so they benefit from sending
2. Split the gains fairly
3. Establish myself as trustworthy for future rounds

If I return \$3.0:

- Sender ends with: $\$2.7 + \$3.0 = \$5.7$ (a gain of \$0.7)
- I end with: $\$6.9 - \$3.0 = \$3.9$

This seems fair - the sender profits from trusting me, and we both benefit from the multiplier effect.

```
'''json
{"return": 3.0}
'''
```

Reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.3 (which is 46% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$6.9 ($2.3 \times 3 = 6.9$)
4. I need to decide how much to return (between \$0 and \$6.9)

The sender's current situation:

- Started with \$5
- Sent \$2.3
- Currently has: $\$5 - \$2.3 = \$2.7$

If I return nothing:

- Sender ends with: \$2.7
- I end with: \$6.9

The sender showed trust by sending nearly half their endowment. A fair approach would be to:

1. Ensure the sender doesn't lose money
2. Split the gains reasonably

Total money in the system now: $\$2.7$ (sender has) + $\$6.9$ (I have) = $\$9.6$

This started from two \$5 endowments = \$10 total, so we have \$9.6 total.

Wait, let me recalculate:

- Sender started with \$5, sent \$2.3, has \$2.7 left
- I received \$6.9 (the \$2.3 multiplied by 3)
- Total in system: $\$2.7 + \$6.9 = \$9.6$

If we split equally: each person would get \$4.8

- For sender to have \$4.8, they need: $\$4.8 - \$2.7 = \$2.1$ returned
- For me to have \$4.8, I'd keep: $\$6.9 - \$2.1 = \$4.8$?

But equal split might not be the best strategy. The sender took a risk by sending money. A common approach in trust games is to:

reward their trust by ensuring they come out ahead of their initial \$5, while also sharing in the multiplier gains. Let me test a few return amounts: returning \$3.0 gives them \$5.7 total (a \$0.7 gain) while I keep \$3.9; returning \$2.5 gives them \$5.2 (a \$0.2 gain) while I keep \$4.4; returning \$2.1 would give them exactly \$4.8 and me \$4.8.

A return around \$3.0-\$3.3 seems optimal-it meaningfully rewards their trust while still letting me benefit from the multiplier effect. At \$3.0, they gain \$0.7 and I keep \$3.9, which feels like a fair split that demonstrates trustworthiness.

The sender's 46% send rate suggests moderate confidence in me, so I should reciprocate proportionally.

```
Usage: {
  "input_tokens": 539,
  "output_tokens": 1314,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T09:16:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_1"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_1": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust with caution since I have no history with this receiver.

If I send money, it gets tripled, creating potential mutual gain. However, the receiver controls how much comes back. The communication noise adds uncertainty.

I'll take a moderately cooperative approach to signal trust while managing risk:

****My decision:****

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return a fair portion, we both benefit. This allows me to learn about their reciprocity while not risking everything in this uncertain first round.

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 to allocate
2. Whatever I send gets tripled
3. The receiver then decides how much to return
4. There's communication noise - amounts may differ slightly from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- This is round 1, so I have no history with this particular receiver
- There's a note about communication noise which could affect trust building

Standard economic reasoning in trust games:

- If I send \$5, it becomes \$15 for the receiver
- A "fair" split might be for them to return \$10, giving us each \$10
- But the receiver might keep everything, leaving me with \$0
- If I send \$0, I guarantee keeping \$5 but miss potential gains

Since this is:

1. Round 1 with no history
2. A multi-round game where reputation/reciprocity could matter across the full run
3. There's communication noise which might complicate trust building

I should consider a moderate opening move that:

- Shows some trust/willingness to cooperate
- Doesn't risk everything
- Allows me to learn about this receiver's behavior
- Sets a cooperative tone

A common strategy in trust games is to send around 50% on the first round as a signal of moderate trust while protecting against complete loss.

I'll send \$2.50 (or \$2 or \$3 since I need to be practical about amounts).

Actually, looking at the format, I should probably stick to whole dollar amounts for simplicity. Let me send \$3 as a cooperative opening move - it's more than half, showing trust, but not everything.

Wait, let me reconsider. The receiver gets \$9 if I send \$3. If they return half (\$4.50), I'd have \$5 - \$3 + \$4.50 = \$6.50, and they'd have \$4.50. ...

```
Usage: {
  "input_tokens": 500,
```

```
"output_tokens": 698,  
"reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_4

Timestamp: 2026-08-12T09:16:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_1"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.77, that is 75.4% of their total endowment. You received \$11.31. How much do you return to the sender? (0-11.31)

Assistant output:

Looking at this situation:

- The sender showed considerable trust by sending \$3.77 (75.4% of their endowment)
- They currently have: $\$5 - \$3.77 = \$1.23$
- I received: \$11.31
- Total money created: $\$1.23 + \$11.31 = \$12.54$

The sender took a significant risk and was quite generous. I want to reciprocate that trust while ensuring we both benefit from cooperation.

If I return \$5.50:

- Sender ends with: $\$1.23 + \$5.50 = \$6.73$
- I end with: $\$11.31 - \$5.50 = \$5.81$
- Both of us gain compared to the initial \$5, rewarding the sender's trust

This seems like a fair reciprocation that rewards their generosity while still allowing both of us to benefit.

{'return': 5.50}

Reasoning:

Let me think about this situation carefully.

I'm the RECEIVER in Round 1.

- The sender started with \$5
- They sent me \$3.77 (which is 75.4% of their \$5 endowment)
- I received \$11.31 (which is $\$3.77 \times 3$)
- I need to decide how much to return (between \$0 and \$11.31)

The sender showed significant trust by sending 75.4% of their endowment. Let me think about fair reciprocity:

If the sender sent \$3.77:

- They kept: $\$5 - \$3.77 = \$1.23$
- I received: \$11.31

For a fair split of the total money created:

- Total money now = $\$1.23$ (sender kept) + $\$11.31$ (I received) = $\$12.54$
- Fair split would be roughly \$6.27 each
- So I should return enough so the sender has ~\$6.27
- Sender currently has \$1.23, so I'd need to return about \$5.04 to get them to \$6.27
- That would leave me with $\$11.31 - \$5.04 = \$6.27$

Alternatively, I could think about matching their generosity:

- They sent 75.4% of their endowment
- If I return 75.4% of what I received: $0.754 \times \$11.31 = \8.53
- That seems quite generous but might be excessive

Another approach - proportional reciprocity:

- They gave me \$3.77 out of their \$5 (75.4%)
- A reasonable return might be to give back around half of what I received, ensuring both benefit
- Half would be \$5.66

Let me think about common strategies in trust games:

1. Return nothing (free-ride)
2. Return what was sent (break even for sender)
3. Return half of what received (both benefit equally from the multiplier)

4. Return enough to equalize final payoffs

Given their generosity, returning \$5-6 seems appropriate. This would give the sender roughly \$6-7 while leaving me with similar amounts, reflecting mutual gain from the exchange without being excessive....

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 1020,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Receiver, Agent_4=Sender, Agent_5=Sender, Agent_6=Sender, Agent_7=Receiver, Agent_8=Receiver

Pairings:

dyad_1: Agent_5=Sender, Agent_3=Receiver
dyad_2: Agent_1=Sender, Agent_2=Receiver
dyad_3: Agent_4=Sender, Agent_8=Receiver
dyad_4: Agent_6=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_5 sent \$2.5; received \$7.5; Agent_3 returned \$6.5; payoffs Agent_5=\$9, Agent_3=\$1; noise/visible: sent visible=\$3.11, received visible=\$9.33, returned visible=\$6.78
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 7.5, "min": 0, "raw": 6.5}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_2: Agent_1 sent \$3.75; received \$11.25; Agent_2 returned \$5.5; payoffs Agent_1=\$6.75, Agent_2=\$5.75; noise/visible: sent visible=\$3.95, received visible=\$11.85, returned visible=\$5.66
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 11.25, "min": 0, "raw": 5.5}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}
dyad_3: Agent_4 sent \$4.5; received \$13.5; Agent_8 returned \$6.5; payoffs Agent_4=\$7, Agent_8=\$7; noise/visible: sent visible=\$4.58, received visible=\$13.74, returned visible=\$7.41
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 13.5, "min": 0, "raw": 6.5}, "sent": {"amount": 4.5, "clamped": false, "max": 5, "min": 0, "raw": 4.5}}
dyad_4: Agent_6 sent \$3; received \$9; Agent_7 returned \$7; payoffs Agent_6=\$9, Agent_7=\$2; noise/visible: sent visible=\$3.8, received visible=\$11.4, returned visible=\$6.42
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 9.0, "min": 0, "raw": 7.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 10.25,
"Agent_2": 10.75,
"Agent_3": 10.0,
"Agent_4": 11.0,
"Agent_5": 15.0,
"Agent_6": 15.0,
"Agent_7": 9.5,
"Agent_8": 12.0

}

Visible balances after round: {

"Agent_1": 12.72,
"Agent_2": 11.160000000000002,
"Agent_3": 12.52,
"Agent_4": 14.25,
"Agent_5": 13.18,
"Agent_6": 12.17,
"Agent_7": 11.909999999999998,
"Agent_8": 12.61

}

LLM CALLS

GAME - Agent_5 - Sender - dyad_1

Timestamp: 2026-08-12T09:17:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_3",


```

"pairing": {
  "agents": [
    "Agent_5",
    "Agent_3"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_3": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.3, that is 46.0% of their total endowment. You received \$6.899999999999995. How much do you return to the sender? (0-6.899999999999995)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$2.3 (46% of their endowment)
- I received \$6.9 after the 3x multiplier
- The sender currently has \$2.7 remaining

I want to:

1. Reward the sender's trust so they benefit from sending
2. Split the gains fairly
3. Establish myself as trustworthy for future rounds

If I return \$3.0:

- Sender ends with: $\$2.7 + \$3.0 = \$5.7$ (a gain of \$0.7)
- I end with: $\$6.9 - \$3.0 = \$3.9$

This seems fair - the sender profits from trusting me, and we both benefit from the multiplier effect.

```
'''json
{"return": 3.0}
'''
```

reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.3 (which is 46% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$6.9 ($2.3 \times 3 = 6.9$)
4. I need to decide how much to return (between \$0 and \$6.9)

The sender's current situation:

- Started with \$5
- Sent \$2.3
- Currently has: $\$5 - \$2.3 = \$2.7$

If I return nothing:

- Sender ends with: \$2.7
- I end with: \$6.9

The sender showed trust by sending nearly half their endowment. A fair approach would be to:

1. Ensure the sender doesn't lose money
2. Split the gains reasonably

Total money in the system now: $\$2.7$ (sender has) + $\$6.9$ (I have) = $\$9.6$

This started from two \$5 endowments = \$10 total, so we have \$9.6 total.

Wait, let me recalculate:

- Sender started with \$5, sent \$2.3, has \$2.7 left
- I received \$6.9 (the \$2.3 multiplied by 3)
- Total in system: $\$2.7 + \$6.9 = \$9.6$

If we split equally: each person would get \$4.8

- For sender to have \$4.8, they need: $\$4.8 - \$2.7 = \$2.1$ returned
- For me to have \$4.8, I'd keep: $\$6.9 - \$2.1 = \$4.8$?

But equal split might not be the best strategy. The sender took a risk by sending money. A common approach in trust games is to:

reward their trust by ensuring they come out ahead of their initial \$5, while also sharing in the multiplier gains. Let me test a few return amounts: returning \$3.0 gives them \$5.7 total (a \$0.7 gain) while I keep \$3.9; returning \$2.5 gives them \$5.2 (a \$0.2 gain) while I keep \$4.4; returning \$2.1 would give them exactly \$4.8 and me \$4.8.

A return around \$3.0-\$3.3 seems optimal-it meaningfully rewards their trust while still letting me benefit from the multiplier effect. At \$3.0, they gain \$0.7 and I keep \$3.9, which feels like a fair split that demonstrates trustworthiness.

The sender's 46% send rate suggests moderate confidence in me, so I should reciprocate proportionally...

usage:

```
{
  "input_tokens": 539,
  "output_tokens": 1314,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.3, it became \$6.899999999999995, and you returned \$3.0. Your visible payoff was \$3.899999999999995.

Your total visible earnings across all rounds are \$3.899999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

- I'm now the sender with a new co-player
- In the previous round, I experienced a sender who sent 46% and I returned fairly
- I want to establish trust with this new partner while being reasonably cautious

Sending \$2.5 (50% of my endowment) seems like a good strategy:

- It shows significant trust
- It's a clear, fair starting point
- If my co-player reciprocates fairly, we both benefit from the 3x multiplier
- It's not overly risky with an unknown partner

```
'''json
{"send": 2.5}
'''
```

Reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.3 (46% of their endowment)
- I received \$6.9
- I returned \$3.0
- My payoff was \$3.9

Now I'm playing against a different agent (since it says "current co-player" and agents are randomly paired each round).

I need to decide how much to send. Key considerations:

1. I don't know anything about this new co-player's behavior
2. The sender in Round 1 sent 46% (\$2.3), which was a reasonable trust signal
3. I returned about 43.5% of what I received (\$3/\$6.9), which was fair
4. I want to build trust but also protect myself since I don't know this opponent

Strategies:

- Sending \$0 would be overly defensive and wouldn't build any relationship
- Sending \$5 (100%) would be too risky with an unknown partner
- Sending around \$2-3 seems reasonable as it shows trust without being reckless

Given that the previous sender sent \$2.3 (46%), I could mirror that behavior. This would be consistent and shows I'm willing to trust at a similar level to what I've experienced.

Alternatively, I could send slightly more to show goodwill, or slightly less to be more cautious.

I think sending \$2.5 (50%) is a good middle ground:

- It's a clear half of my endowment, easy to interpret
- It shows significant trust
- If they return fairly (around 40-45% of \$7.5 = \$3-3.4), we both profit
- It's not too risky

```

Let me calculate the outcomes:
- If I send $2.5, they receive $7.5
- I keep $2.5
...
Usage: {
  "input_tokens": 870,
  "output_tokens": 665,
  "reasoning_tokens": 0
}

```

GAME - Agent_3 - Receiver - dyad_1

Timestamp: 2026-08-12T09:17:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_3"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_3": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round with no prior history, I want to signal cooperation and trust while being somewhat cautious.

In this trust game, sending money creates value (3x multiplier), but requires trusting that my co-player will reciprocate. Since:

- We'll likely play multiple rounds (possibly with different partners)
- Building a cooperative reputation could benefit me when I'm the receiver
- The socially optimal strategy involves sending substantial amounts to maximize total value

I'll send a significant amount to demonstrate trust, but keep a small buffer given the uncertainty.

{"send": 4}

reasoning:

This is the first round of a trust game where I am the SENDER. Let me think through the strategic considerations:

1. ****Game Structure:****

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- My final payoff: \$5 - sent + returned
- Receiver's payoff: 3xsent - returned

2. ****Strategic Considerations:****

- If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
- If I send \$5: Receiver gets \$15, and could return anywhere from \$0-\$15
 - If they return \$0: I get \$0 total (lost \$5)
 - If they return \$5: I get \$5 total (break even)
 - If they return \$7.50: I get \$7.50 (gain \$2.50)
 - If they return \$10: I get \$10 (gain \$5)

3. ****Trust Game Logic:****

- This is a classic trust game
- Sending money shows trust
- The socially optimal outcome is for me to send the maximum (\$5), which creates \$15 total, and then split fairly
 - A fair split might be: I sent \$5, created \$15, so maybe 50/50 of the gain (\$15-\$5=\$10 gain, so \$5 each) meaning they return \$10 and keep \$5
 - Or they could view it as: I took the risk, so 50/50 of total created value: they return \$7.50, we each get \$7.50

4. ****First Round Considerations:****

- This is round 1, so there's no history to build on
- However, there will be multiple rounds with potentially different partners
- Establishing a cooperative reputation early could pay dividends later, especially since I'll alternate between sender and receiver roles
- The communication noise mentioned means amounts might shift slightly, which could com

plicate trust signals

- A moderate to high send amount (\$3-5) demonstrates willingness to cooperate while managing risk, potentially encouraging reciprocal behavior in future rounds

I'm leaning toward sending \$4 as my opening move. This signals meaningful trust-it triples to \$12 for the receiver-while keeping \$1 as a buffer. Even a modest return of \$4-5 puts me ahead, and it establishes the cooperative tone I'd want if positions reverse. Classic trust game research suggests starting with substantial trust and adjusting based on responses works better than cautious opening moves, so I'll go with that approach....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 1038,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.69 to you. Your visible payoff was \$9.69.

Your total visible earnings across all rounds are \$9.69.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

My co-player sent me \$3.11 (62% of their endowment), showing significant trust. I received \$9.33 after the 3x multiplier.

In my previous round, my co-player returned \$8.69 out of \$12 (72%), which was very generous. I want to maintain this cooperative spirit.

A fair approach would be to ensure my co-player profits from their trust while I also benefit from receiving the tripled amount.

Returning approximately \$6.50 would give them a final payoff of about \$8.39 (\$5 - \$3.11 + \$6.50), which rewards their trust handsomely, while I keep \$2.83.

```
{"return": 6.50}
```

Reasoning:

Let me analyze this situation:

1. Last round I was the SENDER:

- I sent \$4.0
- It became \$12.0
- They returned \$8.69
- My payoff: $\$5 - \$4 + \$8.69 = \9.69
- Their payoff: $\$12 - \$8.69 = \$3.31$

2. This round I'm the RECEIVER:

- They sent me \$3.11
- I received $\$3.11 \times 3 = \9.33
- I need to decide how much to return

My co-player sent me \$3.11 out of their \$5, which is 62.2% of their endowment. This shows significant trust.

In the last round, my co-player returned \$8.69 out of \$12 they received, which is 72.4% of what they received. This was quite generous - they gave me more than I sent them.

I should consider:

- Reciprocity: They showed trust by sending \$3.11
- Building reputation: Being fair will help in future rounds
- The note says there may be communication noise, so amounts might differ slightly

A fair return would be:

- They invested \$3.11, I should at least return that amount so they don't lose
- To split gains fairly: They gave up \$3.11, it became \$9.33, so we created \$6.22 in additional value
- If we split that gain: $\$3.11 + \$6.22/2 = \$3.11 + \$3.11 = \$6.22$

However, I also want to be somewhat generous to build trust, similar to how my previous co-player returned 72.4% of what they received.

Let me think about different approaches:

- Return exactly what they sent: \$3.11 (they break even)
- Return to split gains 50/50: $\sim \$6.22$ (they get $\$5 - \$3.11 + \$6.22 = \8.11)
- Return more generously (similar to 72.4% rate): $\sim \$6.75$ (they get $\$5 - \$3.11 + \$6.75 = \8.64)

Given the trust they showed and the precedent from my last game, I'll return around \$6.50-\$7.00, which gives them a good payoff while keeping a reasonable amount for myself. I'll return \$6.50: their final payoff becomes \$8.39 while I keep \$9.

Wait, let me check the math more carefully. If I return \$6.50, my payoff is $\$9.33 - \$6.50 = \$2.83$. That feels...

```
Usage: {
  "input_tokens": 763,
  "output_tokens": 1019,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Sender - dyad_2

Timestamp: 2026-08-12T09:17:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.77, that is 75.4% of their total endowment. You received \$11.31. How much do you return to the sender? (0-11.31)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.77 (75.4% of their endowment)
- They currently have: \$5 - \$3.77 = \$1.23
- I received: \$11.31
- Total money created: \$1.23 + \$11.31 = \$12.54

The sender took a significant risk and was quite generous. I want to reciprocate that trust while ensuring we both benefit from cooperation.

If I return \$5.50:

- Sender ends with: \$1.23 + \$5.50 = \$6.73
- I end with: \$11.31 - \$5.50 = \$5.81
- Both of us gain compared to the initial \$5, rewarding the sender's trust

This seems like a fair reciprocation that rewards their generosity while still allowing both of us to benefit.

{'return': 5.50}

reasoning:

Let me think about this situation carefully.

I'm the RECEIVER in Round 1.

- The sender started with \$5
- They sent me \$3.77 (which is 75.4% of their \$5 endowment)
- I received \$11.31 (which is \$3.77 × 3)
- I need to decide how much to return (between \$0 and \$11.31)

The sender showed significant trust by sending 75.4% of their endowment. Let me think about fair reciprocity:

If the sender sent \$3.77:

- They kept: \$5 - \$3.77 = \$1.23
- I received: \$11.31

For a fair split of the total money created:

- Total money now = \$1.23 (sender kept) + \$11.31 (I received) = \$12.54
- Fair split would be roughly \$6.27 each
- So I should return enough so the sender has ~\$6.27
- Sender currently has \$1.23, so I'd need to return about \$5.04 to get them to \$6.27
- That would leave me with \$11.31 - \$5.04 = \$6.27

Alternatively, I could think about matching their generosity:

- They sent 75.4% of their endowment
- If I return 75.4% of what I received: $0.754 \times \$11.31 = \8.53
- That seems quite generous but might be excessive

Another approach - proportional reciprocity:

- They gave me \$3.77 out of their \$5 (75.4%)
- A reasonable return might be to give back around half of what I received, ensuring both benefit
- Half would be \$5.66

Let me think about common strategies in trust games:

1. Return nothing (free-ride)
2. Return what was sent (break even for sender)
3. Return half of what received (both benefit equally from the multiplier)
4. Return enough to equalize final payoffs

Given their generosity, returning \$5-6 seems appropriate. This would give the sender roughly \$6-7 while leaving me with similar amounts, reflecting mutual gain from the exchange without being excessive....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1020,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.77, it became \$11.31, and you returned \$5.5. Your visible payoff was \$5.8100000000000005.

Your total visible earnings across all rounds are \$5.8100000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the previous round, I saw cooperation work well when the sender showed significant trust by sending 75.4% of their endowment, and I reciprocated fairly.

I'll adopt a similar cooperative strategy with my new co-player. Sending a substantial amount can maximize joint gains if they reciprocate.

```
{'send': 3.75}
```

Reasoning:

I'm now the SENDER in Round 2.

Let me think about this strategically:

1. In the previous round as RECEIVER, I saw someone send \$3.77 (75.4% of their endowment), showing significant trust
2. I returned \$5.50, which gave them \$6.73 total - a good return on their trust

Now I'm the SENDER. What should I consider?

- The person sending \$3.77 showed that cooperation can work well
- I reciprocated fairly, returning enough that they profited significantly
- If my current co-player is similarly cooperative, sending a substantial amount could benefit both of us

Strategy considerations:

- Sending nothing is safe but generates no surplus

- Sending a moderate amount (around \$2-3) tests the waters
- Sending a larger amount (like \$3.77 that I saw) shows trust and can maximize joint gains if reciprocated

Given that:

- I saw cooperation work in the last round
- The game seems to reward trust
- I benefited from someone trusting me and reciprocated fairly
- There may be communication noise (mentioned in instructions), but the pattern of cooperation seemed clear

I think I should send a substantial amount to show trust and potentially establish a cooperative pattern. I'll send around \$3.50-\$4.00.

Let me send \$3.50 (70% of endowment) - this shows significant trust similar to what I received, but slightly less aggressive to be somewhat cautious given this is a new pairing.

Actually, let me reconsider. The previous sender sent 75.4% (\$3.77). I reciprocated well. If I want to encourage cooperation, I should send a similar proportion. Let me send \$3.75 (75% of my endowment) to match that level of trust.

```
Usage: {
  "input_tokens": 874,
  "output_tokens": 494,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_2

Timestamp: 2026-08-12T09:17:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being reasonably cautious since I have no history with my co-player.

The multiplication factor of 3x creates an opportunity for mutual benefit. If I send a moderate amount, we can both potentially end up better off than the initial \$5.

I'll send **\$3**, which gives my co-player \$9 to work with. This shows trust and willingness to cooperate. If they return approximately \$3-4, we'll both be in a good position, and it could establish a positive pattern for future rounds.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit
2. The receiver then decides how much to return
3. There's communication noise mentioned - amounts may appear different than actually sent
4. This is a multi-agent setting with 8 agents total, pairs rotate
5. This is the first round, so I have no history with any player yet

Since this is the first round and I have no information about other players' behavior, I need to consider:

- Sending \$0 guarantees I keep \$5, but no potential for mutual gain
- Sending everything (\$5) maximizes the total pot (\$15 for receiver) but is risky
- A moderate amount balances risk and potential reward

In trust games, common strategies include:

- Sending about half as a signal of conditional cooperation
- The tripling effect means even partial cooperation can benefit both parties

Let me consider sending \$3:

- I send \$3, receiver gets \$9
- If receiver returns \$3, I end up with \$5 (same as starting)
- If receiver returns \$4-5, I'm better off
- If receiver returns less than \$3, I'm worse off
- This shows trust while keeping some insurance

Alternatively, sending \$2.5:

- I send \$2.5, receiver gets \$7.5
- If receiver returns \$2.5, I end up with \$5
- If receiver returns \$3+, I'm better off
- Less risk than \$3

Or sending \$2:

- I send \$2, receiver gets \$6
- If receiver returns \$2, I end up with \$5
- Lower risk, but also lower potential reward

Given this is the first round and I want to establish a cooperative pattern, I'll send \$3.

This shows trust and willingness to cooperate while not being completely vulnerable.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.81 to you. Your visible payoff was \$4.8100000000000005.

Your total visible earnings across all rounds are \$4.8100000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$3.95, so you received \$11.850000000000001. How much do you return? (0-11.850000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$3.95 (79% of their endowment), showing significant trust
- I received \$11.85 total
- They currently have only \$1.05 left

I want to:

1. Reward their trust and cooperation
2. Ensure they end up better than their initial \$5
3. Set a positive tone for potential future interactions

If I return \$5.50:

- They end up with: \$1.05 + \$5.50 = \$6.55
- I end up with: \$11.85 - \$5.50 = \$6.35

Both of us gain from cooperation, and this reciprocates their high level of trust appropriately.

```
'''json
{"return": 5.50}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the SENDER
- I sent \$3.00
- It became \$9.00 for the receiver
- They returned \$2.81
- My payoff: \$5 - \$3 + \$2.81 = \$4.81

Round 2:

- I am now the RECEIVER
- My co-player (a different agent) sent me \$3.95
- I received $\$3.95 \times 3 = \11.85
- I need to decide how much to return

Important considerations:

1. The sender gave me \$3.95 out of their \$5 - that's 79% of their endowment, showing high trust
2. If I return nothing, I keep \$11.85 and they end up with $\$5 - \$3.95 = \$1.05$
3. I should return enough to make cooperation worthwhile for them
4. There's communication noise mentioned, so amounts may differ slightly

What would be fair?

- The sender gave up \$3.95, keeping \$1.05
- I received \$11.85
- If I want them to end up better than their initial \$5, I should return more than \$3.95
- A 50-50 split of total surplus could work: Total = $\$1.05 + \$11.85 = \$12.90$. Half is \$6.45
- If I return \$5.40, sender gets $\$1.05 + \$5.40 = \$6.45$, I get $\$11.85 - \$5.40 = \$6.45$

But let me think about reciprocity:

- In round 1, I sent \$3 (60% of endowment), received \$9, they returned \$2.81 (31% of what they received)
- This round, sender sent \$3.95 (79% of endowment), I received \$11.85
- Using that same 31% return rate would give me \$3.67 to send back

The sender showed significantly more generosity than I did initially, so I should reciprocate with a higher return. Returning \$5 would leave them with \$6.05 total-better than their starting \$5-while I'd keep \$6.85, rewarding their trust and encouraging continued cooperation....

```
Usage: {
  "input_tokens": 793,
  "output_tokens": 977,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_3

Timestamp: 2026-08-12T09:18:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_8"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$4.78, that is 95.6% of their total endowment. You received \$14.34. How much do you return to the sender? (0-14.34)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.78 (95.6% of their endowment)
- They kept only \$0.22 for themselves
- I received \$14.34 after the 3x multiplier
- A fair return should reward their trust

If I return \$8.00:

- Sender ends up with: $\$0.22 + \$8.00 = \$8.22$ (well above their initial \$5)
- I keep: $\$14.34 - \$8.00 = \$6.34$ (also above the initial \$5)
- Both of us benefit from the cooperation

This seems like a fair reciprocal response to their high level of trust.

{'return': 8.00}

reasoning:

Let me understand this situation:

1. The sender had \$5 endowment
2. They sent me \$4.78 (which is 95.6% of their \$5)
3. This was multiplied by 3x, so I received $\$4.78 \times 3 = \14.34

Now I need to decide how much to return.

Key considerations:

- The sender showed a lot of trust by sending 95.6% of their endowment

- They kept only \$5 - $\$4.78 = \0.22 for themselves
- If I return nothing, the sender ends up with just \$0.22
- If I return generously, both of us can benefit from this cooperation

What's fair/reciprocal?

- The sender trusted me significantly
- A common fairness norm would be to split the gains
- Total created value: \$14.34
- Original sender investment feel: \$4.78
- One approach: Return enough so we both end up roughly equal
- Another approach: Return enough so sender gets significantly more than their original \$5 (rewarding trust)

Calculations:

- If I return \$7.17 (about half of what I received):
 - Sender gets: $\$0.22 + \$7.17 = \$7.39$
 - I get: $\$14.34 - \$7.17 = \$7.17$
- If I return \$8:
 - Sender gets: $\$0.22 + \$8 = \$8.22$
 - I get: $\$14.34 - \$8 = \$6.34$
- If I return \$9:
 - Sender gets: $\$0.22 + \$9 = \$9.22$
 - I get: $\$14.34 - \$9 = \$5.34$

Given that the sender showed exceptional trust (sending 95.6%), I should reward this trust generously. A return that gives them significantly more than their original endowment while still keeping a reasonable amount seems appropriate.

Returning \$8 strikes a good balance-it rewards their exceptional trust while ensuring both of us benefit from the cooperation. They'd end up with \$8.22 versus their original \$5, and I'd keep \$6.34. But given how much they risked, maybe I should push toward \$8.50 to give them even more of the gains we created together....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 866,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.78, it became \$14.34, and you returned \$8.0. Your visible payoff was \$6.34.

Your total visible earnings across all rounds are \$6.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

In Round 1, my co-player showed very high trust by sending 95.6% of their endowment, and I reciprocated with a fair return. That cooperative approach worked well for both of us.

Now as the sender with a new co-player, I want to maintain that cooperative spirit. Sending a substantial amount:

- Creates more total value through the 3x multiplier
- Signals trust and willingness to cooperate
- Could encourage reciprocation

I'll send \$4.50 (90% of my endowment), showing strong trust while being slightly more conservative than what I experienced last round.

```
{'send': 4.50}
```

Reasoning:

Let me think about this situation:

Round 1 (just completed):

- I was the RECEIVER
- Sender sent me \$4.78 (95.6% of their endowment) - showing very high trust
- I received \$14.34
- I returned \$8.00
- My payoff: \$6.34
- Sender's payoff would be: $\$5 - \$4.78 + \$8.00 = \8.22

Now in Round 2:

- I'm the SENDER
- I have \$5
- I'm playing against a different co-player (pairings are randomized each round)

The previous sender showed very high trust by sending 95.6%. I reciprocated fairly by returning \$8.00, which gave them a good outcome.

Now I need to decide how much to send. Key considerations:

1. This is a new co-player, so I don't have direct history with them
2. The previous round showed successful cooperation
3. Sending more creates more total value (3x multiplier), but requires trust
4. In a multi-round setting, establishing cooperative behavior can be beneficial

Given that:

- The previous sender sent 95.6% (\$4.78)
- That worked out well for both of us
- I should reciprocate the level of trust shown to me

I think I should send a similar high amount to signal cooperation and trust. Sending around \$4.50-\$4.80 would show similar trust to what I experienced.

Let me send \$4.50 (90% of my endowment). This shows strong trust while being slightly more conservative than the 95.6% I received.

Actually, thinking about it more - since I was treated with high trust and it worked out well, I should pay it forward. Sending \$4.50-\$4.75 seems appropriate.

...

```
Usage: {
  "input_tokens": 824,
  "output_tokens": 591,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_3

Timestamp: 2026-08-12T09:18:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_8"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1 with no prior information about my co-player, I want to demonstrate trust while being reasonably cautious.

If I send \$3:

- My co-player receives \$9 (3×3)
- This shows meaningful cooperation
- If they return a fair portion (say $1/3$ to $1/2$), we both benefit
- My maximum loss is limited to \$3

I'll go with a moderately trusting approach:

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I have \$5 to start with

2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. I keep: \$5 - sent + returned
5. The receiver keeps: received - returned

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There may be communication noise
- I don't know who my co-player is

Strategic thinking:

- If I send \$0, I guarantee keeping \$5, but there's no potential for gain
- If I send everything (\$5), the receiver gets \$15, and if they're cooperative, they might return a fair amount
- A moderate amount (like \$2-3) shows some trust but limits risk

Since this is round 1 and I have no information about my co-player:

- I want to signal cooperation and trust
- But I also want to be somewhat cautious
- Sending around \$3 seems like a reasonable middle ground that:
 - Shows good faith (60% of my endowment)
 - Gives the receiver \$9 to work with
 - If they return fairly (say 1/3 to 1/2 of what they receive), I could end up with \$5-6.
- 5
- Limits my maximum loss to \$3

Let me send \$3 as a cooperative but measured opening move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.37 to you. Your visible payoff was \$5.37.

Your total visible earnings across all rounds are \$5.37.

This round, you are the RECEIVER against your current co-player. They sent you \$4.58, so you received \$13.74. How much do you return? (0-13.74)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

My co-player sent me \$4.58 out of their \$5 (91.6% of their endowment) - showing very high trust.

I received \$13.74. To reciprocate this trust appropriately:

- They kept \$0.42
- If I return around \$6.50, they'd end with \$6.92 (a gain of \$1.92)
- I'd keep \$7.24 (also a solid gain)

This represents returning about 47% of what I received, which rewards their trust while we both benefit substantially.

```
'''json
{"return": 6.50}
'''
```

Reasoning:

Let me analyze this situation:

Round 2 - I'm now the RECEIVER

- My co-player sent me \$4.58
- I received \$13.74 (which is \$4.58 × 3)
- I need to decide how much to return

From Round 1:

- I sent \$3.00, they received \$9.00
- They returned \$3.37 to me
- This is about 37.4% of what they received, which is quite fair/generous

My co-player in this round sent me \$4.58 out of their \$5, which is 91.6% of their endowment - very trusting!

I should consider:

1. They showed high trust by sending almost everything
2. In Round 1, my previous partner returned about 37.4% which seemed fair
3. I want to reciprocate trust and maintain good cooperation

If I return about 40-50% of what I received (\$13.74):

- 40% = \$5.50
- 45% = \$6.18
- 50% = \$6.87

Given they sent almost everything (\$4.58 out of \$5), showing extreme trust, I should be generous in return.

Their final payoff would be:

- They started with \$5
- Sent \$4.58, kept \$0.42
- If I return \$6.00: $\$0.42 + \$6.00 = \$6.42$
- My payoff: $\$13.74 - \$6.00 = \$7.74$

If I return \$6.50:

- Their payoff: $\$0.42 + \$6.50 = \$6.92$
- My payoff: $\$13.74 - \$6.50 = \$7.24$

Given their high trust (sending 91.6% of endowment), I should return a good amount. Returning around \$6.50 seems fair - they end up with \$6.92 (better than their initial \$5), and I keep \$7.24 (also better than my initial \$0).

...

```
Usage: {
  "input_tokens": 755,
  "output_tokens": 653,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_4

Timestamp: 2026-08-12T09:18:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_7"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_6": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.25, that is 45.0% of their total endowment. You received \$6.75. How much do you return to the sender? (0-6.75)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.25 (45% of their endowment)
- This was multiplied to \$6.75 for me
- The sender currently has \$2.75 remaining

I want to reward their trust while maintaining a fair distribution. If I return \$3.00:

- The sender ends up with \$5.75 (a gain of \$0.75)
- I keep \$3.75 (a gain of \$3.75)
- Both of us are better off than if no money had been sent

This seems like a reasonable balance that rewards their moderate trust and encourages future cooperation.

{"return": 3.00}

reasoning:

Let me think through this situation carefully.

I'm the RECEIVER in this round of a trust game.

The sender:

- Started with \$5
- Sent me \$2.25 (which is 45% of their endowment)
- This was multiplied by 3x, so I received \$6.75

Now I need to decide how much to return.

Key considerations:

1. The sender showed moderate trust by sending 45% of their endowment
2. They could have sent nothing, or they could have sent everything
3. If I return nothing, I keep \$6.75 and they keep \$2.75 (\$5 - \$2.25)
4. If I return some amount, we both benefit from the cooperative exchange

Let me think about fair or reasonable return amounts:

- If I return \$3.00: The sender ends up with $\$2.75 + \$3.00 = \$5.75$ (gain of \$0.75)
- I keep: $\$6.75 - \$3.00 = \$3.75$
- Total: $\$5.75 + \$3.75 = \$9.50$
- If I return \$2.25: The sender ends up with $\$2.75 + \$2.25 = \$5.00$ (break even)
- I keep: $\$6.75 - \$2.25 = \$4.50$
- Total: $\$5.00 + \$4.50 = \$9.50$
- If I return \$3.50: The sender ends up with $\$2.75 + \$3.50 = \$6.25$ (gain of \$1.25)
- I keep: $\$6.75 - \$3.50 = \$3.25$
- Total: $\$6.25 + \$3.25 = \$9.50$

Common approaches in trust games:

I could mirror their proportion-returning about 45% of what I received, which would be roughly \$3.04. Alternatively, I could split the cooperative gains equally: the \$4.50 surplus created through the exchange divided in half means returning \$2.25 to give them an even split of the benefit. Or I could aim for a return that ensures they profit from trusting me, somewhere between \$2.25 and \$3.50....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1174,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.25, it became \$6.75, and you returned \$3.0. Your visible payoff was \$3.75.

Your total visible earnings across all rounds are \$3.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

In the previous round as receiver, I experienced moderate trust (45% sent) and reciprocated fairly. Now I'm facing a new co-player as sender.

Since this is a new pairing and I want to establish cooperative behavior while managing risk, I'll send a moderate-to-good amount. Sending \$3.00 (60% of my endowment) signals trust and willingness to cooperate, while still keeping some safety margin.

This will give my co-player \$9.00 to work with, providing them good opportunity to reciprocate if they're inclined to cooperate.

```
{"send": 3.00}
```

Reasoning:

Let me think about this situation:

In Round 1:

- I was the receiver
- The sender sent \$2.25 (45% of their endowment)
- I received \$6.75 and returned \$3.00
- I kept \$3.75
- The sender ended with $\$2.75 + \$3.00 = \$5.75$

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- I'm playing against a different co-player (since pairings are randomized each round)

The previous sender showed moderate trust by sending 45% of their endowment. I returned about 44% of what I received (\$3.00 out of \$6.75), which gave them a positive return.

Now I need to decide how much to send. Considerations:

1. This is a new opponent, so I don't know their behavior yet
2. In multi-round games with changing partners, establishing a reputation for cooperation can be beneficial
3. The previous sender's moderate trust (45%) seemed reasonable
4. There's communication noise, so amounts might differ slightly

I think I should send a moderate-to-good amount to signal trust and cooperation. Sending around 50-60% seems reasonable:

- \$2.50 (50%) or \$3.00 (60%)

Given that the previous sender sent 45% and I want to be at least as cooperative (and perhaps signal willingness to cooperate), I'll send \$3.00 (60%). This will become \$9.00 for the receiver, giving them good opportunity to reciprocate.

Actually, let me reconsider. The previous interaction went well with a 45% send rate. I could match that or go slightly higher. \$3.00 (60%) shows I'm willing to be cooperative and trust my new partner.

```
Usage: {  
  "input_tokens": 806,  
  "output_tokens": 571,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T09:18:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_7"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_6": "investor",  
      "Agent_7": "trustee"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no history with this receiver.

If I send money, it gets tripled, creating potential mutual gain. However, the receiver controls how much comes back. The communication noise adds uncertainty.

I'll take a moderately cooperative approach to signal trust while managing risk:

****My decision:****

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return a fair portion, we both benefit. This allows me to learn about their reciprocity while not risking everything in this uncertain first round.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 to allocate
2. Whatever I send gets tripled

3. The receiver then decides how much to return
4. There's communication noise - amounts may differ slightly from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- This is round 1, so I have no history with this particular receiver
- There's a note about communication noise which could affect trust building

Standard economic reasoning in trust games:

- If I send \$5, it becomes \$15 for the receiver
- A "fair" split might be for them to return \$10, giving us each \$10
- But the receiver might keep everything, leaving me with \$0
- If I send \$0, I guarantee keeping \$5 but miss potential gains

Since this is:

1. Round 1 with no history
2. A multi-round game where reputation/reciprocity could matter across the full run
3. There's communication noise which might complicate trust building

I should consider a moderate opening move that:

- Shows some trust/willingness to cooperate
- Doesn't risk everything
- Allows me to learn about this receiver's behavior
- Sets a cooperative tone

A common strategy in trust games is to send around 50% on the first round as a signal of moderate trust while protecting against complete loss.

I'll send \$2.50 (or \$2 or \$3 since I need to be practical about amounts).

Actually, looking at the format, I should probably stick to whole dollar amounts for simplicity. Let me send \$3 as a cooperative opening move - it's more than half, showing trust, but not everything.

Wait, let me reconsider. The receiver gets \$9 if I send \$3. If they return half (\$4.50), I'd have $\$5 - \$3 + \$4.50 = \6.50 , and they'd have \$4.50. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 698,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.51 to you. Your visible payoff was \$7.51.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****What happened:****

- The sender sent me \$3.8 out of \$5 (76% of their endowment) - showing significant trust
- I received \$11.4 after the 3x multiplier

****My reasoning:****

- In Round 1, when I sent \$3, the receiver returned \$5.51 (about 61% of what they received)
- This sender has shown high trust by sending 76% of their endowment
- I should reciprocate that trust fairly

I'll return an amount that rewards their cooperation while maintaining fairness:


```
'''json
{"return": 7}
'''
```

This gives the sender a total of \$8.2 (\$1.2 kept + \$7 returned) and leaves me with \$4.4. While I keep less, this reciprocates their strong trust and builds cooperation.

Reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$5.51 to me
- My payoff: $\$5 - \$3 + \$5.51 = \7.51
- Their payoff: $\$9 - \$5.51 = \$3.49$

So the receiver returned about 61% of what they received. That's quite generous.

Round 2 (current):

- I am RECEIVER
- Sender sent me \$3.8
- I received \$11.4 (tripled amount)
- I need to decide how much to return

Now, there's communication noise, so the amounts might be slightly different from what was actually sent/returned. But I should work with what I see.

The sender sent me \$3.8 out of \$5, which is 76% of their endowment. That's quite trusting.

If I apply reciprocity based on what happened in Round 1:

- The receiver returned about 61% of what they received
- 61% of \$11.4 = \$6.95

But let me think about fairness:

- Sender keeps: $\$5 - \$3.8 = \$1.2$
- Sender receives back from me: X
- Sender total: $\$1.2 + X$
- I keep: $\$11.4 - X$

For equal split of total value:

- Total created: $\$5 + \$11.4 = \$16.4$
- Equal split: \$8.2 each
- So sender should get: \$8.2 total, meaning I return: $\$8.2 - \$1.2 = \$7$

For proportional reciprocity (they sent 76% of endowment):

- Maybe I return a similar proportion of what I received?
- 76% of \$11.4 = \$8.66

Given their high level of trust, I should match that cooperation. Returning around 55-60% of what I received feels right-it's reciprocal without being excessive. I'll return \$6.50, which gives the sender a total of \$7.70 and leaves me with \$4.90. This mirrors the cooperative spirit they showed....

```
Usage: {
  "input_tokens": 780,
  "output_tokens": 878,
  "reasoning_tokens": 0
}
```

=====

ROUND 3

=====

Roles: Agent_1=Sender, Agent_2=Sender, Agent_3=Sender, Agent_4=Receiver, Agent_5=Receiver, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

- dyad_1: Agent_1=Sender, Agent_5=Receiver
- dyad_2: Agent_7=Sender, Agent_8=Receiver
- dyad_3: Agent_3=Sender, Agent_6=Receiver
- dyad_4: Agent_2=Sender, Agent_4=Receiver

Parsed game decisions:

```

dyad_1: Agent_1 sent $3.75; received $11.25; Agent_5 returned $6; payoffs Agent_1=$7.25,
Agent_5=$5.25; noise/visible: sent visible=$4.17, received visible=$12.51, returned visible=$6.34
  Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 11.25, "min": 0, "raw":
  6.0}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}
dyad_2: Agent_7 sent $3.5; received $10.5; Agent_8 returned $5; payoffs Agent_7=$6.5,
Agent_8=$5.5; noise/visible: sent visible=$3.38, received visible=$10.14, returned visible=$4.82
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 10.5, "min": 0, "raw":
  5.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
dyad_3: Agent_3 sent $5; received $15; Agent_6 returned $10; payoffs Agent_3=$10, Agent_6=$5;
noise/visible: returned visible=$10.29
  Action validation: {"returned": {"amount": 10.0, "clamped": false, "max": 15.0, "min": 0, "raw":
  10.0}, "sent": {"amount": 5.0, "clamped": false, "max": 5, "min": 0, "raw": 5.0}}
dyad_4: Agent_2 sent $3; received $9; Agent_4 returned $6.5; payoffs Agent_2=$8.5, Agent_4=$2.5;
noise/visible: sent visible=$3.8, received visible=$11.4, returned visible=$6.52
  Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 9.0, "min": 0, "raw":
  6.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
Balances after round: {
  "Agent_1": 17.5,
  "Agent_2": 19.25,
  "Agent_3": 20.0,
  "Agent_4": 13.5,
  "Agent_5": 20.25,
  "Agent_6": 20.0,
  "Agent_7": 16.0,
  "Agent_8": 17.5
}
Visible balances after round: {
  "Agent_1": 20.310000000000002,
  "Agent_2": 19.68,
  "Agent_3": 22.81,
  "Agent_4": 19.15,
  "Agent_5": 19.689999999999998,
  "Agent_6": 17.17,
  "Agent_7": 18.229999999999997,
  "Agent_8": 17.75
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:19:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_5"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_5": "trustee"
    },
    "trustee": "Agent_5"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.77, that is 75.4% of their total endowment. You received \$11.31. How much do you return to the sender? (0-11.31)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.77 (75.4% of their endowment)
- They currently have: \$5 - \$3.77 = \$1.23
- I received: \$11.31
- Total money created: \$1.23 + \$11.31 = \$12.54

The sender took a significant risk and was quite generous. I want to reciprocate that trust while ensuring we both benefit from cooperation.

If I return \$5.50:

- Sender ends with: \$1.23 + \$5.50 = \$6.73
- I end with: \$11.31 - \$5.50 = \$5.81
- Both of us gain compared to the initial \$5, rewarding the sender's trust

This seems like a fair reciprocation that rewards their generosity while still allowing both of us to benefit.

{'return': 5.50}

reasoning:

Let me think about this situation carefully.

I'm the RECEIVER in Round 1.

- The sender started with \$5
- They sent me \$3.77 (which is 75.4% of their \$5 endowment)
- I received \$11.31 (which is $\$3.77 \times 3$)
- I need to decide how much to return (between \$0 and \$11.31)

The sender showed significant trust by sending 75.4% of their endowment. Let me think about fair reciprocity:

If the sender sent \$3.77:

- They kept: $\$5 - \$3.77 = \$1.23$
- I received: \$11.31

For a fair split of the total money created:

- Total money now = $\$1.23$ (sender kept) + $\$11.31$ (I received) = $\$12.54$
- Fair split would be roughly \$6.27 each
- So I should return enough so the sender has ~\$6.27
- Sender currently has \$1.23, so I'd need to return about \$5.04 to get them to \$6.27
- That would leave me with $\$11.31 - \$5.04 = \$6.27$

Alternatively, I could think about matching their generosity:

- They sent 75.4% of their endowment
- If I return 75.4% of what I received: $0.754 \times \$11.31 = \8.53
- That seems quite generous but might be excessive

Another approach - proportional reciprocity:

- They gave me \$3.77 out of their \$5 (75.4%)
- A reasonable return might be to give back around half of what I received, ensuring both benefit
- Half would be \$5.66

Let me think about common strategies in trust games:

1. Return nothing (free-ride)
2. Return what was sent (break even for sender)
3. Return half of what received (both benefit equally from the multiplier)
4. Return enough to equalize final payoffs

Given their generosity, returning \$5-6 seems appropriate. This would give the sender roughly \$6-7 while leaving me with similar amounts, reflecting mutual gain from the exchange without being excessive....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1020,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.77, it became \$11.31, and you returned \$5.5. Your visible payoff was \$5.8100000000000005.

Your total visible earnings across all rounds are \$5.8100000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, I saw cooperation work well when the sender showed significant trust by sending 75.4% of their endowment, and I reciprocated fairly.

I'll adopt a similar cooperative strategy with my new co-player. Sending a substantial amount can maximize joint gains if they reciprocate.

```
{'send': 3.75}
```

reasoning:

I'm now the SENDER in Round 2.

Let me think about this strategically:

1. In the previous round as RECEIVER, I saw someone send \$3.77 (75.4% of their endowment), showing significant trust
2. I returned \$5.50, which gave them \$6.73 total - a good return on their trust

Now I'm the SENDER. What should I consider?

- The person sending \$3.77 showed that cooperation can work well
- I reciprocated fairly, returning enough that they profited significantly
- If my current co-player is similarly cooperative, sending a substantial amount could benefit both of us

Strategy considerations:

- Sending nothing is safe but generates no surplus
- Sending a moderate amount (around \$2-3) tests the waters
- Sending a larger amount (like \$3.77 that I saw) shows trust and can maximize joint gains if reciprocated

Given that:

- I saw cooperation work in the last round
- The game seems to reward trust
- I benefited from someone trusting me and reciprocated fairly
- There may be communication noise (mentioned in instructions), but the pattern of cooperation seemed clear

I think I should send a substantial amount to show trust and potentially establish a cooperative pattern. I'll send around \$3.50-\$4.00.

Let me send \$3.50 (70% of endowment) - this shows significant trust similar to what I received, but slightly less aggressive to be somewhat cautious given this is a new pairing.

Actually, let me reconsider. The previous sender sent 75.4% (\$3.77). I reciprocated well. If I want to encourage cooperation, I should send a similar proportion. Let me send \$3.75 (75% of my endowment) to match that level of trust.

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 494,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$5.66 to you. Your visible payoff was \$6.91.

Your total visible earnings across all rounds are \$12.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my recent experience:

- Round 2: I sent \$3.75 (75%), received \$5.66 back, ended with \$6.91 - a good outcome
- Round 1: As receiver, cooperation was rewarded

The cooperative approach has been working well. I'm playing with a new co-player now, so I'll continue signaling trust with a substantial send.

```
{'send': 3.75}
```

Reasoning:

Let me analyze the previous round:

- I sent \$3.75 (75% of my endowment)
- It became \$11.25
- They returned \$5.66
- My payoff was $\$5 - \$3.75 + \$5.66 = \6.91

This worked out well! The receiver reciprocated reasonably (returning about 50% of what they r

received), and I gained \$1.91 above my initial endowment.

This is a new co-player this round. Based on my successful cooperative strategy so far:

- Round 1 as receiver: sender sent 75.4%, I returned fairly
- Round 2 as sender: I sent 75%, receiver returned fairly

The pattern of cooperation seems to be working. I should continue with a similar high-trust strategy. I'll send around 75% again to signal cooperation and trust.

\$3.75 worked well last time. I could send the same amount or similar.

```
Usage: {  
  "input_tokens": 1074,  
  "output_tokens": 323,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_1

Timestamp: 2026-08-12T09:19:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_5"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_5": "trustee"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.3, that is 46.0% of their total endowment. You received \$6.899999999999995. How much do you return to the sender? (0-6.899999999999995)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$2.3 (46% of their endowment)
- I received \$6.9 after the 3x multiplier
- The sender currently has \$2.7 remaining

I want to:

1. Reward the sender's trust so they benefit from sending
2. Split the gains fairly
3. Establish myself as trustworthy for future rounds

If I return \$3.0:

- Sender ends with: $\$2.7 + \$3.0 = \$5.7$ (a gain of \$0.7)
- I end with: $\$6.9 - \$3.0 = \$3.9$

This seems fair - the sender profits from trusting me, and we both benefit from the multiplier effect.

```
'''json
{"return": 3.0}
'''
```

reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.3 (which is 46% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$6.9 ($2.3 \times 3 = 6.9$)
4. I need to decide how much to return (between \$0 and \$6.9)

The sender's current situation:

- Started with \$5
- Sent \$2.3
- Currently has: $\$5 - \$2.3 = \$2.7$

If I return nothing:

- Sender ends with: \$2.7
- I end with: \$6.9

The sender showed trust by sending nearly half their endowment. A fair approach would be to:

1. Ensure the sender doesn't lose money
2. Split the gains reasonably

Total money in the system now: \$2.7 (sender has) + \$6.9 (I have) = \$9.6
This started from two \$5 endowments = \$10 total, so we have \$9.6 total.

Wait, let me recalculate:

- Sender started with \$5, sent \$2.3, has \$2.7 left
- I received \$6.9 (the \$2.3 multiplied by 3)
- Total in system: \$2.7 + \$6.9 = \$9.6

If we split equally: each person would get \$4.8

- For sender to have \$4.8, they need: \$4.8 - \$2.7 = \$2.1 returned
- For me to have \$4.8, I'd keep: \$6.9 - \$2.1 = \$4.8 ?

But equal split might not be the best strategy. The sender took a risk by sending money. A common approach in trust games is to: reward their trust by ensuring they come out ahead of their initial \$5, while also sharing in the multiplier gains. Let me test a few return amounts: returning \$3.0 gives them \$5.7 total (a \$0.7 gain) while I keep \$3.9; returning \$2.5 gives them \$5.2 (a \$0.2 gain) while I keep \$4.4; returning \$2.1 would give them exactly \$4.8 and me \$4.8.

A return around \$3.0-\$3.3 seems optimal-it meaningfully rewards their trust while still letting me benefit from the multiplier effect. At \$3.0, they gain \$0.7 and I keep \$3.9, which feels like a fair split that demonstrates trustworthiness.

The sender's 46% send rate suggests moderate confidence in me, so I should reciprocate proportionally...

usage:

```
{
  "input_tokens": 539,
  "output_tokens": 1314,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.3, it became \$6.899999999999995, and you returned \$3.0. Your visible payoff was \$3.899999999999995.

Your total visible earnings across all rounds are \$3.899999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm now the sender with a new co-player
- In the previous round, I experienced a sender who sent 46% and I returned fairly
- I want to establish trust with this new partner while being reasonably cautious

Sending \$2.5 (50% of my endowment) seems like a good strategy:

- It shows significant trust
- It's a clear, fair starting point
- If my co-player reciprocates fairly, we both benefit from the 3x multiplier
- It's not overly risky with an unknown partner

```
'''json
{"send": 2.5}
'''
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.3 (46% of their endowment)
- I received \$6.9
- I returned \$3.0
- My payoff was \$3.9

Now I'm playing against a different agent (since it says "current co-player" and agents are randomly paired each round).

I need to decide how much to send. Key considerations:

1. I don't know anything about this new co-player's behavior
2. The sender in Round 1 sent 46% (\$2.3), which was a reasonable trust signal
3. I returned about 43.5% of what I received (\$3/\$6.9), which was fair
4. I want to build trust but also protect myself since I don't know this opponent

Strategies:

- Sending \$0 would be overly defensive and wouldn't build any relationship
- Sending \$5 (100%) would be too risky with an unknown partner
- Sending around \$2-3 seems reasonable as it shows trust without being reckless

Given that the previous sender sent \$2.3 (46%), I could mirror that behavior. This would be consistent and shows I'm willing to trust at a similar level to what I've experienced.

Alternatively, I could send slightly more to show goodwill, or slightly less to be more cautious.

I think sending \$2.5 (50%) is a good middle ground:

- It's a clear half of my endowment, easy to interpret
- It shows significant trust
- If they return fairly (around 40-45% of \$7.5 = \$3-3.4), we both profit
- It's not too risky

Let me calculate the outcomes:

- If I send \$2.5, they receive \$7.5
- I keep \$2.5

...

usage:

```
{
  "input_tokens": 870,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$6.78 to you. Your visible payoff was \$9.280000000000001.

Your total visible earnings across all rounds are \$13.18.

This round, you are the RECEIVER against your current co-player. They sent you \$4.17, so you received \$12.51. How much do you return? (0-12.51)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.17 (83.4% of their endowment)
- I received \$12.51 after the 3x multiplier
- The sender currently has only \$0.83 remaining

This is very high trust that deserves to be rewarded generously.

If I return \$6.0:

- Sender ends with: \$0.83 + \$6.0 = \$6.83 (profit of \$1.83)
- I end with: \$12.51 - \$6.0 = \$6.51 (profit of \$6.51)

This gives us nearly equal final payoffs and rewards the sender's exceptional trust with a solid profit.

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- The sender sent \$4.17 (83.4% of their \$5 endowment) - this is very high trust!
- I received \$12.51 (after 3x multiplier)
- The sender currently has \$5 - \$4.17 = \$0.83 remaining

This sender showed enormous trust by sending 83.4% of their endowment. I need to reward this trust generously.

Let me think about what's fair:

- Total in the system: \$0.83 (sender has) + \$12.51 (I have) = \$13.34
- If we split evenly: \$6.67 each

If I return \$5.84:

- Sender ends with: \$0.83 + \$5.84 = \$6.67
- I end with: \$12.51 - \$5.84 = \$6.67
- Perfect 50/50 split

But given they showed such high trust (83.4%), I might want to be even more generous to:

1. Reward their exceptional trust
2. Encourage continued cooperation
3. Establish myself as highly trustworthy

However, I also saw in round 2 that when I sent \$2.5 (50%), I got back \$6.78, which gave me a payoff of \$9.28. That was very generous from my co-player then (they returned more than 90% of what they received!).

Let me recalculate for this round:

- If I return \$6:
 - Sender: \$0.83 + \$6 = \$6.83
 - Me: \$12.51 - \$6 = \$6.51
 - Almost equal, slight advantage to sender
- If I return \$6.5:
 - Sender: \$0.83 + \$6.5 = \$7.33
 - Me: \$12.51 - \$6.5 = \$6.01
 - Sender gets more, rewarding their trust

Given the extreme trust shown here, I should return enough to give them a meaningful advantage while still being reasonable. Returning around \$6 seems appropriate-it creates nearly equal outcomes with a slight edge to the sender who took the bigger risk....

```
Usage: {  
  "input_tokens": 1154,  
  "output_tokens": 955,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-08-12T09:19:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_8"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_7": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",
```

```

"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no history with this receiver.

If I send money, it gets tripled, creating potential mutual gain. However, the receiver controls how much comes back. The communication noise adds uncertainty.

I'll take a moderately cooperative approach to signal trust while managing risk:

****My decision:****

```

'''json
{"send": 3}
'''

```

This sends \$3, which becomes \$9 for the receiver. If they return a fair portion, we both b

enefit. This allows me to learn about their reciprocity while not risking everything in this uncertain first round.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 to allocate
2. Whatever I send gets tripled
3. The receiver then decides how much to return
4. There's communication noise - amounts may differ slightly from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- This is round 1, so I have no history with this particular receiver
- There's a note about communication noise which could affect trust building

Standard economic reasoning in trust games:

- If I send \$5, it becomes \$15 for the receiver
- A "fair" split might be for them to return \$10, giving us each \$10
- But the receiver might keep everything, leaving me with \$0
- If I send \$0, I guarantee keeping \$5 but miss potential gains

Since this is:

1. Round 1 with no history
2. A multi-round game where reputation/reciprocity could matter across the full run
3. There's communication noise which might complicate trust building

I should consider a moderate opening move that:

- Shows some trust/willingness to cooperate
- Doesn't risk everything
- Allows me to learn about this receiver's behavior
- Sets a cooperative tone

A common strategy in trust games is to send around 50% on the first round as a signal of moderate trust while protecting against complete loss.

I'll send \$2.50 (or \$2 or \$3 since I need to be practical about amounts).

Actually, looking at the format, I should probably stick to whole dollar amounts for simplicity. Let me send \$3 as a cooperative opening move - it's more than half, showing trust, but not everything.

Wait, let me reconsider. The receiver gets \$9 if I send \$3. If they return half (\$4.50), I'd have $\$5 - \$3 + \$4.50 = \6.50 , and they'd have \$4.50. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 698,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.51 to you. Your visible payoff was \$7.51.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$3.8 out of \$5 (76% of their endowment) - showing significant trust

- I received \$11.4 after the 3x multiplier

****My reasoning:****

- In Round 1, when I sent \$3, the receiver returned \$5.51 (about 61% of what they received)
- This sender has shown high trust by sending 76% of their endowment
- I should reciprocate that trust fairly

I'll return an amount that rewards their cooperation while maintaining fairness:

```
'''json
{"return": 7}
'''
```

This gives the sender a total of \$8.2 (\$1.2 kept + \$7 returned) and leaves me with \$4.4. While I keep less, this reciprocates their strong trust and builds cooperation.

reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$5.51 to me
- My payoff: $\$5 - \$3 + \$5.51 = \7.51
- Their payoff: $\$9 - \$5.51 = \$3.49$

So the receiver returned about 61% of what they received. That's quite generous.

Round 2 (current):

- I am RECEIVER
- Sender sent me \$3.8
- I received \$11.4 (tripled amount)
- I need to decide how much to return

Now, there's communication noise, so the amounts might be slightly different from what was actually sent/returned. But I should work with what I see.

The sender sent me \$3.8 out of \$5, which is 76% of their endowment. That's quite trusting.

If I apply reciprocity based on what happened in Round 1:

- The receiver returned about 61% of what they received
- 61% of \$11.4 = \$6.95

But let me think about fairness:

- Sender keeps: $\$5 - \$3.8 = \$1.2$
- Sender receives back from me: X
- Sender total: $\$1.2 + X$

- I keep: $\$11.4 - X$

For equal split of total value:

- Total created: $\$5 + \$11.4 = \$16.4$
- Equal split: \$8.2 each
- So sender should get: \$8.2 total, meaning I return: $\$8.2 - \$1.2 = \$7$

For proportional reciprocity (they sent 76% of endowment):

- Maybe I return a similar proportion of what I received?
- 76% of \$11.4 = \$8.66

Given their high level of trust, I should match that cooperation. Returning around 55-60% of what I received feels right-it's reciprocal without being excessive. I'll return \$6.50, which gives the sender a total of \$7.70 and leaves me with \$4.90. This mirrors the cooperative spirit they showed....

usage:

```
{
  "input_tokens": 780,
  "output_tokens": 878,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.8, it became \$11.399999999999999, and you returned \$7.0. Your visible payoff was \$4.399999999999999. Your total visible earnings across all rounds are \$11.909999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this new round with a new co-player:

****My experience so far:****

- Round 1: Sent \$3, received \$5.51 back - good cooperation
- Round 2: Received \$11.4, returned \$7 - fair reciprocation

Since pairings are randomized, this is a new partner with no shared history. However, the cooperative pattern I've experienced suggests trust can be rewarded.

I'll send a moderate-to-high amount to signal cooperation while managing risk:

```
'''json
{"send": 3.5}
'''
```

This sends \$3.5 (70% of my endowment), which becomes \$10.50 for the receiver. It's slightly more than my Round 1 send, reflecting confidence from positive past experiences while remaining balanced.

Reasoning:

Let me analyze the situation:

****My history:****

- Round 1: I sent \$3, received back \$5.51, payoff = \$7.51
- Round 2: I received \$11.4 (they sent \$3.8), I returned \$7, payoff = \$4.4
- Total earnings: \$11.91

****Analysis:****

- In Round 1, my co-player was generous, returning \$5.51 out of \$9 (61%)
- In Round 2, my co-player sent \$3.8 (76% of endowment), showing high trust, and I reciprocated fairly
- Now I'm facing a new co-player in Round 3

Since this is a multi-agent setting with randomized pairings, this Round 3 co-player is likely a different person than either of my previous partners. I don't have direct history with them.

However, I've seen cooperation pay off:

- When I sent \$3, I got back \$5.51 (net gain of \$2.51)
- When someone sent me \$3.8, I returned fairly

Given the overall cooperative pattern I've observed and participated in, I should continue with a cooperative strategy. I'll send a similar amount to what I did in Round 1, or potentially slightly more given the positive experience.

I'll send \$3 again - it's a balanced approach that:

1. Shows trust and willingness to cooperate
2. Doesn't risk everything
3. Creates opportunity for mutual gain (becomes \$9)
4. Has worked well for me in the past

Actually, given that cooperation has been working well, I could send a bit more. Let me send \$3.5, showing slightly increased trust while still being prudent.

```
Usage: {
  "input_tokens": 1109,
  "output_tokens": 583,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_2

Timestamp: 2026-08-12T09:19:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.

- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1 with no prior information about my co-player, I want to demonstrate trust while being reasonably cautious.

If I send \$3:

- My co-player receives \$9 (3×3)
- This shows meaningful cooperation
- If they return a fair portion (say $1/3$ to $1/2$), we both benefit
- My maximum loss is limited to \$3

I'll go with a moderately trusting approach:

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I have \$5 to start with
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. I keep: \$5 - sent + returned
5. The receiver keeps: received - returned

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There may be communication noise
- I don't know who my co-player is

Strategic thinking:

- If I send \$0, I guarantee keeping \$5, but there's no potential for gain
- If I send everything (\$5), the receiver gets \$15, and if they're cooperative, they might return a fair amount
- A moderate amount (like \$2-3) shows some trust but limits risk

Since this is round 1 and I have no information about my co-player:

- I want to signal cooperation and trust
- But I also want to be somewhat cautious
- Sending around \$3 seems like a reasonable middle ground that:
 - Shows good faith (60% of my endowment)
 - Gives the receiver \$9 to work with
 - If they return fairly (say $1/3$ to $1/2$ of what they receive), I could end up with \$5-6.
- 5
- Limits my maximum loss to \$3

Let me send \$3 as a cooperative but measured opening move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.37 to you. Your visible payoff was \$5.37.

Your total visible earnings across all rounds are \$5.37.

This round, you are the RECEIVER against your current co-player. They sent you \$4.58, so you received \$13.74. How much do you return? (0-13.74)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

My co-player sent me \$4.58 out of their \$5 (91.6% of their endowment) - showing very high trust.

I received \$13.74. To reciprocate this trust appropriately:

- They kept \$0.42
- If I return around \$6.50, they'd end with \$6.92 (a gain of \$1.92)
- I'd keep \$7.24 (also a solid gain)

This represents returning about 47% of what I received, which rewards their trust while we both benefit substantially.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 2 - I'm now the RECEIVER

- My co-player sent me \$4.58
- I received \$13.74 (which is $\$4.58 \times 3$)
- I need to decide how much to return

From Round 1:

- I sent \$3.00, they received \$9.00
- They returned \$3.37 to me
- This is about 37.4% of what they received, which is quite fair/generous

My co-player in this round sent me \$4.58 out of their \$5, which is 91.6% of their endowment - very trusting!

I should consider:

1. They showed high trust by sending almost everything
2. In Round 1, my previous partner returned about 37.4% which seemed fair
3. I want to reciprocate trust and maintain good cooperation

If I return about 40-50% of what I received (\$13.74):

- 40% = \$5.50
- 45% = \$6.18
- 50% = \$6.87

Given they sent almost everything (\$4.58 out of \$5), showing extreme trust, I should be generous in return.

Their final payoff would be:

- They started with \$5
- Sent \$4.58, kept \$0.42
- If I return \$6.00: $\$0.42 + \$6.00 = \$6.42$
- My payoff: $\$13.74 - \$6.00 = \$7.74$

If I return \$6.50:

- Their payoff: $\$0.42 + \$6.50 = \$6.92$
- My payoff: $\$13.74 - \$6.50 = \$7.24$

Given their high trust (sending 91.6% of endowment), I should return a good amount. Returning around \$6.50 seems fair - they end up with \$6.92 (better than their initial \$5), and I keep \$7.24 (also better than my initial \$0).

...

usage:

```
{
  "input_tokens": 755,
  "output_tokens": 653,
  "reasoning_tokens": 0
}
```

}

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.58, it became \$13.74, and you returned \$6.5. Your visible payoff was \$7.24.

Your total visible earnings across all rounds are \$12.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.38, so you received \$10.14. How much do you return? (0-10.14)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust.

I received \$10.14.

They kept: \$1.62

To reciprocate fairly:

- If I return \$5.00, they'd end with \$6.62 (a gain of \$1.62)
- I'd keep \$5.14 (also a small gain)

This represents returning about 49% of what I received, which rewards their trust while we both benefit.

```
'''json
{"return": 5.00}
'''
```

Reasoning:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust.

I received \$10.14.

They kept: \$5 - \$3.38 = \$1.62

To be fair and reciprocate their trust:

- If I return around \$5, they'd end with \$1.62 + \$5 = \$6.62 (a gain of \$1.62)
- I'd keep \$10.14 - \$5 = \$5.14 (also a solid gain)

This represents returning about 49% of what I received, which is fair and reciprocates their trust appropriately while we both benefit.

Let me think about this more carefully. They sent \$3.38, showing moderate-to-good trust (68% of endowment).

In Round 2, my co-player sent me \$4.58 (92% of endowment) and I returned \$6.50 (47% of what I received).

For consistency and fairness here with \$3.38 sent (less trust shown):

- Returning around 47-50% of \$10.14 would be \$4.77-\$5.07
- Let me go with \$5.00

That gives them: \$1.62 + \$5.00 = \$6.62 (gain of \$1.62)

I keep: \$10.14 - \$5.00 = \$5.14 (gain of \$0.14)

This seems fair and reciprocal.

Usage: {

```
"input_tokens": 1036,
"output_tokens": 519,
"reasoning_tokens": 0
```

}

GAME - Agent_3 - Sender - dyad_3

Timestamp: 2026-08-12T09:19:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_6"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round with no prior history, I want to signal cooperation and trust while being somewhat cautious.

In this trust game, sending money creates value (3x multiplier), but requires trusting that my co-player will reciprocate. Since:

- We'll likely play multiple rounds (possibly with different partners)
- Building a cooperative reputation could benefit me when I'm the receiver
- The socially optimal strategy involves sending substantial amounts to maximize total value

I'll send a significant amount to demonstrate trust, but keep a small buffer given the uncertainty.

{"send": 4}

reasoning:

This is the first round of a trust game where I am the SENDER. Let me think through the strategic considerations:

1. ****Game Structure:****
 - I have \$5
 - I can send \$0-\$5 to the receiver
 - Whatever I send gets multiplied by 3x
 - The receiver then decides how much to return
 - My final payoff: \$5 - sent + returned
 - Receiver's payoff: 3xsent - returned
2. ****Strategic Considerations:****
 - If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
 - If I send \$5: Receiver gets \$15, and could return anywhere from \$0-\$15
 - If they return \$0: I get \$0 total (lost \$5)
 - If they return \$5: I get \$5 total (break even)
 - If they return \$7.50: I get \$7.50 (gain \$2.50)
 - If they return \$10: I get \$10 (gain \$5)
3. ****Trust Game Logic:****
 - This is a classic trust game
 - Sending money shows trust
 - The socially optimal outcome is for me to send the maximum (\$5), which creates \$15 total, and then split fairly
 - A fair split might be: I sent \$5, created \$15, so maybe 50/50 of the gain (\$15-\$5=\$10 gain, so \$5 each) meaning they return \$10 and keep \$5
 - Or they could view it as: I took the risk, so 50/50 of total created value: they return \$7.50, we each get \$7.50
4. ****First Round Considerations:****
 - This is round 1, so there's no history to build on
 - However, there will be multiple rounds with potentially different partners
 - Establishing a cooperative reputation early could pay dividends later, especially since I'll alternate between sender and receiver roles
 - The communication noise mentioned means amounts might shift slightly, which could complicate trust signals
 - A moderate to high send amount (\$3-5) demonstrates willingness to cooperate while managing risk, potentially encouraging reciprocal behavior in future rounds

I'm leaning toward sending \$4 as my opening move. This signals meaningful trust-it triples to \$12 for the receiver-while keeping \$1 as a buffer. Even a modest return of \$4-5 puts me ahead, and it establishes the cooperative tone I'd want if positions reverse. Classic trust game research suggests starting with substantial trust and adjusting based on responses works better than cautious opening moves, so I'll go with that approach....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 1038,
  "reasoning_tokens": 0
}
```

}

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.69 to you. Your visible payoff was \$9.69.

Your total visible earnings across all rounds are \$9.69.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

My co-player sent me \$3.11 (62% of their endowment), showing significant trust. I received \$9.33 after the 3x multiplier.

In my previous round, my co-player returned \$8.69 out of \$12 (72%), which was very generous. I want to maintain this cooperative spirit.

A fair approach would be to ensure my co-player profits from their trust while I also benefit from receiving the tripled amount.

Returning approximately \$6.50 would give them a final payoff of about \$8.39 (\$5 - \$3.11 + \$6.50), which rewards their trust handsomely, while I keep \$2.83.

{"return": 6.50}

reasoning:

Let me analyze this situation:

1. Last round I was the SENDER:

- I sent \$4.0
- It became \$12.0
- They returned \$8.69
- My payoff: $\$5 - \$4 + \$8.69 = \9.69
- Their payoff: $\$12 - \$8.69 = \$3.31$

2. This round I'm the RECEIVER:

- They sent me \$3.11
- I received $\$3.11 \times 3 = \9.33
- I need to decide how much to return

My co-player sent me \$3.11 out of their \$5, which is 62.2% of their endowment. This shows significant trust.

In the last round, my co-player returned \$8.69 out of \$12 they received, which is 72.4% of what they received. This was quite generous - they gave me more than I sent them.

I should consider:

- Reciprocity: They showed trust by sending \$3.11
- Building reputation: Being fair will help in future rounds
- The note says there may be communication noise, so amounts might differ slightly

A fair return would be:

- They invested \$3.11, I should at least return that amount so they don't lose
- To split gains fairly: They gave up \$3.11, it became \$9.33, so we created \$6.22 in additional value
- If we split that gain: $\$3.11 + \$6.22/2 = \$3.11 + \$3.11 = \$6.22$

However, I also want to be somewhat generous to build trust, similar to how my previous co-player returned 72.4% of what they received.

Let me think about different approaches:

- Return exactly what they sent: \$3.11 (they break even)
- Return to split gains 50/50: $\sim \$6.22$ (they get $\$5 - \$3.11 + \$6.22 = \8.11)
- Return more generously (similar to 72.4% rate): $\sim \$6.75$ (they get $\$5 - \$3.11 + \$6.75 = \8.64)

Given the trust they showed and the precedent from my last game, I'll return around \$6.50-\$7.00, which gives them a good payoff while keeping a reasonable amount for myself. I'll return \$6.50: their final payoff becomes \$8.39 while I keep \$9.

Wait, let me check the math more carefully. If I return \$6.50, my payoff is $\$9.33 - \$6.50 = \$2.83$. That feels...

usage:

```
{
  "input_tokens": 763,
  "output_tokens": 1019,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$6.5. Your visible payoff was \$2.83.

Your total visible earnings across all rounds are \$12.52.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience so far:

- Round 1: Sent \$4, received \$8.69 back (very cooperative partner)
- Round 2: Received from someone who sent \$3.11, I returned \$6.50 (both of us were cooperative)

The environment has been quite cooperative. Sending more creates more value through the 3x multiplier, and past interactions suggest people reciprocate trust.

To maximize value creation and continue building a cooperative reputation, I'll send my full endowment this round.

```
{"send": 5}
```

Reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$4, received \$8.69 back, payoff \$9.69

Round 2: I was RECEIVER, got \$9.33 (from \$3.11 sent), returned \$6.50, payoff \$2.83

Total: \$12.52

Now Round 3: I'm SENDER again with a new co-player.

My experience so far:

- When I sent \$4, I got back \$8.69, which was very generous (72% return rate on the tripled amount)
- When I received from someone who sent \$3.11, I returned \$6.50 (70% return rate)

Both interactions have been quite cooperative. This suggests the environment is reasonably trustworthy.

As a sender this round, I should consider:

1. Past cooperation suggests I can trust my co-players
2. Sending more creates more value overall (3x multiplier)
3. Building a reputation for cooperation may help when I'm receiver
4. The game mentioned communication noise, so exact amounts may vary

Given the positive experiences, I should send a substantial amount. I'll send \$4 again since that worked well before, or maybe even \$5 to maximize value creation.

Actually, let me think about this more strategically. If I send \$5:

- Co-player receives \$15
- If they return even 50%, I get \$7.50 back, netting me \$2.50 profit
- If they're as generous as my Round 1 partner (~72%), I could get ~\$10.80 back

If I send \$4:

- Co-player receives \$12
- At 72% return, I'd get ~\$8.64 back, total payoff ~\$9.64

Given the cooperative environment and to maximize total value, I'll send \$5 this time.

```
Usage: {
  "input_tokens": 1058,
  "output_tokens": 560,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T09:20:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_6"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.25, that is 45.0% of their total endowment. You received \$6.75. How much do you return to the sender? (0-6.75)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.25 (45% of their endowment)
- This was multiplied to \$6.75 for me
- The sender currently has \$2.75 remaining

I want to reward their trust while maintaining a fair distribution. If I return \$3.00:

- The sender ends up with \$5.75 (a gain of \$0.75)
- I keep \$3.75 (a gain of \$3.75)
- Both of us are better off than if no money had been sent

This seems like a reasonable balance that rewards their moderate trust and encourages future cooperation.

{"return": 3.00}

reasoning:

Let me think through this situation carefully.

I'm the RECEIVER in this round of a trust game.

The sender:

- Started with \$5
- Sent me \$2.25 (which is 45% of their endowment)
- This was multiplied by 3x, so I received \$6.75

Now I need to decide how much to return.

Key considerations:

1. The sender showed moderate trust by sending 45% of their endowment
2. They could have sent nothing, or they could have sent everything
3. If I return nothing, I keep \$6.75 and they keep \$2.75 (\$5 - \$2.25)
4. If I return some amount, we both benefit from the cooperative exchange

Let me think about fair or reasonable return amounts:

- If I return \$3.00: The sender ends up with $\$2.75 + \$3.00 = \$5.75$ (gain of \$0.75)
- I keep: $\$6.75 - \$3.00 = \$3.75$
- Total: $\$5.75 + \$3.75 = \$9.50$
- If I return \$2.25: The sender ends up with $\$2.75 + \$2.25 = \$5.00$ (break even)
- I keep: $\$6.75 - \$2.25 = \$4.50$
- Total: $\$5.00 + \$4.50 = \$9.50$
- If I return \$3.50: The sender ends up with $\$2.75 + \$3.50 = \$6.25$ (gain of \$1.25)
- I keep: $\$6.75 - \$3.50 = \$3.25$
- Total: $\$6.25 + \$3.25 = \$9.50$

Common approaches in trust games:

I could mirror their proportion-returning about 45% of what I received, which would be roughly \$3.04. Alternatively, I could split the cooperative gains equally: the \$4.50 surplus

s created through the exchange divided in half means returning \$2.25 to give them an even split of the benefit. Or I could aim for a return that ensures they profit from trusting me, somewhere between \$2.25 and \$3.50...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1174,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.25, it became \$6.75, and you returned \$3.0. Your visible payoff was \$3.75.

Your total visible earnings across all rounds are \$3.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In the previous round as receiver, I experienced moderate trust (45% sent) and reciprocated fairly. Now I'm facing a new co-player as sender.

Since this is a new pairing and I want to establish cooperative behavior while managing risk, I'll send a moderate-to-good amount. Sending \$3.00 (60% of my endowment) signals trust and willingness to cooperate, while still keeping some safety margin.

This will give my co-player \$9.00 to work with, providing them good opportunity to reciprocate if they're inclined to cooperate.

{"send": 3.00}

reasoning:

Let me think about this situation:

In Round 1:

- I was the receiver
- The sender sent \$2.25 (45% of their endowment)
- I received \$6.75 and returned \$3.00
- I kept \$3.75
- The sender ended with $\$2.75 + \$3.00 = \$5.75$

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- I'm playing against a different co-player (since pairings are randomized each round)

The previous sender showed moderate trust by sending 45% of their endowment. I returned about 44% of what I received (\$3.00 out of \$6.75), which gave them a positive return.

Now I need to decide how much to send. Considerations:

1. This is a new opponent, so I don't know their behavior yet
2. In multi-round games with changing partners, establishing a reputation for cooperation can be beneficial
3. The previous sender's moderate trust (45%) seemed reasonable
4. There's communication noise, so amounts might differ slightly

I think I should send a moderate-to-good amount to signal trust and cooperation. Sending a round 50-60% seems reasonable:

- \$2.50 (50%) or \$3.00 (60%)

Given that the previous sender sent 45% and I want to be at least as cooperative (and perhaps signal willingness to cooperate), I'll send \$3.00 (60%). This will become \$9.00 for the receiver, giving them good opportunity to reciprocate.

Actually, let me reconsider. The previous interaction went well with a 45% send rate. I co

uld match that or go slightly higher. \$3.00 (60%) shows I'm willing to be cooperative and trust my new partner.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 571,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.42 to you. Your visible payoff was \$8.42.

Your total visible earnings across all rounds are \$12.17.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player showed **maximum trust** by sending their entire \$5 endowment (100%)
- This became \$15 for me
- They currently have \$0

This exceptional level of trust deserves strong reciprocation. If I return \$10.00:

- The sender ends with \$10.00 (doubling their money, gain of \$5.00)
- I keep \$5.00 (gain of \$5.00)
- We both gain equally from our cooperation

This rewards their maximum trust appropriately and creates a fair outcome where both parties benefit substantially.

```
{"return": 10.00}
```

Reasoning:

Let me analyze this situation:

Previous round:

- I was the SENDER
- I sent \$3.00 (60% of my endowment)
- My co-player received \$9.00
- They returned \$6.42 (71.3% of what they received)
- I ended with \$8.42

This round:

- I am the RECEIVER
- My current co-player sent \$5.00 (100% of their endowment!)
- I received \$15.00
- They currently have \$0
- I need to decide how much to return (0-15)

This is a very strong show of trust - they sent their entire endowment! This deserves a strong reciprocation.

Let me think about fair outcomes:

- If I return \$7.50, they end with \$7.50 (gain of \$2.50), I keep \$7.50 (gain of \$7.50)
- If I return \$10.00, they end with \$10.00 (gain of \$5.00), I keep \$5.00 (gain of \$5.00) - equal split of total gains
- If I return \$12.00, they end with \$12.00 (gain of \$7.00), I keep \$3.00 (gain of \$3.00)

Given they showed maximum trust by sending everything, I should reciprocate generously. An equal split of total gains seems fair - we'd each gain \$5 from the cooperation. That means returning \$10.00.

Actually, let me reconsider. They sent everything (\$5), which became \$15 for me. If I return \$10, they get \$10 total (doubling their money), and I keep \$5 (gaining \$5). Both of us gain equally from the transaction.

This seems like a fair and generous reciprocation for their maximum trust.

```
Usage: {
  "input_tokens": 1082,
  "output_tokens": 580,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Sender - dyad_4

Timestamp: 2026-08-12T09:20:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_4"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being reasonably cautious since I have no history with my co-player.

The multiplication factor of 3x creates an opportunity for mutual benefit. If I send a moderate amount, we can both potentially end up better off than the initial \$5.

I'll send **\$3**, which gives my co-player \$9 to work with. This shows trust and willingness to cooperate. If they return approximately \$3-4, we'll both be in a good position, and it could establish a positive pattern for future rounds.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit
2. The receiver then decides how much to return
3. There's communication noise mentioned - amounts may appear different than actually sent
4. This is a multi-agent setting with 8 agents total, pairs rotate
5. This is the first round, so I have no history with any player yet

Since this is the first round and I have no information about other players' behavior, I need to consider:

- Sending \$0 guarantees I keep \$5, but no potential for mutual gain
- Sending everything (\$5) maximizes the total pot (\$15 for receiver) but is risky
- A moderate amount balances risk and potential reward

In trust games, common strategies include:

- Sending about half as a signal of conditional cooperation
- The tripling effect means even partial cooperation can benefit both parties

Let me consider sending \$3:

- I send \$3, receiver gets \$9
- If receiver returns \$3, I end up with \$5 (same as starting)
- If receiver returns \$4-5, I'm better off
- If receiver returns less than \$3, I'm worse off
- This shows trust while keeping some insurance

Alternatively, sending \$2.5:

- I send \$2.5, receiver gets \$7.5
- If receiver returns \$2.5, I end up with \$5
- If receiver returns \$3+, I'm better off
- Less risk than \$3

Or sending \$2:

- I send \$2, receiver gets \$6
- If receiver returns \$2, I end up with \$5
- Lower risk, but also lower potential reward

Given this is the first round and I want to establish a cooperative pattern, I'll send \$3.

This shows trust and willingness to cooperate while not being completely vulnerable.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.81 to you. Your visible payoff was \$4.8100000000000005.

Your total visible earnings across all rounds are \$4.8100000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$3.95, so you received \$11.850000000000001. How much do you return? (0-11.850000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.95 (79% of their endowment), showing significant trust
- I received \$11.85 total
- They currently have only \$1.05 left

I want to:

1. Reward their trust and cooperation
2. Ensure they end up better than their initial \$5
3. Set a positive tone for potential future interactions

If I return \$5.50:

- They end up with: $\$1.05 + \$5.50 = \$6.55$
- I end up with: $\$11.85 - \$5.50 = \$6.35$

Both of us gain from cooperation, and this reciprocates their high level of trust appropriately.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the SENDER
- I sent \$3.00
- It became \$9.00 for the receiver
- They returned \$2.81
- My payoff: $\$5 - \$3 + \$2.81 = \4.81

Round 2:

- I am now the RECEIVER
- My co-player (a different agent) sent me \$3.95
- I received $\$3.95 \times 3 = \11.85
- I need to decide how much to return

Important considerations:

1. The sender gave me \$3.95 out of their \$5 - that's 79% of their endowment, showing high trust
2. If I return nothing, I keep \$11.85 and they end up with $\$5 - \$3.95 = \$1.05$
3. I should return enough to make cooperation worthwhile for them
4. There's communication noise mentioned, so amounts may differ slightly

What would be fair?

- The sender gave up \$3.95, keeping \$1.05
- I received \$11.85
- If I want them to end up better than their initial \$5, I should return more than \$3.95
- A 50-50 split of total surplus could work: Total = $\$1.05 + \$11.85 = \$12.90$. Half is \$6.45

- If I return \$5.40, sender gets $\$1.05 + \$5.40 = \$6.45$, I get $\$11.85 - \$5.40 = \$6.45$

But let me think about reciprocity:

- In round 1, I sent \$3 (60% of endowment), received \$9, they returned \$2.81 (31% of what they received)
- This round, sender sent \$3.95 (79% of endowment), I received \$11.85
- Using that same 31% return rate would give me \$3.67 to send back

The sender showed significantly more generosity than I did initially, so I should reciprocate with a higher return. Returning \$5 would leave them with \$6.05 total-better than their starting \$5-while I'd keep \$6.85, rewarding their trust and encouraging continued cooperation....

usage:

```
{
  "input_tokens": 793,
  "output_tokens": 977,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.95, it became \$11.850000000000001, and you returned \$5.5. Your visible payoff was \$6.350000000000001.

Your total visible earnings across all rounds are \$11.160000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at Round 3:

I'm paired with a new co-player (pairings are randomized each round), so I have no history with them. My previous experiences showed:

- Round 1: Sent \$3, received \$2.81 back - reasonable cooperation
- Round 2: As receiver, I tried to be generous with a high-trusting sender

For this new pairing, I'll use a similar approach to Round 1 - send a moderate amount that signals cooperation while being reasonably cautious with an unknown partner.

Sending \$3 creates \$9 for them to work with, which gives room for mutual benefit if they reciprocate fairly.

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$3, received back \$2.81, payoff \$4.81

Round 2: I was RECEIVER, got \$11.85 from a \$3.95 send, returned \$5.50, payoff \$6.35

Total earnings: \$11.16

Now Round 3: I'm SENDER again with a new co-player (since pairings are randomized each round).

Key considerations:

- I have no history with this particular co-player
- In Round 1, when I sent \$3, I got back \$2.81 (which is reasonable - about 31% return rate)
- In Round 2, when someone sent me \$3.95, I returned \$5.50 (about 46% return rate)
- The game has communication noise, so exact amounts may differ

Since I'm paired with a new player and have no history with them, I should probably use a similar strategy to Round 1 - establish cooperation but be somewhat cautious.

Given that:

- Sending \$3 worked reasonably well in Round 1
- The multiplication factor of 3x creates good mutual benefit potential
- I want to signal cooperation

I'll send \$3 again. This is a moderate amount that shows willingness to cooperate without being overly risky with a new partner.

```
Usage: {  
  "input_tokens": 1106,  
  "output_tokens": 478,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_4

Timestamp: 2026-08-12T09:20:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_4"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_2": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$4.78, that is 95.6% of their total endowment. You received \$14.34. How much do you return to the sender? (0-14.34)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.78 (95.6% of their endowment)
- They kept only \$0.22 for themselves
- I received \$14.34 after the 3x multiplier
- A fair return should reward their trust

If I return \$8.00:

- Sender ends up with: $\$0.22 + \$8.00 = \$8.22$ (well above their initial \$5)
- I keep: $\$14.34 - \$8.00 = \$6.34$ (also above the initial \$5)
- Both of us benefit from the cooperation

This seems like a fair reciprocal response to their high level of trust.

{'return': 8.00}

reasoning:

Let me understand this situation:

1. The sender had \$5 endowment
2. They sent me \$4.78 (which is 95.6% of their \$5)
3. This was multiplied by 3x, so I received $\$4.78 \times 3 = \14.34

Now I need to decide how much to return.

Key considerations:

- The sender showed a lot of trust by sending 95.6% of their endowment
- They kept only $\$5 - \$4.78 = \$0.22$ for themselves
- If I return nothing, the sender ends up with just \$0.22
- If I return generously, both of us can benefit from this cooperation

What's fair/reciprocal?

- The sender trusted me significantly
- A common fairness norm would be to split the gains
- Total created value: \$14.34
- Original sender investment feel: \$4.78
- One approach: Return enough so we both end up roughly equal
- Another approach: Return enough so sender gets significantly more than their original \$5 (rewarding trust)

Calculations:

- If I return \$7.17 (about half of what I received):
 - Sender gets: $\$0.22 + \$7.17 = \$7.39$
 - I get: $\$14.34 - \$7.17 = \$7.17$
- If I return \$8:
 - Sender gets: $\$0.22 + \$8 = \$8.22$
 - I get: $\$14.34 - \$8 = \$6.34$
- If I return \$9:
 - Sender gets: $\$0.22 + \$9 = \$9.22$
 - I get: $\$14.34 - \$9 = \$5.34$

Given that the sender showed exceptional trust (sending 95.6%), I should reward this trust generously. A return that gives them significantly more than their original endowment while still keeping a reasonable amount seems appropriate. Returning \$8 strikes a good balance-it rewards their exceptional trust while ensuring both of us benefit from the cooperation. They'd end up with \$8.22 versus their original \$5, and I'd keep \$6.34. But given how much they risked, maybe I should push toward \$8.50 to give them even more of the gains we created together....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 866,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.78, it became \$14.34, and you returned \$8.00. Your visible payoff was \$6.34.

Your total visible earnings across all rounds are \$6.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, my co-player showed very high trust by sending 95.6% of their endowment, and I reciprocated with a fair return. That cooperative approach worked well for both of us.

Now as the sender with a new co-player, I want to maintain that cooperative spirit. Sending a substantial amount:

- Creates more total value through the 3x multiplier
- Signals trust and willingness to cooperate
- Could encourage reciprocation

I'll send \$4.50 (90% of my endowment), showing strong trust while being slightly more conservative than what I experienced last round.

```
{'send': 4.50}
```

reasoning:

Let me think about this situation:

Round 1 (just completed):

- I was the RECEIVER
- Sender sent me \$4.78 (95.6% of their endowment) - showing very high trust
- I received \$14.34
- I returned \$8.00
- My payoff: \$6.34
- Sender's payoff would be: $\$5 - \$4.78 + \$8.00 = \8.22

Now in Round 2:

- I'm the SENDER
- I have \$5
- I'm playing against a different co-player (pairings are randomized each round)

The previous sender showed very high trust by sending 95.6%. I reciprocated fairly by returning \$8.00, which gave them a good outcome.

Now I need to decide how much to send. Key considerations:

1. This is a new co-player, so I don't have direct history with them
2. The previous round showed successful cooperation
3. Sending more creates more total value (3x multiplier), but requires trust
4. In a multi-round setting, establishing cooperative behavior can be beneficial

Given that:

- The previous sender sent 95.6% (\$4.78)
- That worked out well for both of us

- I should reciprocate the level of trust shown to me

I think I should send a similar high amount to signal cooperation and trust. Sending around \$4.50-\$4.80 would show similar trust to what I experienced.

Let me send \$4.50 (90% of my endowment). This shows strong trust while being slightly more conservative than the 95.6% I received.

Actually, thinking about it more - since I was treated with high trust and it worked out well, I should pay it forward. Sending \$4.50-\$4.75 seems appropriate.

...

usage:

```
{
  "input_tokens": 824,
  "output_tokens": 591,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.5, it became \$13.5, and you saw them return \$7.41 to you. Your visible payoff was \$7.91.

Your total visible earnings across all rounds are \$14.25.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender showed good trust by sending \$3.80 (76% of their endowment)
- They kept \$1.20 for themselves
- I received \$11.40
- I should reciprocate their trust fairly

Comparing to previous rounds:

- In Round 1, when a sender sent 95.6%, I returned about 56% of what I received
- In Round 2, when I sent 90%, my co-player returned about 55% of what they received

For this sender who sent 76%, a fair return would be around 55-60% of what I received.

$\$11.40 \times 0.57 \approx \6.50

If I return \$6.50:

- Sender ends with: $\$1.20 + \$6.50 = \$7.70$ (good gain)
- I keep: $\$11.40 - \$6.50 = \$4.90$ (reasonable)
- Both benefit from the cooperation

{'return': 6.50}

Reasoning:

Let me analyze this situation:

Round 2 recap:

- I was the sender and sent \$4.50
- It became \$13.50 for the receiver
- They returned \$7.41 to me
- My payoff: $\$5 - \$4.50 + \$7.41 = \7.91

Round 3 current situation:

- I'm the receiver
- Sender sent \$3.80 (76% of their endowment)
- I received \$11.40 (approximately, due to floating point)
- They kept \$1.20

This sender showed good trust by sending 76% of their endowment. Let me think about a fair return:

If I return \$6.00:

- Sender ends up with: $\$1.20 + \$6.00 = \$7.20$
- I keep: $\$11.40 - \$6.00 = \$5.40$
- Both benefit, sender gains more

If I return \$7.00:

- Sender ends up with: $\$1.20 + \$7.00 = \$8.20$
- I keep: $\$11.40 - \$7.00 = \$4.40$
- Sender gains significantly more

In Round 1, when I received \$14.34 from a sender who sent \$4.78 (95.6%), I returned \$8.00.

- That sender ended with \$8.22 (from \$0.22)
- I kept \$6.34
- Return ratio: $\$8.00/\$14.34 = 55.9\%$

In Round 2, when I sent \$4.50 and it became \$13.50, they returned \$7.41

- Return ratio: $\$7.41/\$13.50 = 54.9\%$

For consistency, if I use a similar return ratio (around 55%):

- $\$11.40 \times 0.55 = \6.27

Let me round to \$6.50 to be slightly more generous:

...

```
Usage: {
  "input_tokens": 1120,
  "output_tokens": 820,
  "reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1=Receiver, Agent_2=Receiver, Agent_3=Receiver, Agent_4=Sender, Agent_5=Sender, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_4=Sender, Agent_7=Receiver
 dyad_2: Agent_6=Sender, Agent_1=Receiver
 dyad_3: Agent_8=Sender, Agent_3=Receiver
 dyad_4: Agent_5=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_4 sent \$4; received \$12; Agent_7 returned \$5.5; payoffs Agent_4=\$6.5, Agent_7=\$6.5;
 noise/visible: sent visible=\$3.11, received visible=\$9.33, returned visible=\$4.58
 Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 12.0, "min": 0, "raw": 5.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
 dyad_2: Agent_6 sent \$4; received \$12; Agent_1 returned \$6.5; payoffs Agent_6=\$7.5, Agent_1=\$5.5;
 noise/visible: sent visible=\$4.09, received visible=\$12.27, returned visible=\$6.64
 Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 12.0, "min": 0, "raw": 6.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
 dyad_3: Agent_8 sent \$3.5; received \$10.5; Agent_3 returned \$7.5; payoffs Agent_8=\$9, Agent_3=\$3;
 noise/visible: sent visible=\$3.57, received visible=\$10.71, returned visible=\$7.22
 Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 10.5, "min": 0, "raw": 7.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
 dyad_4: Agent_5 sent \$3; received \$9; Agent_2 returned \$5; payoffs Agent_5=\$7, Agent_2=\$4;
 noise/visible: sent visible=\$3.36, received visible=\$10.08, returned visible=\$5.22
 Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 23.0,
 "Agent_2": 23.25,
 "Agent_3": 23.0,
 "Agent_4": 20.0,
 "Agent_5": 27.25,
 "Agent_6": 27.5,
 "Agent_7": 22.5,
 "Agent_8": 26.5

}

Visible balances after round: {

"Agent_1": 26.080000000000002,
 "Agent_2": 24.759999999999998,
 "Agent_3": 26.019999999999996,
 "Agent_4": 24.729999999999997,

```

"Agent_5": 26.909999999999997,
"Agent_6": 24.810000000000002,
"Agent_7": 22.059999999999995,
"Agent_8": 26.47
}

```

LLM CALLS

GAME - Agent_4 - Sender - dyad_1

Timestamp: 2026-08-12T09:20:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_7"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$4.78, that is 95.6% of their total endowment. You received \$14.34. How much do you return to the sender? (0-14.34)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.78 (95.6% of their endowment)
- They kept only \$0.22 for themselves
- I received \$14.34 after the 3x multiplier
- A fair return should reward their trust

If I return \$8.00:

- Sender ends up with: $\$0.22 + \$8.00 = \$8.22$ (well above their initial \$5)
- I keep: $\$14.34 - \$8.00 = \$6.34$ (also above the initial \$5)
- Both of us benefit from the cooperation

This seems like a fair reciprocal response to their high level of trust.

{'return': 8.00}

reasoning:

Let me understand this situation:

1. The sender had \$5 endowment
2. They sent me \$4.78 (which is 95.6% of their \$5)
3. This was multiplied by 3x, so I received $\$4.78 \times 3 = \14.34

Now I need to decide how much to return.

Key considerations:

- The sender showed a lot of trust by sending 95.6% of their endowment
- They kept only $\$5 - \$4.78 = \$0.22$ for themselves
- If I return nothing, the sender ends up with just \$0.22
- If I return generously, both of us can benefit from this cooperation

What's fair/reciprocal?

- The sender trusted me significantly
- A common fairness norm would be to split the gains
- Total created value: \$14.34
- Original sender investment feel: \$4.78
- One approach: Return enough so we both end up roughly equal
- Another approach: Return enough so sender gets significantly more than their original \$5 (rewarding trust)

Calculations:

- If I return \$7.17 (about half of what I received):
 - Sender gets: $\$0.22 + \$7.17 = \$7.39$
 - I get: $\$14.34 - \$7.17 = \$7.17$
- If I return \$8:
 - Sender gets: $\$0.22 + \$8 = \$8.22$
 - I get: $\$14.34 - \$8 = \$6.34$
- If I return \$9:
 - Sender gets: $\$0.22 + \$9 = \$9.22$
 - I get: $\$14.34 - \$9 = \$5.34$

Given that the sender showed exceptional trust (sending 95.6%), I should reward this trust generously. A return that gives them significantly more than their original endowment while still keeping a reasonable amount seems appropriate. Returning \$8 strikes a good balance-it rewards their exceptional trust while ensuring both of us benefit from the cooperation. They'd end up with \$8.22 versus their original \$5, and I'd keep \$6.34. But given how much they risked, maybe I should push toward \$8.50 to give them even more of the gains we created together....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 866,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.78, it became \$14.34, and you returned \$8.00. Your visible payoff was \$6.34.

Your total visible earnings across all rounds are \$6.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, my co-player showed very high trust by sending 95.6% of their endowment, and I reciprocated with a fair return. That cooperative approach worked well for both of us.

Now as the sender with a new co-player, I want to maintain that cooperative spirit. Sending a substantial amount:

- Creates more total value through the 3x multiplier
- Signals trust and willingness to cooperate
- Could encourage reciprocation

I'll send \$4.50 (90% of my endowment), showing strong trust while being slightly more conservative than what I experienced last round.

```
{'send': 4.50}
```

reasoning:

Let me think about this situation:

Round 1 (just completed):

- I was the RECEIVER
- Sender sent me \$4.78 (95.6% of their endowment) - showing very high trust
- I received \$14.34
- I returned \$8.00
- My payoff: \$6.34
- Sender's payoff would be: $\$5 - \$4.78 + \$8.00 = \8.22

Now in Round 2:

- I'm the SENDER
- I have \$5
- I'm playing against a different co-player (pairings are randomized each round)

The previous sender showed very high trust by sending 95.6%. I reciprocated fairly by returning \$8.00, which gave them a good outcome.

Now I need to decide how much to send. Key considerations:

1. This is a new co-player, so I don't have direct history with them
2. The previous round showed successful cooperation
3. Sending more creates more total value (3x multiplier), but requires trust
4. In a multi-round setting, establishing cooperative behavior can be beneficial

Given that:

- The previous sender sent 95.6% (\$4.78)
- That worked out well for both of us

- I should reciprocate the level of trust shown to me

I think I should send a similar high amount to signal cooperation and trust. Sending around \$4.50-\$4.80 would show similar trust to what I experienced.

Let me send \$4.50 (90% of my endowment). This shows strong trust while being slightly more conservative than the 95.6% I received.

Actually, thinking about it more - since I was treated with high trust and it worked out well, I should pay it forward. Sending \$4.50-\$4.75 seems appropriate.

...

usage:

```
{
  "input_tokens": 824,
  "output_tokens": 591,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.5, it became \$13.5, and you saw them return \$7.41 to you. Your visible payoff was \$7.91.

Your total visible earnings across all rounds are \$14.25.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed good trust by sending \$3.80 (76% of their endowment)
- They kept \$1.20 for themselves
- I received \$11.40
- I should reciprocate their trust fairly

Comparing to previous rounds:

- In Round 1, when a sender sent 95.6%, I returned about 56% of what I received
- In Round 2, when I sent 90%, my co-player returned about 55% of what they received

For this sender who sent 76%, a fair return would be around 55-60% of what I received.

$\$11.40 \times 0.57 \approx \6.50

If I return \$6.50:

- Sender ends with: $\$1.20 + \$6.50 = \$7.70$ (good gain)
- I keep: $\$11.40 - \$6.50 = \$4.90$ (reasonable)
- Both benefit from the cooperation

{'return': 6.50}

reasoning:

Let me analyze this situation:

Round 2 recap:

- I was the sender and sent \$4.50
- It became \$13.50 for the receiver
- They returned \$7.41 to me
- My payoff: $\$5 - \$4.50 + \$7.41 = \7.91

Round 3 current situation:

- I'm the receiver
- Sender sent \$3.80 (76% of their endowment)
- I received \$11.40 (approximately, due to floating point)
- They kept \$1.20

This sender showed good trust by sending 76% of their endowment. Let me think about a fair return:

If I return \$6.00:

- Sender ends up with: $\$1.20 + \$6.00 = \$7.20$
- I keep: $\$11.40 - \$6.00 = \$5.40$
- Both benefit, sender gains more

If I return \$7.00:

- Sender ends up with: $\$1.20 + \$7.00 = \$8.20$
- I keep: $\$11.40 - \$7.00 = \$4.40$
- Sender gains significantly more

In Round 1, when I received \$14.34 from a sender who sent \$4.78 (95.6%), I returned \$8.00.

- That sender ended with \$8.22 (from \$0.22)
- I kept \$6.34
- Return ratio: $\$8.00/\$14.34 = 55.9\%$

In Round 2, when I sent \$4.50 and it became \$13.50, they returned \$7.41

- Return ratio: $\$7.41/\$13.50 = 54.9\%$

For consistency, if I use a similar return ratio (around 55%):

- $\$11.40 \times 0.55 = \6.27

Let me round to \$6.50 to be slightly more generous:

...

usage:

```
{
  "input_tokens": 1120,
  "output_tokens": 820,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.8, it became \$11.399999999999999, and you returned \$6.5. Your visible payoff was \$4.899999999999999.

Your total visible earnings across all rounds are \$19.15.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience so far:

- Round 1: As receiver, someone sent high trust (95.6%), I reciprocated fairly
- Round 2: I sent \$4.50 (90%), got back \$7.41 - good cooperation
- Round 3: As receiver, returned fairly when sent \$3.80

The pattern shows that high trust tends to be reciprocated with fair returns (around 55% return rate). My cooperative strategy has been working well - averaging over \$6 per round.

I'll continue with a strong but slightly varied approach. Sending \$4.00 (80%):

- Shows substantial trust
- Creates \$12.00 for the receiver through the 3x multiplier
- Should encourage reciprocation based on the established pattern
- Slightly more conservative than my last sending round to manage risk

```
{'send': 4.00}
```

Reasoning:

Let me analyze my history:

Round 1 (RECEIVER): Sender sent \$4.78 (95.6%), I received \$14.34, returned \$8.00, kept \$6.34

Round 2 (SENDER): I sent \$4.50 (90%), became \$13.50, received back \$7.41, ended with \$7.91

Round 3 (RECEIVER): Sender sent \$3.80 (76%), I received \$11.40, returned \$6.50, kept \$4.90

Now Round 4, I'm the SENDER again with a new co-player.

Looking at the pattern:

- When I sent \$4.50 (90%), I got back \$7.41 from \$13.50 (about 55% return rate)
- When someone sent \$4.78 (95.6%), I returned \$8.00 from \$14.34 (about 56% return rate)

- When someone sent \$3.80 (76%), I returned \$6.50 from \$11.40 (about 57% return rate)

The return rates seem fairly consistent around 55-57%. This suggests cooperation is being reciprocated.

I've been doing well with high cooperation levels. My total earnings are \$19.15 across 3 rounds, averaging about \$6.38 per round, which is well above the baseline of \$5.

Given this positive experience with cooperation, I should continue sending a substantial amount. I'll send \$4.00 (80%), which is:

- Still showing significant trust
- Creating good value through the 3x multiplier (\$12.00 received)
- Likely to be reciprocated based on the pattern
- Slightly more conservative than my previous \$4.50 to diversify my strategy a bit

If they return around 55%, I'd get back about \$6.60, ending with \$1.00 + \$6.60 = \$7.60, which would be good.

```
Usage: {  
  "input_tokens": 1488,  
  "output_tokens": 656,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Receiver - dyad_1

Timestamp: 2026-08-12T09:20:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_7"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_4": "investor",  
      "Agent_7": "trustee"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no history with this receiver.

If I send money, it gets tripled, creating potential mutual gain. However, the receiver controls how much comes back. The communication noise adds uncertainty.

I'll take a moderately cooperative approach to signal trust while managing risk:

****My decision:****

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return a fair portion, we both benefit. This allows me to learn about their reciprocity while not risking everything in this uncertain first round.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 to allocate
2. Whatever I send gets tripled
3. The receiver then decides how much to return
4. There's communication noise - amounts may differ slightly from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- This is round 1, so I have no history with this particular receiver
- There's a note about communication noise which could affect trust building

Standard economic reasoning in trust games:

- If I send \$5, it becomes \$15 for the receiver
- A "fair" split might be for them to return \$10, giving us each \$10
- But the receiver might keep everything, leaving me with \$0
- If I send \$0, I guarantee keeping \$5 but miss potential gains

Since this is:

1. Round 1 with no history
2. A multi-round game where reputation/reciprocity could matter across the full run
3. There's communication noise which might complicate trust building

I should consider a moderate opening move that:

- Shows some trust/willingness to cooperate
- Doesn't risk everything
- Allows me to learn about this receiver's behavior
- Sets a cooperative tone

A common strategy in trust games is to send around 50% on the first round as a signal of moderate trust while protecting against complete loss.

I'll send \$2.50 (or \$2 or \$3 since I need to be practical about amounts).

Actually, looking at the format, I should probably stick to whole dollar amounts for simplicity. Let me send \$3 as a cooperative opening move - it's more than half, showing trust, but not everything.

Wait, let me reconsider. The receiver gets \$9 if I send \$3. If they return half (\$4.50), I'd have $\$5 - \$3 + \$4.50 = \6.50 , and they'd have \$4.50. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 698,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.51 to you. Your visible payoff was \$7.51.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$3.8 out of \$5 (76% of their endowment) - showing significant trust
- I received \$11.4 after the 3x multiplier

****My reasoning:****

- In Round 1, when I sent \$3, the receiver returned \$5.51 (about 61% of what they received)
- This sender has shown high trust by sending 76% of their endowment
- I should reciprocate that trust fairly

I'll return an amount that rewards their cooperation while maintaining fairness:

```
'''json
{"return": 7}
'''
```

This gives the sender a total of \$8.2 (\$1.2 kept + \$7 returned) and leaves me with \$4.4. While I keep less, this reciprocates their strong trust and builds cooperation.

reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$5.51 to me
- My payoff: $\$5 - \$3 + \$5.51 = \7.51

- Their payoff: $\$9 - \$5.51 = \$3.49$

So the receiver returned about 61% of what they received. That's quite generous.

Round 2 (current):

- I am RECEIVER
- Sender sent me \$3.8
- I received \$11.4 (tripled amount)
- I need to decide how much to return

Now, there's communication noise, so the amounts might be slightly different from what was actually sent/returned. But I should work with what I see.

The sender sent me \$3.8 out of \$5, which is 76% of their endowment. That's quite trusting.

If I apply reciprocity based on what happened in Round 1:

- The receiver returned about 61% of what they received
- 61% of \$11.4 = \$6.95

But let me think about fairness:

- Sender keeps: $\$5 - \$3.8 = \$1.2$
- Sender receives back from me: X
- Sender total: $\$1.2 + X$

- I keep: $\$11.4 - X$

For equal split of total value:

- Total created: $\$5 + \$11.4 = \$16.4$
- Equal split: \$8.2 each
- So sender should get: \$8.2 total, meaning I return: $\$8.2 - \$1.2 = \$7$

For proportional reciprocity (they sent 76% of endowment):

- Maybe I return a similar proportion of what I received?
- 76% of \$11.4 = \$8.66

Given their high level of trust, I should match that cooperation. Returning around 55-60% of what I received feels right-it's reciprocal without being excessive. I'll return \$6.50, which gives the sender a total of \$7.70 and leaves me with \$4.90. This mirrors the cooperative spirit they showed....

usage:

```
{
  "input_tokens": 780,
  "output_tokens": 878,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.8, it became \$11.399999999999999, and you returned \$7.0. Your visible payoff was \$4.399999999999999.

Your total visible earnings across all rounds are \$11.909999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this new round with a new co-player:

****My experience so far:****

- Round 1: Sent \$3, received \$5.51 back - good cooperation
- Round 2: Received \$11.4, returned \$7 - fair reciprocation

Since pairings are randomized, this is a new partner with no shared history. However, the cooperative pattern I've experienced suggests trust can be rewarded.

I'll send a moderate-to-high amount to signal cooperation while managing risk:

```
'''json
{"send": 3.5}
'''
```

This sends \$3.5 (70% of my endowment), which becomes \$10.50 for the receiver. It's slightly more than my Round 1 send, reflecting confidence from positive past experiences while remaining balanced.

reasoning:

Let me analyze the situation:

****My history:****

- Round 1: I sent \$3, received back \$5.51, payoff = \$7.51
- Round 2: I received \$11.4 (they sent \$3.8), I returned \$7, payoff = \$4.4
- Total earnings: \$11.91

****Analysis:****

- In Round 1, my co-player was generous, returning \$5.51 out of \$9 (61%)
- In Round 2, my co-player sent \$3.8 (76% of endowment), showing high trust, and I reciprocated fairly
- Now I'm facing a new co-player in Round 3

Since this is a multi-agent setting with randomized pairings, this Round 3 co-player is likely a different person than either of my previous partners. I don't have direct history with them.

However, I've seen cooperation pay off:

- When I sent \$3, I got back \$5.51 (net gain of \$2.51)
- When someone sent me \$3.8, I returned fairly

Given the overall cooperative pattern I've observed and participated in, I should continue with a cooperative strategy. I'll send a similar amount to what I did in Round 1, or potentially slightly more given the positive experience.

I'll send \$3 again - it's a balanced approach that:

1. Shows trust and willingness to cooperate
2. Doesn't risk everything
3. Creates opportunity for mutual gain (becomes \$9)
4. Has worked well for me in the past

Actually, given that cooperation has been working well, I could send a bit more. Let me send \$3.5, showing slightly increased trust while still being prudent.

usage:

```
{
  "input_tokens": 1109,
  "output_tokens": 583,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.82 to you. Your visible payoff was \$6.32.

Your total visible earnings across all rounds are \$18.229999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****What happened:****

- The sender sent me \$3.11 out of \$5 (62.2% of their endowment) - showing moderate trust
- I received \$9.33 after the 3x multiplier

****My analysis:****

- In Round 3, when I sent \$3.5, I received back \$4.82 (only 46% of what they received) - less cooperative
- In Round 2, when I received \$11.4, I returned \$7 (61%)

- This sender showed moderate trust, deserving fair reciprocation

I'll return a fair amount consistent with their trust level:

```
'''json
{"return": 5.5}
'''
```

This gives the sender a total of \$7.39 (\$1.89 kept + \$5.50 returned) and leaves me with \$3.83.

It's a balanced return that rewards their trust while being fair to both of us.

Reasoning:

Let me analyze this situation:

****Round 4:****

- I'm the RECEIVER
- They sent \$3.11 out of \$5 (62.2% of their endowment)
- I received \$9.33 after the 3x multiplier

****Past experience:****

- Round 1: I sent \$3, got back \$5.51 (they returned 61% of \$9)
- Round 2: I received \$11.4, returned \$7 (61% of what I received)
- Round 3: I sent \$3.5, got back \$4.82 (they returned 46% of \$10.5)

The sender this round sent me \$3.11, showing moderate trust (62.2% of their endowment).

In Round 3, when I sent \$3.5, I only got back \$4.82, which was less than half of what they received. That was somewhat disappointing.

For this round, the sender sent \$3.11. I should be fair but also consider that:

1. They showed moderate trust (62.2%)
2. My past returns have been around 61% when receiving
3. I want to maintain cooperation

If I return around 55-60% of \$9.33, that would be:

- 55% = \$5.13
- 60% = \$5.60

This would give them:

- At \$5.13: \$1.89 kept + \$5.13 = \$7.02 total
- At \$5.60: \$1.89 kept + \$5.60 = \$7.49 total

And I would keep:

- At \$5.13: \$4.20
- At \$5.60: \$3.73

I'll return around \$5.5 to be fair and reciprocate their moderate trust.

```
Usage: {
  "input_tokens": 1428,
  "output_tokens": 648,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_2

Timestamp: 2026-08-12T09:21:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_1"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_1": "trustee",
      "Agent_6": "investor"
    }
  }
}
```

```

    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.25, that is 45.0% of their total endowment. You received \$6.75. How much do you return to the sender? (0-6.75)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.25 (45% of their endowment)
- This was multiplied to \$6.75 for me
- The sender currently has \$2.75 remaining

I want to reward their trust while maintaining a fair distribution. If I return \$3.00:

- The sender ends up with \$5.75 (a gain of \$0.75)

- I keep \$3.75 (a gain of \$3.75)
- Both of us are better off than if no money had been sent

This seems like a reasonable balance that rewards their moderate trust and encourages future cooperation.

```
{"return": 3.00}
```

reasoning:

Let me think through this situation carefully.

I'm the RECEIVER in this round of a trust game.

The sender:

- Started with \$5
- Sent me \$2.25 (which is 45% of their endowment)
- This was multiplied by 3x, so I received \$6.75

Now I need to decide how much to return.

Key considerations:

1. The sender showed moderate trust by sending 45% of their endowment
2. They could have sent nothing, or they could have sent everything
3. If I return nothing, I keep \$6.75 and they keep \$2.75 (\$5 - \$2.25)
4. If I return some amount, we both benefit from the cooperative exchange

Let me think about fair or reasonable return amounts:

- If I return \$3.00: The sender ends up with \$2.75 + \$3.00 = \$5.75 (gain of \$0.75)
- I keep: \$6.75 - \$3.00 = \$3.75
- Total: \$5.75 + \$3.75 = \$9.50
- If I return \$2.25: The sender ends up with \$2.75 + \$2.25 = \$5.00 (break even)
- I keep: \$6.75 - \$2.25 = \$4.50
- Total: \$5.00 + \$4.50 = \$9.50
- If I return \$3.50: The sender ends up with \$2.75 + \$3.50 = \$6.25 (gain of \$1.25)
- I keep: \$6.75 - \$3.50 = \$3.25
- Total: \$6.25 + \$3.25 = \$9.50

Common approaches in trust games:

I could mirror their proportion-returning about 45% of what I received, which would be roughly \$3.04. Alternatively, I could split the cooperative gains equally: the \$4.50 surplus created through the exchange divided in half means returning \$2.25 to give them an even split of the benefit. Or I could aim for a return that ensures they profit from trusting me, somewhere between \$2.25 and \$3.50...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1174,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.25, it became \$6.75, and you returned \$3.0. Your visible payoff was \$3.75.

Your total visible earnings across all rounds are \$3.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In the previous round as receiver, I experienced moderate trust (45% sent) and reciprocated fairly. Now I'm facing a new co-player as sender.

Since this is a new pairing and I want to establish cooperative behavior while managing risk, I'll send a moderate-to-good amount. Sending \$3.00 (60% of my endowment) signals trust and willingness to cooperate, while still keeping some safety margin.

This will give my co-player \$9.00 to work with, providing them good opportunity to reciprocate if they're inclined to cooperate.

```
{"send": 3.00}
```

reasoning:

Let me think about this situation:

In Round 1:

- I was the receiver
- The sender sent \$2.25 (45% of their endowment)
- I received \$6.75 and returned \$3.00
- I kept \$3.75
- The sender ended with $\$2.75 + \$3.00 = \$5.75$

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- I'm playing against a different co-player (since pairings are randomized each round)

The previous sender showed moderate trust by sending 45% of their endowment. I returned about 44% of what I received (\$3.00 out of \$6.75), which gave them a positive return.

Now I need to decide how much to send. Considerations:

1. This is a new opponent, so I don't know their behavior yet
2. In multi-round games with changing partners, establishing a reputation for cooperation can be beneficial
3. The previous sender's moderate trust (45%) seemed reasonable
4. There's communication noise, so amounts might differ slightly

I think I should send a moderate-to-good amount to signal trust and cooperation. Sending a round 50-60% seems reasonable:

- \$2.50 (50%) or \$3.00 (60%)

Given that the previous sender sent 45% and I want to be at least as cooperative (and perhaps signal willingness to cooperate), I'll send \$3.00 (60%). This will become \$9.00 for the receiver, giving them good opportunity to reciprocate.

Actually, let me reconsider. The previous interaction went well with a 45% send rate. I could match that or go slightly higher. \$3.00 (60%) shows I'm willing to be cooperative and trust my new partner.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 571,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.42 to you. Your visible payoff was \$8.42.

Your total visible earnings across all rounds are \$12.17.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player showed ****maximum trust**** by sending their entire \$5 endowment (100%)
- This became \$15 for me
- They currently have \$0

This exceptional level of trust deserves strong reciprocation. If I return \$10.00:

- The sender ends with \$10.00 (doubling their money, gain of \$5.00)
- I keep \$5.00 (gain of \$5.00)
- We both gain equally from our cooperation

This rewards their maximum trust appropriately and creates a fair outcome where both parties benefit substantially.

```
{"return": 10.00}
```

reasoning:

Let me analyze this situation:

Previous round:

- I was the SENDER
- I sent \$3.00 (60% of my endowment)
- My co-player received \$9.00
- They returned \$6.42 (71.3% of what they received)
- I ended with \$8.42

This round:

- I am the RECEIVER
- My current co-player sent \$5.00 (100% of their endowment!)
- I received \$15.00
- They currently have \$0
- I need to decide how much to return (0-15)

This is a very strong show of trust - they sent their entire endowment! This deserves a strong reciprocation.

Let me think about fair outcomes:

- If I return \$7.50, they end with \$7.50 (gain of \$2.50), I keep \$7.50 (gain of \$7.50)
- If I return \$10.00, they end with \$10.00 (gain of \$5.00), I keep \$5.00 (gain of \$5.00) - equal split of total gains
- If I return \$12.00, they end with \$12.00 (gain of \$7.00), I keep \$3.00 (gain of \$3.00)

Given they showed maximum trust by sending everything, I should reciprocate generously. An equal split of total gains seems fair - we'd each gain \$5 from the cooperation. That means returning \$10.00.

Actually, let me reconsider. They sent everything (\$5), which became \$15 for me. If I return \$10, they get \$10 total (doubling their money), and I keep \$5 (gaining \$5). Both of us gain equally from the transaction.

This seems like a fair and generous reciprocation for their maximum trust.

usage:

```
{
  "input_tokens": 1082,
  "output_tokens": 580,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$5, it became \$15, and you returned \$10.0. Your visible payoff was \$5.0. Your total visible earnings across all rounds are \$17.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience so far:

- Round 2 (sender): Sent \$3.00 (60%) and received excellent reciprocation (\$6.42)
- Round 3 (receiver): Received maximum trust (\$5.00 sent) and reciprocated generously

The pattern so far shows strong cooperation in this game. Since this is a new co-player, I want to signal trust and encourage cooperation. Given the positive experiences, I'll send a high amount while keeping a small buffer.

Sending \$4.00 (80% of my endowment) demonstrates strong trust and gives my co-player \$12.00 to work with, creating good opportunity for mutual benefit.

```
{"send": 4.00}
```

Reasoning:

Let me analyze the situation:

Previous rounds:

1. Round 1: RECEIVER - saw \$2.25 sent, received \$6.75, returned \$3.00, kept \$3.75
2. Round 2: SENDER - sent \$3.00, became \$9.00, saw \$6.42 returned, payoff \$8.42
3. Round 3: RECEIVER - saw \$5.00 sent, received \$15.00, returned \$10.00, kept \$5.00

Total visible earnings: \$17.17

Now Round 4: I'm the SENDER against a new co-player.

Key observations:

- In Round 2, I sent \$3.00 (60%) and got very good reciprocation (\$6.42 return on \$9.00)
- In Round 3, someone sent me everything (\$5.00) and I reciprocated generously (\$10.00 return)

I've seen good cooperation so far. My previous experience as sender (Round 2) with moderate-high trust level worked well.

For this round:

- This is a new co-player (different from previous rounds)
- I don't know their behavior yet
- But the overall pattern in this game seems cooperative
- I've been both generous and received generosity

Given the positive experiences so far, I should maintain a cooperative strategy. I'll send a substantial amount again. Sending \$3.50 (70%) shows strong trust while keeping a small safety margin.

Actually, let me reconsider. The game seems quite cooperative overall. In Round 3, someone trusted me completely with \$5.00, and in Round 2 when I sent \$3.00 I got excellent reciprocation.

I could send \$4.00 (80%) to signal strong cooperation, or even \$5.00 to match the maximum trust I received in Round 3.

Given the high cooperation levels I've experienced, I'll send \$4.00 (80% of endowment) - this is a strong signal of trust without going to the absolute maximum.

...

```
Usage: {  
  "input_tokens": 1343,  
  "output_tokens": 646,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_2

Timestamp: 2026-08-12T09:21:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_1"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_1"
```

```

},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.77, that is 75.4% of their total endowment. You received \$11.31. How much do you return to the sender? (0-11.31)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.77 (75.4% of their endowment)
- They currently have: \$5 - \$3.77 = \$1.23
- I received: \$11.31
- Total money created: \$1.23 + \$11.31 = \$12.54

The sender took a significant risk and was quite generous. I want to reciprocate that trust while ensuring we both benefit from cooperation.

If I return \$5.50:

- Sender ends with: $\$1.23 + \$5.50 = \$6.73$
- I end with: $\$11.31 - \$5.50 = \$5.81$
- Both of us gain compared to the initial \$5, rewarding the sender's trust

This seems like a fair reciprocation that rewards their generosity while still allowing both of us to benefit.

```
{'return': 5.50}
```

reasoning:

Let me think about this situation carefully.

I'm the RECEIVER in Round 1.

- The sender started with \$5
- They sent me \$3.77 (which is 75.4% of their \$5 endowment)
- I received \$11.31 (which is $\$3.77 \times 3$)
- I need to decide how much to return (between \$0 and \$11.31)

The sender showed significant trust by sending 75.4% of their endowment. Let me think about fair reciprocity:

If the sender sent \$3.77:

- They kept: $\$5 - \$3.77 = \$1.23$
- I received: \$11.31

For a fair split of the total money created:

- Total money now = $\$1.23$ (sender kept) + $\$11.31$ (I received) = $\$12.54$
- Fair split would be roughly \$6.27 each
- So I should return enough so the sender has ~\$6.27
- Sender currently has \$1.23, so I'd need to return about \$5.04 to get them to \$6.27
- That would leave me with $\$11.31 - \$5.04 = \$6.27$

Alternatively, I could think about matching their generosity:

- They sent 75.4% of their endowment
- If I return 75.4% of what I received: $0.754 \times \$11.31 = \8.53
- That seems quite generous but might be excessive

Another approach - proportional reciprocity:

- They gave me \$3.77 out of their \$5 (75.4%)
- A reasonable return might be to give back around half of what I received, ensuring both benefit
- Half would be \$5.66

Let me think about common strategies in trust games:

1. Return nothing (free-ride)
2. Return what was sent (break even for sender)
3. Return half of what received (both benefit equally from the multiplier)
4. Return enough to equalize final payoffs

Given their generosity, returning \$5-6 seems appropriate. This would give the sender roughly \$6-7 while leaving me with similar amounts, reflecting mutual gain from the exchange without being excessive....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1020,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.77, it became \$11.31, and you returned \$5.5. Your visible payoff was \$5.8100000000000005.

Your total visible earnings across all rounds are \$5.8100000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, I saw cooperation work well when the sender showed significant trust by sending 75.4% of their endowment, and I reciprocated fairly.

I'll adopt a similar cooperative strategy with my new co-player. Sending a substantial amount can maximize joint gains if they reciprocate.

{'send': 3.75}

reasoning:

I'm now the SENDER in Round 2.

Let me think about this strategically:

1. In the previous round as RECEIVER, I saw someone send \$3.77 (75.4% of their endowment), showing significant trust
2. I returned \$5.50, which gave them \$6.73 total - a good return on their trust

Now I'm the SENDER. What should I consider?

- The person sending \$3.77 showed that cooperation can work well
- I reciprocated fairly, returning enough that they profited significantly
- If my current co-player is similarly cooperative, sending a substantial amount could benefit both of us

Strategy considerations:

- Sending nothing is safe but generates no surplus
- Sending a moderate amount (around \$2-3) tests the waters
- Sending a larger amount (like \$3.77 that I saw) shows trust and can maximize joint gains if reciprocated

Given that:

- I saw cooperation work in the last round
- The game seems to reward trust
- I benefited from someone trusting me and reciprocated fairly
- There may be communication noise (mentioned in instructions), but the pattern of cooperation seemed clear

I think I should send a substantial amount to show trust and potentially establish a cooperative pattern. I'll send around \$3.50-\$4.00.

Let me send \$3.50 (70% of endowment) - this shows significant trust similar to what I received, but slightly less aggressive to be somewhat cautious given this is a new pairing.

Actually, let me reconsider. The previous sender sent 75.4% (\$3.77). I reciprocated well. If I want to encourage cooperation, I should send a similar proportion. Let me send \$3.75 (75% of my endowment) to match that level of trust.

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 494,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$5.66 to you. Your visible payoff was \$6.91.

Your total visible earnings across all rounds are \$12.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience:

- Round 2: I sent \$3.75 (75%), received \$5.66 back, ended with \$6.91 - a good outcome
- Round 1: As receiver, cooperation was rewarded

The cooperative approach has been working well. I'm playing with a new co-player now, so I'll continue signaling trust with a substantial send.

```
{'send': 3.75}
```

reasoning:

Let me analyze the previous round:

- I sent \$3.75 (75% of my endowment)
- It became \$11.25
- They returned \$5.66
- My payoff was $\$5 - \$3.75 + \$5.66 = \6.91

This worked out well! The receiver reciprocated reasonably (returning about 50% of what they received), and I gained \$1.91 above my initial endowment.

This is a new co-player this round. Based on my successful cooperative strategy so far:

- Round 1 as receiver: sender sent 75.4%, I returned fairly
- Round 2 as sender: I sent 75%, receiver returned fairly

The pattern of cooperation seems to be working. I should continue with a similar high-trust strategy. I'll send around 75% again to signal cooperation and trust.

\$3.75 worked well last time. I could send the same amount or similar.

usage:

```
{
  "input_tokens": 1074,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$6.34 to you. Your visible payoff was \$7.59.

Your total visible earnings across all rounds are \$20.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$4.09, so you received \$12.27. How much do you return? (0-12.27)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.09 (81.8% of their endowment)
- They currently have: $\$5 - \$4.09 = \$0.91$
- I received: \$12.27
- Total created: \$13.18

This is even more trust than I saw in Round 1 (when sender sent 75.4%). I want to reward this high level of cooperation appropriately.

If I return \$6.50:

- Sender ends with: $\$0.91 + \$6.50 = \$7.41$ (gain of \$2.41)
- I end with: $\$12.27 - \$6.50 = \$5.77$ (gain of \$0.77)

Both of us benefit, and the sender's exceptional trust is well-rewarded.

```
{'return': 6.50}
```

Reasoning:

Let me analyze this situation:

- The sender sent \$4.09 out of \$5, which is 81.8% of their endowment - very high trust!
- They now have: $\$5 - \$4.09 = \$0.91$
- I received: \$12.27
- Total money in the system: $\$0.91 + \$12.27 = \$13.18$

This sender showed even more trust than my previous partners. I want to reciprocate appropriately.

Looking at my previous behavior:

- Round 1: As receiver, got \$11.31, returned \$5.50 (about 48.6%)
- The sender had sent 75.4%
- That seemed fair and both parties benefited

This sender sent 81.8%, even more generous. I should reciprocate with a good return.

If I return \$6.50:

- Sender ends with: $\$0.91 + \$6.50 = \$7.41$
- I end with: $\$12.27 - \$6.50 = \$5.77$
- Sender gains \$2.41 over initial \$5
- I gain \$0.77 over initial \$5

Both benefit, and the sender's higher risk/trust is rewarded with higher gains.

Actually, let me think about this differently. The sender sent 81.8% which is very high. In Round 1, sender sent 75.4% and I returned 48.6% of what I received.

For 81.8% send rate, maybe I should return around 50-53% to reward the extra trust?

$$\$12.27 * 0.52 = \$6.38$$

If I return \$6.40:

- Sender ends with: $\$0.91 + \$6.40 = \$7.31$
- I end with: $\$12.27 - \$6.40 = \$5.87$

This seems fair - both gain, sender gets rewarded for high trust.

...

```
Usage: {
  "input_tokens": 1310,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_3

Timestamp: 2026-08-12T09:21:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_3"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_3": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1 with no prior information about my co-player, I want to demonstrate trust while being reasonably cautious.

If I send \$3:

- My co-player receives \$9 (3×3)
- This shows meaningful cooperation
- If they return a fair portion (say $1/3$ to $1/2$), we both benefit
- My maximum loss is limited to \$3

I'll go with a moderately trusting approach:

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I have \$5 to start with
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. I keep: \$5 - sent + returned
5. The receiver keeps: received - returned

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There may be communication noise

- I don't know who my co-player is

Strategic thinking:

- If I send \$0, I guarantee keeping \$5, but there's no potential for gain
- If I send everything (\$5), the receiver gets \$15, and if they're cooperative, they might return a fair amount
- A moderate amount (like \$2-3) shows some trust but limits risk

Since this is round 1 and I have no information about my co-player:

- I want to signal cooperation and trust
- But I also want to be somewhat cautious
- Sending around \$3 seems like a reasonable middle ground that:
 - Shows good faith (60% of my endowment)
 - Gives the receiver \$9 to work with
 - If they return fairly (say 1/3 to 1/2 of what they receive), I could end up with \$5-6.
- Limits my maximum loss to \$3

Let me send \$3 as a cooperative but measured opening move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.37 to you. Your visible payoff was \$5.37.

Your total visible earnings across all rounds are \$5.37.

This round, you are the RECEIVER against your current co-player. They sent you \$4.58, so you received \$13.74. How much do you return? (0-13.74)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

My co-player sent me \$4.58 out of their \$5 (91.6% of their endowment) - showing very high trust.

I received \$13.74. To reciprocate this trust appropriately:

- They kept \$0.42
- If I return around \$6.50, they'd end with \$6.92 (a gain of \$1.92)
- I'd keep \$7.24 (also a solid gain)

This represents returning about 47% of what I received, which rewards their trust while we both benefit substantially.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 2 - I'm now the RECEIVER

- My co-player sent me \$4.58
- I received \$13.74 (which is 4.58×3)
- I need to decide how much to return

From Round 1:

- I sent \$3.00, they received \$9.00
- They returned \$3.37 to me
- This is about 37.4% of what they received, which is quite fair/generous

My co-player in this round sent me \$4.58 out of their \$5, which is 91.6% of their endowment - very trusting!

I should consider:

1. They showed high trust by sending almost everything
2. In Round 1, my previous partner returned about 37.4% which seemed fair
3. I want to reciprocate trust and maintain good cooperation

If I return about 40-50% of what I received (\$13.74):

- 40% = \$5.50
- 45% = \$6.18
- 50% = \$6.87

Given they sent almost everything (\$4.58 out of \$5), showing extreme trust, I should be generous in return.

Their final payoff would be:

- They started with \$5
- Sent \$4.58, kept \$0.42
- If I return \$6.00: $\$0.42 + \$6.00 = \$6.42$
- My payoff: $\$13.74 - \$6.00 = \$7.74$

If I return \$6.50:

- Their payoff: $\$0.42 + \$6.50 = \$6.92$
- My payoff: $\$13.74 - \$6.50 = \$7.24$

Given their high trust (sending 91.6% of endowment), I should return a good amount. Returning around \$6.50 seems fair - they end up with \$6.92 (better than their initial \$5), and I keep \$7.24 (also better than my initial \$0).

...

usage:

```
{
  "input_tokens": 755,
  "output_tokens": 653,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.58, it became \$13.74, and you returned \$6.5. Your visible payoff was \$7.24.

Your total visible earnings across all rounds are \$12.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.38, so you received \$10.14. How much do you return? (0-10.14)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust.

I received \$10.14.

They kept: \$1.62

To reciprocate fairly:

- If I return \$5.00, they'd end with \$6.62 (a gain of \$1.62)
- I'd keep \$5.14 (also a small gain)

This represents returning about 49% of what I received, which rewards their trust while we both benefit.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust

I received \$10.14.

They kept: $\$5 - \$3.38 = \$1.62$

To be fair and reciprocate their trust:

- If I return around \$5, they'd end with $\$1.62 + \$5 = \$6.62$ (a gain of \$1.62)
- I'd keep $\$10.14 - \$5 = \$5.14$ (also a solid gain)

This represents returning about 49% of what I received, which is fair and reciprocates the trust appropriately while we both benefit.

Let me think about this more carefully. They sent \$3.38, showing moderate-to-good trust (68% of endowment).

In Round 2, my co-player sent me \$4.58 (92% of endowment) and I returned \$6.50 (47% of what I received).

For consistency and fairness here with \$3.38 sent (less trust shown):

- Returning around 47-50% of \$10.14 would be \$4.77-\$5.07
- Let me go with \$5.00

That gives them: $\$1.62 + \$5.00 = \$6.62$ (gain of \$1.62)

I keep: $\$10.14 - \$5.00 = \$5.14$ (gain of \$0.14)

This seems fair and reciprocal.

usage:

```
{
  "input_tokens": 1036,
  "output_tokens": 519,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.38, it became \$10.14, and you returned \$5.0. Your visible payoff was \$5.140000000000001.

Your total visible earnings across all rounds are \$17.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience so far:

Round 1: As SENDER, I sent \$3 and received \$3.37 back - a reasonable return.

Rounds 2-3: As RECEIVER, others showed good trust, and I reciprocated fairly.

This is a new co-player (random pairing), so I don't have specific information about them. However, the overall pattern suggests players are reasonably cooperative.

I'll send \$3.50 - slightly more than my first round to show increasing trust while remaining measured:

```
'''json
{"send": 3.5}
'''
```

Reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$3, received back \$3.37, payoff \$5.37

Round 2: I was RECEIVER, received \$13.74 (from \$4.58), returned \$6.50, payoff \$7.24

Round 3: I was RECEIVER, received \$10.14 (from \$3.38), returned \$5.00, payoff \$5.14

Round 4: I am SENDER

My total earnings: \$17.75

Looking at the pattern:

- When I sent \$3, I got back \$3.37 (a bit more than 1:1 return on sent amount)
- When others sent to me, they showed high trust (\$4.58 and \$3.38 out of \$5)
- I've been fairly reciprocal as a receiver, returning roughly half of what I received

Now I'm a sender again. Since:

1. My first send of \$3 was met with a decent return (\$3.37)
2. Others have been trusting with me
3. I've been fair in return

I should continue with moderate to high trust. I'll send \$3.5 this time - slightly more than my first send to show I'm willing to build cooperation.

Actually, let me reconsider. The co-player this round is new (pairings are randomized). I don't know their history. In Round 1, I sent \$3 and got a reasonable return. Let me stick with a similar strategy - send \$3 again as a baseline cooperative strategy.

Actually, looking at the overall pattern, players seem reasonably cooperative. I could try \$3.5 to be slightly more trusting.

Let me go with \$3.5 - showing increased trust while not overcommitting.

```
Usage: {  
  "input_tokens": 1303,  
  "output_tokens": 551,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Receiver - dyad_3

Timestamp: 2026-08-12T09:21:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_3"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round with no prior history, I want to signal cooperation and trust while being somewhat cautious.

In this trust game, sending money creates value (3x multiplier), but requires trusting that my co-player will reciprocate. Since:

- We'll likely play multiple rounds (possibly with different partners)
- Building a cooperative reputation could benefit me when I'm the receiver
- The socially optimal strategy involves sending substantial amounts to maximize total value

I'll send a significant amount to demonstrate trust, but keep a small buffer given the uncertainty.

{"send": 4}

reasoning:

This is the first round of a trust game where I am the SENDER. Let me think through the strategic considerations:

1. ****Game Structure:****

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- My final payoff: \$5 - sent + returned
- Receiver's payoff: 3xsent - returned

2. ****Strategic Considerations:****

- If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
- If I send \$5: Receiver gets \$15, and could return anywhere from \$0-\$15
 - If they return \$0: I get \$0 total (lost \$5)
 - If they return \$5: I get \$5 total (break even)
 - If they return \$7.50: I get \$7.50 (gain \$2.50)
 - If they return \$10: I get \$10 (gain \$5)

3. ****Trust Game Logic:****
- This is a classic trust game
 - Sending money shows trust
 - The socially optimal outcome is for me to send the maximum (\$5), which creates \$15 to tal, and then split fairly
 - A fair split might be: I sent \$5, created \$15, so maybe 50/50 of the gain (\$15-\$5=\$10 gain, so \$5 each) meaning they return \$10 and keep \$5
 - Or they could view it as: I took the risk, so 50/50 of total created value: they return \$7.50, we each get \$7.50

4. ****First Round Considerations:****
- This is round 1, so there's no history to build on
 - However, there will be multiple rounds with potentially different partners
 - Establishing a cooperative reputation early could pay dividends later, especially since I'll alternate between sender and receiver roles
 - The communication noise mentioned means amounts might shift slightly, which could complicate trust signals
 - A moderate to high send amount (\$3-5) demonstrates willingness to cooperate while managing risk, potentially encouraging reciprocal behavior in future rounds

I'm leaning toward sending \$4 as my opening move. This signals meaningful trust-it triples to \$12 for the receiver-while keeping \$1 as a buffer. Even a modest return of \$4-5 puts me ahead, and it establishes the cooperative tone I'd want if positions reverse. Classic trust game research suggests starting with substantial trust and adjusting based on responses works better than cautious opening moves, so I'll go with that approach....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 1038,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.69 to you. Your visible payoff was \$9.69.

Your total visible earnings across all rounds are \$9.69.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

My co-player sent me \$3.11 (62% of their endowment), showing significant trust. I received \$9.33 after the 3x multiplier.

In my previous round, my co-player returned \$8.69 out of \$12 (72%), which was very generous. I want to maintain this cooperative spirit.

A fair approach would be to ensure my co-player profits from their trust while I also benefit from receiving the tripled amount.

Returning approximately \$6.50 would give them a final payoff of about \$8.39 (\$5 - \$3.11 + \$6.50), which rewards their trust handsomely, while I keep \$2.83.

```
{"return": 6.50}
```

reasoning:

Let me analyze this situation:

1. Last round I was the SENDER:

- I sent \$4.0
- It became \$12.0
- They returned \$8.69
- My payoff: \$5 - \$4 + \$8.69 = \$9.69
- Their payoff: \$12 - \$8.69 = \$3.31

2. This round I'm the RECEIVER:
- They sent me \$3.11
 - I received $\$3.11 \times 3 = \9.33
 - I need to decide how much to return

My co-player sent me \$3.11 out of their \$5, which is 62.2% of their endowment. This shows significant trust.

In the last round, my co-player returned \$8.69 out of \$12 they received, which is 72.4% of what they received. This was quite generous - they gave me more than I sent them.

I should consider:

- Reciprocity: They showed trust by sending \$3.11
- Building reputation: Being fair will help in future rounds
- The note says there may be communication noise, so amounts might differ slightly

A fair return would be:

- They invested \$3.11, I should at least return that amount so they don't lose
- To split gains fairly: They gave up \$3.11, it became \$9.33, so we created \$6.22 in additional value
- If we split that gain: $\$3.11 + \$6.22/2 = \$3.11 + \$3.11 = \$6.22$

However, I also want to be somewhat generous to build trust, similar to how my previous co-player returned 72.4% of what they received.

Let me think about different approaches:

- Return exactly what they sent: \$3.11 (they break even)
- Return to split gains 50/50: $\sim \$6.22$ (they get $\$5 - \$3.11 + \$6.22 = \8.11)
- Return more generously (similar to 72.4% rate): $\sim \$6.75$ (they get $\$5 - \$3.11 + \$6.75 = \8.64)

Given the trust they showed and the precedent from my last game, I'll return around \$6.50-\$7.00, which gives them a good payoff while keeping a reasonable amount for myself. I'll return \$6.50: their final payoff becomes \$8.39 while I keep \$9.

Wait, let me check the math more carefully. If I return \$6.50, my payoff is $\$9.33 - \$6.50 = \$2.83$. That feels...

usage:

```
{
  "input_tokens": 763,
  "output_tokens": 1019,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$6.5. Your visible payoff was \$2.83.

Your total visible earnings across all rounds are \$12.52.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

- Round 1: Sent \$4, received \$8.69 back (very cooperative partner)
- Round 2: Received from someone who sent \$3.11, I returned \$6.50 (both of us were cooperative)

The environment has been quite cooperative. Sending more creates more value through the 3x multiplier, and past interactions suggest people reciprocate trust.

To maximize value creation and continue building a cooperative reputation, I'll send my full endowment this round.

```
{"send": 5}
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$4, received \$8.69 back, payoff \$9.69

Round 2: I was RECEIVER, got \$9.33 (from \$3.11 sent), returned \$6.50, payoff \$2.83

Total: \$12.52

Now Round 3: I'm SENDER again with a new co-player.

My experience so far:

- When I sent \$4, I got back \$8.69, which was very generous (72% return rate on the tripled amount)
- When I received from someone who sent \$3.11, I returned \$6.50 (70% return rate)

Both interactions have been quite cooperative. This suggests the environment is reasonably trustworthy.

As a sender this round, I should consider:

1. Past cooperation suggests I can trust my co-players
2. Sending more creates more value overall (3x multiplier)
3. Building a reputation for cooperation may help when I'm receiver
4. The game mentioned communication noise, so exact amounts may vary

Given the positive experiences, I should send a substantial amount. I'll send \$4 again since that worked well before, or maybe even \$5 to maximize value creation.

Actually, let me think about this more strategically. If I send \$5:

- Co-player receives \$15
- If they return even 50%, I get \$7.50 back, netting me \$2.50 profit
- If they're as generous as my Round 1 partner (~72%), I could get ~\$10.80 back

If I send \$4:

- Co-player receives \$12
- At 72% return, I'd get ~\$8.64 back, total payoff ~\$9.64

Given the cooperative environment and to maximize total value, I'll send \$5 this time.

usage:

```
{
  "input_tokens": 1058,
  "output_tokens": 560,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$10.29 to you. Your visible payoff was \$10.29.

Your total visible earnings across all rounds are \$22.81.

This round, you are the RECEIVER against your current co-player. They sent you \$3.57, so you received \$10.709999999999999. How much do you return? (0-10.709999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

My co-player sent \$3.57 (71.4% of their endowment), showing strong trust. I received \$10.71.

The game has been very cooperative so far:

- Round 1: I sent \$4, got back \$8.69
- Round 3: I sent \$5, got back \$10.29
- Round 2: When I was receiver, I returned \$6.50 out of \$9.33

My co-player trusted me significantly. A generous return would be around 70% of what I received, which would give them a strong profit while still rewarding me for receiving.

Returning \$7.50 would:

- Give them: \$5 - \$3.57 + \$7.50 = \$8.93 total
- Leave me with: \$10.71 - \$7.50 = \$3.21

This maintains the cooperative pattern.

```
{"return": 7.50}
```

Reasoning:

Let me analyze this situation:

My co-player sent me \$3.57 out of \$5 (71.4% of their endowment), showing strong trust.
I received \$10.71 after the 3x multiplier.

In my previous round as sender:

- I sent \$5 (100% of endowment)
- They returned \$10.29 out of \$15 (68.6%)
- This was very generous, giving me \$10.29 total payoff

The pattern has been quite cooperative. Let me think about a fair return:

If I return around \$7-7.50:

- They would get: $\$5 - \$3.57 + \$7 = \8.43 (if I return \$7)
- They would get: $\$5 - \$3.57 + \$7.50 = \8.93 (if I return \$7.50)
- I would keep: $\$10.71 - \$7 = \$3.71$ (if I return \$7)
- I would keep: $\$10.71 - \$7.50 = \$3.21$ (if I return \$7.50)

Given they sent 71.4% of their endowment, a generous return would be appropriate. The previous receiver returned about 68.6% when I sent 100%.

Returning about \$7.20 would:

- Give them \$8.63 total ($\$5 - \$3.57 + \7.20)
- Leave me with \$3.51
- This rewards their trust well

Actually, let me be a bit more generous to match the cooperative environment. Returning \$7.50:

- Gives them \$8.93 total (a \$3.93 profit on their \$5)
- Leaves me with \$3.21 (also a good profit)
- Return rate: 70% of what I received

...

```
Usage: {  
  "input_tokens": 1326,  
  "output_tokens": 675,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-08-12T09:22:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_2"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_2": "trustee",  
      "Agent_5": "investor"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user**content:**

Round 1: You are the RECEIVER. the sender sent \$2.3, that is 46.0% of their total endowment. You received \$6.899999999999995. How much do you return to the sender? (0-6.899999999999995)

Message 3 role=assistant**content:**

Looking at this situation:

- The sender showed trust by sending \$2.3 (46% of their endowment)
- I received \$6.9 after the 3x multiplier
- The sender currently has \$2.7 remaining

I want to:

1. Reward the sender's trust so they benefit from sending
2. Split the gains fairly
3. Establish myself as trustworthy for future rounds

If I return \$3.0:

- Sender ends with: $\$2.7 + \$3.0 = \$5.7$ (a gain of \$0.7)
- I end with: $\$6.9 - \$3.0 = \$3.9$

This seems fair - the sender profits from trusting me, and we both benefit from the multiplier effect.

```
'''json
{"return": 3.0}
'''
```

reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.3 (which is 46% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$6.9 ($2.3 \times 3 = 6.9$)
4. I need to decide how much to return (between \$0 and \$6.9)

The sender's current situation:

- Started with \$5
- Sent \$2.3
- Currently has: $\$5 - \$2.3 = \$2.7$

If I return nothing:

- Sender ends with: \$2.7
- I end with: \$6.9

The sender showed trust by sending nearly half their endowment. A fair approach would be to:

1. Ensure the sender doesn't lose money
2. Split the gains reasonably

Total money in the system now: \$2.7 (sender has) + \$6.9 (I have) = \$9.6

This started from two \$5 endowments = \$10 total, so we have \$9.6 total.

Wait, let me recalculate:

- Sender started with \$5, sent \$2.3, has \$2.7 left
- I received \$6.9 (the \$2.3 multiplied by 3)
- Total in system: $\$2.7 + \$6.9 = \$9.6$

If we split equally: each person would get \$4.8

- For sender to have \$4.8, they need: $\$4.8 - \$2.7 = \$2.1$ returned
- For me to have \$4.8, I'd keep: $\$6.9 - \$2.1 = \$4.8$?

But equal split might not be the best strategy. The sender took a risk by sending money. A common approach in trust games is to:

reward their trust by ensuring they come out ahead of their initial \$5, while also sharing in the multiplier gains. Let me test a few return amounts: returning \$3.0 gives them \$5.7 total (a \$0.7 gain) while I keep \$3.9; returning \$2.5 gives them \$5.2 (a \$0.2 gain) while I keep \$4.4; returning \$2.1 would give them exactly \$4.8 and me \$4.8.

A return around \$3.0-\$3.3 seems optimal-it meaningfully rewards their trust while still letting me benefit from the multiplier effect. At \$3.0, they gain \$0.7 and I keep \$3.9, which feels like a fair split that demonstrates trustworthiness.

The sender's 46% send rate suggests moderate confidence in me, so I should reciprocate proportionally...

usage:

```
{
  "input_tokens": 539,
  "output_tokens": 1314,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.3, it became \$6.899999999999995, and you returned \$3.0. Your visible payoff was \$3.899999999999995.

Your total visible earnings across all rounds are \$3.899999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm now the sender with a new co-player
- In the previous round, I experienced a sender who sent 46% and I returned fairly
- I want to establish trust with this new partner while being reasonably cautious

Sending \$2.5 (50% of my endowment) seems like a good strategy:

- It shows significant trust
- It's a clear, fair starting point
- If my co-player reciprocates fairly, we both benefit from the 3x multiplier
- It's not overly risky with an unknown partner

```
'''json
{"send": 2.5}
'''
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.3 (46% of their endowment)
- I received \$6.9
- I returned \$3.0
- My payoff was \$3.9

Now I'm playing against a different agent (since it says "current co-player" and agents are randomly paired each round).

I need to decide how much to send. Key considerations:

1. I don't know anything about this new co-player's behavior
2. The sender in Round 1 sent 46% (\$2.3), which was a reasonable trust signal
3. I returned about 43.5% of what I received (\$3/\$6.9), which was fair
4. I want to build trust but also protect myself since I don't know this opponent

Strategies:

- Sending \$0 would be overly defensive and wouldn't build any relationship
- Sending \$5 (100%) would be too risky with an unknown partner
- Sending around \$2-3 seems reasonable as it shows trust without being reckless

Given that the previous sender sent \$2.3 (46%), I could mirror that behavior. This would be consistent and shows I'm willing to trust at a similar level to what I've experienced.

Alternatively, I could send slightly more to show goodwill, or slightly less to be more cautious.

I think sending \$2.5 (50%) is a good middle ground:

- It's a clear half of my endowment, easy to interpret
- It shows significant trust
- If they return fairly (around 40-45% of \$7.5 = \$3-3.4), we both profit
- It's not too risky

Let me calculate the outcomes:

- If I send \$2.5, they receive \$7.5
- I keep \$2.5

...

usage:

```
{
  "input_tokens": 870,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$6.78 to you. Your visible payoff was \$9.280000000000001.

Your total visible earnings across all rounds are \$13.18.

This round, you are the RECEIVER against your current co-player. They sent you \$4.17, so you received \$12.51. How much do you return? (0-12.51)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.17 (83.4% of their endowment)
- I received \$12.51 after the 3x multiplier
- The sender currently has only \$0.83 remaining

This is very high trust that deserves to be rewarded generously.

If I return \$6.0:

- Sender ends with: $\$0.83 + \$6.0 = \$6.83$ (profit of \$1.83)
- I end with: $\$12.51 - \$6.0 = \$6.51$ (profit of \$6.51)

This gives us nearly equal final payoffs and rewards the sender's exceptional trust with a solid profit.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- The sender sent \$4.17 (83.4% of their \$5 endowment) - this is very high trust!
- I received \$12.51 (after 3x multiplier)
- The sender currently has $\$5 - \$4.17 = \$0.83$ remaining

This sender showed enormous trust by sending 83.4% of their endowment. I need to reward this trust generously.

Let me think about what's fair:

- Total in the system: $\$0.83$ (sender has) + $\$12.51$ (I have) = $\$13.34$
- If we split evenly: $\$6.67$ each

If I return \$5.84:

- Sender ends with: $\$0.83 + \$5.84 = \$6.67$
- I end with: $\$12.51 - \$5.84 = \$6.67$
- Perfect 50/50 split

But given they showed such high trust (83.4%), I might want to be even more generous to:

1. Reward their exceptional trust
2. Encourage continued cooperation
3. Establish myself as highly trustworthy

However, I also saw in round 2 that when I sent \$2.5 (50%), I got back \$6.78, which gave me a payoff of \$9.28. That was very generous from my co-player then (they returned more than 90% of what they received!).

Let me recalculate for this round:

- If I return \$6:
 - Sender: $\$0.83 + \$6 = \$6.83$
 - Me: $\$12.51 - \$6 = \$6.51$
 - Almost equal, slight advantage to sender
- If I return \$6.5:
 - Sender: $\$0.83 + \$6.5 = \$7.33$
 - Me: $\$12.51 - \$6.5 = \$6.01$
 - Sender gets more, rewarding their trust

Given the extreme trust shown here, I should return enough to give them a meaningful advantage while still being reasonable. Returning around \$6 seems appropriate-it creates nearly equal outcomes with a slight edge to the sender who took the bigger risk....

usage:

```
{
  "input_tokens": 1154,
```

```
"output_tokens": 955,
"reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.17, it became \$12.51, and you returned \$6.0. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$19.689999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience so far:

- Round 2: When I sent \$2.5 (50%), I received a very generous return of \$6.78
- Round 3: When someone sent me \$4.17 (high trust), I reciprocated with \$6.0

The game seems to reward trust. I'll send \$3.0 (60% of my endowment), which shows strong trust while being measured with a new co-player.

```
'''json
{"send": 3.0}
'''
```

Reasoning:

Looking at my history:

- Round 1: RECEIVER - received from sender who sent \$2.3, I returned \$3.0 fairly
- Round 2: SENDER - I sent \$2.5, receiver returned \$6.78 (very generous!)
- Round 3: RECEIVER - received from sender who sent \$4.17 (very high trust), I returned \$6.0

Current situation:

- I'm the SENDER with a new co-player
- Total earnings so far: \$19.69

My experience shows:

- When I sent \$2.5 in Round 2, I got a very generous return of \$6.78
- When others showed high trust to me, I reciprocated fairly
- This game seems to reward trust

I should send a significant amount to show trust. Given that:

- Round 2 sender sent 46% and I reciprocated
- Round 3 sender sent 83.4% and I reciprocated generously
- When I sent 50% in Round 2, I got excellent returns

I think sending around \$3.0 (60%) would be a good balance - showing strong trust without being reckless with an unknown partner.

Actually, let me think about this more strategically. I've seen that:

- Lower trust (46%) got fair returns
- Medium trust (50% from me) got very generous returns
- High trust (83.4%) got generous returns

Sending \$3.0 seems like a good middle-high trust level that should encourage reciprocation.

Usage: {

```
"input_tokens": 1461,
"output_tokens": 474,
"reasoning_tokens": 0
```

}

GAME - Agent_2 - Receiver - dyad_4

Timestamp: 2026-08-12T09:22:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 7,
```

```

"opponent_id": "Agent_5",
"pairing": {
  "agents": [
    "Agent_5",
    "Agent_2"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_2": "trustee",
    "Agent_5": "investor"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being reasonably cautious since I have no history with my co-player.

The multiplication factor of 3x creates an opportunity for mutual benefit. If I send a moderate amount, we can both potentially end up better off than the initial \$5.

I'll send **\$3**, which gives my co-player \$9 to work with. This shows trust and willingness to cooperate. If they return approximately \$3-4, we'll both be in a good position, and it could establish a positive pattern for future rounds.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit
2. The receiver then decides how much to return
3. There's communication noise mentioned - amounts may appear different than actually sent
4. This is a multi-agent setting with 8 agents total, pairs rotate
5. This is the first round, so I have no history with any player yet

Since this is the first round and I have no information about other players' behavior, I need to consider:

- Sending \$0 guarantees I keep \$5, but no potential for mutual gain
- Sending everything (\$5) maximizes the total pot (\$15 for receiver) but is risky
- A moderate amount balances risk and potential reward

In trust games, common strategies include:

- Sending about half as a signal of conditional cooperation
- The tripling effect means even partial cooperation can benefit both parties

Let me consider sending \$3:

- I send \$3, receiver gets \$9
- If receiver returns \$3, I end up with \$5 (same as starting)
- If receiver returns \$4-5, I'm better off
- If receiver returns less than \$3, I'm worse off
- This shows trust while keeping some insurance

Alternatively, sending \$2.5:

- I send \$2.5, receiver gets \$7.5
- If receiver returns \$2.5, I end up with \$5
- If receiver returns \$3+, I'm better off
- Less risk than \$3

Or sending \$2:

- I send \$2, receiver gets \$6
- If receiver returns \$2, I end up with \$5
- Lower risk, but also lower potential reward

Given this is the first round and I want to establish a cooperative pattern, I'll send \$3.

This shows trust and willingness to cooperate while not being completely vulnerable.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 4 role=user**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.81 to you. Your visible payoff was \$4.8100000000000005.

Your total visible earnings across all rounds are \$4.8100000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$3.95, so you received \$11.850000000000001. How much do you return? (0-11.850000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.95 (79% of their endowment), showing significant trust
- I received \$11.85 total
- They currently have only \$1.05 left

I want to:

1. Reward their trust and cooperation
2. Ensure they end up better than their initial \$5
3. Set a positive tone for potential future interactions

If I return \$5.50:

- They end up with: $\$1.05 + \$5.50 = \$6.55$
- I end up with: $\$11.85 - \$5.50 = \$6.35$

Both of us gain from cooperation, and this reciprocates their high level of trust appropriately.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the SENDER
- I sent \$3.00
- It became \$9.00 for the receiver
- They returned \$2.81
- My payoff: $\$5 - \$3 + \$2.81 = \4.81

Round 2:

- I am now the RECEIVER
- My co-player (a different agent) sent me \$3.95
- I received $\$3.95 \times 3 = \11.85
- I need to decide how much to return

Important considerations:

1. The sender gave me \$3.95 out of their \$5 - that's 79% of their endowment, showing high trust
2. If I return nothing, I keep \$11.85 and they end up with $\$5 - \$3.95 = \$1.05$
3. I should return enough to make cooperation worthwhile for them
4. There's communication noise mentioned, so amounts may differ slightly

What would be fair?

- The sender gave up \$3.95, keeping \$1.05
- I received \$11.85
- If I want them to end up better than their initial \$5, I should return more than \$3.95
- A 50-50 split of total surplus could work: Total = $\$1.05 + \$11.85 = \$12.90$. Half is \$6.45
- If I return \$5.40, sender gets $\$1.05 + \$5.40 = \$6.45$, I get $\$11.85 - \$5.40 = \$6.45$

But let me think about reciprocity:

- In round 1, I sent \$3 (60% of endowment), received \$9, they returned \$2.81 (31% of what they received)
- This round, sender sent \$3.95 (79% of endowment), I received \$11.85
- Using that same 31% return rate would give me \$3.67 to send back

The sender showed significantly more generosity than I did initially, so I should reciprocate with a higher return. Returning \$5 would leave them with \$6.05 total-better than their starting \$5-while I'd keep \$6.85, rewarding their trust and encouraging continued cooperation....

usage:

```
{
  "input_tokens": 793,
```

```

        "output_tokens": 977,
        "reasoning_tokens": 0
    }
Message 6 role=user
    content:
        Round 3

        In your most recent previous game round, you were the RECEIVER against another agent. You
        saw them send $3.95, it became $11.850000000000001, and you returned $5.5. Your visible
        payoff was $6.350000000000001.
        Your total visible earnings across all rounds are $11.160000000000002.

        This round, you are the SENDER against your current co-player. You have $5. How much do yo
        u send? (0-5)
        Respond exactly as JSON: {'send': <amount>}
Message 7 role=assistant
    content:
        Looking at Round 3:

        I'm paired with a new co-player (pairings are randomized each round), so I have no history
        with them. My previous experiences showed:
        - Round 1: Sent $3, received $2.81 back - reasonable cooperation
        - Round 2: As receiver, I tried to be generous with a high-trusting sender

        For this new pairing, I'll use a similar approach to Round 1 - send a moderate amount that
        signals cooperation while being reasonably cautious with an unknown partner.

        Sending $3 creates $9 for them to work with, which gives room for mutual benefit if they r
        eciprocate fairly.

        '''json
        {"send": 3}
        '''
    reasoning:
        Let me analyze the situation:

        Round 1: I was SENDER, sent $3, received back $2.81, payoff $4.81
        Round 2: I was RECEIVER, got $11.85 from a $3.95 send, returned $5.50, payoff $6.35
        Total earnings: $11.16

        Now Round 3: I'm SENDER again with a new co-player (since pairings are randomized each rou
        nd).

        Key considerations:
        - I have no history with this particular co-player
        - In Round 1, when I sent $3, I got back $2.81 (which is reasonable - about 31% return rat
        e)
        - In Round 2, when someone sent me $3.95, I returned $5.50 (about 46% return rate)
        - The game has communication noise, so exact amounts may differ

        Since I'm paired with a new player and have no history with them, I should probably use a
        similar strategy to Round 1 - establish cooperation but be somewhat cautious.

        Given that:
        - Sending $3 worked reasonably well in Round 1
        - The multiplication factor of 3x creates good mutual benefit potential
        - I want to signal cooperation

        I'll send $3 again. This is a moderate amount that shows willingness to cooperate without
        being overly risky with a new partner.
    usage:
        {
            "input_tokens": 1106,
            "output_tokens": 478,
            "reasoning_tokens": 0
        }
Message 8 role=user
    content:
        Round 4

```

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.52 to you. Your visible payoff was \$8.52.

Your total visible earnings across all rounds are \$19.68.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$3.36 (67% of their endowment), showing solid trust
- I received \$10.08 total
- They currently have \$1.64 left

I want to reciprocate their trust fairly. If I return \$5.00:

- They end up with: \$1.64 + \$5.00 = \$6.64 (gain of \$1.64)
- I end up with: \$10.08 - \$5.00 = \$5.08 (gain of \$0.08)

Both of us benefit from cooperation, and this rewards their trust appropriately.

```
'''json
{"return": 5.00}
'''
```

Reasoning:

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- My co-player sent \$3.36 (67.2% of their endowment)
- I received \$10.08 total
- They currently have: \$5 - \$3.36 = \$1.64 left

My previous round (Round 3):

- I sent \$3, received \$6.52 back - very generous return!

This co-player sent \$3.36, showing good trust. I should reciprocate appropriately.

If I return \$5.00:

- They end up with: \$1.64 + \$5.00 = \$6.64
- I end up with: \$10.08 - \$5.00 = \$5.08

Both of us gain. This is a fair reciprocation that rewards their trust.

Alternatively, if I return \$5.50:

- They end up with: \$1.64 + \$5.50 = \$7.14
- I end up with: \$10.08 - \$5.50 = \$4.58

This would be even more generous, similar to how I acted in Round 2.

Given that they sent about 67% of their endowment (compared to 79% in Round 2), I think returning around \$5 is appropriate and fair. It ensures they profit significantly from their trust, and we both end up better off than the initial \$5.

I'll go with \$5.00 as a fair reciprocation.

```
Usage: {
  "input_tokens": 1402,
  "output_tokens": 522,
  "reasoning_tokens": 0
}
```

=====

ROUND 5

=====

Roles: Agent_1=Sender, Agent_2=Sender, Agent_3=Sender, Agent_4=Receiver, Agent_5=Receiver, Agent_6=Receiver, Agent_7=Receiver, Agent_8=Sender

Pairings:

- dyad_1: Agent_8=Sender, Agent_6=Receiver
- dyad_2: Agent_3=Sender, Agent_4=Receiver
- dyad_3: Agent_1=Sender, Agent_5=Receiver
- dyad_4: Agent_2=Sender, Agent_7=Receiver

Parsed game decisions:

```
dyad_1: Agent_8 sent $4; received $12; Agent_6 returned $7; payoffs Agent_8=$8, Agent_6=$5;
noise/visible: sent visible=$3.67, received visible=$11.01, returned visible=$6.98
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw":
7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_3 sent $5; received $15; Agent_4 returned $8; payoffs Agent_3=$8, Agent_4=$7;
noise/visible: sent visible=$4.73, received visible=$14.19, returned visible=$8.2
Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 15.0, "min": 0, "raw":
8.0}, "sent": {"amount": 5.0, "clamped": false, "max": 5, "min": 0, "raw": 5.0}}
dyad_3: Agent_1 sent $3.75; received $11.25; Agent_5 returned $6; payoffs Agent_1=$7.25,
Agent_5=$5.25; noise/visible: sent visible=$3.84, received visible=$11.52, returned visible=$6.54
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 11.25, "min": 0, "raw":
6.0}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}
dyad_4: Agent_2 sent $3; received $9; Agent_7 returned $3.5; payoffs Agent_2=$5.5, Agent_7=$5.5;
noise/visible: sent visible=$2.26, received visible=$6.78, returned visible=$3.15
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw":
3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
```

Balances after round: {

```
"Agent_1": 30.25,
"Agent_2": 28.75,
"Agent_3": 31.0,
"Agent_4": 27.0,
"Agent_5": 32.5,
"Agent_6": 32.5,
"Agent_7": 28.0,
"Agent_8": 34.5
```

}

Visible balances after round: {

```
"Agent_1": 33.870000000000005,
"Agent_2": 29.909999999999997,
"Agent_3": 34.22,
"Agent_4": 30.919999999999998,
"Agent_5": 32.429999999999999,
"Agent_6": 28.82,
"Agent_7": 25.339999999999996,
"Agent_8": 34.45
```

}

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

Timestamp: 2026-08-12T09:22:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_6",
"pairing": {
  "agents": [
    "Agent_8",
    "Agent_6"
  ],
"investor": "Agent_8",
"roles": {
  "Agent_6": "trustee",
  "Agent_8": "investor"
},
"trustee": "Agent_6"
```

}

```
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1 with no prior information about my co-player, I want to demonstrate trust while being reasonably cautious.

If I send \$3:

- My co-player receives \$9 (3×3)
- This shows meaningful cooperation
- If they return a fair portion (say $1/3$ to $1/2$), we both benefit
- My maximum loss is limited to \$3

I'll go with a moderately trusting approach:

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I have \$5 to start with
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return

4. I keep: \$5 - sent + returned
5. The receiver keeps: received - returned

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There may be communication noise
- I don't know who my co-player is

Strategic thinking:

- If I send \$0, I guarantee keeping \$5, but there's no potential for gain
- If I send everything (\$5), the receiver gets \$15, and if they're cooperative, they might return a fair amount
- A moderate amount (like \$2-3) shows some trust but limits risk

Since this is round 1 and I have no information about my co-player:

- I want to signal cooperation and trust
- But I also want to be somewhat cautious
- Sending around \$3 seems like a reasonable middle ground that:
 - Shows good faith (60% of my endowment)
 - Gives the receiver \$9 to work with
 - If they return fairly (say 1/3 to 1/2 of what they receive), I could end up with \$5-6.
- 5
- Limits my maximum loss to \$3

Let me send \$3 as a cooperative but measured opening move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.37 to you. Your visible payoff was \$5.37.

Your total visible earnings across all rounds are \$5.37.

This round, you are the RECEIVER against your current co-player. They sent you \$4.58, so you received \$13.74. How much do you return? (0-13.74)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

My co-player sent me \$4.58 out of their \$5 (91.6% of their endowment) - showing very high trust.

I received \$13.74. To reciprocate this trust appropriately:

- They kept \$0.42
- If I return around \$6.50, they'd end with \$6.92 (a gain of \$1.92)
- I'd keep \$7.24 (also a solid gain)

This represents returning about 47% of what I received, which rewards their trust while we both benefit substantially.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 2 - I'm now the RECEIVER

- My co-player sent me \$4.58
- I received \$13.74 (which is 4.58×3)
- I need to decide how much to return

From Round 1:

- I sent \$3.00, they received \$9.00
- They returned \$3.37 to me
- This is about 37.4% of what they received, which is quite fair/generous

My co-player in this round sent me \$4.58 out of their \$5, which is 91.6% of their endowment - very trusting!

I should consider:

1. They showed high trust by sending almost everything
2. In Round 1, my previous partner returned about 37.4% which seemed fair
3. I want to reciprocate trust and maintain good cooperation

If I return about 40-50% of what I received (\$13.74):

- 40% = \$5.50
- 45% = \$6.18
- 50% = \$6.87

Given they sent almost everything (\$4.58 out of \$5), showing extreme trust, I should be generous in return.

Their final payoff would be:

- They started with \$5
- Sent \$4.58, kept \$0.42
- If I return \$6.00: $\$0.42 + \$6.00 = \$6.42$
- My payoff: $\$13.74 - \$6.00 = \$7.74$

If I return \$6.50:

- Their payoff: $\$0.42 + \$6.50 = \$6.92$
- My payoff: $\$13.74 - \$6.50 = \$7.24$

Given their high trust (sending 91.6% of endowment), I should return a good amount. Returning around \$6.50 seems fair - they end up with \$6.92 (better than their initial \$5), and I keep \$7.24 (also better than my initial \$0).

...

usage:

```
{
  "input_tokens": 755,
  "output_tokens": 653,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.58, it became \$13.74, and you returned \$6.5. Your visible payoff was \$7.24.

Your total visible earnings across all rounds are \$12.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.38, so you received \$10.14. How much do you return? (0-10.14)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust.

I received \$10.14.

They kept: \$1.62

To reciprocate fairly:

- If I return \$5.00, they'd end with \$6.62 (a gain of \$1.62)
- I'd keep \$5.14 (also a small gain)

This represents returning about 49% of what I received, which rewards their trust while we both benefit.


```
'''json
{"return": 5.00}
'''
```

reasoning:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust
.

I received \$10.14.

They kept: \$5 - \$3.38 = \$1.62

To be fair and reciprocate their trust:

- If I return around \$5, they'd end with \$1.62 + \$5 = \$6.62 (a gain of \$1.62)
- I'd keep \$10.14 - \$5 = \$5.14 (also a solid gain)

This represents returning about 49% of what I received, which is fair and reciprocates the
ir trust appropriately while we both benefit.

Let me think about this more carefully. They sent \$3.38, showing moderate-to-good trust (6
8% of endowment).

In Round 2, my co-player sent me \$4.58 (92% of endowment) and I returned \$6.50 (47% of wha
t I received).

For consistency and fairness here with \$3.38 sent (less trust shown):

- Returning around 47-50% of \$10.14 would be \$4.77-\$5.07
- Let me go with \$5.00

That gives them: \$1.62 + \$5.00 = \$6.62 (gain of \$1.62)

I keep: \$10.14 - \$5.00 = \$5.14 (gain of \$0.14)

This seems fair and reciprocal.

usage:

```
{
  "input_tokens": 1036,
  "output_tokens": 519,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You
saw them send \$3.38, it became \$10.14, and you returned \$5.0. Your visible payoff was \$5
.1400000000000001.

Your total visible earnings across all rounds are \$17.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do yo
u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

Round 1: As SENDER, I sent \$3 and received \$3.37 back - a reasonable return.

Rounds 2-3: As RECEIVER, others showed good trust, and I reciprocated fairly.

This is a new co-player (random pairing), so I don't have specific information about them.
However, the overall pattern suggests players are reasonably cooperative.

I'll send \$3.50 - slightly more than my first round to show increasing trust while remaini
ng measured:

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$3, received back \$3.37, payoff \$5.37
Round 2: I was RECEIVER, received \$13.74 (from \$4.58), returned \$6.50, payoff \$7.24
Round 3: I was RECEIVER, received \$10.14 (from \$3.38), returned \$5.00, payoff \$5.14
Round 4: I am SENDER

My total earnings: \$17.75

Looking at the pattern:

- When I sent \$3, I got back \$3.37 (a bit more than 1:1 return on sent amount)
- When others sent to me, they showed high trust (\$4.58 and \$3.38 out of \$5)
- I've been fairly reciprocal as a receiver, returning roughly half of what I received

Now I'm a sender again. Since:

1. My first send of \$3 was met with a decent return (\$3.37)
2. Others have been trusting with me
3. I've been fair in return

I should continue with moderate to high trust. I'll send \$3.5 this time - slightly more than my first send to show I'm willing to build cooperation.

Actually, let me reconsider. The co-player this round is new (pairings are randomized). I don't know their history. In Round 1, I sent \$3 and got a reasonable return. Let me stick with a similar strategy - send \$3 again as a baseline cooperative strategy.

Actually, looking at the overall pattern, players seem reasonably cooperative. I could try \$3.5 to be slightly more trusting.

Let me go with \$3.5 - showing increased trust while not overcommitting.

usage:

```
{
  "input_tokens": 1303,
  "output_tokens": 551,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$7.22 to you. Your visible payoff was \$8.719999999999999.

Your total visible earnings across all rounds are \$26.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my recent experience:

Round 4: I sent \$3.50 and received \$7.22 back - excellent cooperation!

The overall pattern shows players are being reasonably cooperative. While this is a new co-player (random pairing), I'll show increased trust based on positive experiences.

Sending \$4:

- Demonstrates strong trust (80% of my endowment)
- Creates \$12 for them to work with
- Encourages reciprocal cooperation

```
'''json
{"send": 4}
'''
```

Reasoning:

Round 4 recap: I sent \$3.50, they returned \$7.22 - excellent return! That's about 69% of what they received, very generous.

This is Round 5, and I'm the SENDER again against a new co-player (random pairing each round).

So far my experience:

- Round 1 SENDER: sent \$3, got \$3.37 back (reasonable)
- Round 2 RECEIVER: received \$13.74, returned \$6.50
- Round 3 RECEIVER: received \$10.14, returned \$5.00
- Round 4 SENDER: sent \$3.50, got \$7.22 back (very good!)

The last round showed great cooperation. Players seem generally trustworthy. This is a new co-player though, so I don't know their specific behavior.

I should probably continue with a trusting approach. I could:

- Send \$4 (80% of endowment) - shows high trust
- Send \$3.5 (70%) - similar to last round
- Send \$3 (60%) - more cautious

Given the positive experiences, I'll increase slightly to \$4 to show strong trust.

```
Usage: {
  "input_tokens": 1562,
  "output_tokens": 402,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_1

Timestamp: 2026-08-12T09:22:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_6"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_6": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.25, that is 45.0% of their total endowment. You received \$6.75. How much do you return to the sender? (0-6.75)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.25 (45% of their endowment)
- This was multiplied to \$6.75 for me
- The sender currently has \$2.75 remaining

I want to reward their trust while maintaining a fair distribution. If I return \$3.00:

- The sender ends up with \$5.75 (a gain of \$0.75)
- I keep \$3.75 (a gain of \$3.75)
- Both of us are better off than if no money had been sent

This seems like a reasonable balance that rewards their moderate trust and encourages future cooperation.

```
{"return": 3.00}
```

reasoning:

Let me think through this situation carefully.

I'm the RECEIVER in this round of a trust game.

The sender:

- Started with \$5
- Sent me \$2.25 (which is 45% of their endowment)
- This was multiplied by 3x, so I received \$6.75

Now I need to decide how much to return.

Key considerations:

1. The sender showed moderate trust by sending 45% of their endowment
2. They could have sent nothing, or they could have sent everything
3. If I return nothing, I keep \$6.75 and they keep \$2.75 (\$5 - \$2.25)
4. If I return some amount, we both benefit from the cooperative exchange

Let me think about fair or reasonable return amounts:

- If I return \$3.00: The sender ends up with \$2.75 + \$3.00 = \$5.75 (gain of \$0.75)
- I keep: \$6.75 - \$3.00 = \$3.75
- Total: \$5.75 + \$3.75 = \$9.50
- If I return \$2.25: The sender ends up with \$2.75 + \$2.25 = \$5.00 (break even)

- I keep: $\$6.75 - \$2.25 = \$4.50$
- Total: $\$5.00 + \$4.50 = \$9.50$
- If I return $\$3.50$: The sender ends up with $\$2.75 + \$3.50 = \$6.25$ (gain of $\$1.25$)
- I keep: $\$6.75 - \$3.50 = \$3.25$
- Total: $\$6.25 + \$3.25 = \$9.50$

Common approaches in trust games:

I could mirror their proportion—returning about 45% of what I received, which would be roughly $\$3.04$. Alternatively, I could split the cooperative gains equally: the $\$4.50$ surplus created through the exchange divided in half means returning $\$2.25$ to give them an even split of the benefit. Or I could aim for a return that ensures they profit from trusting me, somewhere between $\$2.25$ and $\$3.50$

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1174,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send $\$2.25$, it became $\$6.75$, and you returned $\$3.0$. Your visible payoff was $\$3.75$.

Your total visible earnings across all rounds are $\$3.75$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In the previous round as receiver, I experienced moderate trust (45% sent) and reciprocated fairly. Now I'm facing a new co-player as sender.

Since this is a new pairing and I want to establish cooperative behavior while managing risk, I'll send a moderate-to-good amount. Sending $\$3.00$ (60% of my endowment) signals trust and willingness to cooperate, while still keeping some safety margin.

This will give my co-player $\$9.00$ to work with, providing them good opportunity to reciprocate if they're inclined to cooperate.

```
{"send": 3.00}
```

reasoning:

Let me think about this situation:

In Round 1:

- I was the receiver
- The sender sent $\$2.25$ (45% of their endowment)
- I received $\$6.75$ and returned $\$3.00$
- I kept $\$3.75$
- The sender ended with $\$2.75 + \$3.00 = \$5.75$

Now in Round 2:

- I'm the sender
- I have $\$5$ to potentially send
- I'm playing against a different co-player (since pairings are randomized each round)

The previous sender showed moderate trust by sending 45% of their endowment. I returned about 44% of what I received ($\$3.00$ out of $\$6.75$), which gave them a positive return.

Now I need to decide how much to send. Considerations:

1. This is a new opponent, so I don't know their behavior yet
2. In multi-round games with changing partners, establishing a reputation for cooperation can be beneficial
3. The previous sender's moderate trust (45%) seemed reasonable
4. There's communication noise, so amounts might differ slightly

I think I should send a moderate-to-good amount to signal trust and cooperation. Sending a round 50-60% seems reasonable:

- \$2.50 (50%) or \$3.00 (60%)

Given that the previous sender sent 45% and I want to be at least as cooperative (and perhaps signal willingness to cooperate), I'll send \$3.00 (60%). This will become \$9.00 for the receiver, giving them good opportunity to reciprocate.

Actually, let me reconsider. The previous interaction went well with a 45% send rate. I could match that or go slightly higher. \$3.00 (60%) shows I'm willing to be cooperative and trust my new partner.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 571,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.42 to you. Your visible payoff was \$8.42.

Your total visible earnings across all rounds are \$12.17.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player showed ****maximum trust**** by sending their entire \$5 endowment (100%)
- This became \$15 for me
- They currently have \$0

This exceptional level of trust deserves strong reciprocation. If I return \$10.00:

- The sender ends with \$10.00 (doubling their money, gain of \$5.00)
- I keep \$5.00 (gain of \$5.00)
- We both gain equally from our cooperation

This rewards their maximum trust appropriately and creates a fair outcome where both parties benefit substantially.

```
{"return": 10.00}
```

reasoning:

Let me analyze this situation:

Previous round:

- I was the SENDER
- I sent \$3.00 (60% of my endowment)
- My co-player received \$9.00
- They returned \$6.42 (71.3% of what they received)
- I ended with \$8.42

This round:

- I am the RECEIVER
- My current co-player sent \$5.00 (100% of their endowment!)
- I received \$15.00
- They currently have \$0
- I need to decide how much to return (0-15)

This is a very strong show of trust - they sent their entire endowment! This deserves a strong reciprocation.

Let me think about fair outcomes:

- If I return \$7.50, they end with \$7.50 (gain of \$2.50), I keep \$7.50 (gain of \$7.50)
- If I return \$10.00, they end with \$10.00 (gain of \$5.00), I keep \$5.00 (gain of \$5.00) -

equal split of total gains
- If I return \$12.00, they end with \$12.00 (gain of \$7.00), I keep \$3.00 (gain of \$3.00)

Given they showed maximum trust by sending everything, I should reciprocate generously. An equal split of total gains seems fair - we'd each gain \$5 from the cooperation. That means returning \$10.00.

Actually, let me reconsider. They sent everything (\$5), which became \$15 for me. If I return \$10, they get \$10 total (doubling their money), and I keep \$5 (gaining \$5). Both of us gain equally from the transaction.

This seems like a fair and generous reciprocation for their maximum trust.

usage:

```
{
  "input_tokens": 1082,
  "output_tokens": 580,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$5, it became \$15, and you returned \$10.0. Your visible payoff was \$5.0. Your total visible earnings across all rounds are \$17.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

- Round 2 (sender): Sent \$3.00 (60%) and received excellent reciprocation (\$6.42)
- Round 3 (receiver): Received maximum trust (\$5.00 sent) and reciprocated generously

The pattern so far shows strong cooperation in this game. Since this is a new co-player, I want to signal trust and encourage cooperation. Given the positive experiences, I'll send a high amount while keeping a small buffer.

Sending \$4.00 (80% of my endowment) demonstrates strong trust and gives my co-player \$12.00 to work with, creating good opportunity for mutual benefit.

```
{"send": 4.00}
```

reasoning:

Let me analyze the situation:

Previous rounds:

1. Round 1: RECEIVER - saw \$2.25 sent, received \$6.75, returned \$3.00, kept \$3.75
2. Round 2: SENDER - sent \$3.00, became \$9.00, saw \$6.42 returned, payoff \$8.42
3. Round 3: RECEIVER - saw \$5.00 sent, received \$15.00, returned \$10.00, kept \$5.00

Total visible earnings: \$17.17

Now Round 4: I'm the SENDER against a new co-player.

Key observations:

- In Round 2, I sent \$3.00 (60%) and got very good reciprocation (\$6.42 return on \$9.00)
- In Round 3, someone sent me everything (\$5.00) and I reciprocated generously (\$10.00 return)

I've seen good cooperation so far. My previous experience as sender (Round 2) with moderate-high trust level worked well.

For this round:

- This is a new co-player (different from previous rounds)
- I don't know their behavior yet
- But the overall pattern in this game seems cooperative
- I've been both generous and received generosity

Given the positive experiences so far, I should maintain a cooperative strategy. I'll send a substantial amount again. Sending \$3.50 (70%) shows strong trust while keeping a small safety margin.

Actually, let me reconsider. The game seems quite cooperative overall. In Round 3, someone trusted me completely with \$5.00, and in Round 2 when I sent \$3.00 I got excellent reciprocation.

I could send \$4.00 (80%) to signal strong cooperation, or even \$5.00 to match the maximum trust I received in Round 3.

Given the high cooperation levels I've experienced, I'll send \$4.00 (80% of endowment) - this is a strong signal of trust without going to the absolute maximum.

...

usage:

```
{
  "input_tokens": 1343,
  "output_tokens": 646,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.64 to you. Your visible payoff was \$7.64.

Your total visible earnings across all rounds are \$24.810000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.67, so you received \$11.01. How much do you return? (0-11.01)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$3.67 (73.4% of their endowment) - this shows strong trust
- This became \$11.01 for me
- They currently have \$1.33 remaining

Their strong trust (sending nearly 3/4 of their endowment) deserves good reciprocation. If I return \$7.00:

- The sender ends with \$8.33 (gain of \$3.33)
- I keep \$4.01 (gain of \$4.01)
- Both of us benefit, with returns roughly proportional to their trust level

This balances rewarding their trust while maintaining fairness.

{"return": 7.00}

Reasoning:

Let me analyze this situation:

- My co-player sent \$3.67 (73.4% of their endowment) - this is strong trust
- This became \$11.01 for me
- They currently have \$1.33 remaining

Looking at my pattern of reciprocation:

- Round 1: Received \$6.75, returned \$3.00 (44.4% return rate)
- Round 3: Received \$15.00, returned \$10.00 (66.7% return rate) - but they sent everything

And when I've been sender:

- Round 2: Sent \$3.00 (60%), received \$6.42 back (71.3% return rate from receiver)
- Round 4: Sent \$4.00 (80%), received \$6.64 back (55.3% return rate from receiver)

The sender showed strong trust at 73.4%. I should reciprocate well, but need to be fair.

If I return \$6.00:

- Sender ends with \$1.33 + \$6.00 = \$7.33 (gain of \$2.33)
- I keep \$11.01 - \$6.00 = \$5.01 (gain of \$5.01)

This is about a 54.5% return rate, which seems reasonable given their strong (but not maximum)

trust.

If I return \$7.00:

- Sender ends with $\$1.33 + \$7.00 = \$8.33$ (gain of \$3.33)
- I keep $\$11.01 - \$7.00 = \$4.01$ (gain of \$4.01)

This is about a 63.6% return rate, which rewards their strong trust more generously.

Given they sent 73.4% of their endowment (strong trust), I think \$7.00 is appropriate. It's more generous than my Round 1 reciprocation (where they sent 45%) but less than my Round 3 reciprocation (where they sent ...

```
Usage: {
  "input_tokens": 1645,
  "output_tokens": 659,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T09:22:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$
- You keep: $\text{received} - \text{returned}$
- Response format: `{'return': <amount>}`

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round with no prior history, I want to signal cooperation and trust while being somewhat cautious.

In this trust game, sending money creates value (3x multiplier), but requires trusting that my co-player will reciprocate. Since:

- We'll likely play multiple rounds (possibly with different partners)
- Building a cooperative reputation could benefit me when I'm the receiver
- The socially optimal strategy involves sending substantial amounts to maximize total value

I'll send a significant amount to demonstrate trust, but keep a small buffer given the uncertainty.

```
{"send": 4}
```

reasoning:

This is the first round of a trust game where I am the SENDER. Let me think through the strategic considerations:

- 1. **Game Structure:****
 - I have \$5
 - I can send \$0-\$5 to the receiver
 - Whatever I send gets multiplied by 3x
 - The receiver then decides how much to return
 - My final payoff: \$5 - sent + returned
 - Receiver's payoff: 3xsent - returned
- 2. **Strategic Considerations:****
 - If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
 - If I send \$5: Receiver gets \$15, and could return anywhere from \$0-\$15
 - If they return \$0: I get \$0 total (lost \$5)
 - If they return \$5: I get \$5 total (break even)
 - If they return \$7.50: I get \$7.50 (gain \$2.50)
 - If they return \$10: I get \$10 (gain \$5)
- 3. **Trust Game Logic:****
 - This is a classic trust game
 - Sending money shows trust
 - The socially optimal outcome is for me to send the maximum (\$5), which creates \$15 total, and then split fairly
 - A fair split might be: I sent \$5, created \$15, so maybe 50/50 of the gain (\$15-\$5=\$10 gain, so \$5 each) meaning they return \$10 and keep \$5
 - Or they could view it as: I took the risk, so 50/50 of total created value: they return \$7.50, we each get \$7.50

4. ****First Round Considerations:****

- This is round 1, so there's no history to build on
- However, there will be multiple rounds with potentially different partners
- Establishing a cooperative reputation early could pay dividends later, especially since I'll alternate between sender and receiver roles
- The communication noise mentioned means amounts might shift slightly, which could complicate trust signals
- A moderate to high send amount (\$3-5) demonstrates willingness to cooperate while managing risk, potentially encouraging reciprocal behavior in future rounds

I'm leaning toward sending \$4 as my opening move. This signals meaningful trust-it triples to \$12 for the receiver-while keeping \$1 as a buffer. Even a modest return of \$4-5 puts me ahead, and it establishes the cooperative tone I'd want if positions reverse. Classic trust game research suggests starting with substantial trust and adjusting based on responses works better than cautious opening moves, so I'll go with that approach....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 1038,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.69 to you. Your visible payoff was \$9.69.

Your total visible earnings across all rounds are \$9.69.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

My co-player sent me \$3.11 (62% of their endowment), showing significant trust. I received \$9.33 after the 3x multiplier.

In my previous round, my co-player returned \$8.69 out of \$12 (72%), which was very generous. I want to maintain this cooperative spirit.

A fair approach would be to ensure my co-player profits from their trust while I also benefit from receiving the tripled amount.

Returning approximately \$6.50 would give them a final payoff of about \$8.39 (\$5 - \$3.11 + \$6.50), which rewards their trust handsomely, while I keep \$2.83.

```
{"return": 6.50}
```

reasoning:

Let me analyze this situation:

1. Last round I was the SENDER:

- I sent \$4.0
- It became \$12.0
- They returned \$8.69
- My payoff: \$5 - \$4 + \$8.69 = \$9.69
- Their payoff: \$12 - \$8.69 = \$3.31

2. This round I'm the RECEIVER:

- They sent me \$3.11
- I received $\$3.11 \times 3 = \9.33
- I need to decide how much to return

My co-player sent me \$3.11 out of their \$5, which is 62.2% of their endowment. This shows significant trust.

In the last round, my co-player returned \$8.69 out of \$12 they received, which is 72.4% of what they received. This was quite generous - they gave me more than I sent them.

I should consider:

- Reciprocity: They showed trust by sending \$3.11
- Building reputation: Being fair will help in future rounds
- The note says there may be communication noise, so amounts might differ slightly

A fair return would be:

- They invested \$3.11, I should at least return that amount so they don't lose
- To split gains fairly: They gave up \$3.11, it became \$9.33, so we created \$6.22 in additional value
- If we split that gain: $\$3.11 + \$6.22/2 = \$3.11 + \$3.11 = \$6.22$

However, I also want to be somewhat generous to build trust, similar to how my previous co-player returned 72.4% of what they received.

Let me think about different approaches:

- Return exactly what they sent: \$3.11 (they break even)
- Return to split gains 50/50: ~\$6.22 (they get \$5 - \$3.11 + \$6.22 = \$8.11)
- Return more generously (similar to 72.4% rate): ~\$6.75 (they get \$5 - \$3.11 + \$6.75 = \$8.64)

Given the trust they showed and the precedent from my last game, I'll return around \$6.50-\$7.00, which gives them a good payoff while keeping a reasonable amount for myself. I'll return \$6.50: their final payoff becomes \$8.39 while I keep \$9.

Wait, let me check the math more carefully. If I return \$6.50, my payoff is $\$9.33 - \$6.50 = \$2.83$. That feels...

usage:

```
{
  "input_tokens": 763,
  "output_tokens": 1019,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$6.5. Your visible payoff was \$2.83.

Your total visible earnings across all rounds are \$12.52.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

- Round 1: Sent \$4, received \$8.69 back (very cooperative partner)
- Round 2: Received from someone who sent \$3.11, I returned \$6.50 (both of us were cooperative)

The environment has been quite cooperative. Sending more creates more value through the 3x multiplier, and past interactions suggest people reciprocate trust.

To maximize value creation and continue building a cooperative reputation, I'll send my full endowment this round.

```
{"send": 5}
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$4, received \$8.69 back, payoff \$9.69

Round 2: I was RECEIVER, got \$9.33 (from \$3.11 sent), returned \$6.50, payoff \$2.83

Total: \$12.52

Now Round 3: I'm SENDER again with a new co-player.

My experience so far:

- When I sent \$4, I got back \$8.69, which was very generous (72% return rate on the triple

d amount)

- When I received from someone who sent \$3.11, I returned \$6.50 (70% return rate)

Both interactions have been quite cooperative. This suggests the environment is reasonably trustworthy.

As a sender this round, I should consider:

1. Past cooperation suggests I can trust my co-players
2. Sending more creates more value overall (3x multiplier)
3. Building a reputation for cooperation may help when I'm receiver
4. The game mentioned communication noise, so exact amounts may vary

Given the positive experiences, I should send a substantial amount. I'll send \$4 again since that worked well before, or maybe even \$5 to maximize value creation.

Actually, let me think about this more strategically. If I send \$5:

- Co-player receives \$15
- If they return even 50%, I get \$7.50 back, netting me \$2.50 profit
- If they're as generous as my Round 1 partner (~72%), I could get ~\$10.80 back

If I send \$4:

- Co-player receives \$12
- At 72% return, I'd get ~\$8.64 back, total payoff ~\$9.64

Given the cooperative environment and to maximize total value, I'll send \$5 this time.

usage:

```
{
  "input_tokens": 1058,
  "output_tokens": 560,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$10.29 to you. Your visible payoff was \$10.29.

Your total visible earnings across all rounds are \$22.81.

This round, you are the RECEIVER against your current co-player. They sent you \$3.57, so you received \$10.709999999999999. How much do you return? (0-10.709999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

My co-player sent \$3.57 (71.4% of their endowment), showing strong trust. I received \$10.71.

The game has been very cooperative so far:

- Round 1: I sent \$4, got back \$8.69
- Round 3: I sent \$5, got back \$10.29
- Round 2: When I was receiver, I returned \$6.50 out of \$9.33

My co-player trusted me significantly. A generous return would be around 70% of what I received, which would give them a strong profit while still rewarding me for receiving.

Returning \$7.50 would:

- Give them: \$5 - \$3.57 + \$7.50 = \$8.93 total
- Leave me with: \$10.71 - \$7.50 = \$3.21

This maintains the cooperative pattern.

```
{"return": 7.50}
```

reasoning:

Let me analyze this situation:

My co-player sent me \$3.57 out of \$5 (71.4% of their endowment), showing strong trust. I received \$10.71 after the 3x multiplier.

In my previous round as sender:

- I sent \$5 (100% of endowment)
- They returned \$10.29 out of \$15 (68.6%)
- This was very generous, giving me \$10.29 total payoff

The pattern has been quite cooperative. Let me think about a fair return:

If I return around \$7-7.50:

- They would get: $\$5 - \$3.57 + \$7 = \8.43 (if I return \$7)
- They would get: $\$5 - \$3.57 + \$7.50 = \8.93 (if I return \$7.50)
- I would keep: $\$10.71 - \$7 = \$3.71$ (if I return \$7)
- I would keep: $\$10.71 - \$7.50 = \$3.21$ (if I return \$7.50)

Given they sent 71.4% of their endowment, a generous return would be appropriate. The previous receiver returned about 68.6% when I sent 100%.

Returning about \$7.20 would:

- Give them \$8.63 total ($\$5 - \$3.57 + \7.20)
- Leave me with \$3.51
- This rewards their trust well

Actually, let me be a bit more generous to match the cooperative environment. Returning \$7.50:

- Gives them \$8.93 total (a \$3.93 profit on their \$5)
- Leaves me with \$3.21 (also a good profit)
- Return rate: 70% of what I received

...

usage:

```
{
  "input_tokens": 1326,
  "output_tokens": 675,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.57, it became \$10.709999999999999, and you returned \$7.5. Your visible payoff was \$3.2099999999999999.

Your total visible earnings across all rounds are \$26.019999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

- As SENDER: Sent \$4 (got \$8.69), Sent \$5 (got \$10.29)
- As RECEIVER: Returned generously both times

The environment is highly cooperative. Sending the full amount has consistently yielded very good returns for me. My average earnings per round (\$6.50) significantly exceed the \$5 baseline, showing that cooperation pays off.

I'll continue the cooperative strategy and send my full endowment.

```
{"send": 5}
```

Reasoning:

Looking at my history:

- Round 1 (SENDER): Sent \$4, got back \$8.69 - great outcome
- Round 2 (RECEIVER): Got \$9.33, returned \$6.50
- Round 3 (SENDER): Sent \$5, got back \$10.29 - excellent outcome
- Round 4 (RECEIVER): Got \$10.71, returned \$7.50

The pattern is very clear - I've experienced highly cooperative behavior. When I've been the sender and sent significant amounts, I've gotten generous returns. When I've been the receiver, I've also returned generously.

My cumulative earnings are \$26.02 across 4 rounds, averaging \$6.50 per round, which is well ab

ove the \$5 baseline if I had just kept everything.

The optimal strategy in this cooperative environment is to continue sending high amounts:

1. It creates value through the 3x multiplier
2. Past experience shows I get good returns
3. It builds my reputation for future rounds

I should send \$5 again to maximize value creation and maintain my cooperative reputation.

```
Usage: {  
  "input_tokens": 1678,  
  "output_tokens": 377,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T09:22:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$4.78, that is 95.6% of their total endowment. You received \$14.34. How much do you return to the sender? (0-14.34)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.78 (95.6% of their endowment)
- They kept only \$0.22 for themselves
- I received \$14.34 after the 3x multiplier
- A fair return should reward their trust

If I return \$8.00:

- Sender ends up with: $\$0.22 + \$8.00 = \$8.22$ (well above their initial \$5)
- I keep: $\$14.34 - \$8.00 = \$6.34$ (also above the initial \$5)
- Both of us benefit from the cooperation

This seems like a fair reciprocal response to their high level of trust.

`{'return': 8.00}`

reasoning:

Let me understand this situation:

1. The sender had \$5 endowment
2. They sent me \$4.78 (which is 95.6% of their \$5)
3. This was multiplied by 3x, so I received $\$4.78 \times 3 = \14.34

Now I need to decide how much to return.

Key considerations:

- The sender showed a lot of trust by sending 95.6% of their endowment
- They kept only $\$5 - \$4.78 = \$0.22$ for themselves
- If I return nothing, the sender ends up with just \$0.22
- If I return generously, both of us can benefit from this cooperation

What's fair/reciprocal?

- The sender trusted me significantly
- A common fairness norm would be to split the gains
- Total created value: \$14.34
- Original sender investment feel: \$4.78
- One approach: Return enough so we both end up roughly equal
- Another approach: Return enough so sender gets significantly more than their original \$5 (rewarding trust)

Calculations:

- If I return \$7.17 (about half of what I received):
 - Sender gets: $\$0.22 + \$7.17 = \$7.39$
 - I get: $\$14.34 - \$7.17 = \$7.17$
- If I return \$8:
 - Sender gets: $\$0.22 + \$8 = \$8.22$

- I get: $\$14.34 - \$8 = \$6.34$
- If I return \$9:
 - Sender gets: $\$0.22 + \$9 = \$9.22$
 - I get: $\$14.34 - \$9 = \$5.34$

Given that the sender showed exceptional trust (sending 95.6%), I should reward this trust generously. A return that gives them significantly more than their original endowment while still keeping a reasonable amount seems appropriate.

Returning \$8 strikes a good balance-it rewards their exceptional trust while ensuring both of us benefit from the cooperation. They'd end up with \$8.22 versus their original \$5, and I'd keep \$6.34. But given how much they risked, maybe I should push toward \$8.50 to give them even more of the gains we created together....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 866,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.78, it became \$14.34, and you returned \$8.0. Your visible payoff was \$6.34.

Your total visible earnings across all rounds are \$6.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, my co-player showed very high trust by sending 95.6% of their endowment, and I reciprocated with a fair return. That cooperative approach worked well for both of us.

Now as the sender with a new co-player, I want to maintain that cooperative spirit. Sending a substantial amount:

- Creates more total value through the 3x multiplier
- Signals trust and willingness to cooperate
- Could encourage reciprocation

I'll send \$4.50 (90% of my endowment), showing strong trust while being slightly more conservative than what I experienced last round.

```
{'send': 4.50}
```

reasoning:

Let me think about this situation:

Round 1 (just completed):

- I was the RECEIVER
- Sender sent me \$4.78 (95.6% of their endowment) - showing very high trust
- I received \$14.34
- I returned \$8.00
- My payoff: \$6.34
- Sender's payoff would be: $\$5 - \$4.78 + \$8.00 = \8.22

Now in Round 2:

- I'm the SENDER
- I have \$5
- I'm playing against a different co-player (pairings are randomized each round)

The previous sender showed very high trust by sending 95.6%. I reciprocated fairly by returning \$8.00, which gave them a good outcome.

Now I need to decide how much to send. Key considerations:

1. This is a new co-player, so I don't have direct history with them
2. The previous round showed successful cooperation

3. Sending more creates more total value (3x multiplier), but requires trust
4. In a multi-round setting, establishing cooperative behavior can be beneficial

Given that:

- The previous sender sent 95.6% (\$4.78)
- That worked out well for both of us
- I should reciprocate the level of trust shown to me

I think I should send a similar high amount to signal cooperation and trust. Sending around \$4.50-\$4.80 would show similar trust to what I experienced.

Let me send \$4.50 (90% of my endowment). This shows strong trust while being slightly more conservative than the 95.6% I received.

Actually, thinking about it more - since I was treated with high trust and it worked out well, I should pay it forward. Sending \$4.50-\$4.75 seems appropriate.

...

usage:

```
{
  "input_tokens": 824,
  "output_tokens": 591,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.5, it became \$13.5, and you saw them return \$7.41 to you. Your visible payoff was \$7.91.

Your total visible earnings across all rounds are \$14.25.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed good trust by sending \$3.80 (76% of their endowment)
- They kept \$1.20 for themselves
- I received \$11.40
- I should reciprocate their trust fairly

Comparing to previous rounds:

- In Round 1, when a sender sent 95.6%, I returned about 56% of what I received
- In Round 2, when I sent 90%, my co-player returned about 55% of what they received

For this sender who sent 76%, a fair return would be around 55-60% of what I received.

$\$11.40 \times 0.57 \approx \6.50

If I return \$6.50:

- Sender ends with: $\$1.20 + \$6.50 = \$7.70$ (good gain)
- I keep: $\$11.40 - \$6.50 = \$4.90$ (reasonable)
- Both benefit from the cooperation

{ 'return': 6.50 }

reasoning:

Let me analyze this situation:

Round 2 recap:

- I was the sender and sent \$4.50
- It became \$13.50 for the receiver
- They returned \$7.41 to me
- My payoff: $\$5 - \$4.50 + \$7.41 = \7.91

Round 3 current situation:

- I'm the receiver
- Sender sent \$3.80 (76% of their endowment)

- I received \$11.40 (approximately, due to floating point)
- They kept \$1.20

This sender showed good trust by sending 76% of their endowment. Let me think about a fair return:

If I return \$6.00:

- Sender ends up with: $\$1.20 + \$6.00 = \$7.20$
- I keep: $\$11.40 - \$6.00 = \$5.40$
- Both benefit, sender gains more

If I return \$7.00:

- Sender ends up with: $\$1.20 + \$7.00 = \$8.20$
- I keep: $\$11.40 - \$7.00 = \$4.40$
- Sender gains significantly more

In Round 1, when I received \$14.34 from a sender who sent \$4.78 (95.6%), I returned \$8.00.

- That sender ended with \$8.22 (from \$0.22)
- I kept \$6.34
- Return ratio: $\$8.00/\$14.34 = 55.9\%$

In Round 2, when I sent \$4.50 and it became \$13.50, they returned \$7.41

- Return ratio: $\$7.41/\$13.50 = 54.9\%$

For consistency, if I use a similar return ratio (around 55%):

- $\$11.40 \times 0.55 = \6.27

Let me round to \$6.50 to be slightly more generous:

...

usage:

```
{
  "input_tokens": 1120,
  "output_tokens": 820,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.8, it became \$11.399999999999999, and you returned \$6.5. Your visible payoff was \$4.899999999999999.

Your total visible earnings across all rounds are \$19.15.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

- Round 1: As receiver, someone sent high trust (95.6%), I reciprocated fairly
- Round 2: I sent \$4.50 (90%), got back \$7.41 - good cooperation
- Round 3: As receiver, returned fairly when sent \$3.80

The pattern shows that high trust tends to be reciprocated with fair returns (around 55% return rate). My cooperative strategy has been working well - averaging over \$6 per round.

I'll continue with a strong but slightly varied approach. Sending \$4.00 (80%):

- Shows substantial trust
- Creates \$12.00 for the receiver through the 3x multiplier
- Should encourage reciprocation based on the established pattern
- Slightly more conservative than my last sending round to manage risk

```
{'send': 4.00}
```

reasoning:

Let me analyze my history:

Round 1 (RECEIVER): Sender sent \$4.78 (95.6%), I received \$14.34, returned \$8.00, kept \$6.

Round 2 (SENDER): I sent \$4.50 (90%), became \$13.50, received back \$7.41, ended with \$7.91
 Round 3 (RECEIVER): Sender sent \$3.80 (76%), I received \$11.40, returned \$6.50, kept \$4.90

Now Round 4, I'm the SENDER again with a new co-player.

Looking at the pattern:

- When I sent \$4.50 (90%), I got back \$7.41 from \$13.50 (about 55% return rate)
- When someone sent \$4.78 (95.6%), I returned \$8.00 from \$14.34 (about 56% return rate)
- When someone sent \$3.80 (76%), I returned \$6.50 from \$11.40 (about 57% return rate)

The return rates seem fairly consistent around 55-57%. This suggests cooperation is being reciprocated.

I've been doing well with high cooperation levels. My total earnings are \$19.15 across 3 rounds, averaging about \$6.38 per round, which is well above the baseline of \$5.

Given this positive experience with cooperation, I should continue sending a substantial amount. I'll send \$4.00 (80%), which is:

- Still showing significant trust
- Creating good value through the 3x multiplier (\$12.00 received)
- Likely to be reciprocated based on the pattern
- Slightly more conservative than my previous \$4.50 to diversify my strategy a bit

If they return around 55%, I'd get back about \$6.60, ending with \$1.00 + \$6.60 = \$7.60, which would be good.

usage:

```
{
  "input_tokens": 1488,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$4.58 to you. Your visible payoff was \$5.58.

Your total visible earnings across all rounds are \$24.729999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.73 (94.6% of their endowment)
- They kept only \$0.27 for themselves
- I received \$14.19
- This is very similar to Round 1, where someone sent 95.6%

In Round 1, when faced with similarly high trust, I returned \$8.00 out of \$14.34. That worked well as a fair reciprocation.

For this round with \$14.19 received, returning \$8.00:

- Sender ends with: \$0.27 + \$8.00 = \$8.27 (strong return on their trust)
- I keep: \$14.19 - \$8.00 = \$6.19
- Both benefit significantly from the cooperation

Consistency with my previous fair reciprocation of high trust seems appropriate.

```
{'return': 8.00}
```

Reasoning:

Let me analyze this situation:

Round 5 - I'm the RECEIVER

- Sender sent \$4.73 (94.6% of their endowment) - very high trust!
- They kept only \$0.27
- I received \$14.19 after the 3x multiplier

This is similar to Round 1 where someone sent \$4.78 (95.6%) and I returned \$8.00.

The sender is showing exceptional trust, almost sending everything. I should reciprocate fairly.

If I return \$8.00:

- Sender ends with: $\$0.27 + \$8.00 = \$8.27$ (good gain from initial \$5)
- I keep: $\$14.19 - \$8.00 = \$6.19$
- Both benefit, sender gets rewarded for high trust

This seems consistent with my Round 1 behavior where I also returned \$8.00 for a similar high-trust send.

Actually, let me think about this more carefully. The sender sent 94.6%. In Round 1, when sender sent 95.6%, I returned \$8.00 out of \$14.34, which is about 56%.

$\$14.19 \times 0.56 \approx \7.95

So around \$8.00 seems right. Let me stick with that.

```
Usage: {  
  "input_tokens": 1831,  
  "output_tokens": 528,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Sender - dyad_3

Timestamp: 2026-08-12T09:22:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_5"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_5": "trustee"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.77, that is 75.4% of their total endowment. You received \$11.31. How much do you return to the sender? (0-11.31)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.77 (75.4% of their endowment)
- They currently have: \$5 - \$3.77 = \$1.23
- I received: \$11.31
- Total money created: \$1.23 + \$11.31 = \$12.54

The sender took a significant risk and was quite generous. I want to reciprocate that trust while ensuring we both benefit from cooperation.

If I return \$5.50:

- Sender ends with: \$1.23 + \$5.50 = \$6.73
- I end with: \$11.31 - \$5.50 = \$5.81
- Both of us gain compared to the initial \$5, rewarding the sender's trust

This seems like a fair reciprocation that rewards their generosity while still allowing both of us to benefit.

{'return': 5.50}

reasoning:

Let me think about this situation carefully.

I'm the RECEIVER in Round 1.

- The sender started with \$5
- They sent me \$3.77 (which is 75.4% of their \$5 endowment)
- I received \$11.31 (which is \$3.77 × 3)
- I need to decide how much to return (between \$0 and \$11.31)

The sender showed significant trust by sending 75.4% of their endowment. Let me think about fair reciprocity:

If the sender sent \$3.77:

- They kept: \$5 - \$3.77 = \$1.23
- I received: \$11.31

For a fair split of the total money created:

- Total money now = \$1.23 (sender kept) + \$11.31 (I received) = \$12.54
- Fair split would be roughly \$6.27 each
- So I should return enough so the sender has ~\$6.27
- Sender currently has \$1.23, so I'd need to return about \$5.04 to get them to \$6.27
- That would leave me with \$11.31 - \$5.04 = \$6.27

Alternatively, I could think about matching their generosity:

- They sent 75.4% of their endowment
- If I return 75.4% of what I received: $0.754 \times \$11.31 = \8.53
- That seems quite generous but might be excessive

Another approach - proportional reciprocity:

- They gave me \$3.77 out of their \$5 (75.4%)
- A reasonable return might be to give back around half of what I received, ensuring both benefit
- Half would be \$5.66

Let me think about common strategies in trust games:

1. Return nothing (free-ride)
2. Return what was sent (break even for sender)
3. Return half of what received (both benefit equally from the multiplier)
4. Return enough to equalize final payoffs

Given their generosity, returning \$5-6 seems appropriate. This would give the sender roughly \$6-7 while leaving me with similar amounts, reflecting mutual gain from the exchange without being excessive....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1020,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.77, it became \$11.31, and you returned \$5.5. Your visible payoff was \$5.8100000000000005.

Your total visible earnings across all rounds are \$5.8100000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, I saw cooperation work well when the sender showed significant trust by sending 75.4% of their endowment, and I reciprocated fairly.

I'll adopt a similar cooperative strategy with my new co-player. Sending a substantial amount can maximize joint gains if they reciprocate.

```
{'send': 3.75}
```

reasoning:

I'm now the SENDER in Round 2.

Let me think about this strategically:

1. In the previous round as RECEIVER, I saw someone send \$3.77 (75.4% of their endowment), showing significant trust
2. I returned \$5.50, which gave them \$6.73 total - a good return on their trust

Now I'm the SENDER. What should I consider?

- The person sending \$3.77 showed that cooperation can work well
- I reciprocated fairly, returning enough that they profited significantly
- If my current co-player is similarly cooperative, sending a substantial amount could benefit both of us

Strategy considerations:

- Sending nothing is safe but generates no surplus
- Sending a moderate amount (around \$2-3) tests the waters
- Sending a larger amount (like \$3.77 that I saw) shows trust and can maximize joint gains if reciprocated

Given that:

- I saw cooperation work in the last round
- The game seems to reward trust
- I benefited from someone trusting me and reciprocated fairly
- There may be communication noise (mentioned in instructions), but the pattern of cooperation seemed clear

I think I should send a substantial amount to show trust and potentially establish a cooperative pattern. I'll send around \$3.50-\$4.00.

Let me send \$3.50 (70% of endowment) - this shows significant trust similar to what I received, but slightly less aggressive to be somewhat cautious given this is a new pairing.

Actually, let me reconsider. The previous sender sent 75.4% (\$3.77). I reciprocated well. If I want to encourage cooperation, I should send a similar proportion. Let me send \$3.75 (75% of my endowment) to match that level of trust.

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 494,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$5.66 to you. Your visible payoff was \$6.91.

Your total visible earnings across all rounds are \$12.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience:

- Round 2: I sent \$3.75 (75%), received \$5.66 back, ended with \$6.91 - a good outcome
- Round 1: As receiver, cooperation was rewarded

The cooperative approach has been working well. I'm playing with a new co-player now, so I'll continue signaling trust with a substantial send.

```
{'send': 3.75}
```

reasoning:

Let me analyze the previous round:

- I sent \$3.75 (75% of my endowment)
- It became \$11.25
- They returned \$5.66
- My payoff was $\$5 - \$3.75 + \$5.66 = \6.91

This worked out well! The receiver reciprocated reasonably (returning about 50% of what they received), and I gained \$1.91 above my initial endowment.

This is a new co-player this round. Based on my successful cooperative strategy so far:

- Round 1 as receiver: sender sent 75.4%, I returned fairly
- Round 2 as sender: I sent 75%, receiver returned fairly

The pattern of cooperation seems to be working. I should continue with a similar high-trust strategy. I'll send around 75% again to signal cooperation and trust.

\$3.75 worked well last time. I could send the same amount or similar.

usage:

```
{
```



```
"input_tokens": 1074,  
"output_tokens": 323,  
"reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$6.34 to you. Your visible payoff was \$7.59.

Your total visible earnings across all rounds are \$20.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$4.09, so you received \$12.27. How much do you return? (0-12.27)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.09 (81.8% of their endowment)
- They currently have: \$5 - \$4.09 = \$0.91
- I received: \$12.27
- Total created: \$13.18

This is even more trust than I saw in Round 1 (when sender sent 75.4%). I want to reward this high level of cooperation appropriately.

If I return \$6.50:

- Sender ends with: \$0.91 + \$6.50 = \$7.41 (gain of \$2.41)
- I end with: \$12.27 - \$6.50 = \$5.77 (gain of \$0.77)

Both of us benefit, and the sender's exceptional trust is well-rewarded.

{'return': 6.50}

reasoning:

Let me analyze this situation:

- The sender sent \$4.09 out of \$5, which is 81.8% of their endowment - very high trust!
- They now have: \$5 - \$4.09 = \$0.91
- I received: \$12.27
- Total money in the system: \$0.91 + \$12.27 = \$13.18

This sender showed even more trust than my previous partners. I want to reciprocate appropriately.

Looking at my previous behavior:

- Round 1: As receiver, got \$11.31, returned \$5.50 (about 48.6%)
- The sender had sent 75.4%
- That seemed fair and both parties benefited

This sender sent 81.8%, even more generous. I should reciprocate with a good return.

If I return \$6.50:

- Sender ends with: \$0.91 + \$6.50 = \$7.41
- I end with: \$12.27 - \$6.50 = \$5.77
- Sender gains \$2.41 over initial \$5
- I gain \$0.77 over initial \$5

Both benefit, and the sender's higher risk/trust is rewarded with higher gains.

Actually, let me think about this differently. The sender sent 81.8% which is very high. In Round 1, sender sent 75.4% and I returned 48.6% of what I received.

For 81.8% send rate, maybe I should return around 50-53% to reward the extra trust?

$\$12.27 * 0.52 = \6.38

If I return \$6.40:

- Sender ends with: \$0.91 + \$6.40 = \$7.31

- I end with: \$12.27 - \$6.40 = \$5.87

This seems fair - both gain, sender gets rewarded for high trust.

...

usage:

```
{
  "input_tokens": 1310,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.09, it became \$12.27, and you returned \$6.5. Your visible payoff was \$5.77.

Your total visible earnings across all rounds are \$26.080000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

- As sender: Sent \$3.75 twice, received good returns (\$5.66 and \$6.34)
- As receiver: Responded fairly to cooperative senders

The high-trust strategy continues to work well across different co-players. I'll maintain this approach.

{'send': 3.75}

Reasoning:

Let me review my history:

- Round 1: Receiver - sender sent \$3.77, I returned \$5.50, payoff \$5.81
- Round 2: Sender - I sent \$3.75, got back \$5.66, payoff \$6.91
- Round 3: Sender - I sent \$3.75, got back \$6.34, payoff \$7.59
- Round 4: Receiver - sender sent \$4.09, I returned \$6.50, payoff \$5.77

Total visible earnings: \$26.08

My sending strategy has been working well - sending around \$3.75 (75%) has led to good returns. In Round 4, I saw an even higher send amount (\$4.09, 81.8%) and I rewarded it well.

Now I'm facing a new co-player as sender. The cooperative strategy has been consistently working across different opponents. I should continue with a similar high-trust signal.

I'll send \$3.75 again, which has proven successful.

Usage: {

```
"input_tokens": 1637,
"output_tokens": 324,
"reasoning_tokens": 0
```

}

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T09:23:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_1",
    "Agent_5"
  ],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
```

```

    "Agent_5": "trustee"
  },
  "trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.3, that is 46.0% of their total endowment. You received \$6.899999999999995. How much do you return to the sender? (0-6.899999999999995)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$2.3 (46% of their endowment)
- I received \$6.9 after the 3x multiplier
- The sender currently has \$2.7 remaining

I want to:

1. Reward the sender's trust so they benefit from sending
2. Split the gains fairly
3. Establish myself as trustworthy for future rounds

If I return \$3.0:

- Sender ends with: $\$2.7 + \$3.0 = \$5.7$ (a gain of \$0.7)
- I end with: $\$6.9 - \$3.0 = \$3.9$

This seems fair - the sender profits from trusting me, and we both benefit from the multiplier effect.

```
'''json
{"return": 3.0}
'''
```

reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.3 (which is 46% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$6.9 ($2.3 \times 3 = 6.9$)
4. I need to decide how much to return (between \$0 and \$6.9)

The sender's current situation:

- Started with \$5
- Sent \$2.3
- Currently has: $\$5 - \$2.3 = \$2.7$

If I return nothing:

- Sender ends with: \$2.7
- I end with: \$6.9

The sender showed trust by sending nearly half their endowment. A fair approach would be to:

1. Ensure the sender doesn't lose money
2. Split the gains reasonably

Total money in the system now: $\$2.7$ (sender has) + $\$6.9$ (I have) = $\$9.6$

This started from two \$5 endowments = \$10 total, so we have \$9.6 total.

Wait, let me recalculate:

- Sender started with \$5, sent \$2.3, has \$2.7 left
- I received \$6.9 (the \$2.3 multiplied by 3)
- Total in system: $\$2.7 + \$6.9 = \$9.6$

If we split equally: each person would get \$4.8

- For sender to have \$4.8, they need: $\$4.8 - \$2.7 = \$2.1$ returned
- For me to have \$4.8, I'd keep: $\$6.9 - \$2.1 = \$4.8$?

But equal split might not be the best strategy. The sender took a risk by sending money. A common approach in trust games is to:

reward their trust by ensuring they come out ahead of their initial \$5, while also sharing in the multiplier gains. Let me test a few return amounts: returning \$3.0 gives them \$5.7 total (a \$0.7 gain) while I keep \$3.9; returning \$2.5 gives them \$5.2 (a \$0.2 gain) while I keep \$4.4; returning \$2.1 would give them exactly \$4.8 and me \$4.8.

A return around \$3.0-\$3.3 seems optimal-it meaningfully rewards their trust while still letting me benefit from the multiplier effect. At \$3.0, they gain \$0.7 and I keep \$3.9, which feels like a fair split that demonstrates trustworthiness.

The sender's 46% send rate suggests moderate confidence in me, so I should reciprocate proportionally.

usage:

```
{
  "input_tokens": 539,
  "output_tokens": 1314,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.3, it became \$6.899999999999995, and you returned \$3.0. Your visible payoff was \$3.899999999999995.

Your total visible earnings across all rounds are \$3.899999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm now the sender with a new co-player
- In the previous round, I experienced a sender who sent 46% and I returned fairly
- I want to establish trust with this new partner while being reasonably cautious

Sending \$2.5 (50% of my endowment) seems like a good strategy:

- It shows significant trust
- It's a clear, fair starting point
- If my co-player reciprocates fairly, we both benefit from the 3x multiplier
- It's not overly risky with an unknown partner

```
'''json
{"send": 2.5}
'''
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.3 (46% of their endowment)
- I received \$6.9
- I returned \$3.0
- My payoff was \$3.9

Now I'm playing against a different agent (since it says "current co-player" and agents are randomly paired each round).

I need to decide how much to send. Key considerations:

1. I don't know anything about this new co-player's behavior
2. The sender in Round 1 sent 46% (\$2.3), which was a reasonable trust signal
3. I returned about 43.5% of what I received (\$3/\$6.9), which was fair
4. I want to build trust but also protect myself since I don't know this opponent

Strategies:

- Sending \$0 would be overly defensive and wouldn't build any relationship
- Sending \$5 (100%) would be too risky with an unknown partner
- Sending around \$2-3 seems reasonable as it shows trust without being reckless

Given that the previous sender sent \$2.3 (46%), I could mirror that behavior. This would be consistent and shows I'm willing to trust at a similar level to what I've experienced.

Alternatively, I could send slightly more to show goodwill, or slightly less to be more cautious.

I think sending \$2.5 (50%) is a good middle ground:

- It's a clear half of my endowment, easy to interpret
- It shows significant trust
- If they return fairly (around 40-45% of \$7.5 = \$3-3.4), we both profit
- It's not too risky

Let me calculate the outcomes:

- If I send \$2.5, they receive \$7.5
- I keep \$2.5

...

usage:

```
{
```

```
"input_tokens": 870,  
"output_tokens": 665,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$6.78 to you. Your visible payoff was \$9.280000000000001.

Your total visible earnings across all rounds are \$13.18.

This round, you are the RECEIVER against your current co-player. They sent you \$4.17, so you received \$12.51. How much do you return? (0-12.51)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.17 (83.4% of their endowment)
- I received \$12.51 after the 3x multiplier
- The sender currently has only \$0.83 remaining

This is very high trust that deserves to be rewarded generously.

If I return \$6.0:

- Sender ends with: $\$0.83 + \$6.0 = \$6.83$ (profit of \$1.83)
- I end with: $\$12.51 - \$6.0 = \$6.51$ (profit of \$6.51)

This gives us nearly equal final payoffs and rewards the sender's exceptional trust with a solid profit.

```
'''json  
{"return": 6.0}  
'''
```

reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- The sender sent \$4.17 (83.4% of their \$5 endowment) - this is very high trust!
- I received \$12.51 (after 3x multiplier)
- The sender currently has $\$5 - \$4.17 = \$0.83$ remaining

This sender showed enormous trust by sending 83.4% of their endowment. I need to reward this trust generously.

Let me think about what's fair:

- Total in the system: $\$0.83$ (sender has) + $\$12.51$ (I have) = $\$13.34$
- If we split evenly: $\$6.67$ each

If I return \$5.84:

- Sender ends with: $\$0.83 + \$5.84 = \$6.67$
- I end with: $\$12.51 - \$5.84 = \$6.67$
- Perfect 50/50 split

But given they showed such high trust (83.4%), I might want to be even more generous to:

1. Reward their exceptional trust
2. Encourage continued cooperation
3. Establish myself as highly trustworthy

However, I also saw in round 2 that when I sent \$2.5 (50%), I got back \$6.78, which gave me a payoff of \$9.28. That was very generous from my co-player then (they returned more than 90% of what they received!).

Let me recalculate for this round:

- If I return \$6:
 - Sender: $\$0.83 + \$6 = \$6.83$
 - Me: $\$12.51 - \$6 = \$6.51$
 - Almost equal, slight advantage to sender

- If I return \$6.5:
- Sender: $\$0.83 + \$6.5 = \$7.33$
- Me: $\$12.51 - \$6.5 = \$6.01$
- Sender gets more, rewarding their trust

Given the extreme trust shown here, I should return enough to give them a meaningful advantage while still being reasonable. Returning around \$6 seems appropriate-it creates nearly equal outcomes with a slight edge to the sender who took the bigger risk...

usage:

```
{
  "input_tokens": 1154,
  "output_tokens": 955,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.17, it became \$12.51, and you returned \$6.0. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$19.689999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

- Round 2: When I sent \$2.5 (50%), I received a very generous return of \$6.78
- Round 3: When someone sent me \$4.17 (high trust), I reciprocated with \$6.0

The game seems to reward trust. I'll send \$3.0 (60% of my endowment), which shows strong trust while being measured with a new co-player.

```
'''json
{"send": 3.0}
'''
```

reasoning:

Looking at my history:

- Round 1: RECEIVER - received from sender who sent \$2.3, I returned \$3.0 fairly
- Round 2: SENDER - I sent \$2.5, receiver returned \$6.78 (very generous!)
- Round 3: RECEIVER - received from sender who sent \$4.17 (very high trust), I returned \$6.0

Current situation:

- I'm the SENDER with a new co-player
- Total earnings so far: \$19.69

My experience shows:

- When I sent \$2.5 in Round 2, I got a very generous return of \$6.78
- When others showed high trust to me, I reciprocated fairly
- This game seems to reward trust

I should send a significant amount to show trust. Given that:

- Round 2 sender sent 46% and I reciprocated
- Round 3 sender sent 83.4% and I reciprocated generously
- When I sent 50% in Round 2, I got excellent returns

I think sending around \$3.0 (60%) would be a good balance - showing strong trust without being reckless with an unknown partner.

Actually, let me think about this more strategically. I've seen that:

- Lower trust (46%) got fair returns
- Medium trust (50% from me) got very generous returns
- High trust (83.4%) got generous returns

Sending \$3.0 seems like a good middle-high trust level that should encourage reciprocation

usage:

```
{
  "input_tokens": 1461,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.22 to you. Your visible payoff was \$7.22.

Your total visible earnings across all rounds are \$26.909999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.84, so you received \$11.52. How much do you return? (0-11.52)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender showed high trust by sending \$3.84 (76.8% of their endowment)
- I received \$11.52 after the 3x multiplier
- The sender currently has only \$1.16 remaining

This level of trust deserves a generous return.

If I return \$6.0:

- Sender ends with: $\$1.16 + \$6.0 = \$7.16$ (profit of \$2.16)
- I end with: $\$11.52 - \$6.0 = \$5.52$

This rewards their high trust with a good profit and keeps our final payoffs relatively balanced.

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Looking at this situation:

- The sender sent \$3.84, which is 76.8% of their endowment - very high trust
- I received \$11.52 after the 3x multiplier
- The sender currently has $\$5 - \$3.84 = \$1.16$ remaining

This is high trust that should be rewarded well.

Let me consider different return amounts:

If I return \$5.5:

- Sender ends with: $\$1.16 + \$5.5 = \$6.66$ (profit of \$1.66)
- I end with: $\$11.52 - \$5.5 = \$6.02$

If I return \$6.0:

- Sender ends with: $\$1.16 + \$6.0 = \$7.16$ (profit of \$2.16)
- I end with: $\$11.52 - \$6.0 = \$5.52$

Given the high trust (76.8%), I should return generously. \$6.0 seems fair and rewards their trust well.

Usage: {

```
"input_tokens": 1718,
"output_tokens": 421,
"reasoning_tokens": 0
}
```

GAME - Agent_2 - Sender - dyad_4

Timestamp: 2026-08-12T09:23:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {


```

"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 6,
"opponent_id": "Agent_7",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_7"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the **SENDER**:

- You receive \$5 each round
- You decide how much to **SEND** to the **RECEIVER** (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to **RETURN**
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the **RECEIVER**:

- Sender receives \$5
- Sender decides how much to **SEND** to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to **RETURN**
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being reasonably cautious since I have no history with my co-player.

The multiplication factor of 3x creates an opportunity for mutual benefit. If I send a moderate amount, we can both potentially end up better off than the initial \$5.

I'll send **\$3**, which gives my co-player \$9 to work with. This shows trust and willingness to cooperate. If they return approximately \$3-4, we'll both be in a good position, and it could establish a positive pattern for future rounds.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit
2. The receiver then decides how much to return
3. There's communication noise mentioned - amounts may appear different than actually sent
4. This is a multi-agent setting with 8 agents total, pairs rotate
5. This is the first round, so I have no history with any player yet

Since this is the first round and I have no information about other players' behavior, I need to consider:

- Sending \$0 guarantees I keep \$5, but no potential for mutual gain
- Sending everything (\$5) maximizes the total pot (\$15 for receiver) but is risky
- A moderate amount balances risk and potential reward

In trust games, common strategies include:

- Sending about half as a signal of conditional cooperation
- The tripling effect means even partial cooperation can benefit both parties

Let me consider sending \$3:

- I send \$3, receiver gets \$9
- If receiver returns \$3, I end up with \$5 (same as starting)
- If receiver returns \$4-5, I'm better off
- If receiver returns less than \$3, I'm worse off
- This shows trust while keeping some insurance

Alternatively, sending \$2.5:

- I send \$2.5, receiver gets \$7.5
- If receiver returns \$2.5, I end up with \$5
- If receiver returns \$3+, I'm better off
- Less risk than \$3

Or sending \$2:

- I send \$2, receiver gets \$6
- If receiver returns \$2, I end up with \$5
- Lower risk, but also lower potential reward

Given this is the first round and I want to establish a cooperative pattern, I'll send \$3.

This shows trust and willingness to cooperate while not being completely vulnerable.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.81 to you. Your visible payoff was \$

4.8100000000000005.

Your total visible earnings across all rounds are \$4.8100000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$3.95, so you received \$11.850000000000001. How much do you return? (0-11.850000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.95 (79% of their endowment), showing significant trust
- I received \$11.85 total
- They currently have only \$1.05 left

I want to:

1. Reward their trust and cooperation
2. Ensure they end up better than their initial \$5
3. Set a positive tone for potential future interactions

If I return \$5.50:

- They end up with: $\$1.05 + \$5.50 = \$6.55$
- I end up with: $\$11.85 - \$5.50 = \$6.35$

Both of us gain from cooperation, and this reciprocates their high level of trust appropriately.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the SENDER
- I sent \$3.00
- It became \$9.00 for the receiver
- They returned \$2.81
- My payoff: $\$5 - \$3 + \$2.81 = \4.81

Round 2:

- I am now the RECEIVER
- My co-player (a different agent) sent me \$3.95
- I received $\$3.95 \times 3 = \11.85
- I need to decide how much to return

Important considerations:

1. The sender gave me \$3.95 out of their \$5 - that's 79% of their endowment, showing high trust
2. If I return nothing, I keep \$11.85 and they end up with $\$5 - \$3.95 = \$1.05$
3. I should return enough to make cooperation worthwhile for them
4. There's communication noise mentioned, so amounts may differ slightly

What would be fair?

- The sender gave up \$3.95, keeping \$1.05
- I received \$11.85
- If I want them to end up better than their initial \$5, I should return more than \$3.95
- A 50-50 split of total surplus could work: Total = $\$1.05 + \$11.85 = \$12.90$. Half is \$6.45
- If I return \$5.40, sender gets $\$1.05 + \$5.40 = \$6.45$, I get $\$11.85 - \$5.40 = \$6.45$

But let me think about reciprocity:

- In round 1, I sent \$3 (60% of endowment), received \$9, they returned \$2.81 (31% of what they received)
- This round, sender sent \$3.95 (79% of endowment), I received \$11.85
- Using that same 31% return rate would give me \$3.67 to send back

The sender showed significantly more generosity than I did initially, so I should reciprocate with a higher return. Returning \$5 would leave them with \$6.05 total-better than their starting \$5-while I'd keep \$6.85, rewarding their trust and encouraging continued cooperation....

usage:

```
{
  "input_tokens": 793,
  "output_tokens": 977,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.95, it became \$11.850000000000001, and you returned \$5.5. Your visible payoff was \$6.350000000000001.

Your total visible earnings across all rounds are \$11.160000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at Round 3:

I'm paired with a new co-player (pairings are randomized each round), so I have no history with them. My previous experiences showed:

- Round 1: Sent \$3, received \$2.81 back - reasonable cooperation
- Round 2: As receiver, I tried to be generous with a high-trusting sender

For this new pairing, I'll use a similar approach to Round 1 - send a moderate amount that signals cooperation while being reasonably cautious with an unknown partner.

Sending \$3 creates \$9 for them to work with, which gives room for mutual benefit if they reciprocate fairly.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$3, received back \$2.81, payoff \$4.81

Round 2: I was RECEIVER, got \$11.85 from a \$3.95 send, returned \$5.50, payoff \$6.35

Total earnings: \$11.16

Now Round 3: I'm SENDER again with a new co-player (since pairings are randomized each round).

Key considerations:

- I have no history with this particular co-player
- In Round 1, when I sent \$3, I got back \$2.81 (which is reasonable - about 31% return rate)
- In Round 2, when someone sent me \$3.95, I returned \$5.50 (about 46% return rate)
- The game has communication noise, so exact amounts may differ

Since I'm paired with a new player and have no history with them, I should probably use a similar strategy to Round 1 - establish cooperation but be somewhat cautious.

Given that:

- Sending \$3 worked reasonably well in Round 1
- The multiplication factor of 3x creates good mutual benefit potential
- I want to signal cooperation

I'll send \$3 again. This is a moderate amount that shows willingness to cooperate without being overly risky with a new partner.

usage:

```
{
  "input_tokens": 1106,
  "output_tokens": 478,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.52 to you. Your visible payoff was \$8.52.
Your total visible earnings across all rounds are \$19.68.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant**content:**

Looking at this situation:

- My co-player sent \$3.36 (67% of their endowment), showing solid trust
- I received \$10.08 total
- They currently have \$1.64 left

I want to reciprocate their trust fairly. If I return \$5.00:

- They end up with: $\$1.64 + \$5.00 = \$6.64$ (gain of \$1.64)
- I end up with: $\$10.08 - \$5.00 = \$5.08$ (gain of \$0.08)

Both of us benefit from cooperation, and this rewards their trust appropriately.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- My co-player sent \$3.36 (67.2% of their endowment)
- I received \$10.08 total
- They currently have: $\$5 - \$3.36 = \$1.64$ left

My previous round (Round 3):

- I sent \$3, received \$6.52 back - very generous return!

This co-player sent \$3.36, showing good trust. I should reciprocate appropriately.

If I return \$5.00:

- They end up with: $\$1.64 + \$5.00 = \$6.64$
- I end up with: $\$10.08 - \$5.00 = \$5.08$

Both of us gain. This is a fair reciprocation that rewards their trust.

Alternatively, if I return \$5.50:

- They end up with: $\$1.64 + \$5.50 = \$7.14$
- I end up with: $\$10.08 - \$5.50 = \$4.58$

This would be even more generous, similar to how I acted in Round 2.

Given that they sent about 67% of their endowment (compared to 79% in Round 2), I think returning around \$5 is appropriate and fair. It ensures they profit significantly from their trust, and we both end up better off than the initial \$5.

I'll go with \$5.00 as a fair reciprocation.

usage:

```
{
  "input_tokens": 1402,
  "output_tokens": 522,
  "reasoning_tokens": 0
}
```

Message 10 role=user**content:**

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$5.0. Your visible payoff was \$5

.08.

Your total visible earnings across all rounds are \$24.759999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at Round 5:

I'm with another new co-player. My experience so far shows:

- Sending \$3 has worked well in previous rounds, getting returns of \$2.81 and \$6.52
- The cooperative approach has generally been rewarded

I'll continue with the same strategy - send \$3 to signal trust and cooperation while being reasonably cautious with an unknown partner.

```
'''json
{"send": 3}
'''
```

Reasoning:

Round 5, I'm the sender with a new co-player (random pairings each round).

Looking at my history:

- Round 1 (Sender): Sent \$3, got back \$2.81 - decent cooperation
- Round 2 (Receiver): Received \$11.85, returned \$5.50 - I was generous
- Round 3 (Sender): Sent \$3, got back \$6.52 - excellent return!
- Round 4 (Receiver): Received \$10.08, returned \$5.00 - fair reciprocation

So far, sending \$3 has worked well - I got \$2.81 and \$6.52 back in my two sender rounds. The average return has been positive.

For this new pairing, I should continue with a cooperative strategy. Sending \$3 seems to be working as a good middle-ground approach that encourages cooperation.

```
'''json
{"send": 3}
'''
```

Usage: {

```
"input_tokens": 1690,
"output_tokens": 332,
"reasoning_tokens": 0
```

}

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T09:23:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_7"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_2": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no history with this receiver.

If I send money, it gets tripled, creating potential mutual gain. However, the receiver controls how much comes back. The communication noise adds uncertainty.

I'll take a moderately cooperative approach to signal trust while managing risk:

****My decision:****

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return a fair portion, we both benefit. This allows me to learn about their reciprocity while not risking everything in this uncertain first round.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategy.

ategic considerations:

1. I have \$5 to allocate
2. Whatever I send gets tripled
3. The receiver then decides how much to return
4. There's communication noise - amounts may differ slightly from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- This is round 1, so I have no history with this particular receiver
- There's a note about communication noise which could affect trust building

Standard economic reasoning in trust games:

- If I send \$5, it becomes \$15 for the receiver
- A "fair" split might be for them to return \$10, giving us each \$10
- But the receiver might keep everything, leaving me with \$0
- If I send \$0, I guarantee keeping \$5 but miss potential gains

Since this is:

1. Round 1 with no history
2. A multi-round game where reputation/reciprocity could matter across the full run
3. There's communication noise which might complicate trust building

I should consider a moderate opening move that:

- Shows some trust/willingness to cooperate
- Doesn't risk everything
- Allows me to learn about this receiver's behavior
- Sets a cooperative tone

A common strategy in trust games is to send around 50% on the first round as a signal of moderate trust while protecting against complete loss.

I'll send \$2.50 (or \$2 or \$3 since I need to be practical about amounts).

Actually, looking at the format, I should probably stick to whole dollar amounts for simplicity. Let me send \$3 as a cooperative opening move - it's more than half, showing trust, but not everything.

Wait, let me reconsider. The receiver gets \$9 if I send \$3. If they return half (\$4.50), I'd have $\$5 - \$3 + \$4.50 = \6.50 , and they'd have \$4.50. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 698,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.51 to you. Your visible payoff was \$7.51.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$3.8 out of \$5 (76% of their endowment) - showing significant trust
- I received \$11.4 after the 3x multiplier

****My reasoning:****

- In Round 1, when I sent \$3, the receiver returned \$5.51 (about 61% of what they received)

-)
- This sender has shown high trust by sending 76% of their endowment
- I should reciprocate that trust fairly

I'll return an amount that rewards their cooperation while maintaining fairness:

```
'''json
{"return": 7}
'''
```

This gives the sender a total of \$8.2 (\$1.2 kept + \$7 returned) and leaves me with \$4.4. While I keep less, this reciprocates their strong trust and builds cooperation.

reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$5.51 to me
- My payoff: $\$5 - \$3 + \$5.51 = \7.51
- Their payoff: $\$9 - \$5.51 = \$3.49$

So the receiver returned about 61% of what they received. That's quite generous.

Round 2 (current):

- I am RECEIVER
- Sender sent me \$3.8
- I received \$11.4 (tripled amount)
- I need to decide how much to return

Now, there's communication noise, so the amounts might be slightly different from what was actually sent/returned. But I should work with what I see.

The sender sent me \$3.8 out of \$5, which is 76% of their endowment. That's quite trusting.

If I apply reciprocity based on what happened in Round 1:

- The receiver returned about 61% of what they received
- 61% of \$11.4 = \$6.95

But let me think about fairness:

- Sender keeps: $\$5 - \$3.8 = \$1.2$
- Sender receives back from me: X
- Sender total: $\$1.2 + X$

- I keep: $\$11.4 - X$

For equal split of total value:

- Total created: $\$5 + \$11.4 = \$16.4$
- Equal split: \$8.2 each
- So sender should get: \$8.2 total, meaning I return: $\$8.2 - \$1.2 = \$7$

For proportional reciprocity (they sent 76% of endowment):

- Maybe I return a similar proportion of what I received?
- 76% of \$11.4 = \$8.66

Given their high level of trust, I should match that cooperation. Returning around 55-60% of what I received feels right-it's reciprocal without being excessive. I'll return \$6.50, which gives the sender a total of \$7.70 and leaves me with \$4.90. This mirrors the cooperative spirit they showed....

usage:

```
{
  "input_tokens": 780,
  "output_tokens": 878,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You

saw them send \$3.8, it became \$11.399999999999999, and you returned \$7.0. Your visible payoff was \$4.399999999999999.
Your total visible earnings across all rounds are \$11.909999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this new round with a new co-player:

****My experience so far:****

- Round 1: Sent \$3, received \$5.51 back - good cooperation
- Round 2: Received \$11.4, returned \$7 - fair reciprocation

Since pairings are randomized, this is a new partner with no shared history. However, the cooperative pattern I've experienced suggests trust can be rewarded.

I'll send a moderate-to-high amount to signal cooperation while managing risk:

```
'''json
{"send": 3.5}
'''
```

This sends \$3.5 (70% of my endowment), which becomes \$10.50 for the receiver. It's slightly more than my Round 1 send, reflecting confidence from positive past experiences while remaining balanced.

reasoning:

Let me analyze the situation:

****My history:****

- Round 1: I sent \$3, received back \$5.51, payoff = \$7.51
- Round 2: I received \$11.4 (they sent \$3.8), I returned \$7, payoff = \$4.4
- Total earnings: \$11.91

****Analysis:****

- In Round 1, my co-player was generous, returning \$5.51 out of \$9 (61%)
- In Round 2, my co-player sent \$3.8 (76% of endowment), showing high trust, and I reciprocated fairly
- Now I'm facing a new co-player in Round 3

Since this is a multi-agent setting with randomized pairings, this Round 3 co-player is likely a different person than either of my previous partners. I don't have direct history with them.

However, I've seen cooperation pay off:

- When I sent \$3, I got back \$5.51 (net gain of \$2.51)
- When someone sent me \$3.8, I returned fairly

Given the overall cooperative pattern I've observed and participated in, I should continue with a cooperative strategy. I'll send a similar amount to what I did in Round 1, or potentially slightly more given the positive experience.

I'll send \$3 again - it's a balanced approach that:

1. Shows trust and willingness to cooperate
2. Doesn't risk everything
3. Creates opportunity for mutual gain (becomes \$9)
4. Has worked well for me in the past

Actually, given that cooperation has been working well, I could send a bit more. Let me send \$3.5, showing slightly increased trust while still being prudent.

usage:

```
{
  "input_tokens": 1109,
  "output_tokens": 583,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.82 to you. Your visible payoff was \$6.32.

Your total visible earnings across all rounds are \$18.229999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$3.11 out of \$5 (62.2% of their endowment) - showing moderate trust
- I received \$9.33 after the 3x multiplier

****My analysis:****

- In Round 3, when I sent \$3.5, I received back \$4.82 (only 46% of what they received) - less cooperative
- In Round 2, when I received \$11.4, I returned \$7 (61%)
- This sender showed moderate trust, deserving fair reciprocation

I'll return a fair amount consistent with their trust level:

```
'''json
{"return": 5.5}
'''
```

This gives the sender a total of \$7.39 (\$1.89 kept + \$5.50 returned) and leaves me with \$3.83. It's a balanced return that rewards their trust while being fair to both of us.

reasoning:

Let me analyze this situation:

****Round 4:****

- I'm the RECEIVER
- They sent \$3.11 out of \$5 (62.2% of their endowment)
- I received \$9.33 after the 3x multiplier

****Past experience:****

- Round 1: I sent \$3, got back \$5.51 (they returned 61% of \$9)
- Round 2: I received \$11.4, returned \$7 (61% of what I received)
- Round 3: I sent \$3.5, got back \$4.82 (they returned 46% of \$10.5)

The sender this round sent me \$3.11, showing moderate trust (62.2% of their endowment).

In Round 3, when I sent \$3.5, I only got back \$4.82, which was less than half of what they received. That was somewhat disappointing.

For this round, the sender sent \$3.11. I should be fair but also consider that:

1. They showed moderate trust (62.2%)
2. My past returns have been around 61% when receiving
3. I want to maintain cooperation

If I return around 55-60% of \$9.33, that would be:

- 55% = \$5.13
- 60% = \$5.60

This would give them:

- At \$5.13: \$1.89 kept + \$5.13 = \$7.02 total
- At \$5.60: \$1.89 kept + \$5.60 = \$7.49 total

And I would keep:

- At \$5.13: \$4.20
- At \$5.60: \$3.73

I'll return around \$5.5 to be fair and reciprocate their moderate trust.

usage:

```
{
  "input_tokens": 1428,
```

```
"output_tokens": 648,
"reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$5.5. Your visible payoff was \$3.83.

Your total visible earnings across all rounds are \$22.059999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.26, so you received \$6.779999999999999. How much do you return? (0-6.779999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****What happened:****

- The sender sent me \$2.26 out of \$5 (45.2% of their endowment) - showing cautious/modest trust
- I received \$6.78 after the 3x multiplier

****My analysis:****

- This sender showed less trust than previous partners (45% vs 62-76%)
- However, I should still reciprocate fairly to maintain cooperation
- My previous return rates have been around 59-61%

Given their more cautious approach, I'll return a fair but proportionally adjusted amount:

```
'''json
{"return": 3.5}
'''
```

This gives the sender a total of \$6.24 (\$2.74 kept + \$3.50 returned), a gain of \$1.24, and leaves me with \$3.28. It's a balanced return that rewards their trust while acknowledging their more conservative send.

Reasoning:

Looking at this situation:

****What happened:****

- The sender sent me \$2.26 out of \$5 (45.2% of their endowment) - showing modest/cautious trust
- I received \$6.78 after the 3x multiplier

****My historical pattern:****

- Round 2: Received \$11.4, returned \$7 (61% of received)
- Round 4: Received \$9.33, returned \$5.5 (59% of received)

This sender showed less trust than previous senders (only 45% vs 62-76%). I should still reciprocate fairly, but proportionally to their lower trust level.

A return rate around 50-55% seems appropriate given their more cautious approach:

- 50% would be \$3.39
- 55% would be \$3.73
- 60% would be \$4.07

Let me think about what's fair:

- They sent \$2.26, keeping \$2.74
- If I return \$3.5, they'd have \$5.76 total (a gain of \$0.76)
- I'd keep \$3.28

Actually, let me reconsider. Even though they sent less, I should maintain a cooperative stance. Let me return around 50-55%:

If I return \$3.5:

- Sender gets: \$2.74 (kept) + \$3.50 (returned) = \$6.24 total
- I keep: \$6.78 - \$3.50 = \$3.28

This seems balanced - they gain \$1.24, I gain \$3.28. It's fair while still rewarding their trust.

st, even if it was more cautious.

Let me go with \$3.5 (about 52% of what I received).

```
Usage: {
  "input_tokens": 1789,
  "output_tokens": 630,
  "reasoning_tokens": 0
}
```

ROUND 6

Roles: Agent_1=Receiver, Agent_2=Receiver, Agent_3=Receiver, Agent_4=Sender, Agent_5=Sender, Agent_6=Sender, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_6=Sender, Agent_2=Receiver
dyad_2: Agent_7=Sender, Agent_8=Receiver
dyad_3: Agent_4=Sender, Agent_1=Receiver
dyad_4: Agent_5=Sender, Agent_3=Receiver

Parsed game decisions:

dyad_1: Agent_6 sent \$3.5; received \$10.5; Agent_2 returned \$6.5; payoffs Agent_6=\$8, Agent_2=\$4;
noise/visible: sent visible=\$4.47, received visible=\$13.41, returned visible=\$6.3
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 10.5, "min": 0, "raw": 6.5}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}
dyad_2: Agent_7 sent \$3; received \$9; Agent_8 returned \$4; payoffs Agent_7=\$6, Agent_8=\$5;
noise/visible: sent visible=\$2.9, received visible=\$8.7, returned visible=\$4.44
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_4 sent \$4.5; received \$13.5; Agent_1 returned \$6.5; payoffs Agent_4=\$7, Agent_1=\$7;
noise/visible: sent visible=\$4.1, received visible=\$12.3, returned visible=\$5.77
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 13.5, "min": 0, "raw": 6.5}, "sent": {"amount": 4.5, "clamped": false, "max": 5, "min": 0, "raw": 4.5}}
dyad_4: Agent_5 sent \$3.5; received \$10.5; Agent_3 returned \$7; payoffs Agent_5=\$8.5, Agent_3=\$3.5; noise/visible: sent visible=\$3.41, received visible=\$10.23, returned visible=\$6.22
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 10.5, "min": 0, "raw": 7.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

"Agent_1": 37.25,
"Agent_2": 32.75,
"Agent_3": 34.5,
"Agent_4": 34.0,
"Agent_5": 41.0,
"Agent_6": 40.5,
"Agent_7": 34.0,
"Agent_8": 39.5

}

Visible balances after round: {

"Agent_1": 39.67,
"Agent_2": 36.819999999999999,
"Agent_3": 37.45,
"Agent_4": 37.19,
"Agent_5": 40.149999999999999,
"Agent_6": 36.62,
"Agent_7": 31.779999999999998,
"Agent_8": 39.150000000000006

}

LLM CALLS

GAME - Agent_6 - Sender - dyad_1

Timestamp: 2026-08-12T09:23:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [

```

    "Agent_6",
    "Agent_2"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_2": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.25, that is 45.0% of their total endowment. You received \$6.75. How much do you return to the sender? (0-6.75)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.25 (45% of their endowment)
- This was multiplied to \$6.75 for me
- The sender currently has \$2.75 remaining

I want to reward their trust while maintaining a fair distribution. If I return \$3.00:

- The sender ends up with \$5.75 (a gain of \$0.75)
- I keep \$3.75 (a gain of \$3.75)
- Both of us are better off than if no money had been sent

This seems like a reasonable balance that rewards their moderate trust and encourages future cooperation.

```
{"return": 3.00}
```

reasoning:

Let me think through this situation carefully.

I'm the RECEIVER in this round of a trust game.

The sender:

- Started with \$5
- Sent me \$2.25 (which is 45% of their endowment)
- This was multiplied by 3x, so I received \$6.75

Now I need to decide how much to return.

Key considerations:

1. The sender showed moderate trust by sending 45% of their endowment
2. They could have sent nothing, or they could have sent everything
3. If I return nothing, I keep \$6.75 and they keep \$2.75 (\$5 - \$2.25)
4. If I return some amount, we both benefit from the cooperative exchange

Let me think about fair or reasonable return amounts:

- If I return \$3.00: The sender ends up with \$2.75 + \$3.00 = \$5.75 (gain of \$0.75)
- I keep: \$6.75 - \$3.00 = \$3.75
- Total: \$5.75 + \$3.75 = \$9.50
- If I return \$2.25: The sender ends up with \$2.75 + \$2.25 = \$5.00 (break even)
- I keep: \$6.75 - \$2.25 = \$4.50
- Total: \$5.00 + \$4.50 = \$9.50
- If I return \$3.50: The sender ends up with \$2.75 + \$3.50 = \$6.25 (gain of \$1.25)
- I keep: \$6.75 - \$3.50 = \$3.25
- Total: \$6.25 + \$3.25 = \$9.50

Common approaches in trust games:

I could mirror their proportion-returning about 45% of what I received, which would be roughly \$3.04. Alternatively, I could split the cooperative gains equally: the \$4.50 surplus created through the exchange divided in half means returning \$2.25 to give them an even split of the benefit. Or I could aim for a return that ensures they profit from trusting me, somewhere between \$2.25 and \$3.50....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1174,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.25, it became \$6.75, and you returned \$3.0. Your visible payoff was \$3.75.

Your total visible earnings across all rounds are \$3.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In the previous round as receiver, I experienced moderate trust (45% sent) and reciprocated fairly. Now I'm facing a new co-player as sender.

Since this is a new pairing and I want to establish cooperative behavior while managing risk, I'll send a moderate-to-good amount. Sending \$3.00 (60% of my endowment) signals trust and willingness to cooperate, while still keeping some safety margin.

This will give my co-player \$9.00 to work with, providing them good opportunity to reciprocate if they're inclined to cooperate.

```
{"send": 3.00}
```

reasoning:

Let me think about this situation:

In Round 1:

- I was the receiver
- The sender sent \$2.25 (45% of their endowment)
- I received \$6.75 and returned \$3.00
- I kept \$3.75
- The sender ended with $\$2.75 + \$3.00 = \$5.75$

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- I'm playing against a different co-player (since pairings are randomized each round)

The previous sender showed moderate trust by sending 45% of their endowment. I returned about 44% of what I received (\$3.00 out of \$6.75), which gave them a positive return.

Now I need to decide how much to send. Considerations:

1. This is a new opponent, so I don't know their behavior yet
2. In multi-round games with changing partners, establishing a reputation for cooperation can be beneficial
3. The previous sender's moderate trust (45%) seemed reasonable
4. There's communication noise, so amounts might differ slightly

I think I should send a moderate-to-good amount to signal trust and cooperation. Sending around 50-60% seems reasonable:

- \$2.50 (50%) or \$3.00 (60%)

Given that the previous sender sent 45% and I want to be at least as cooperative (and perhaps signal willingness to cooperate), I'll send \$3.00 (60%). This will become \$9.00 for the receiver, giving them good opportunity to reciprocate.

Actually, let me reconsider. The previous interaction went well with a 45% send rate. I could match that or go slightly higher. \$3.00 (60%) shows I'm willing to be cooperative and trust my new partner.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 571,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.42 to you. Your visible payoff was \$8.42.

Your total visible earnings across all rounds are \$12.17.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player showed **maximum trust** by sending their entire \$5 endowment (100%)
- This became \$15 for me
- They currently have \$0

This exceptional level of trust deserves strong reciprocation. If I return \$10.00:

- The sender ends with \$10.00 (doubling their money, gain of \$5.00)
- I keep \$5.00 (gain of \$5.00)
- We both gain equally from our cooperation

This rewards their maximum trust appropriately and creates a fair outcome where both parties benefit substantially.

```
{"return": 10.00}
```

reasoning:

Let me analyze this situation:

Previous round:

- I was the SENDER
- I sent \$3.00 (60% of my endowment)
- My co-player received \$9.00
- They returned \$6.42 (71.3% of what they received)
- I ended with \$8.42

This round:

- I am the RECEIVER
- My current co-player sent \$5.00 (100% of their endowment!)
- I received \$15.00
- They currently have \$0
- I need to decide how much to return (0-15)

This is a very strong show of trust - they sent their entire endowment! This deserves a strong reciprocation.

Let me think about fair outcomes:

- If I return \$7.50, they end with \$7.50 (gain of \$2.50), I keep \$7.50 (gain of \$7.50)
- If I return \$10.00, they end with \$10.00 (gain of \$5.00), I keep \$5.00 (gain of \$5.00) - equal split of total gains
- If I return \$12.00, they end with \$12.00 (gain of \$7.00), I keep \$3.00 (gain of \$3.00)

Given they showed maximum trust by sending everything, I should reciprocate generously. An equal split of total gains seems fair - we'd each gain \$5 from the cooperation. That means returning \$10.00.

Actually, let me reconsider. They sent everything (\$5), which became \$15 for me. If I return \$10, they get \$10 total (doubling their money), and I keep \$5 (gaining \$5). Both of us gain equally from the transaction.

This seems like a fair and generous reciprocation for their maximum trust.

usage:

```
{
  "input_tokens": 1082,
  "output_tokens": 580,
  "reasoning_tokens": 0
}
```

Message 8 role=user**content:**

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$5, it became \$15, and you returned \$10.0. Your visible payoff was \$5.0. Your total visible earnings across all rounds are \$17.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant**content:**

Looking at my experience so far:

- Round 2 (sender): Sent \$3.00 (60%) and received excellent reciprocation (\$6.42)
- Round 3 (receiver): Received maximum trust (\$5.00 sent) and reciprocated generously

The pattern so far shows strong cooperation in this game. Since this is a new co-player, I want to signal trust and encourage cooperation. Given the positive experiences, I'll send a high amount while keeping a small buffer.

Sending \$4.00 (80% of my endowment) demonstrates strong trust and gives my co-player \$12.00 to work with, creating good opportunity for mutual benefit.

```
{"send": 4.00}
```

reasoning:

Let me analyze the situation:

Previous rounds:

1. Round 1: RECEIVER - saw \$2.25 sent, received \$6.75, returned \$3.00, kept \$3.75
2. Round 2: SENDER - sent \$3.00, became \$9.00, saw \$6.42 returned, payoff \$8.42
3. Round 3: RECEIVER - saw \$5.00 sent, received \$15.00, returned \$10.00, kept \$5.00

Total visible earnings: \$17.17

Now Round 4: I'm the SENDER against a new co-player.

Key observations:

- In Round 2, I sent \$3.00 (60%) and got very good reciprocation (\$6.42 return on \$9.00)
- In Round 3, someone sent me everything (\$5.00) and I reciprocated generously (\$10.00 return)

I've seen good cooperation so far. My previous experience as sender (Round 2) with moderate-high trust level worked well.

For this round:

- This is a new co-player (different from previous rounds)
- I don't know their behavior yet
- But the overall pattern in this game seems cooperative
- I've been both generous and received generosity

Given the positive experiences so far, I should maintain a cooperative strategy. I'll send a substantial amount again. Sending \$3.50 (70%) shows strong trust while keeping a small safety margin.

Actually, let me reconsider. The game seems quite cooperative overall. In Round 3, someone trusted me completely with \$5.00, and in Round 2 when I sent \$3.00 I got excellent reciprocation.

I could send \$4.00 (80%) to signal strong cooperation, or even \$5.00 to match the maximum trust I received in Round 3.

Given the high cooperation levels I've experienced, I'll send \$4.00 (80% of endowment) - this is a strong signal of trust without going to the absolute maximum.

...

usage:

```
{
  "input_tokens": 1343,
  "output_tokens": 646,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.64 to you. Your visible payoff was \$7.64.

Your total visible earnings across all rounds are \$24.810000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.67, so you received \$11.01. How much do you return? (0-11.01)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.67 (73.4% of their endowment) - this shows strong trust
- This became \$11.01 for me
- They currently have \$1.33 remaining

Their strong trust (sending nearly 3/4 of their endowment) deserves good reciprocation. If

I return \$7.00:

- The sender ends with \$8.33 (gain of \$3.33)
- I keep \$4.01 (gain of \$4.01)
- Both of us benefit, with returns roughly proportional to their trust level

This balances rewarding their trust while maintaining fairness.

{"return": 7.00}

reasoning:

Let me analyze this situation:

- My co-player sent \$3.67 (73.4% of their endowment) - this is strong trust
- This became \$11.01 for me
- They currently have \$1.33 remaining

Looking at my pattern of reciprocation:

- Round 1: Received \$6.75, returned \$3.00 (44.4% return rate)
- Round 3: Received \$15.00, returned \$10.00 (66.7% return rate) - but they sent everything

And when I've been sender:

- Round 2: Sent \$3.00 (60%), received \$6.42 back (71.3% return rate from receiver)
- Round 4: Sent \$4.00 (80%), received \$6.64 back (55.3% return rate from receiver)

The sender showed strong trust at 73.4%. I should reciprocate well, but need to be fair.

If I return \$6.00:

- Sender ends with \$1.33 + \$6.00 = \$7.33 (gain of \$2.33)
- I keep \$11.01 - \$6.00 = \$5.01 (gain of \$5.01)

This is about a 54.5% return rate, which seems reasonable given their strong (but not maximum) trust.

If I return \$7.00:

- Sender ends with \$1.33 + \$7.00 = \$8.33 (gain of \$3.33)
- I keep \$11.01 - \$7.00 = \$4.01 (gain of \$4.01)

This is about a 63.6% return rate, which rewards their strong trust more generously.

Given they sent 73.4% of their endowment (strong trust), I think \$7.00 is appropriate. It's more generous than my Round 1 reciprocation (where they sent 45%) but less than my Round 3 reciprocation (where they sent ...

usage:

```
{
  "input_tokens": 1645,
  "output_tokens": 659,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.67, it became \$11.01, and you returned \$7.0. Your visible payoff was \$4.01.

Your total visible earnings across all rounds are \$28.82.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience as sender:

- Round 2: Sent \$3.00 (60%) ? received \$6.42 back, strong return
- Round 4: Sent \$4.00 (80%) ? received \$6.64 back, good return

Both times I've been sender, trust has been rewarded. This is a new co-player, and I want to continue building cooperative patterns while being reasonably cautious.

Sending \$3.50 (70% of my endowment) strikes a good balance - it shows trust and gives my co-player \$10.50 to work with, creating opportunity for mutual benefit.

```
{"send": 3.50}
```

Reasoning:

Looking at my history:

- Round 2 (sender): Sent \$3.00 (60%), received \$6.42 back, payoff \$8.42
- Round 3 (receiver): Received \$5.00 (100%), returned \$10.00, payoff \$5.00
- Round 4 (sender): Sent \$4.00 (80%), received \$6.64 back, payoff \$7.64
- Round 5 (receiver): Received \$3.67 (73.4%), returned \$7.00, payoff \$4.01

As a sender, I've been quite trusting (60% and 80%) and have received good returns on both occasions. As a receiver, I've been fairly generous as well.

This is Round 6, and I'm facing a new co-player. The overall pattern suggests trust pays off in this game. I should continue with a moderate-to-high trust level.

Given my successful experiences, I'll send a good amount. Sending \$3.50 (70%) seems like a balanced choice - shows trust, not too conservative, but also not going all-in like some co-players have done.

Actually, let me reconsider. My sending history:

- \$3.00 (60%) - got back \$6.42, net gain \$3.42
- \$4.00 (80%) - got back \$6.64, net gain \$2.64

Both were profitable. The 60% send actually gave me a better net gain. Let me go with something around 70% to balance trust and caution.

\$3.50 seems reasonable.

Usage: {

```
"input_tokens": 1928,  
"output_tokens": 534,  
"reasoning_tokens": 0
```

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:23:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_1",  
"move_index": 1,  
"opponent_id": "Agent_6",  
"pairing": {  
  "agents": [  
    "Agent_6",  
    "Agent_2"  
  ],  
  "investor": "Agent_6",  
  "roles": {  
    "Agent_2": "trustee",  
    "Agent_6": "investor"  
  },  
  "trustee": "Agent_2"  
},  
"role": "trustee",  
"role_label": "Receiver",  
"round": 6,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being reasonably cautious since I have no history with my co-player.

The multiplication factor of 3x creates an opportunity for mutual benefit. If I send a moderate amount, we can both potentially end up better off than the initial \$5.

I'll send **\$3**, which gives my co-player \$9 to work with. This shows trust and willingness to cooperate. If they return approximately \$3-4, we'll both be in a good position, and it could establish a positive pattern for future rounds.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit
2. The receiver then decides how much to return
3. There's communication noise mentioned - amounts may appear different than actually sent
4. This is a multi-agent setting with 8 agents total, pairs rotate
5. This is the first round, so I have no history with any player yet

Since this is the first round and I have no information about other players' behavior, I need to consider:

- Sending \$0 guarantees I keep \$5, but no potential for mutual gain
- Sending everything (\$5) maximizes the total pot (\$15 for receiver) but is risky
- A moderate amount balances risk and potential reward

In trust games, common strategies include:

- Sending about half as a signal of conditional cooperation
- The tripling effect means even partial cooperation can benefit both parties

Let me consider sending \$3:

- I send \$3, receiver gets \$9
- If receiver returns \$3, I end up with \$5 (same as starting)
- If receiver returns \$4-5, I'm better off
- If receiver returns less than \$3, I'm worse off
- This shows trust while keeping some insurance

Alternatively, sending \$2.5:

- I send \$2.5, receiver gets \$7.5
- If receiver returns \$2.5, I end up with \$5
- If receiver returns \$3+, I'm better off
- Less risk than \$3

Or sending \$2:

- I send \$2, receiver gets \$6
- If receiver returns \$2, I end up with \$5
- Lower risk, but also lower potential reward

Given this is the first round and I want to establish a cooperative pattern, I'll send \$3.

This shows trust and willingness to cooperate while not being completely vulnerable.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.81 to you. Your visible payoff was \$4.8100000000000005.

Your total visible earnings across all rounds are \$4.8100000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$3.95, so you received \$11.850000000000001. How much do you return? (0-11.850000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.95 (79% of their endowment), showing significant trust
- I received \$11.85 total
- They currently have only \$1.05 left

I want to:

1. Reward their trust and cooperation
2. Ensure they end up better than their initial \$5
3. Set a positive tone for potential future interactions

If I return \$5.50:

- They end up with: \$1.05 + \$5.50 = \$6.55
- I end up with: \$11.85 - \$5.50 = \$6.35

Both of us gain from cooperation, and this reciprocates their high level of trust appropriately.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the SENDER
- I sent \$3.00
- It became \$9.00 for the receiver
- They returned \$2.81
- My payoff: $\$5 - \$3 + \$2.81 = \4.81

Round 2:

- I am now the RECEIVER
- My co-player (a different agent) sent me \$3.95
- I received $\$3.95 \times 3 = \11.85
- I need to decide how much to return

Important considerations:

1. The sender gave me \$3.95 out of their \$5 - that's 79% of their endowment, showing high trust
2. If I return nothing, I keep \$11.85 and they end up with $\$5 - \$3.95 = \$1.05$
3. I should return enough to make cooperation worthwhile for them
4. There's communication noise mentioned, so amounts may differ slightly

What would be fair?

- The sender gave up \$3.95, keeping \$1.05
- I received \$11.85
- If I want them to end up better than their initial \$5, I should return more than \$3.95
- A 50-50 split of total surplus could work: Total = $\$1.05 + \$11.85 = \$12.90$. Half is \$6.45
- If I return \$5.40, sender gets $\$1.05 + \$5.40 = \$6.45$, I get $\$11.85 - \$5.40 = \$6.45$

But let me think about reciprocity:

- In round 1, I sent \$3 (60% of endowment), received \$9, they returned \$2.81 (31% of what they received)
- This round, sender sent \$3.95 (79% of endowment), I received \$11.85
- Using that same 31% return rate would give me \$3.67 to send back

The sender showed significantly more generosity than I did initially, so I should reciprocate with a higher return. Returning \$5 would leave them with \$6.05 total-better than their starting \$5-while I'd keep \$6.85, rewarding their trust and encouraging continued cooperation....

usage:

```
{
  "input_tokens": 793,
  "output_tokens": 977,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.95, it became \$11.850000000000001, and you returned \$5.5. Your visible payoff was \$6.350000000000001.

Your total visible earnings across all rounds are \$11.160000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at Round 3:

I'm paired with a new co-player (pairings are randomized each round), so I have no history with them. My previous experiences showed:

- Round 1: Sent \$3, received \$2.81 back - reasonable cooperation
- Round 2: As receiver, I tried to be generous with a high-trusting sender

For this new pairing, I'll use a similar approach to Round 1 - send a moderate amount that signals cooperation while being reasonably cautious with an unknown partner.

Sending \$3 creates \$9 for them to work with, which gives room for mutual benefit if they reciprocate fairly.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$3, received back \$2.81, payoff \$4.81

Round 2: I was RECEIVER, got \$11.85 from a \$3.95 send, returned \$5.50, payoff \$6.35

Total earnings: \$11.16

Now Round 3: I'm SENDER again with a new co-player (since pairings are randomized each round).

Key considerations:

- I have no history with this particular co-player
- In Round 1, when I sent \$3, I got back \$2.81 (which is reasonable - about 31% return rate)
- In Round 2, when someone sent me \$3.95, I returned \$5.50 (about 46% return rate)
- The game has communication noise, so exact amounts may differ

Since I'm paired with a new player and have no history with them, I should probably use a similar strategy to Round 1 - establish cooperation but be somewhat cautious.

Given that:

- Sending \$3 worked reasonably well in Round 1
- The multiplication factor of 3x creates good mutual benefit potential
- I want to signal cooperation

I'll send \$3 again. This is a moderate amount that shows willingness to cooperate without being overly risky with a new partner.

usage:

```
{
  "input_tokens": 1106,
  "output_tokens": 478,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.52 to you. Your visible payoff was \$8.52.

Your total visible earnings across all rounds are \$19.68.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.36 (67% of their endowment), showing solid trust
- I received \$10.08 total
- They currently have \$1.64 left

I want to reciprocate their trust fairly. If I return \$5.00:

- They end up with: \$1.64 + \$5.00 = \$6.64 (gain of \$1.64)
- I end up with: \$10.08 - \$5.00 = \$5.08 (gain of \$0.08)

Both of us benefit from cooperation, and this rewards their trust appropriately.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- My co-player sent \$3.36 (67.2% of their endowment)
- I received \$10.08 total
- They currently have: $\$5 - \$3.36 = \$1.64$ left

My previous round (Round 3):

- I sent \$3, received \$6.52 back - very generous return!

This co-player sent \$3.36, showing good trust. I should reciprocate appropriately.

If I return \$5.00:

- They end up with: $\$1.64 + \$5.00 = \$6.64$
- I end up with: $\$10.08 - \$5.00 = \$5.08$

Both of us gain. This is a fair reciprocation that rewards their trust.

Alternatively, if I return \$5.50:

- They end up with: $\$1.64 + \$5.50 = \$7.14$
- I end up with: $\$10.08 - \$5.50 = \$4.58$

This would be even more generous, similar to how I acted in Round 2.

Given that they sent about 67% of their endowment (compared to 79% in Round 2), I think returning around \$5 is appropriate and fair. It ensures they profit significantly from their trust, and we both end up better off than the initial \$5.

I'll go with \$5.00 as a fair reciprocation.

usage:

```
{
  "input_tokens": 1402,
  "output_tokens": 522,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$5.0. Your visible payoff was \$5.08.

Your total visible earnings across all rounds are \$24.759999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at Round 5:

I'm with another new co-player. My experience so far shows:

- Sending \$3 has worked well in previous rounds, getting returns of \$2.81 and \$6.52
- The cooperative approach has generally been rewarded

I'll continue with the same strategy - send \$3 to signal trust and cooperation while being reasonably cautious with an unknown partner.

```
'''json
{"send": 3}
'''
```

reasoning:

Round 5, I'm the sender with a new co-player (random pairings each round).

Looking at my history:

- Round 1 (Sender): Sent \$3, got back \$2.81 - decent cooperation
- Round 2 (Receiver): Received \$11.85, returned \$5.50 - I was generous
- Round 3 (Sender): Sent \$3, got back \$6.52 - excellent return!
- Round 4 (Receiver): Received \$10.08, returned \$5.00 - fair reciprocation

So far, sending \$3 has worked well - I got \$2.81 and \$6.52 back in my two sender rounds. The average return has been positive.

For this new pairing, I should continue with a cooperative strategy. Sending \$3 seems to be working as a good middle-ground approach that encourages cooperation.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1690,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.15 to you. Your visible payoff was \$5.15.

Your total visible earnings across all rounds are \$29.909999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.47, so you received \$13.41. How much do you return? (0-13.41)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$4.47 (89% of their endowment), showing very high trust
- I received \$13.41 total
- They currently have only \$0.53 left

This is a very trusting sender, and I should reciprocate generously. If I return \$6.50:

- They end up with: $\$0.53 + \$6.50 = \$7.03$ (gain of \$2.03)
- I end up with: $\$13.41 - \$6.50 = \$6.91$ (gain of \$1.91)

Both of us gain approximately \$2, which appropriately rewards their exceptional trust and maintains strong cooperation.

```
'''json
{"return": 6.50}
'''
```

Reasoning:

Looking at this situation:

- My co-player sent \$4.47 (89% of their endowment), showing very high trust
- I received \$13.41 total
- They currently have \$0.53 left

This is a very trusting sender. I should reciprocate generously.

If I return \$6.50:

- They end up with: $\$0.53 + \$6.50 = \$7.03$ (gain of \$2.03)
- I end up with: $\$13.41 - \$6.50 = \$6.91$ (gain of \$1.91)

Both benefit well, and this appropriately rewards their very high level of trust.

Let me think about this more. They sent \$4.47 out of \$5, leaving themselves with only \$0.53. That's 89.4% of their endowment. This is extremely trusting.

I should return enough to:

1. Ensure they profit significantly from their trust
2. Show I value cooperation
3. Both of us should end up better than our starting \$5

If I return \$6.50:

- Them: $\$0.53 + \$6.50 = \$7.03$
- Me: $\$13.41 - \$6.50 = \$6.91$

Both gain roughly \$2, which seems fair and reciprocates their high trust.

```
Usage: {
  "input_tokens": 1931,
  "output_tokens": 502,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-08-12T09:23:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: `{'send': <amount>}`

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: $\$5 - \text{sent} + \text{returned}$
- You keep: $\text{received} - \text{returned}$
- Response format: `{'return': <amount>}`

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no history with this receiver.

If I send money, it gets tripled, creating potential mutual gain. However, the receiver controls how much comes back. The communication noise adds uncertainty.

I'll take a moderately cooperative approach to signal trust while managing risk:

****My decision:****

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return a fair portion, we both benefit. This allows me to learn about their reciprocity while not risking everything in this uncertain first round.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 to allocate
2. Whatever I send gets tripled
3. The receiver then decides how much to return
4. There's communication noise - amounts may differ slightly from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- This is round 1, so I have no history with this particular receiver
- There's a note about communication noise which could affect trust building

Standard economic reasoning in trust games:

- If I send \$5, it becomes \$15 for the receiver
- A "fair" split might be for them to return \$10, giving us each \$10
- But the receiver might keep everything, leaving me with \$0
- If I send \$0, I guarantee keeping \$5 but miss potential gains

Since this is:

1. Round 1 with no history
2. A multi-round game where reputation/reciprocity could matter across the full run
3. There's communication noise which might complicate trust building

I should consider a moderate opening move that:

- Shows some trust/willingness to cooperate
- Doesn't risk everything

- Allows me to learn about this receiver's behavior
- Sets a cooperative tone

A common strategy in trust games is to send around 50% on the first round as a signal of moderate trust while protecting against complete loss.

I'll send \$2.50 (or \$2 or \$3 since I need to be practical about amounts).

Actually, looking at the format, I should probably stick to whole dollar amounts for simplicity. Let me send \$3 as a cooperative opening move - it's more than half, showing trust, but not everything.

Wait, let me reconsider. The receiver gets \$9 if I send \$3. If they return half (\$4.50), I'd have $\$5 - \$3 + \$4.50 = \6.50 , and they'd have \$4.50. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 698,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.51 to you. Your visible payoff was \$7.51.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$3.8 out of \$5 (76% of their endowment) - showing significant trust
- I received \$11.4 after the 3x multiplier

****My reasoning:****

- In Round 1, when I sent \$3, the receiver returned \$5.51 (about 61% of what they received)
- This sender has shown high trust by sending 76% of their endowment
- I should reciprocate that trust fairly

I'll return an amount that rewards their cooperation while maintaining fairness:

```
'''json
{"return": 7}
'''
```

This gives the sender a total of \$8.2 (\$1.2 kept + \$7 returned) and leaves me with \$4.4. While I keep less, this reciprocates their strong trust and builds cooperation.

reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$5.51 to me
- My payoff: $\$5 - \$3 + \$5.51 = \7.51
- Their payoff: $\$9 - \$5.51 = \$3.49$

So the receiver returned about 61% of what they received. That's quite generous.

Round 2 (current):

- I am RECEIVER
- Sender sent me \$3.8
- I received \$11.4 (tripled amount)

- I need to decide how much to return

Now, there's communication noise, so the amounts might be slightly different from what was actually sent/returned. But I should work with what I see.

The sender sent me \$3.8 out of \$5, which is 76% of their endowment. That's quite trusting.

If I apply reciprocity based on what happened in Round 1:

- The receiver returned about 61% of what they received
- 61% of \$11.4 = \$6.95

But let me think about fairness:

- Sender keeps: \$5 - \$3.8 = \$1.2
- Sender receives back from me: X
- Sender total: \$1.2 + X

- I keep: \$11.4 - X

For equal split of total value:

- Total created: \$5 + \$11.4 = \$16.4
- Equal split: \$8.2 each
- So sender should get: \$8.2 total, meaning I return: \$8.2 - \$1.2 = \$7

For proportional reciprocity (they sent 76% of endowment):

- Maybe I return a similar proportion of what I received?
- 76% of \$11.4 = \$8.66

Given their high level of trust, I should match that cooperation. Returning around 55-60% of what I received feels right-it's reciprocal without being excessive. I'll return \$6.50, which gives the sender a total of \$7.70 and leaves me with \$4.90. This mirrors the cooperative spirit they showed....

usage:

```
{
  "input_tokens": 780,
  "output_tokens": 878,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.8, it became \$11.399999999999999, and you returned \$7.0. Your visible payoff was \$4.399999999999999.

Your total visible earnings across all rounds are \$11.909999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this new round with a new co-player:

****My experience so far:****

- Round 1: Sent \$3, received \$5.51 back - good cooperation
- Round 2: Received \$11.4, returned \$7 - fair reciprocation

Since pairings are randomized, this is a new partner with no shared history. However, the cooperative pattern I've experienced suggests trust can be rewarded.

I'll send a moderate-to-high amount to signal cooperation while managing risk:

```
'''json
{"send": 3.5}
'''
```

This sends \$3.5 (70% of my endowment), which becomes \$10.50 for the receiver. It's slightly more than my Round 1 send, reflecting confidence from positive past experiences while remaining balanced.

reasoning:

Let me analyze the situation:

****My history:****

- Round 1: I sent \$3, received back \$5.51, payoff = \$7.51
- Round 2: I received \$11.4 (they sent \$3.8), I returned \$7, payoff = \$4.4
- Total earnings: \$11.91

****Analysis:****

- In Round 1, my co-player was generous, returning \$5.51 out of \$9 (61%)
- In Round 2, my co-player sent \$3.8 (76% of endowment), showing high trust, and I reciprocated fairly
- Now I'm facing a new co-player in Round 3

Since this is a multi-agent setting with randomized pairings, this Round 3 co-player is likely a different person than either of my previous partners. I don't have direct history with them.

However, I've seen cooperation pay off:

- When I sent \$3, I got back \$5.51 (net gain of \$2.51)
- When someone sent me \$3.8, I returned fairly

Given the overall cooperative pattern I've observed and participated in, I should continue with a cooperative strategy. I'll send a similar amount to what I did in Round 1, or potentially slightly more given the positive experience.

I'll send \$3 again - it's a balanced approach that:

1. Shows trust and willingness to cooperate
2. Doesn't risk everything
3. Creates opportunity for mutual gain (becomes \$9)
4. Has worked well for me in the past

Actually, given that cooperation has been working well, I could send a bit more. Let me send \$3.5, showing slightly increased trust while still being prudent.

usage:

```
{
  "input_tokens": 1109,
  "output_tokens": 583,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.82 to you. Your visible payoff was \$6.32.

Your total visible earnings across all rounds are \$18.229999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$3.11 out of \$5 (62.2% of their endowment) - showing moderate trust
- I received \$9.33 after the 3x multiplier

****My analysis:****

- In Round 3, when I sent \$3.5, I received back \$4.82 (only 46% of what they received) - less cooperative
- In Round 2, when I received \$11.4, I returned \$7 (61%)
- This sender showed moderate trust, deserving fair reciprocation

I'll return a fair amount consistent with their trust level:

```
'''json
{"return": 5.5}
'''
```

This gives the sender a total of \$7.39 (\$1.89 kept + \$5.50 returned) and leaves me with \$3.83. It's a balanced return that rewards their trust while being fair to both of us.

reasoning:

Let me analyze this situation:

****Round 4:****

- I'm the RECEIVER
- They sent \$3.11 out of \$5 (62.2% of their endowment)
- I received \$9.33 after the 3x multiplier

****Past experience:****

- Round 1: I sent \$3, got back \$5.51 (they returned 61% of \$9)
- Round 2: I received \$11.4, returned \$7 (61% of what I received)
- Round 3: I sent \$3.5, got back \$4.82 (they returned 46% of \$10.5)

The sender this round sent me \$3.11, showing moderate trust (62.2% of their endowment).

In Round 3, when I sent \$3.5, I only got back \$4.82, which was less than half of what they received. That was somewhat disappointing.

For this round, the sender sent \$3.11. I should be fair but also consider that:

1. They showed moderate trust (62.2%)
2. My past returns have been around 61% when receiving
3. I want to maintain cooperation

If I return around 55-60% of \$9.33, that would be:

- 55% = \$5.13
- 60% = \$5.60

This would give them:

- At \$5.13: \$1.89 kept + \$5.13 = \$7.02 total
- At \$5.60: \$1.89 kept + \$5.60 = \$7.49 total

And I would keep:

- At \$5.13: \$4.20
- At \$5.60: \$3.73

I'll return around \$5.5 to be fair and reciprocate their moderate trust.

usage:

```
{
  "input_tokens": 1428,
  "output_tokens": 648,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$5.5. Your visible payoff was \$3.83.

Your total visible earnings across all rounds are \$22.059999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.26, so you received \$6.779999999999999. How much do you return? (0-6.779999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$2.26 out of \$5 (45.2% of their endowment) - showing cautious/modest trust
- I received \$6.78 after the 3x multiplier

****My analysis:****

- This sender showed less trust than previous partners (45% vs 62-76%)
- However, I should still reciprocate fairly to maintain cooperation
- My previous return rates have been around 59-61%

Given their more cautious approach, I'll return a fair but proportionally adjusted amount:

```
'''json
{"return": 3.5}
'''
```

This gives the sender a total of \$6.24 (\$2.74 kept + \$3.50 returned), a gain of \$1.24, and leaves me with \$3.28. It's a balanced return that rewards their trust while acknowledging their more conservative send.

reasoning:

Looking at this situation:

****What happened:****

- The sender sent me \$2.26 out of \$5 (45.2% of their endowment) - showing modest/cautious trust
- I received \$6.78 after the 3x multiplier

****My historical pattern:****

- Round 2: Received \$11.4, returned \$7 (61% of received)
- Round 4: Received \$9.33, returned \$5.5 (59% of received)

This sender showed less trust than previous senders (only 45% vs 62-76%). I should still reciprocate fairly, but proportionally to their lower trust level.

A return rate around 50-55% seems appropriate given their more cautious approach:

- 50% would be \$3.39
- 55% would be \$3.73
- 60% would be \$4.07

Let me think about what's fair:

- They sent \$2.26, keeping \$2.74
- If I return \$3.5, they'd have \$5.76 total (a gain of \$0.76)
- I'd keep \$3.28

Actually, let me reconsider. Even though they sent less, I should maintain a cooperative stance. Let me return around 50-55%:

If I return \$3.5:

- Sender gets: \$2.74 (kept) + \$3.50 (returned) = \$6.24 total
- I keep: \$6.78 - \$3.50 = \$3.28

This seems balanced - they gain \$1.24, I gain \$3.28. It's fair while still rewarding their trust, even if it was more cautious.

Let me go with \$3.5 (about 52% of what I received).

usage:

```
{
  "input_tokens": 1789,
  "output_tokens": 630,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.26, it became \$6.779999999999999, and you returned \$3.5. Your visible payoff was \$3.2799999999999994.

Your total visible earnings across all rounds are \$25.339999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this new round with a new co-player:

****My experience so far as sender:****

- Round 1: Sent \$3 (60%), received \$5.51 back - good return
- Round 3: Sent \$3.5 (70%), received \$4.82 back - lower return rate

****My observations:****

- Sending more didn't necessarily yield better outcomes
- As a receiver, I've been fairly cooperative
- This is a new partner, so I have no history with them

I'll send a moderate amount that signals cooperation while managing risk:

```
'''json
{"send": 3}
'''
```

This sends \$3 (60% of endowment), which becomes \$9 for the receiver. It's a balanced approach that worked well in Round 1 and allows me to gauge this new partner's reciprocity.

Reasoning:

Looking at my history:

Round 1 (Sender): Sent \$3, received back \$5.51, payoff \$7.51
Round 2 (Receiver): Received \$11.4 (from \$3.8 send), returned \$7, payoff \$4.4
Round 3 (Sender): Sent \$3.5, received back \$4.82, payoff \$6.32
Round 4 (Receiver): Received \$9.33 (from \$3.11 send), returned \$5.5, payoff \$3.83
Round 5 (Receiver): Received \$6.78 (from \$2.26 send), returned \$3.5, payoff \$3.28

Now I'm a sender again in Round 6 with a new co-player.

My experience as sender:

- Round 1: Sent \$3 (60%), got back \$5.51 (61% of what they received)
- Round 3: Sent \$3.5 (70%), got back \$4.82 (46% of what they received)

When I send more, I don't necessarily get better returns. The Round 3 experience was less favorable.

My experience as receiver shows I've been fairly cooperative:

- Returned 61%, 59%, and 52% of what I received

Since this is a new partner and there's communication noise, I should:

1. Signal cooperation but not overcommit
2. Learn about this specific partner's behavior

I'll send a moderate amount around \$3 to balance risk and cooperation potential.

```
Usage: {
  "input_tokens": 2129,
  "output_tokens": 563,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_2

Timestamp: 2026-08-12T09:24:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
```

```

"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1 with no prior information about my co-player, I want to demonstrate trust while being reasonably cautious.

If I send \$3:

- My co-player receives \$9 (3×3)
- This shows meaningful cooperation
- If they return a fair portion (say $1/3$ to $1/2$), we both benefit
- My maximum loss is limited to \$3

I'll go with a moderately trusting approach:

```

'''json
{"send": 3}
'''

```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I have \$5 to start with
2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. I keep: \$5 - sent + returned
5. The receiver keeps: received - returned

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There may be communication noise
- I don't know who my co-player is

Strategic thinking:

- If I send \$0, I guarantee keeping \$5, but there's no potential for gain
- If I send everything (\$5), the receiver gets \$15, and if they're cooperative, they might return a fair amount
- A moderate amount (like \$2-3) shows some trust but limits risk

Since this is round 1 and I have no information about my co-player:

- I want to signal cooperation and trust
- But I also want to be somewhat cautious
- Sending around \$3 seems like a reasonable middle ground that:
 - Shows good faith (60% of my endowment)
 - Gives the receiver \$9 to work with
 - If they return fairly (say 1/3 to 1/2 of what they receive), I could end up with \$5-6.
- 5
- Limits my maximum loss to \$3

Let me send \$3 as a cooperative but measured opening move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.37 to you. Your visible payoff was \$5.37.

Your total visible earnings across all rounds are \$5.37.

This round, you are the RECEIVER against your current co-player. They sent you \$4.58, so you received \$13.74. How much do you return? (0-13.74)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

My co-player sent me \$4.58 out of their \$5 (91.6% of their endowment) - showing very high trust.

I received \$13.74. To reciprocate this trust appropriately:

- They kept \$0.42
- If I return around \$6.50, they'd end with \$6.92 (a gain of \$1.92)
- I'd keep \$7.24 (also a solid gain)

This represents returning about 47% of what I received, which rewards their trust while we both benefit substantially.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 2 - I'm now the RECEIVER

- My co-player sent me \$4.58
- I received \$13.74 (which is $\$4.58 \times 3$)
- I need to decide how much to return

From Round 1:

- I sent \$3.00, they received \$9.00
- They returned \$3.37 to me
- This is about 37.4% of what they received, which is quite fair/generous

My co-player in this round sent me \$4.58 out of their \$5, which is 91.6% of their endowment - very trusting!

I should consider:

1. They showed high trust by sending almost everything
2. In Round 1, my previous partner returned about 37.4% which seemed fair
3. I want to reciprocate trust and maintain good cooperation

If I return about 40-50% of what I received (\$13.74):

- 40% = \$5.50
- 45% = \$6.18
- 50% = \$6.87

Given they sent almost everything (\$4.58 out of \$5), showing extreme trust, I should be generous in return.

Their final payoff would be:

- They started with \$5
- Sent \$4.58, kept \$0.42
- If I return \$6.00: $\$0.42 + \$6.00 = \$6.42$
- My payoff: $\$13.74 - \$6.00 = \$7.74$

If I return \$6.50:

- Their payoff: $\$0.42 + \$6.50 = \$6.92$
- My payoff: $\$13.74 - \$6.50 = \$7.24$

Given their high trust (sending 91.6% of endowment), I should return a good amount. Returning around \$6.50 seems fair - they end up with \$6.92 (better than their initial \$5), and I keep \$7.24 (also better than my initial \$0).

...

usage:

```
{
  "input_tokens": 755,
  "output_tokens": 653,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.58, it became \$13.74, and you returned \$6.5. Your visible payoff was \$7.24.

Your total visible earnings across all rounds are \$12.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.38, so you received \$10.14. How much do you return? (0-10.14)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust.

I received \$10.14.

They kept: \$1.62

To reciprocate fairly:

- If I return \$5.00, they'd end with \$6.62 (a gain of \$1.62)
- I'd keep \$5.14 (also a small gain)

This represents returning about 49% of what I received, which rewards their trust while we both benefit.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust

I received \$10.14.

They kept: \$5 - \$3.38 = \$1.62

To be fair and reciprocate their trust:

- If I return around \$5, they'd end with \$1.62 + \$5 = \$6.62 (a gain of \$1.62)
- I'd keep \$10.14 - \$5 = \$5.14 (also a solid gain)

This represents returning about 49% of what I received, which is fair and reciprocates their trust appropriately while we both benefit.

Let me think about this more carefully. They sent \$3.38, showing moderate-to-good trust (68% of endowment).

In Round 2, my co-player sent me \$4.58 (92% of endowment) and I returned \$6.50 (47% of what I received).

For consistency and fairness here with \$3.38 sent (less trust shown):

- Returning around 47-50% of \$10.14 would be \$4.77-\$5.07
- Let me go with \$5.00

That gives them: \$1.62 + \$5.00 = \$6.62 (gain of \$1.62)

I keep: \$10.14 - \$5.00 = \$5.14 (gain of \$0.14)

This seems fair and reciprocal.

usage:

```
{
  "input_tokens": 1036,
  "output_tokens": 519,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.38, it became \$10.14, and you returned \$5.0. Your visible payoff was \$5.140000000000001.

Your total visible earnings across all rounds are \$17.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

Round 1: As SENDER, I sent \$3 and received \$3.37 back - a reasonable return.
Rounds 2-3: As RECEIVER, others showed good trust, and I reciprocated fairly.

This is a new co-player (random pairing), so I don't have specific information about them. However, the overall pattern suggests players are reasonably cooperative.

I'll send \$3.50 - slightly more than my first round to show increasing trust while remaining measured:

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$3, received back \$3.37, payoff \$5.37
Round 2: I was RECEIVER, received \$13.74 (from \$4.58), returned \$6.50, payoff \$7.24
Round 3: I was RECEIVER, received \$10.14 (from \$3.38), returned \$5.00, payoff \$5.14
Round 4: I am SENDER

My total earnings: \$17.75

Looking at the pattern:

- When I sent \$3, I got back \$3.37 (a bit more than 1:1 return on sent amount)
- When others sent to me, they showed high trust (\$4.58 and \$3.38 out of \$5)
- I've been fairly reciprocal as a receiver, returning roughly half of what I received

Now I'm a sender again. Since:

1. My first send of \$3 was met with a decent return (\$3.37)
2. Others have been trusting with me
3. I've been fair in return

I should continue with moderate to high trust. I'll send \$3.5 this time - slightly more than my first send to show I'm willing to build cooperation.

Actually, let me reconsider. The co-player this round is new (pairings are randomized). I don't know their history. In Round 1, I sent \$3 and got a reasonable return. Let me stick with a similar strategy - send \$3 again as a baseline cooperative strategy.

Actually, looking at the overall pattern, players seem reasonably cooperative. I could try \$3.5 to be slightly more trusting.

Let me go with \$3.5 - showing increased trust while not overcommitting.

usage:

```
{
  "input_tokens": 1303,
  "output_tokens": 551,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$7.22 to you. Your visible payoff was \$8.719999999999999.

Your total visible earnings across all rounds are \$26.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my recent experience:

Round 4: I sent \$3.50 and received \$7.22 back - excellent cooperation!

The overall pattern shows players are being reasonably cooperative. While this is a new co-player (random pairing), I'll show increased trust based on positive experiences.

Sending \$4:

- Demonstrates strong trust (80% of my endowment)
- Creates \$12 for them to work with
- Encourages reciprocal cooperation

```
'''json
{"send": 4}
'''
```

reasoning:

Round 4 recap: I sent \$3.50, they returned \$7.22 - excellent return! That's about 69% of what they received, very generous.

This is Round 5, and I'm the SENDER again against a new co-player (random pairing each round).

So far my experience:

- Round 1 SENDER: sent \$3, got \$3.37 back (reasonable)
- Round 2 RECEIVER: received \$13.74, returned \$6.50
- Round 3 RECEIVER: received \$10.14, returned \$5.00
- Round 4 SENDER: sent \$3.50, got \$7.22 back (very good!)

The last round showed great cooperation. Players seem generally trustworthy. This is a new co-player though, so I don't know their specific behavior.

I should probably continue with a trusting approach. I could:

- Send \$4 (80% of endowment) - shows high trust
- Send \$3.5 (70%) - similar to last round
- Send \$3 (60%) - more cautious

Given the positive experiences, I'll increase slightly to \$4 to show strong trust.

usage:

```
{
  "input_tokens": 1562,
  "output_tokens": 402,
  "reasoning_tokens": 0
}
```

Message 12 role=user**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.98 to you. Your visible payoff was \$7.98.

Your total visible earnings across all rounds are \$34.45.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

My co-player sent me \$2.9 out of their \$5 (58% of their endowment) - showing moderate trust.

I received \$8.7.

They kept: \$2.1

To reciprocate fairly:

- If I return \$4.00, they'd end with \$6.10 (a gain of \$1.10)
- I'd keep \$4.70 (also a decent gain)

This represents returning about 46% of what I received, which is proportional to the moderate trust shown and ensures we both benefit.

```
'''json
{"return": 4.00}
'''
```

Reasoning:

Looking at this situation:

My co-player sent me \$2.9 out of their \$5 (58% of their endowment) - showing moderate trust.

I received \$8.7.

They kept: \$2.1

To reciprocate fairly:

- If I return around \$4.00-\$4.50, they'd end with \$6.10-\$6.60 (a gain of \$1.10-\$1.60)
- I'd keep \$4.70-\$4.20

This represents returning about 46-52% of what I received, which rewards their trust while we both benefit.

Let me think about what's fair. They sent \$2.9, showing moderate but not high trust. I should return something proportional.

In previous rounds as receiver:

- Round 2: Received \$13.74, returned \$6.50 (47%)
- Round 3: Received \$10.14, returned \$5.00 (49%)

So I've been returning around 47-49% of what I receive.

For \$8.7, returning around 47-49% would be:

- 47%: \$4.09
- 49%: \$4.26

Let me go with \$4.25, which is about 49% and gives them a total of \$6.35 (gain of \$1.35), while I keep \$4.45 (also a decent gain).

Actually, let me reconsider. They sent \$2.9 (58% of endowment), which is moderate trust. In Round 1, someone sent me \$4.58 (91.6%) and I returned \$6.50. In Round 3, someone sent \$3.38 (67.6%) and I returned \$5.00.

The person who sent \$3.38 (67.6%) got back \$5.00 from \$10.14 received.

...

```
Usage: {
  "input_tokens": 1817,
  "output_tokens": 654,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_3

Timestamp: 2026-08-12T09:24:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_1"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_1": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$4.78, that is 95.6% of their total endowment. You received \$14.34. How much do you return to the sender? (0-14.34)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.78 (95.6% of their endowment)
- They kept only \$0.22 for themselves
- I received \$14.34 after the 3x multiplier
- A fair return should reward their trust

If I return \$8.00:

- Sender ends up with: $\$0.22 + \$8.00 = \$8.22$ (well above their initial \$5)
- I keep: $\$14.34 - \$8.00 = \$6.34$ (also above the initial \$5)
- Both of us benefit from the cooperation

This seems like a fair reciprocal response to their high level of trust.

{'return': 8.00}

reasoning:

Let me understand this situation:

1. The sender had \$5 endowment
2. They sent me \$4.78 (which is 95.6% of their \$5)
3. This was multiplied by 3x, so I received $\$4.78 \times 3 = \14.34

Now I need to decide how much to return.

Key considerations:

- The sender showed a lot of trust by sending 95.6% of their endowment
- They kept only \$5 - \$4.78 = \$0.22 for themselves
- If I return nothing, the sender ends up with just \$0.22
- If I return generously, both of us can benefit from this cooperation

What's fair/reciprocal?

- The sender trusted me significantly
- A common fairness norm would be to split the gains
- Total created value: \$14.34
- Original sender investment fee: \$4.78
- One approach: Return enough so we both end up roughly equal
- Another approach: Return enough so sender gets significantly more than their original \$5 (rewarding trust)

Calculations:

- If I return \$7.17 (about half of what I received):
 - Sender gets: $\$0.22 + \$7.17 = \$7.39$
 - I get: $\$14.34 - \$7.17 = \$7.17$
- If I return \$8:
 - Sender gets: $\$0.22 + \$8 = \$8.22$
 - I get: $\$14.34 - \$8 = \$6.34$
- If I return \$9:
 - Sender gets: $\$0.22 + \$9 = \$9.22$
 - I get: $\$14.34 - \$9 = \$5.34$

Given that the sender showed exceptional trust (sending 95.6%), I should reward this trust generously. A return that gives them significantly more than their original endowment while still keeping a reasonable amount seems appropriate.

Returning \$8 strikes a good balance—it rewards their exceptional trust while ensuring both of us benefit from the cooperation. They'd end up with \$8.22 versus their original \$5, and I'd keep \$6.34. But given how much they risked, maybe I should push toward \$8.50 to give them even more of the gains we created together....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 866,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.78, it became \$14.34, and you returned \$8.0. Your visible payoff was \$6.34.

Your total visible earnings across all rounds are \$6.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, my co-player showed very high trust by sending 95.6% of their endowment, and I reciprocated with a fair return. That cooperative approach worked well for both of us.

Now as the sender with a new co-player, I want to maintain that cooperative spirit. Sending a substantial amount:

- Creates more total value through the 3x multiplier
- Signals trust and willingness to cooperate
- Could encourage reciprocation

I'll send \$4.50 (90% of my endowment), showing strong trust while being slightly more conservative than what I experienced last round.

```
{'send': 4.50}
```

reasoning:

Let me think about this situation:

Round 1 (just completed):

- I was the RECEIVER

- Sender sent me \$4.78 (95.6% of their endowment) - showing very high trust
- I received \$14.34
- I returned \$8.00
- My payoff: \$6.34
- Sender's payoff would be: $\$5 - \$4.78 + \$8.00 = \8.22

Now in Round 2:

- I'm the SENDER
- I have \$5
- I'm playing against a different co-player (pairings are randomized each round)

The previous sender showed very high trust by sending 95.6%. I reciprocated fairly by returning \$8.00, which gave them a good outcome.

Now I need to decide how much to send. Key considerations:

1. This is a new co-player, so I don't have direct history with them
2. The previous round showed successful cooperation
3. Sending more creates more total value (3x multiplier), but requires trust
4. In a multi-round setting, establishing cooperative behavior can be beneficial

Given that:

- The previous sender sent 95.6% (\$4.78)
- That worked out well for both of us
- I should reciprocate the level of trust shown to me

I think I should send a similar high amount to signal cooperation and trust. Sending around \$4.50-\$4.80 would show similar trust to what I experienced.

Let me send \$4.50 (90% of my endowment). This shows strong trust while being slightly more conservative than the 95.6% I received.

Actually, thinking about it more - since I was treated with high trust and it worked out well, I should pay it forward. Sending \$4.50-\$4.75 seems appropriate.

...

usage:

```
{
  "input_tokens": 824,
  "output_tokens": 591,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.5, it became \$13.5, and you saw them return \$7.41 to you. Your visible payoff was \$7.91.

Your total visible earnings across all rounds are \$14.25.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed good trust by sending \$3.80 (76% of their endowment)
- They kept \$1.20 for themselves
- I received \$11.40
- I should reciprocate their trust fairly

Comparing to previous rounds:

- In Round 1, when a sender sent 95.6%, I returned about 56% of what I received
- In Round 2, when I sent 90%, my co-player returned about 55% of what they received

For this sender who sent 76%, a fair return would be around 55-60% of what I received.

$\$11.40 \times 0.57 \approx \6.50

If I return \$6.50:

- Sender ends with: $\$1.20 + \$6.50 = \$7.70$ (good gain)
- I keep: $\$11.40 - \$6.50 = \$4.90$ (reasonable)
- Both benefit from the cooperation

{'return': 6.50}

reasoning:

Let me analyze this situation:

Round 2 recap:

- I was the sender and sent \$4.50
- It became \$13.50 for the receiver
- They returned \$7.41 to me
- My payoff: $\$5 - \$4.50 + \$7.41 = \7.91

Round 3 current situation:

- I'm the receiver
- Sender sent \$3.80 (76% of their endowment)
- I received \$11.40 (approximately, due to floating point)
- They kept \$1.20

This sender showed good trust by sending 76% of their endowment. Let me think about a fair return:

If I return \$6.00:

- Sender ends up with: $\$1.20 + \$6.00 = \$7.20$
- I keep: $\$11.40 - \$6.00 = \$5.40$
- Both benefit, sender gains more

If I return \$7.00:

- Sender ends up with: $\$1.20 + \$7.00 = \$8.20$
- I keep: $\$11.40 - \$7.00 = \$4.40$
- Sender gains significantly more

In Round 1, when I received \$14.34 from a sender who sent \$4.78 (95.6%), I returned \$8.00.

- That sender ended with \$8.22 (from \$0.22)
- I kept \$6.34
- Return ratio: $\$8.00/\$14.34 = 55.9\%$

In Round 2, when I sent \$4.50 and it became \$13.50, they returned \$7.41

- Return ratio: $\$7.41/\$13.50 = 54.9\%$

For consistency, if I use a similar return ratio (around 55%):

- $\$11.40 \times 0.55 = \6.27

Let me round to \$6.50 to be slightly more generous:

...

usage:

```
{
  "input_tokens": 1120,
  "output_tokens": 820,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.8, it became \$11.399999999999999, and you returned \$6.5. Your visible payoff was \$4.899999999999999.

Your total visible earnings across all rounds are \$19.15.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

- Round 1: As receiver, someone sent high trust (95.6%), I reciprocated fairly
- Round 2: I sent \$4.50 (90%), got back \$7.41 - good cooperation

- Round 3: As receiver, returned fairly when sent \$3.80

The pattern shows that high trust tends to be reciprocated with fair returns (around 55% return rate). My cooperative strategy has been working well - averaging over \$6 per round.

I'll continue with a strong but slightly varied approach. Sending \$4.00 (80%):

- Shows substantial trust
- Creates \$12.00 for the receiver through the 3x multiplier
- Should encourage reciprocation based on the established pattern
- Slightly more conservative than my last sending round to manage risk

```
{'send': 4.00}
```

reasoning:

Let me analyze my history:

Round 1 (RECEIVER): Sender sent \$4.78 (95.6%), I received \$14.34, returned \$8.00, kept \$6.34

Round 2 (SENDER): I sent \$4.50 (90%), became \$13.50, received back \$7.41, ended with \$7.91

Round 3 (RECEIVER): Sender sent \$3.80 (76%), I received \$11.40, returned \$6.50, kept \$4.90

Now Round 4, I'm the SENDER again with a new co-player.

Looking at the pattern:

- When I sent \$4.50 (90%), I got back \$7.41 from \$13.50 (about 55% return rate)
- When someone sent \$4.78 (95.6%), I returned \$8.00 from \$14.34 (about 56% return rate)
- When someone sent \$3.80 (76%), I returned \$6.50 from \$11.40 (about 57% return rate)

The return rates seem fairly consistent around 55-57%. This suggests cooperation is being reciprocated.

I've been doing well with high cooperation levels. My total earnings are \$19.15 across 3 rounds, averaging about \$6.38 per round, which is well above the baseline of \$5.

Given this positive experience with cooperation, I should continue sending a substantial amount. I'll send \$4.00 (80%), which is:

- Still showing significant trust
- Creating good value through the 3x multiplier (\$12.00 received)
- Likely to be reciprocated based on the pattern
- Slightly more conservative than my previous \$4.50 to diversify my strategy a bit

If they return around 55%, I'd get back about \$6.60, ending with \$1.00 + \$6.60 = \$7.60, which would be good.

usage:

```
{
  "input_tokens": 1488,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$4.58 to you. Your visible payoff was \$5.58.

Your total visible earnings across all rounds are \$24.729999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.73 (94.6% of their endowment)
- They kept only \$0.27 for themselves
- I received \$14.19
- This is very similar to Round 1, where someone sent 95.6%

In Round 1, when faced with similarly high trust, I returned \$8.00 out of \$14.34. That worked well as a fair reciprocation.

For this round with \$14.19 received, returning \$8.00:

- Sender ends with: $\$0.27 + \$8.00 = \$8.27$ (strong return on their trust)
- I keep: $\$14.19 - \$8.00 = \$6.19$
- Both benefit significantly from the cooperation

Consistency with my previous fair reciprocation of high trust seems appropriate.

```
{'return': 8.00}
```

reasoning:

Let me analyze this situation:

Round 5 - I'm the RECEIVER

- Sender sent \$4.73 (94.6% of their endowment) - very high trust!
- They kept only \$0.27
- I received \$14.19 after the 3x multiplier

This is similar to Round 1 where someone sent \$4.78 (95.6%) and I returned \$8.00.

The sender is showing exceptional trust, almost sending everything. I should reciprocate fairly.

If I return \$8.00:

- Sender ends with: $\$0.27 + \$8.00 = \$8.27$ (good gain from initial \$5)
- I keep: $\$14.19 - \$8.00 = \$6.19$
- Both benefit, sender gets rewarded for high trust

This seems consistent with my Round 1 behavior where I also returned \$8.00 for a similar high-trust send.

Actually, let me think about this more carefully. The sender sent 94.6%. In Round 1, when sender sent 95.6%, I returned \$8.00 out of \$14.34, which is about 56%.

$\$14.19 \times 0.56 \approx \7.95

So around \$8.00 seems right. Let me stick with that.

usage:

```
{
  "input_tokens": 1831,
  "output_tokens": 528,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.73, it became \$14.190000000000001, and you returned \$8.0. Your visible payoff was \$6.190000000000001.

Your total visible earnings across all rounds are \$30.919999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience as a sender:

- Round 2: Sent \$4.50 (90%), received \$7.41 back - good cooperation
- Round 4: Sent \$4.00 (80%), received \$4.58 back - lower reciprocation

The pattern suggests that higher trust levels tend to get better reciprocation. I'm averaging around \$6.18 per round with my cooperative strategy, which is working well.

As a receiver, I've been consistent in returning around 56% of what I receive, which has seemed fair to my co-players.

I'll send \$4.50 again (90% of endowment):

- Shows strong, consistent trust

- Previously resulted in good reciprocation
- Creates \$13.50 for the receiver
- Maintains my cooperative approach

```
{'send': 4.50}
```

Reasoning:

Let me review the pattern of my play:

Round 1 (Receiver): Received \$14.34 from sender who sent \$4.78 (95.6%), returned \$8.00 ? kept \$6.34
 Round 2 (Sender): Sent \$4.50 (90%), received back \$7.41 ? total \$7.91
 Round 3 (Receiver): Received \$11.40 from sender who sent \$3.80 (76%), returned \$6.50 ? kept \$4.90
 Round 4 (Sender): Sent \$4.00 (80%), received back \$4.58 ? total \$5.58
 Round 5 (Receiver): Received \$14.19 from sender who sent \$4.73 (94.6%), returned \$8.00 ? kept \$6.19

My total visible earnings: \$30.92

As a sender:

- Round 2: Sent \$4.50, got back \$7.41 (55% return rate)
- Round 4: Sent \$4.00, got back \$4.58 (38% return rate)

As a receiver:

- Round 1: Returned \$8.00 out of \$14.34 (56% return rate) when sender sent 95.6%
- Round 3: Returned \$6.50 out of \$11.40 (57% return rate) when sender sent 76%
- Round 5: Returned \$8.00 out of \$14.19 (56% return rate) when sender sent 94.6%

I've been consistent in my returns - around 56% when I'm the receiver.

When I've been the sender:

- Higher sends seem to get better returns (Round 2 with \$4.50 got 55% back)

...

```
Usage: {
  "input_tokens": 2173,
  "output_tokens": 794,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_3

Timestamp: 2026-08-12T09:24:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_1"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_1": "trustee",
      "Agent_4": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.77, that is 75.4% of their total endowment. You received \$11.31. How much do you return to the sender? (0-11.31)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.77 (75.4% of their endowment)
- They currently have: \$5 - \$3.77 = \$1.23
- I received: \$11.31
- Total money created: \$1.23 + \$11.31 = \$12.54

The sender took a significant risk and was quite generous. I want to reciprocate that trust while ensuring we both benefit from cooperation.

If I return \$5.50:

- Sender ends with: \$1.23 + \$5.50 = \$6.73
- I end with: \$11.31 - \$5.50 = \$5.81
- Both of us gain compared to the initial \$5, rewarding the sender's trust

This seems like a fair reciprocation that rewards their generosity while still allowing both of us to benefit.

{ 'return': 5.50 }

reasoning:

Let me think about this situation carefully.

I'm the RECEIVER in Round 1.

- The sender started with \$5
- They sent me \$3.77 (which is 75.4% of their \$5 endowment)
- I received \$11.31 (which is $\$3.77 \times 3$)
- I need to decide how much to return (between \$0 and \$11.31)

The sender showed significant trust by sending 75.4% of their endowment. Let me think about fair reciprocity:

If the sender sent \$3.77:

- They kept: $\$5 - \$3.77 = \$1.23$
- I received: \$11.31

For a fair split of the total money created:

- Total money now = $\$1.23$ (sender kept) + $\$11.31$ (I received) = $\$12.54$
- Fair split would be roughly \$6.27 each
- So I should return enough so the sender has ~\$6.27
- Sender currently has \$1.23, so I'd need to return about \$5.04 to get them to \$6.27
- That would leave me with $\$11.31 - \$5.04 = \$6.27$

Alternatively, I could think about matching their generosity:

- They sent 75.4% of their endowment
- If I return 75.4% of what I received: $0.754 \times \$11.31 = \8.53
- That seems quite generous but might be excessive

Another approach - proportional reciprocity:

- They gave me \$3.77 out of their \$5 (75.4%)
- A reasonable return might be to give back around half of what I received, ensuring both benefit
- Half would be \$5.66

Let me think about common strategies in trust games:

1. Return nothing (free-ride)
2. Return what was sent (break even for sender)
3. Return half of what received (both benefit equally from the multiplier)
4. Return enough to equalize final payoffs

Given their generosity, returning \$5-6 seems appropriate. This would give the sender roughly \$6-7 while leaving me with similar amounts, reflecting mutual gain from the exchange without being excessive....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1020,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.77, it became \$11.31, and you returned \$5.5. Your visible payoff was \$5.8100000000000005.

Your total visible earnings across all rounds are \$5.8100000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, I saw cooperation work well when the sender showed significant trust by sending 75.4% of their endowment, and I reciprocated fairly.

I'll adopt a similar cooperative strategy with my new co-player. Sending a substantial amount can maximize joint gains if they reciprocate.

```
{'send': 3.75}
```

reasoning:

I'm now the SENDER in Round 2.

Let me think about this strategically:

1. In the previous round as RECEIVER, I saw someone send \$3.77 (75.4% of their endowment), showing significant trust
2. I returned \$5.50, which gave them \$6.73 total - a good return on their trust

Now I'm the SENDER. What should I consider?

- The person sending \$3.77 showed that cooperation can work well
- I reciprocated fairly, returning enough that they profited significantly
- If my current co-player is similarly cooperative, sending a substantial amount could benefit both of us

Strategy considerations:

- Sending nothing is safe but generates no surplus
- Sending a moderate amount (around \$2-3) tests the waters
- Sending a larger amount (like \$3.77 that I saw) shows trust and can maximize joint gains if reciprocated

Given that:

- I saw cooperation work in the last round
- The game seems to reward trust
- I benefited from someone trusting me and reciprocated fairly
- There may be communication noise (mentioned in instructions), but the pattern of cooperation seemed clear

I think I should send a substantial amount to show trust and potentially establish a cooperative pattern. I'll send around \$3.50-\$4.00.

Let me send \$3.50 (70% of endowment) - this shows significant trust similar to what I received, but slightly less aggressive to be somewhat cautious given this is a new pairing.

Actually, let me reconsider. The previous sender sent 75.4% (\$3.77). I reciprocated well. If I want to encourage cooperation, I should send a similar proportion. Let me send \$3.75 (75% of my endowment) to match that level of trust.

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 494,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$5.66 to you. Your visible payoff was \$6.91.

Your total visible earnings across all rounds are \$12.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience:

- Round 2: I sent \$3.75 (75%), received \$5.66 back, ended with \$6.91 - a good outcome
- Round 1: As receiver, cooperation was rewarded

The cooperative approach has been working well. I'm playing with a new co-player now, so I'll continue signaling trust with a substantial send.

```
{'send': 3.75}
```

reasoning:

Let me analyze the previous round:

- I sent \$3.75 (75% of my endowment)
- It became \$11.25
- They returned \$5.66
- My payoff was \$5 - \$3.75 + \$5.66 = \$6.91

This worked out well! The receiver reciprocated reasonably (returning about 50% of what they received), and I gained \$1.91 above my initial endowment.

This is a new co-player this round. Based on my successful cooperative strategy so far:

- Round 1 as receiver: sender sent 75.4%, I returned fairly
- Round 2 as sender: I sent 75%, receiver returned fairly

The pattern of cooperation seems to be working. I should continue with a similar high-trust strategy. I'll send around 75% again to signal cooperation and trust.

\$3.75 worked well last time. I could send the same amount or similar.

usage:

```
{
  "input_tokens": 1074,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$6.34 to you. Your visible payoff was \$7.59.

Your total visible earnings across all rounds are \$20.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$4.09, so you received \$12.27. How much do you return? (0-12.27)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.09 (81.8% of their endowment)
- They currently have: \$5 - \$4.09 = \$0.91
- I received: \$12.27
- Total created: \$13.18

This is even more trust than I saw in Round 1 (when sender sent 75.4%). I want to reward this high level of cooperation appropriately.

If I return \$6.50:

- Sender ends with: \$0.91 + \$6.50 = \$7.41 (gain of \$2.41)
- I end with: \$12.27 - \$6.50 = \$5.77 (gain of \$0.77)

Both of us benefit, and the sender's exceptional trust is well-rewarded.

{'return': 6.50}

reasoning:

Let me analyze this situation:

- The sender sent \$4.09 out of \$5, which is 81.8% of their endowment - very high trust!
- They now have: \$5 - \$4.09 = \$0.91
- I received: \$12.27
- Total money in the system: \$0.91 + \$12.27 = \$13.18

This sender showed even more trust than my previous partners. I want to reciprocate appropriately.

Looking at my previous behavior:

- Round 1: As receiver, got \$11.31, returned \$5.50 (about 48.6%)
- The sender had sent 75.4%
- That seemed fair and both parties benefited

This sender sent 81.8%, even more generous. I should reciprocate with a good return.

If I return \$6.50:

- Sender ends with: \$0.91 + \$6.50 = \$7.41
- I end with: \$12.27 - \$6.50 = \$5.77
- Sender gains \$2.41 over initial \$5

- I gain \$0.77 over initial \$5

Both benefit, and the sender's higher risk/trust is rewarded with higher gains.

Actually, let me think about this differently. The sender sent 81.8% which is very high. In Round 1, sender sent 75.4% and I returned 48.6% of what I received.

For 81.8% send rate, maybe I should return around 50-53% to reward the extra trust?

$\$12.27 * 0.52 = \6.38

If I return \$6.40:

- Sender ends with: $\$0.91 + \$6.40 = \$7.31$

- I end with: $\$12.27 - \$6.40 = \$5.87$

This seems fair - both gain, sender gets rewarded for high trust.

...

usage:

```
{
  "input_tokens": 1310,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.09, it became \$12.27, and you returned \$6.5. Your visible payoff was \$5.77.

Your total visible earnings across all rounds are \$26.080000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my track record:

- As sender: Sent \$3.75 twice, received good returns (\$5.66 and \$6.34)

- As receiver: Responded fairly to cooperative senders

The high-trust strategy continues to work well across different co-players. I'll maintain this approach.

{'send': 3.75}

reasoning:

Let me review my history:

- Round 1: Receiver - sender sent \$3.77, I returned \$5.50, payoff \$5.81

- Round 2: Sender - I sent \$3.75, got back \$5.66, payoff \$6.91

- Round 3: Sender - I sent \$3.75, got back \$6.34, payoff \$7.59

- Round 4: Receiver - sender sent \$4.09, I returned \$6.50, payoff \$5.77

Total visible earnings: \$26.08

My sending strategy has been working well - sending around \$3.75 (75%) has led to good returns. In Round 4, I saw an even higher send amount (\$4.09, 81.8%) and I rewarded it well.

Now I'm facing a new co-player as sender. The cooperative strategy has been consistently working across different opponents. I should continue with a similar high-trust signal.

I'll send \$3.75 again, which has proven successful.

usage:

```
{
  "input_tokens": 1637,
  "output_tokens": 324,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$6.54 to you. Your visible payoff was \$7.79.

Your total visible earnings across all rounds are \$33.870000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$4.1, so you received \$12.299999999999999. How much do you return? (0-12.299999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.10 (82% of their endowment)
- They currently have: \$5 - \$4.10 = \$0.90
- I received: \$12.30
- Total created: \$13.20

This is very similar to Round 4 where the sender sent \$4.09 and I returned \$6.50, which worked well for both parties.

If I return \$6.50:

- Sender ends with: \$0.90 + \$6.50 = \$7.40
- I end with: \$12.30 - \$6.50 = \$5.80

Both of us benefit substantially, rewarding the sender's high trust.

{'return': 6.50}

Reasoning:

The sender sent \$4.10, which is 82% of their endowment - very high trust again!

They currently have: \$5 - \$4.10 = \$0.90

I received: \$12.30

Total created: \$13.20

This is similar to Round 4 where sender sent \$4.09. I returned \$6.50 then.

Let me be consistent and reward this high trust appropriately.

If I return \$6.50:

- Sender ends with: \$0.90 + \$6.50 = \$7.40
- I end with: \$12.30 - \$6.50 = \$5.80

Both benefit, sender gets a good reward for their trust.

Usage: {

"input_tokens": 1863,

"output_tokens": 354,

"reasoning_tokens": 0

}

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-08-12T09:24:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_4",

"move_index": 6,

"opponent_id": "Agent_3",

"pairing": {

"agents": [

"Agent_5",

"Agent_3"

],

"investor": "Agent_5",

"roles": {

"Agent_3": "trustee",

"Agent_5": "investor"

},

"trustee": "Agent_3"

```

},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.3, that is 46.0% of their total endowment. You received \$6.899999999999995. How much do you return to the sender? (0-6.899999999999995)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$2.3 (46% of their endowment)
- I received \$6.9 after the 3x multiplier
- The sender currently has \$2.7 remaining

I want to:

1. Reward the sender's trust so they benefit from sending
2. Split the gains fairly

3. Establish myself as trustworthy for future rounds

If I return \$3.0:

- Sender ends with: $\$2.7 + \$3.0 = \$5.7$ (a gain of \$0.7)
- I end with: $\$6.9 - \$3.0 = \$3.9$

This seems fair - the sender profits from trusting me, and we both benefit from the multiplier effect.

```
'''json
{"return": 3.0}
'''
```

reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.3 (which is 46% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$6.9 ($2.3 \times 3 = 6.9$)
4. I need to decide how much to return (between \$0 and \$6.9)

The sender's current situation:

- Started with \$5
- Sent \$2.3
- Currently has: $\$5 - \$2.3 = \$2.7$

If I return nothing:

- Sender ends with: \$2.7
- I end with: \$6.9

The sender showed trust by sending nearly half their endowment. A fair approach would be to:

1. Ensure the sender doesn't lose money
2. Split the gains reasonably

Total money in the system now: $\$2.7$ (sender has) + $\$6.9$ (I have) = $\$9.6$

This started from two \$5 endowments = \$10 total, so we have \$9.6 total.

Wait, let me recalculate:

- Sender started with \$5, sent \$2.3, has \$2.7 left
- I received \$6.9 (the \$2.3 multiplied by 3)
- Total in system: $\$2.7 + \$6.9 = \$9.6$

If we split equally: each person would get \$4.8

- For sender to have \$4.8, they need: $\$4.8 - \$2.7 = \$2.1$ returned
- For me to have \$4.8, I'd keep: $\$6.9 - \$2.1 = \$4.8$?

But equal split might not be the best strategy. The sender took a risk by sending money. A common approach in trust games is to:

reward their trust by ensuring they come out ahead of their initial \$5, while also sharing in the multiplier gains. Let me test a few return amounts: returning \$3.0 gives them \$5.7 total (a \$0.7 gain) while I keep \$3.9; returning \$2.5 gives them \$5.2 (a \$0.2 gain) while I keep \$4.4; returning \$2.1 would give them exactly \$4.8 and me \$4.8.

A return around \$3.0-\$3.3 seems optimal-it meaningfully rewards their trust while still letting me benefit from the multiplier effect. At \$3.0, they gain \$0.7 and I keep \$3.9, which feels like a fair split that demonstrates trustworthiness.

The sender's 46% send rate suggests moderate confidence in me, so I should reciprocate proportionally...

usage:

```
{
  "input_tokens": 539,
  "output_tokens": 1314,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You

saw them send \$2.3, it became \$6.899999999999995, and you returned \$3.0. Your visible payoff was \$3.899999999999995.
Your total visible earnings across all rounds are \$3.899999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm now the sender with a new co-player
- In the previous round, I experienced a sender who sent 46% and I returned fairly
- I want to establish trust with this new partner while being reasonably cautious

Sending \$2.5 (50% of my endowment) seems like a good strategy:

- It shows significant trust
- It's a clear, fair starting point
- If my co-player reciprocates fairly, we both benefit from the 3x multiplier
- It's not overly risky with an unknown partner

```
'''json
{"send": 2.5}
'''
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.3 (46% of their endowment)
- I received \$6.9
- I returned \$3.0
- My payoff was \$3.9

Now I'm playing against a different agent (since it says "current co-player" and agents are randomly paired each round).

I need to decide how much to send. Key considerations:

1. I don't know anything about this new co-player's behavior
2. The sender in Round 1 sent 46% (\$2.3), which was a reasonable trust signal
3. I returned about 43.5% of what I received (\$3/\$6.9), which was fair
4. I want to build trust but also protect myself since I don't know this opponent

Strategies:

- Sending \$0 would be overly defensive and wouldn't build any relationship
- Sending \$5 (100%) would be too risky with an unknown partner
- Sending around \$2-3 seems reasonable as it shows trust without being reckless

Given that the previous sender sent \$2.3 (46%), I could mirror that behavior. This would be consistent and shows I'm willing to trust at a similar level to what I've experienced.

Alternatively, I could send slightly more to show goodwill, or slightly less to be more cautious.

I think sending \$2.5 (50%) is a good middle ground:

- It's a clear half of my endowment, easy to interpret
- It shows significant trust
- If they return fairly (around 40-45% of \$7.5 = \$3-3.4), we both profit
- It's not too risky

Let me calculate the outcomes:

- If I send \$2.5, they receive \$7.5
- I keep \$2.5

...

usage:

```
{
  "input_tokens": 870,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

}

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$6.78 to you. Your visible payoff was \$9.280000000000001.

Your total visible earnings across all rounds are \$13.18.

This round, you are the RECEIVER against your current co-player. They sent you \$4.17, so you received \$12.51. How much do you return? (0-12.51)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.17 (83.4% of their endowment)
- I received \$12.51 after the 3x multiplier
- The sender currently has only \$0.83 remaining

This is very high trust that deserves to be rewarded generously.

If I return \$6.0:

- Sender ends with: $\$0.83 + \$6.0 = \$6.83$ (profit of \$1.83)
- I end with: $\$12.51 - \$6.0 = \$6.51$ (profit of \$6.51)

This gives us nearly equal final payoffs and rewards the sender's exceptional trust with a solid profit.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- The sender sent \$4.17 (83.4% of their \$5 endowment) - this is very high trust!
- I received \$12.51 (after 3x multiplier)
- The sender currently has $\$5 - \$4.17 = \$0.83$ remaining

This sender showed enormous trust by sending 83.4% of their endowment. I need to reward this trust generously.

Let me think about what's fair:

- Total in the system: $\$0.83$ (sender has) + $\$12.51$ (I have) = $\$13.34$
- If we split evenly: $\$6.67$ each

If I return \$5.84:

- Sender ends with: $\$0.83 + \$5.84 = \$6.67$
- I end with: $\$12.51 - \$5.84 = \$6.67$
- Perfect 50/50 split

But given they showed such high trust (83.4%), I might want to be even more generous to:

1. Reward their exceptional trust
2. Encourage continued cooperation
3. Establish myself as highly trustworthy

However, I also saw in round 2 that when I sent \$2.5 (50%), I got back \$6.78, which gave me a payoff of \$9.28. That was very generous from my co-player then (they returned more than 90% of what they received!).

Let me recalculate for this round:

- If I return \$6:
 - Sender: $\$0.83 + \$6 = \$6.83$
 - Me: $\$12.51 - \$6 = \$6.51$
 - Almost equal, slight advantage to sender
- If I return \$6.5:
 - Sender: $\$0.83 + \$6.5 = \$7.33$

- Me: $\$12.51 - \$6.5 = \$6.01$
- Sender gets more, rewarding their trust

Given the extreme trust shown here, I should return enough to give them a meaningful advantage while still being reasonable. Returning around \$6 seems appropriate-it creates nearly equal outcomes with a slight edge to the sender who took the bigger risk...

usage:

```
{
  "input_tokens": 1154,
  "output_tokens": 955,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.17, it became \$12.51, and you returned \$6.0. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$19.689999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

- Round 2: When I sent \$2.5 (50%), I received a very generous return of \$6.78
- Round 3: When someone sent me \$4.17 (high trust), I reciprocated with \$6.0

The game seems to reward trust. I'll send \$3.0 (60% of my endowment), which shows strong trust while being measured with a new co-player.

```
'''json
{"send": 3.0}
'''
```

reasoning:

Looking at my history:

- Round 1: RECEIVER - received from sender who sent \$2.3, I returned \$3.0 fairly
- Round 2: SENDER - I sent \$2.5, receiver returned \$6.78 (very generous!)
- Round 3: RECEIVER - received from sender who sent \$4.17 (very high trust), I returned \$6.0

Current situation:

- I'm the SENDER with a new co-player
- Total earnings so far: \$19.69

My experience shows:

- When I sent \$2.5 in Round 2, I got a very generous return of \$6.78
- When others showed high trust to me, I reciprocated fairly
- This game seems to reward trust

I should send a significant amount to show trust. Given that:

- Round 2 sender sent 46% and I reciprocated
- Round 3 sender sent 83.4% and I reciprocated generously
- When I sent 50% in Round 2, I got excellent returns

I think sending around \$3.0 (60%) would be a good balance - showing strong trust without being reckless with an unknown partner.

Actually, let me think about this more strategically. I've seen that:

- Lower trust (46%) got fair returns
- Medium trust (50% from me) got very generous returns
- High trust (83.4%) got generous returns

Sending \$3.0 seems like a good middle-high trust level that should encourage reciprocation

usage:

```
{
```

```
"input_tokens": 1461,
"output_tokens": 474,
"reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.22 to you. Your visible payoff was \$7.22.

Your total visible earnings across all rounds are \$26.909999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.84, so you received \$11.52. How much do you return? (0-11.52)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender showed high trust by sending \$3.84 (76.8% of their endowment)
- I received \$11.52 after the 3x multiplier
- The sender currently has only \$1.16 remaining

This level of trust deserves a generous return.

If I return \$6.0:

- Sender ends with: \$1.16 + \$6.0 = \$7.16 (profit of \$2.16)
- I end with: \$11.52 - \$6.0 = \$5.52

This rewards their high trust with a good profit and keeps our final payoffs relatively balanced.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$3.84, which is 76.8% of their endowment - very high trust
- I received \$11.52 after the 3x multiplier
- The sender currently has \$5 - \$3.84 = \$1.16 remaining

This is high trust that should be rewarded well.

Let me consider different return amounts:

If I return \$5.5:

- Sender ends with: \$1.16 + \$5.5 = \$6.66 (profit of \$1.66)
- I end with: \$11.52 - \$5.5 = \$6.02

If I return \$6.0:

- Sender ends with: \$1.16 + \$6.0 = \$7.16 (profit of \$2.16)
- I end with: \$11.52 - \$6.0 = \$5.52

Given the high trust (76.8%), I should return generously. \$6.0 seems fair and rewards their trust well.

usage:

```
{
  "input_tokens": 1718,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.84, it became \$11.52, and you returned \$6.0. Your visible payoff was \$5.52.

Your total visible earnings across all rounds are \$32.42999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience:

- Round 2: Sent \$2.5 (50%), received \$6.78 back - great return
- Round 4: Sent \$3.0 (60%), received \$5.22 back - solid return

The game has been rewarding trust. I'll continue to show strong trust with this new co-player by sending \$3.5 (70% of my endowment).

```
'''json
{"send": 3.5}
'''
```

Reasoning:

Let me analyze my experience:

Round 1 (Receiver): Received \$6.9 from sender sending \$2.3, returned \$3.0. Payoff: \$3.9
Round 2 (Sender): Sent \$2.5, received \$6.78 back. Payoff: \$9.28
Round 3 (Receiver): Received \$12.51 from sender sending \$4.17, returned \$6.0. Payoff: \$6.51
Round 4 (Sender): Sent \$3.0, received \$5.22 back. Payoff: \$7.22
Round 5 (Receiver): Received \$11.52 from sender sending \$3.84, returned \$6.0. Payoff: \$5.52

So far:

- As a sender, I've sent \$2.5 and \$3.0 (50% and 60%)
- Both times I received generous returns (\$6.78 and \$5.22)
- As a receiver, I've been fair/generous in my returns

The game continues to reward trust. I've been gradually increasing my sending amounts. Let me continue this trend and send \$3.5 (70% of endowment) to show strong trust with this new co-player.

```
Usage: {
  "input_tokens": 2009,
  "output_tokens": 417,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_4

Timestamp: 2026-08-12T09:24:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_3"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_3": "trustee",
      "Agent_5": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system
content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round with no prior history, I want to signal cooperation and trust while being somewhat cautious.

In this trust game, sending money creates value (3x multiplier), but requires trusting that my co-player will reciprocate. Since:

- We'll likely play multiple rounds (possibly with different partners)
- Building a cooperative reputation could benefit me when I'm the receiver
- The socially optimal strategy involves sending substantial amounts to maximize total value

I'll send a significant amount to demonstrate trust, but keep a small buffer given the uncertainty.

```
{"send": 4}
```

reasoning:

This is the first round of a trust game where I am the SENDER. Let me think through the strategic considerations:

1. **Game Structure:**

- I have \$5
- I can send \$0-\$5 to the receiver

- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- My final payoff: \$5 - sent + returned
- Receiver's payoff: 3×sent - returned

2. ****Strategic Considerations:****

- If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
- If I send \$5: Receiver gets \$15, and could return anywhere from \$0-\$15
 - If they return \$0: I get \$0 total (lost \$5)
 - If they return \$5: I get \$5 total (break even)
 - If they return \$7.50: I get \$7.50 (gain \$2.50)
 - If they return \$10: I get \$10 (gain \$5)

3. ****Trust Game Logic:****

- This is a classic trust game
- Sending money shows trust
- The socially optimal outcome is for me to send the maximum (\$5), which creates \$15 total, and then split fairly
- A fair split might be: I sent \$5, created \$15, so maybe 50/50 of the gain (\$15-\$5=\$10 gain, so \$5 each) meaning they return \$10 and keep \$5
- Or they could view it as: I took the risk, so 50/50 of total created value: they return \$7.50, we each get \$7.50

4. ****First Round Considerations:****

- This is round 1, so there's no history to build on
- However, there will be multiple rounds with potentially different partners
- Establishing a cooperative reputation early could pay dividends later, especially since I'll alternate between sender and receiver roles
- The communication noise mentioned means amounts might shift slightly, which could complicate trust signals
- A moderate to high send amount (\$3-5) demonstrates willingness to cooperate while managing risk, potentially encouraging reciprocal behavior in future rounds

I'm leaning toward sending \$4 as my opening move. This signals meaningful trust-it triples to \$12 for the receiver-while keeping \$1 as a buffer. Even a modest return of \$4-5 puts me ahead, and it establishes the cooperative tone I'd want if positions reverse. Classic trust game research suggests starting with substantial trust and adjusting based on responses works better than cautious opening moves, so I'll go with that approach....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 1038,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.69 to you. Your visible payoff was \$9.69.

Your total visible earnings across all rounds are \$9.69.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

My co-player sent me \$3.11 (62% of their endowment), showing significant trust. I received \$9.33 after the 3x multiplier.

In my previous round, my co-player returned \$8.69 out of \$12 (72%), which was very generous. I want to maintain this cooperative spirit.

A fair approach would be to ensure my co-player profits from their trust while I also benefit from receiving the tripled amount.

Returning approximately \$6.50 would give them a final payoff of about \$8.39 (\$5 - \$3.11 +

\$6.50), which rewards their trust handsomely, while I keep \$2.83.

```
{"return": 6.50}
```

reasoning:

Let me analyze this situation:

1. Last round I was the SENDER:
 - I sent \$4.0
 - It became \$12.0
 - They returned \$8.69
 - My payoff: $\$5 - \$4 + \$8.69 = \9.69
 - Their payoff: $\$12 - \$8.69 = \$3.31$
2. This round I'm the RECEIVER:
 - They sent me \$3.11
 - I received $\$3.11 \times 3 = \9.33
 - I need to decide how much to return

My co-player sent me \$3.11 out of their \$5, which is 62.2% of their endowment. This shows significant trust.

In the last round, my co-player returned \$8.69 out of \$12 they received, which is 72.4% of what they received. This was quite generous - they gave me more than I sent them.

I should consider:

- Reciprocity: They showed trust by sending \$3.11
- Building reputation: Being fair will help in future rounds
- The note says there may be communication noise, so amounts might differ slightly

A fair return would be:

- They invested \$3.11, I should at least return that amount so they don't lose
- To split gains fairly: They gave up \$3.11, it became \$9.33, so we created \$6.22 in additional value
- If we split that gain: $\$3.11 + \$6.22/2 = \$3.11 + \$3.11 = \$6.22$

However, I also want to be somewhat generous to build trust, similar to how my previous co-player returned 72.4% of what they received.

Let me think about different approaches:

- Return exactly what they sent: \$3.11 (they break even)
- Return to split gains 50/50: $\sim \$6.22$ (they get $\$5 - \$3.11 + \$6.22 = \8.11)
- Return more generously (similar to 72.4% rate): $\sim \$6.75$ (they get $\$5 - \$3.11 + \$6.75 = \8.64)

Given the trust they showed and the precedent from my last game, I'll return around \$6.50-\$7.00, which gives them a good payoff while keeping a reasonable amount for myself. I'll return \$6.50: their final payoff becomes \$8.39 while I keep \$9.

Wait, let me check the math more carefully. If I return \$6.50, my payoff is $\$9.33 - \$6.50 = \$2.83$. That feels...

usage:

```
{
  "input_tokens": 763,
  "output_tokens": 1019,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$6.5. Your visible payoff was \$2.83.

Your total visible earnings across all rounds are \$12.52.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

- Round 1: Sent \$4, received \$8.69 back (very cooperative partner)
- Round 2: Received from someone who sent \$3.11, I returned \$6.50 (both of us were cooperative)

The environment has been quite cooperative. Sending more creates more value through the 3x multiplier, and past interactions suggest people reciprocate trust.

To maximize value creation and continue building a cooperative reputation, I'll send my full endowment this round.

```
{"send": 5}
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$4, received \$8.69 back, payoff \$9.69

Round 2: I was RECEIVER, got \$9.33 (from \$3.11 sent), returned \$6.50, payoff \$2.83

Total: \$12.52

Now Round 3: I'm SENDER again with a new co-player.

My experience so far:

- When I sent \$4, I got back \$8.69, which was very generous (72% return rate on the triple d amount)
- When I received from someone who sent \$3.11, I returned \$6.50 (70% return rate)

Both interactions have been quite cooperative. This suggests the environment is reasonably trustworthy.

As a sender this round, I should consider:

1. Past cooperation suggests I can trust my co-players
2. Sending more creates more value overall (3x multiplier)
3. Building a reputation for cooperation may help when I'm receiver
4. The game mentioned communication noise, so exact amounts may vary

Given the positive experiences, I should send a substantial amount. I'll send \$4 again since that worked well before, or maybe even \$5 to maximize value creation.

Actually, let me think about this more strategically. If I send \$5:

- Co-player receives \$15
- If they return even 50%, I get \$7.50 back, netting me \$2.50 profit
- If they're as generous as my Round 1 partner (~72%), I could get ~\$10.80 back

If I send \$4:

- Co-player receives \$12
- At 72% return, I'd get ~\$8.64 back, total payoff ~\$9.64

Given the cooperative environment and to maximize total value, I'll send \$5 this time.

usage:

```
{
  "input_tokens": 1058,
  "output_tokens": 560,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$10.29 to you. Your visible payoff was \$10.29.

Your total visible earnings across all rounds are \$22.81.

This round, you are the RECEIVER against your current co-player. They sent you \$3.57, so you received \$10.709999999999999. How much do you return? (0-10.709999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

My co-player sent \$3.57 (71.4% of their endowment), showing strong trust. I received \$10.71.

The game has been very cooperative so far:

- Round 1: I sent \$4, got back \$8.69
- Round 3: I sent \$5, got back \$10.29
- Round 2: When I was receiver, I returned \$6.50 out of \$9.33

My co-player trusted me significantly. A generous return would be around 70% of what I received, which would give them a strong profit while still rewarding me for receiving.

Returning \$7.50 would:

- Give them: $\$5 - \$3.57 + \$7.50 = \8.93 total
- Leave me with: $\$10.71 - \$7.50 = \$3.21$

This maintains the cooperative pattern.

```
{"return": 7.50}
```

reasoning:

Let me analyze this situation:

My co-player sent me \$3.57 out of \$5 (71.4% of their endowment), showing strong trust. I received \$10.71 after the 3x multiplier.

In my previous round as sender:

- I sent \$5 (100% of endowment)
- They returned \$10.29 out of \$15 (68.6%)
- This was very generous, giving me \$10.29 total payoff

The pattern has been quite cooperative. Let me think about a fair return:

If I return around \$7-7.50:

- They would get: $\$5 - \$3.57 + \$7 = \8.43 (if I return \$7)
- They would get: $\$5 - \$3.57 + \$7.50 = \8.93 (if I return \$7.50)
- I would keep: $\$10.71 - \$7 = \$3.71$ (if I return \$7)
- I would keep: $\$10.71 - \$7.50 = \$3.21$ (if I return \$7.50)

Given they sent 71.4% of their endowment, a generous return would be appropriate. The previous receiver returned about 68.6% when I sent 100%.

Returning about \$7.20 would:

- Give them \$8.63 total ($\$5 - \$3.57 + \7.20)
- Leave me with \$3.51
- This rewards their trust well

Actually, let me be a bit more generous to match the cooperative environment. Returning \$7.50:

- Gives them \$8.93 total (a \$3.93 profit on their \$5)
- Leaves me with \$3.21 (also a good profit)
- Return rate: 70% of what I received

...

usage:

```
{
  "input_tokens": 1326,
  "output_tokens": 675,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.57, it became \$10.709999999999999, and you returned \$7.5. Your visible payoff was \$3.2099999999999999.

Your total visible earnings across all rounds are \$26.019999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my track record:

- As SENDER: Sent \$4 (got \$8.69), Sent \$5 (got \$10.29)
- As RECEIVER: Returned generously both times

The environment is highly cooperative. Sending the full amount has consistently yielded very good returns for me. My average earnings per round (\$6.50) significantly exceed the \$5 baseline, showing that cooperation pays off.

I'll continue the cooperative strategy and send my full endowment.

```
{"send": 5}
```

reasoning:

Looking at my history:

- Round 1 (SENDER): Sent \$4, got back \$8.69 - great outcome
- Round 2 (RECEIVER): Got \$9.33, returned \$6.50
- Round 3 (SENDER): Sent \$5, got back \$10.29 - excellent outcome
- Round 4 (RECEIVER): Got \$10.71, returned \$7.50

The pattern is very clear - I've experienced highly cooperative behavior. When I've been the sender and sent significant amounts, I've gotten generous returns. When I've been the receiver, I've also returned generously.

My cumulative earnings are \$26.02 across 4 rounds, averaging \$6.50 per round, which is well above the \$5 baseline if I had just kept everything.

The optimal strategy in this cooperative environment is to continue sending high amounts:

1. It creates value through the 3x multiplier
2. Past experience shows I get good returns
3. It builds my reputation for future rounds

I should send \$5 again to maximize value creation and maintain my cooperative reputation.

usage:

```
{
  "input_tokens": 1678,
  "output_tokens": 377,
  "reasoning_tokens": 0
}
```

Message 12 role=user**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$8.2 to you. Your visible payoff was \$8.2.

Your total visible earnings across all rounds are \$34.22.

This round, you are the RECEIVER against your current co-player. They sent you \$3.41, so you received \$10.23. How much do you return? (0-10.23)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

My co-player sent \$3.41 (68% of their endowment), showing solid trust. I received \$10.23.

My pattern as receiver has been:

- Round 2: Returned \$6.50 out of \$9.33 (~70%)
- Round 4: Returned \$7.50 out of \$10.71 (~70%)

Maintaining this cooperative approach, returning around \$7 would:

- Give them: \$5 - \$3.41 + \$7 = \$8.59 total
- Leave me with: \$10.23 - \$7 = \$3.23

This rewards their trust while staying consistent with my cooperative strategy.

```
{"return": 7.00}
```

Reasoning:

My co-player sent \$3.41 (68.2% of their endowment), showing good trust. I received \$10.23.

Looking at my history:

- As SENDER: Sent \$4 (got \$8.69), Sent \$5 (got \$10.29), Sent \$5 (got \$8.2)
- As RECEIVER: Returned \$6.50 out of \$9.33, Returned \$7.50 out of \$10.71

The pattern has been very cooperative. When I receive around \$10, I've been returning between \$6.50-\$7.50.

For this round with \$10.23 received:

- Returning around \$7 would give my co-player: $\$5 - \$3.41 + \$7 = \8.59
- I'd keep: $\$10.23 - \$7 = \$3.23$

This seems fair and maintains the cooperative pattern. I'll return \$7.

Actually, let me reconsider. My previous returns as receiver:

- Round 2: Received \$9.33, returned \$6.50 (69.7%)
- Round 4: Received \$10.71, returned \$7.50 (70%)

So I've been returning around 70% of what I received. 70% of \$10.23 = \$7.16

I'll return \$7.00 to stay consistent with my cooperative pattern.

```
Usage: {
  "input_tokens": 1931,
  "output_tokens": 513,
  "reasoning_tokens": 0
}
```

ROUND 7

Roles: Agent_1=Receiver, Agent_2=Receiver, Agent_3=Sender, Agent_4=Sender, Agent_5=Sender, Agent_6=Sender, Agent_7=Receiver, Agent_8=Receiver

Pairings:

dyad_1: Agent_3=Sender, Agent_2=Receiver
 dyad_2: Agent_6=Sender, Agent_8=Receiver
 dyad_3: Agent_4=Sender, Agent_1=Receiver
 dyad_4: Agent_5=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_3 sent \$5; received \$15; Agent_2 returned \$7.5; payoffs Agent_3=\$7.5, Agent_2=\$7.5;
 noise/visible: returned visible=\$7.64

Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 15.0, "min": 0, "raw": 7.5}, "sent": {"amount": 5.0, "clamped": false, "max": 5, "min": 0, "raw": 5.0}}

dyad_2: Agent_6 sent \$3.5; received \$10.5; Agent_8 returned \$4.1; payoffs Agent_6=\$5.6, Agent_8=\$6.4; noise/visible: sent visible=\$2.99, received visible=\$8.97, returned visible=\$4.47

Action validation: {"returned": {"amount": 4.1, "clamped": false, "max": 10.5, "min": 0, "raw": 4.1}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

dyad_3: Agent_4 sent \$4; received \$12; Agent_1 returned \$8; payoffs Agent_4=\$9, Agent_1=\$4; noise/visible: sent visible=\$5, received visible=\$15, returned visible=\$8.87

Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 12.0, "min": 0, "raw": 8.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad_4: Agent_5 sent \$4; received \$12; Agent_7 returned \$8; payoffs Agent_5=\$9, Agent_7=\$4; noise/visible: sent visible=\$4.47, received visible=\$13.41, returned visible=\$8.61

Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 12.0, "min": 0, "raw": 8.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 41.25,
 "Agent_2": 40.25,
 "Agent_3": 42.0,
 "Agent_4": 43.0,
 "Agent_5": 50.0,
 "Agent_6": 46.1,
 "Agent_7": 38.0,
 "Agent_8": 45.9

}

Visible balances after round: {

"Agent_1": 46.67,
 "Agent_2": 44.319999999999999,
 "Agent_3": 45.09,
 "Agent_4": 47.059999999999995,
 "Agent_5": 49.759999999999999,
 "Agent_6": 42.589999999999996,
 "Agent_7": 37.19,

```
"Agent_8": 44.02000000000001
}
```

LLM CALLS

GAME - Agent_3 - Sender - dyad_1

Timestamp: 2026-08-12T09:24:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_2"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_2": "trustee",
      "Agent_3": "investor"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths
, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round with no prior history, I want to signal cooperation and trust while being somewhat cautious.

In this trust game, sending money creates value (3x multiplier), but requires trusting that my co-player will reciprocate. Since:

- We'll likely play multiple rounds (possibly with different partners)
- Building a cooperative reputation could benefit me when I'm the receiver
- The socially optimal strategy involves sending substantial amounts to maximize total value

I'll send a significant amount to demonstrate trust, but keep a small buffer given the uncertainty.

{"send": 4}

reasoning:

This is the first round of a trust game where I am the SENDER. Let me think through the strategic considerations:

1. ****Game Structure:****

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return
- My final payoff: \$5 - sent + returned
- Receiver's payoff: 3xsent - returned

2. ****Strategic Considerations:****

- If I send \$0: I keep \$5, receiver gets \$0. Safe but no potential gain.
- If I send \$5: Receiver gets \$15, and could return anywhere from \$0-\$15
 - If they return \$0: I get \$0 total (lost \$5)
 - If they return \$5: I get \$5 total (break even)
 - If they return \$7.50: I get \$7.50 (gain \$2.50)
 - If they return \$10: I get \$10 (gain \$5)

3. ****Trust Game Logic:****

- This is a classic trust game
- Sending money shows trust
- The socially optimal outcome is for me to send the maximum (\$5), which creates \$15 total, and then split fairly
 - A fair split might be: I sent \$5, created \$15, so maybe 50/50 of the gain (\$15-\$5=\$10 gain, so \$5 each) meaning they return \$10 and keep \$5
 - Or they could view it as: I took the risk, so 50/50 of total created value: they return \$7.50, we each get \$7.50

4. ****First Round Considerations:****

- This is round 1, so there's no history to build on
- However, there will be multiple rounds with potentially different partners
- Establishing a cooperative reputation early could pay dividends later, especially since I'll alternate between sender and receiver roles
- The communication noise mentioned means amounts might shift slightly, which could complicate trust signals
- A moderate to high send amount (\$3-5) demonstrates willingness to cooperate while managing risk, potentially encouraging reciprocal behavior in future rounds

I'm leaning toward sending \$4 as my opening move. This signals meaningful trust-it triples to \$12 for the receiver-while keeping \$1 as a buffer. Even a modest return of \$4-5 puts

me ahead, and it establishes the cooperative tone I'd want if positions reverse. Classic trust game research suggests starting with substantial trust and adjusting based on responses works better than cautious opening moves, so I'll go with that approach....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 1038,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.69 to you. Your visible payoff was \$9.69.

Your total visible earnings across all rounds are \$9.69.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

My co-player sent me \$3.11 (62% of their endowment), showing significant trust. I received \$9.33 after the 3x multiplier.

In my previous round, my co-player returned \$8.69 out of \$12 (72%), which was very generous. I want to maintain this cooperative spirit.

A fair approach would be to ensure my co-player profits from their trust while I also benefit from receiving the tripled amount.

Returning approximately \$6.50 would give them a final payoff of about \$8.39 (\$5 - \$3.11 + \$6.50), which rewards their trust handsomely, while I keep \$2.83.

```
{"return": 6.50}
```

reasoning:

Let me analyze this situation:

1. Last round I was the SENDER:

- I sent \$4.0
- It became \$12.0
- They returned \$8.69
- My payoff: \$5 - \$4 + \$8.69 = \$9.69
- Their payoff: \$12 - \$8.69 = \$3.31

2. This round I'm the RECEIVER:

- They sent me \$3.11
- I received \$3.11 × 3 = \$9.33
- I need to decide how much to return

My co-player sent me \$3.11 out of their \$5, which is 62.2% of their endowment. This shows significant trust.

In the last round, my co-player returned \$8.69 out of \$12 they received, which is 72.4% of what they received. This was quite generous - they gave me more than I sent them.

I should consider:

- Reciprocity: They showed trust by sending \$3.11
- Building reputation: Being fair will help in future rounds
- The note says there may be communication noise, so amounts might differ slightly

A fair return would be:

- They invested \$3.11, I should at least return that amount so they don't lose
- To split gains fairly: They gave up \$3.11, it became \$9.33, so we created \$6.22 in additional value
- If we split that gain: \$3.11 + \$6.22/2 = \$3.11 + \$3.11 = \$6.22

However, I also want to be somewhat generous to build trust, similar to how my previous co-player returned 72.4% of what they received.

Let me think about different approaches:

- Return exactly what they sent: \$3.11 (they break even)
- Return to split gains 50/50: ~\$6.22 (they get \$5 - \$3.11 + \$6.22 = \$8.11)
- Return more generously (similar to 72.4% rate): ~\$6.75 (they get \$5 - \$3.11 + \$6.75 = \$8.64)

Given the trust they showed and the precedent from my last game, I'll return around \$6.50-\$7.00, which gives them a good payoff while keeping a reasonable amount for myself. I'll return \$6.50: their final payoff becomes \$8.39 while I keep \$9.

Wait, let me check the math more carefully. If I return \$6.50, my payoff is \$9.33 - \$6.50 = \$2.83. That feels...

usage:

```
{
  "input_tokens": 763,
  "output_tokens": 1019,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$6.5. Your visible payoff was \$2.83.

Your total visible earnings across all rounds are \$12.52.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

- Round 1: Sent \$4, received \$8.69 back (very cooperative partner)
- Round 2: Received from someone who sent \$3.11, I returned \$6.50 (both of us were cooperative)

The environment has been quite cooperative. Sending more creates more value through the 3x multiplier, and past interactions suggest people reciprocate trust.

To maximize value creation and continue building a cooperative reputation, I'll send my full endowment this round.

```
{"send": 5}
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$4, received \$8.69 back, payoff \$9.69

Round 2: I was RECEIVER, got \$9.33 (from \$3.11 sent), returned \$6.50, payoff \$2.83

Total: \$12.52

Now Round 3: I'm SENDER again with a new co-player.

My experience so far:

- When I sent \$4, I got back \$8.69, which was very generous (72% return rate on the tripled amount)
- When I received from someone who sent \$3.11, I returned \$6.50 (70% return rate)

Both interactions have been quite cooperative. This suggests the environment is reasonably trustworthy.

As a sender this round, I should consider:

1. Past cooperation suggests I can trust my co-players
2. Sending more creates more value overall (3x multiplier)
3. Building a reputation for cooperation may help when I'm receiver
4. The game mentioned communication noise, so exact amounts may vary

Given the positive experiences, I should send a substantial amount. I'll send \$4 again since that worked well before, or maybe even \$5 to maximize value creation.

Actually, let me think about this more strategically. If I send \$5:

- Co-player receives \$15
- If they return even 50%, I get \$7.50 back, netting me \$2.50 profit
- If they're as generous as my Round 1 partner (~72%), I could get ~\$10.80 back

If I send \$4:

- Co-player receives \$12
- At 72% return, I'd get ~\$8.64 back, total payoff ~\$9.64

Given the cooperative environment and to maximize total value, I'll send \$5 this time.

usage:

```
{
  "input_tokens": 1058,
  "output_tokens": 560,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$10.29 to you. Your visible payoff was \$10.29.

Your total visible earnings across all rounds are \$22.81.

This round, you are the RECEIVER against your current co-player. They sent you \$3.57, so you received \$10.709999999999999. How much do you return? (0-10.709999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

My co-player sent \$3.57 (71.4% of their endowment), showing strong trust. I received \$10.71.

The game has been very cooperative so far:

- Round 1: I sent \$4, got back \$8.69
- Round 3: I sent \$5, got back \$10.29
- Round 2: When I was receiver, I returned \$6.50 out of \$9.33

My co-player trusted me significantly. A generous return would be around 70% of what I received, which would give them a strong profit while still rewarding me for receiving.

Returning \$7.50 would:

- Give them: \$5 - \$3.57 + \$7.50 = \$8.93 total
- Leave me with: \$10.71 - \$7.50 = \$3.21

This maintains the cooperative pattern.

```
{"return": 7.50}
```

reasoning:

Let me analyze this situation:

My co-player sent me \$3.57 out of \$5 (71.4% of their endowment), showing strong trust. I received \$10.71 after the 3x multiplier.

In my previous round as sender:

- I sent \$5 (100% of endowment)
- They returned \$10.29 out of \$15 (68.6%)
- This was very generous, giving me \$10.29 total payoff

The pattern has been quite cooperative. Let me think about a fair return:

If I return around \$7-7.50:

- They would get: \$5 - \$3.57 + \$7 = \$8.43 (if I return \$7)
- They would get: \$5 - \$3.57 + \$7.50 = \$8.93 (if I return \$7.50)
- I would keep: \$10.71 - \$7 = \$3.71 (if I return \$7)

- I would keep: $\$10.71 - \$7.50 = \$3.21$ (if I return $\$7.50$)

Given they sent 71.4% of their endowment, a generous return would be appropriate. The previous receiver returned about 68.6% when I sent 100%.

Returning about $\$7.20$ would:

- Give them $\$8.63$ total ($\$5 - \$3.57 + \$7.20$)
- Leave me with $\$3.51$
- This rewards their trust well

Actually, let me be a bit more generous to match the cooperative environment. Returning $\$7.50$:

- Gives them $\$8.93$ total (a $\$3.93$ profit on their $\$5$)
- Leaves me with $\$3.21$ (also a good profit)
- Return rate: 70% of what I received

...

usage:

```
{
  "input_tokens": 1326,
  "output_tokens": 675,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send $\$3.57$, it became $\$10.709999999999999$, and you returned $\$7.5$. Your visible payoff was $\$3.2099999999999999$.

Your total visible earnings across all rounds are $\$26.019999999999996$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my track record:

- As SENDER: Sent $\$4$ (got $\$8.69$), Sent $\$5$ (got $\$10.29$)
- As RECEIVER: Returned generously both times

The environment is highly cooperative. Sending the full amount has consistently yielded very good returns for me. My average earnings per round ($\$6.50$) significantly exceed the $\$5$ baseline, showing that cooperation pays off.

I'll continue the cooperative strategy and send my full endowment.

```
{"send": 5}
```

reasoning:

Looking at my history:

- Round 1 (SENDER): Sent $\$4$, got back $\$8.69$ - great outcome
- Round 2 (RECEIVER): Got $\$9.33$, returned $\$6.50$
- Round 3 (SENDER): Sent $\$5$, got back $\$10.29$ - excellent outcome
- Round 4 (RECEIVER): Got $\$10.71$, returned $\$7.50$

The pattern is very clear - I've experienced highly cooperative behavior. When I've been the sender and sent significant amounts, I've gotten generous returns. When I've been the receiver, I've also returned generously.

My cumulative earnings are $\$26.02$ across 4 rounds, averaging $\$6.50$ per round, which is well above the $\$5$ baseline if I had just kept everything.

The optimal strategy in this cooperative environment is to continue sending high amounts:

1. It creates value through the 3x multiplier
2. Past experience shows I get good returns
3. It builds my reputation for future rounds

I should send $\$5$ again to maximize value creation and maintain my cooperative reputation.

usage:

```
{
  "input_tokens": 1678,
```

```

        "output_tokens": 377,
        "reasoning_tokens": 0
    }
Message 12 role=user
  content:
    Round 6

    In your most recent previous game round, you were the SENDER against another agent. You sent $5.0, it became $15.0, and you saw them return $8.2 to you. Your visible payoff was $8.2.
    Your total visible earnings across all rounds are $34.22.

    This round, you are the RECEIVER against your current co-player. They sent you $3.41, so you received $10.23. How much do you return? (0-10.23)
    Respond exactly as JSON: {'return': <amount>}
Message 13 role=assistant
  content:
    Looking at this situation:

    My co-player sent $3.41 (68% of their endowment), showing solid trust. I received $10.23.

    My pattern as receiver has been:
    - Round 2: Returned $6.50 out of $9.33 (~70%)
    - Round 4: Returned $7.50 out of $10.71 (~70%)

    Maintaining this cooperative approach, returning around $7 would:
    - Give them: $5 - $3.41 + $7 = $8.59 total
    - Leave me with: $10.23 - $7 = $3.23

    This rewards their trust while staying consistent with my cooperative strategy.

    {"return": 7.00}
  reasoning:
    My co-player sent $3.41 (68.2% of their endowment), showing good trust. I received $10.23.

    Looking at my history:
    - As SENDER: Sent $4 (got $8.69), Sent $5 (got $10.29), Sent $5 (got $8.2)
    - As RECEIVER: Returned $6.50 out of $9.33, Returned $7.50 out of $10.71

    The pattern has been very cooperative. When I receive around $10, I've been returning between $6.50-$7.50.

    For this round with $10.23 received:
    - Returning around $7 would give my co-player: $5 - $3.41 + $7 = $8.59
    - I'd keep: $10.23 - $7 = $3.23

    This seems fair and maintains the cooperative pattern. I'll return $7.

    Actually, let me reconsider. My previous returns as receiver:
    - Round 2: Received $9.33, returned $6.50 (69.7%)
    - Round 4: Received $10.71, returned $7.50 (70%)

    So I've been returning around 70% of what I received. 70% of $10.23 = $7.16

    I'll return $7.00 to stay consistent with my cooperative pattern.
  usage:
    {
      "input_tokens": 1931,
      "output_tokens": 513,
      "reasoning_tokens": 0
    }
Message 14 role=user
  content:
    Round 7

    In your most recent previous game round, you were the RECEIVER against another agent. You saw them send $3.41, it became $10.23, and you returned $7.0. Your visible payoff was $3.2300000000000004.
    Your total visible earnings across all rounds are $37.45.

```

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience:

As SENDER:

- Sent \$4, received \$8.69
- Sent \$5, received \$10.29
- Sent \$5, received \$8.20

All my sending decisions have been profitable, and the environment remains cooperative. Sending the full \$5 maximizes potential gains through the 3x multiplier and has consistently yielded good returns.

I'll continue with the cooperative strategy.

```
{"send": 5}
```

Reasoning:

Looking at my history:

- As SENDER:
 - * Round 1: Sent \$4, got back \$8.69
 - * Round 3: Sent \$5, got back \$10.29
 - * Round 5: Sent \$5, got back \$8.20
- As RECEIVER:
 - * Round 2: Received \$9.33, returned \$6.50
 - * Round 4: Received \$10.71, returned \$7.50
 - * Round 6: Received \$10.23, returned \$7.00

The environment has been consistently cooperative. All three times I sent money as a sender, I received generous returns. When I send \$5, I've gotten back \$10.29 and \$8.20 (averaging about \$9.25), both of which are profitable.

I'm earning well above the \$5 baseline per round. The cooperative strategy is working. I should continue sending my full endowment to maximize value creation and maintain the cooperative equilibrium.

```
Usage: {  
  "input_tokens": 2228,  
  "output_tokens": 341,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:25:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_2"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_2": "trustee",  
      "Agent_3": "investor"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to establish a cooperative approach while being reasonably cautious since I have no history with my co-player.

The multiplication factor of 3x creates an opportunity for mutual benefit. If I send a moderate amount, we can both potentially end up better off than the initial \$5.

I'll send **\$3**, which gives my co-player \$9 to work with. This shows trust and willingness to cooperate. If they return approximately \$3-4, we'll both be in a good position, and it could establish a positive pattern for future rounds.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about the strategic considerations:

1. If I send money, it gets tripled, which creates potential for mutual benefit

2. The receiver then decides how much to return
3. There's communication noise mentioned - amounts may appear different than actually sent
4. This is a multi-agent setting with 8 agents total, pairs rotate
5. This is the first round, so I have no history with any player yet

Since this is the first round and I have no information about other players' behavior, I need to consider:

- Sending \$0 guarantees I keep \$5, but no potential for mutual gain
- Sending everything (\$5) maximizes the total pot (\$15 for receiver) but is risky
- A moderate amount balances risk and potential reward

In trust games, common strategies include:

- Sending about half as a signal of conditional cooperation
- The tripling effect means even partial cooperation can benefit both parties

Let me consider sending \$3:

- I send \$3, receiver gets \$9
- If receiver returns \$3, I end up with \$5 (same as starting)
- If receiver returns \$4-5, I'm better off
- If receiver returns less than \$3, I'm worse off
- This shows trust while keeping some insurance

Alternatively, sending \$2.5:

- I send \$2.5, receiver gets \$7.5
- If receiver returns \$2.5, I end up with \$5
- If receiver returns \$3+, I'm better off
- Less risk than \$3

Or sending \$2:

- I send \$2, receiver gets \$6
- If receiver returns \$2, I end up with \$5
- Lower risk, but also lower potential reward

Given this is the first round and I want to establish a cooperative pattern, I'll send \$3.

This shows trust and willingness to cooperate while not being completely vulnerable.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 589,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.81 to you. Your visible payoff was \$4.8100000000000005.

Your total visible earnings across all rounds are \$4.8100000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$3.95, so you received \$11.850000000000001. How much do you return? (0-11.850000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.95 (79% of their endowment), showing significant trust
- I received \$11.85 total
- They currently have only \$1.05 left

I want to:

1. Reward their trust and cooperation
2. Ensure they end up better than their initial \$5
3. Set a positive tone for potential future interactions

If I return \$5.50:

- They end up with: \$1.05 + \$5.50 = \$6.55
- I end up with: \$11.85 - \$5.50 = \$6.35

Both of us gain from cooperation, and this reciprocates their high level of trust appropriately.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the SENDER
- I sent \$3.00
- It became \$9.00 for the receiver
- They returned \$2.81
- My payoff: $\$5 - \$3 + \$2.81 = \4.81

Round 2:

- I am now the RECEIVER
- My co-player (a different agent) sent me \$3.95
- I received $\$3.95 \times 3 = \11.85
- I need to decide how much to return

Important considerations:

1. The sender gave me \$3.95 out of their \$5 - that's 79% of their endowment, showing high trust
2. If I return nothing, I keep \$11.85 and they end up with $\$5 - \$3.95 = \$1.05$
3. I should return enough to make cooperation worthwhile for them
4. There's communication noise mentioned, so amounts may differ slightly

What would be fair?

- The sender gave up \$3.95, keeping \$1.05
- I received \$11.85
- If I want them to end up better than their initial \$5, I should return more than \$3.95
- A 50-50 split of total surplus could work: Total = $\$1.05 + \$11.85 = \$12.90$. Half is \$6.45
- If I return \$5.40, sender gets $\$1.05 + \$5.40 = \$6.45$, I get $\$11.85 - \$5.40 = \$6.45$

But let me think about reciprocity:

- In round 1, I sent \$3 (60% of endowment), received \$9, they returned \$2.81 (31% of what they received)
- This round, sender sent \$3.95 (79% of endowment), I received \$11.85
- Using that same 31% return rate would give me \$3.67 to send back

The sender showed significantly more generosity than I did initially, so I should reciprocate with a higher return. Returning \$5 would leave them with \$6.05 total-better than the ir starting \$5-while I'd keep \$6.85, rewarding their trust and encouraging continued cooperation....

usage:

```
{
  "input_tokens": 793,
  "output_tokens": 977,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.95, it became \$11.850000000000001, and you returned \$5.5. Your visible payoff was \$6.350000000000001.

Your total visible earnings across all rounds are \$11.160000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at Round 3:

I'm paired with a new co-player (pairings are randomized each round), so I have no history

with them. My previous experiences showed:

- Round 1: Sent \$3, received \$2.81 back - reasonable cooperation
- Round 2: As receiver, I tried to be generous with a high-trusting sender

For this new pairing, I'll use a similar approach to Round 1 - send a moderate amount that signals cooperation while being reasonably cautious with an unknown partner.

Sending \$3 creates \$9 for them to work with, which gives room for mutual benefit if they reciprocate fairly.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$3, received back \$2.81, payoff \$4.81

Round 2: I was RECEIVER, got \$11.85 from a \$3.95 send, returned \$5.50, payoff \$6.35

Total earnings: \$11.16

Now Round 3: I'm SENDER again with a new co-player (since pairings are randomized each round).

Key considerations:

- I have no history with this particular co-player
- In Round 1, when I sent \$3, I got back \$2.81 (which is reasonable - about 31% return rate)
- In Round 2, when someone sent me \$3.95, I returned \$5.50 (about 46% return rate)
- The game has communication noise, so exact amounts may differ

Since I'm paired with a new player and have no history with them, I should probably use a similar strategy to Round 1 - establish cooperation but be somewhat cautious.

Given that:

- Sending \$3 worked reasonably well in Round 1
- The multiplication factor of 3x creates good mutual benefit potential
- I want to signal cooperation

I'll send \$3 again. This is a moderate amount that shows willingness to cooperate without being overly risky with a new partner.

usage:

```
{
  "input_tokens": 1106,
  "output_tokens": 478,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.52 to you. Your visible payoff was \$8.52.

Your total visible earnings across all rounds are \$19.68.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.36 (67% of their endowment), showing solid trust
- I received \$10.08 total
- They currently have \$1.64 left

I want to reciprocate their trust fairly. If I return \$5.00:

- They end up with: \$1.64 + \$5.00 = \$6.64 (gain of \$1.64)
- I end up with: \$10.08 - \$5.00 = \$5.08 (gain of \$0.08)

Both of us benefit from cooperation, and this rewards their trust appropriately.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- My co-player sent \$3.36 (67.2% of their endowment)
- I received \$10.08 total
- They currently have: \$5 - \$3.36 = \$1.64 left

My previous round (Round 3):

- I sent \$3, received \$6.52 back - very generous return!

This co-player sent \$3.36, showing good trust. I should reciprocate appropriately.

If I return \$5.00:

- They end up with: \$1.64 + \$5.00 = \$6.64
- I end up with: \$10.08 - \$5.00 = \$5.08

Both of us gain. This is a fair reciprocation that rewards their trust.

Alternatively, if I return \$5.50:

- They end up with: \$1.64 + \$5.50 = \$7.14
- I end up with: \$10.08 - \$5.50 = \$4.58

This would be even more generous, similar to how I acted in Round 2.

Given that they sent about 67% of their endowment (compared to 79% in Round 2), I think returning around \$5 is appropriate and fair. It ensures they profit significantly from their trust, and we both end up better off than the initial \$5.

I'll go with \$5.00 as a fair reciprocation.

usage:

```
{
  "input_tokens": 1402,
  "output_tokens": 522,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$5.0. Your visible payoff was \$5.08.

Your total visible earnings across all rounds are \$24.759999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at Round 5:

I'm with another new co-player. My experience so far shows:

- Sending \$3 has worked well in previous rounds, getting returns of \$2.81 and \$6.52
- The cooperative approach has generally been rewarded

I'll continue with the same strategy - send \$3 to signal trust and cooperation while being reasonably cautious with an unknown partner.

```
'''json
{"send": 3}
'''
```

reasoning:

Round 5, I'm the sender with a new co-player (random pairings each round).

Looking at my history:

- Round 1 (Sender): Sent \$3, got back \$2.81 - decent cooperation
- Round 2 (Receiver): Received \$11.85, returned \$5.50 - I was generous
- Round 3 (Sender): Sent \$3, got back \$6.52 - excellent return!
- Round 4 (Receiver): Received \$10.08, returned \$5.00 - fair reciprocation

So far, sending \$3 has worked well - I got \$2.81 and \$6.52 back in my two sender rounds. The average return has been positive.

For this new pairing, I should continue with a cooperative strategy. Sending \$3 seems to be working as a good middle-ground approach that encourages cooperation.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1690,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.15 to you. Your visible payoff was \$5.15.

Your total visible earnings across all rounds are \$29.909999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.47, so you received \$13.41. How much do you return? (0-13.41)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.47 (89% of their endowment), showing very high trust
- I received \$13.41 total
- They currently have only \$0.53 left

This is a very trusting sender, and I should reciprocate generously. If I return \$6.50:

- They end up with: $\$0.53 + \$6.50 = \$7.03$ (gain of \$2.03)
- I end up with: $\$13.41 - \$6.50 = \$6.91$ (gain of \$1.91)

Both of us gain approximately \$2, which appropriately rewards their exceptional trust and maintains strong cooperation.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Looking at this situation:

- My co-player sent \$4.47 (89% of their endowment), showing very high trust
- I received \$13.41 total
- They currently have \$0.53 left

This is a very trusting sender. I should reciprocate generously.

If I return \$6.50:

- They end up with: $\$0.53 + \$6.50 = \$7.03$ (gain of \$2.03)
- I end up with: $\$13.41 - \$6.50 = \$6.91$ (gain of \$1.91)

Both benefit well, and this appropriately rewards their very high level of trust.

Let me think about this more. They sent \$4.47 out of \$5, leaving themselves with only \$0.53. That's 89.4% of their endowment. This is extremely trusting.

I should return enough to:

1. Ensure they profit significantly from their trust
2. Show I value cooperation
3. Both of us should end up better than our starting \$5

If I return \$6.50:

- Them: $\$0.53 + \$6.50 = \$7.03$
- Me: $\$13.41 - \$6.50 = \$6.91$

Both gain roughly \$2, which seems fair and reciprocates their high trust.

usage:

```
{
  "input_tokens": 1931,
  "output_tokens": 502,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.47, it became \$13.41, and you returned \$6.5. Your visible payoff was \$6.91.

Your total visible earnings across all rounds are \$36.81999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent the maximum \$5 (100% of their endowment), showing complete trust
- I received \$15 total
- They currently have \$0 left

This is the highest level of trust possible. I should reciprocate very generously. If I return \$7.50:

- They end up with: $\$0 + \$7.50 = \$7.50$ (gain of \$2.50)
- I end up with: $\$15 - \$7.50 = \$7.50$ (gain of \$2.50)

We both gain equally from this maximum cooperation, which appropriately rewards their complete trust.

```
'''json
{"return": 7.50}
'''
```

Reasoning:

This is Round 7, and I'm the RECEIVER.

My co-player sent the maximum amount: \$5 (100% of their endowment), showing complete trust.

- I received \$15 total
- They currently have \$0 left

This sender showed maximum trust. I should reciprocate very generously.

If I return \$7.50:

- They end up with: $\$0 + \$7.50 = \$7.50$ (gain of \$2.50)
- I end up with: $\$15 - \$7.50 = \$7.50$ (gain of \$2.50)

This gives us both equal gains and rewards their complete trust. This seems fair and reciprocates their maximum trust appropriately.

Actually, let me think about this more. They sent everything (\$5), so they're showing maximum trust. I want to reward that very well.

If I return \$8:

- They end up with: $\$0 + \$8 = \$8$ (gain of \$3)
- I end up with: $\$15 - \$8 = \$7$ (gain of \$2)

This gives them a slightly better outcome for taking the risk of sending everything.

Let me go with \$7.50 as a good middle ground - we both gain equally, which seems fair when the y showed maximum trust.

```
Usage: {
  "input_tokens": 2240,
  "output_tokens": 476,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_2

Timestamp: 2026-08-12T09:25:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_8"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_6": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.25, that is 45.0% of their total endowment. You received \$6.75. How much do you return to the sender? (0-6.75)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate trust by sending \$2.25 (45% of their endowment)
- This was multiplied to \$6.75 for me
- The sender currently has \$2.75 remaining

I want to reward their trust while maintaining a fair distribution. If I return \$3.00:

- The sender ends up with \$5.75 (a gain of \$0.75)
- I keep \$3.75 (a gain of \$3.75)
- Both of us are better off than if no money had been sent

This seems like a reasonable balance that rewards their moderate trust and encourages future cooperation.

{"return": 3.00}

reasoning:

Let me think through this situation carefully.

I'm the RECEIVER in this round of a trust game.

The sender:

- Started with \$5
- Sent me \$2.25 (which is 45% of their endowment)
- This was multiplied by 3x, so I received \$6.75

Now I need to decide how much to return.

Key considerations:

1. The sender showed moderate trust by sending 45% of their endowment
2. They could have sent nothing, or they could have sent everything
3. If I return nothing, I keep \$6.75 and they keep \$2.75 (\$5 - \$2.25)
4. If I return some amount, we both benefit from the cooperative exchange

Let me think about fair or reasonable return amounts:

- If I return \$3.00: The sender ends up with $\$2.75 + \$3.00 = \$5.75$ (gain of \$0.75)
- I keep: $\$6.75 - \$3.00 = \$3.75$
- Total: $\$5.75 + \$3.75 = \$9.50$
- If I return \$2.25: The sender ends up with $\$2.75 + \$2.25 = \$5.00$ (break even)
- I keep: $\$6.75 - \$2.25 = \$4.50$
- Total: $\$5.00 + \$4.50 = \$9.50$
- If I return \$3.50: The sender ends up with $\$2.75 + \$3.50 = \$6.25$ (gain of \$1.25)
- I keep: $\$6.75 - \$3.50 = \$3.25$
- Total: $\$6.25 + \$3.25 = \$9.50$

Common approaches in trust games:

I could mirror their proportion-returning about 45% of what I received, which would be roughly \$3.04. Alternatively, I could split the cooperative gains equally: the \$4.50 surplus created through the exchange divided in half means returning \$2.25 to give them an even split of the benefit. Or I could aim for a return that ensures they profit from trust.

ng me, somewhere between \$2.25 and \$3.50....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1174,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.25, it became \$6.75, and you returned \$3.0. Your visible payoff was \$3.75.

Your total visible earnings across all rounds are \$3.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In the previous round as receiver, I experienced moderate trust (45% sent) and reciprocated fairly. Now I'm facing a new co-player as sender.

Since this is a new pairing and I want to establish cooperative behavior while managing risk, I'll send a moderate-to-good amount. Sending \$3.00 (60% of my endowment) signals trust and willingness to cooperate, while still keeping some safety margin.

This will give my co-player \$9.00 to work with, providing them good opportunity to reciprocate if they're inclined to cooperate.

```
{"send": 3.00}
```

reasoning:

Let me think about this situation:

In Round 1:

- I was the receiver
- The sender sent \$2.25 (45% of their endowment)
- I received \$6.75 and returned \$3.00
- I kept \$3.75
- The sender ended with $\$2.75 + \$3.00 = \$5.75$

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- I'm playing against a different co-player (since pairings are randomized each round)

The previous sender showed moderate trust by sending 45% of their endowment. I returned about 44% of what I received (\$3.00 out of \$6.75), which gave them a positive return.

Now I need to decide how much to send. Considerations:

1. This is a new opponent, so I don't know their behavior yet
2. In multi-round games with changing partners, establishing a reputation for cooperation can be beneficial
3. The previous sender's moderate trust (45%) seemed reasonable
4. There's communication noise, so amounts might differ slightly

I think I should send a moderate-to-good amount to signal trust and cooperation. Sending a round 50-60% seems reasonable:

- \$2.50 (50%) or \$3.00 (60%)

Given that the previous sender sent 45% and I want to be at least as cooperative (and perhaps signal willingness to cooperate), I'll send \$3.00 (60%). This will become \$9.00 for the receiver, giving them good opportunity to reciprocate.

Actually, let me reconsider. The previous interaction went well with a 45% send rate. I could match that or go slightly higher. \$3.00 (60%) shows I'm willing to be cooperative and trust my new partner.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 571,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.42 to you. Your visible payoff was \$8.42.

Your total visible earnings across all rounds are \$12.17.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player showed **maximum trust** by sending their entire \$5 endowment (100%)
- This became \$15 for me
- They currently have \$0

This exceptional level of trust deserves strong reciprocation. If I return \$10.00:

- The sender ends with \$10.00 (doubling their money, gain of \$5.00)
- I keep \$5.00 (gain of \$5.00)
- We both gain equally from our cooperation

This rewards their maximum trust appropriately and creates a fair outcome where both parties benefit substantially.

```
{"return": 10.00}
```

reasoning:

Let me analyze this situation:

Previous round:

- I was the SENDER
- I sent \$3.00 (60% of my endowment)
- My co-player received \$9.00
- They returned \$6.42 (71.3% of what they received)
- I ended with \$8.42

This round:

- I am the RECEIVER
- My current co-player sent \$5.00 (100% of their endowment!)
- I received \$15.00
- They currently have \$0
- I need to decide how much to return (0-15)

This is a very strong show of trust - they sent their entire endowment! This deserves a strong reciprocation.

Let me think about fair outcomes:

- If I return \$7.50, they end with \$7.50 (gain of \$2.50), I keep \$7.50 (gain of \$7.50)
- If I return \$10.00, they end with \$10.00 (gain of \$5.00), I keep \$5.00 (gain of \$5.00) - equal split of total gains
- If I return \$12.00, they end with \$12.00 (gain of \$7.00), I keep \$3.00 (gain of \$3.00)

Given they showed maximum trust by sending everything, I should reciprocate generously. An equal split of total gains seems fair - we'd each gain \$5 from the cooperation. That means returning \$10.00.

Actually, let me reconsider. They sent everything (\$5), which became \$15 for me. If I return \$10, they get \$10 total (doubling their money), and I keep \$5 (gaining \$5). Both of us gain equally from the transaction.

This seems like a fair and generous reciprocation for their maximum trust.

usage:

```
{
  "input_tokens": 1082,
  "output_tokens": 580,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$5, it became \$15, and you returned \$10.0. Your visible payoff was \$5.0. Your total visible earnings across all rounds are \$17.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

- Round 2 (sender): Sent \$3.00 (60%) and received excellent reciprocation (\$6.42)
- Round 3 (receiver): Received maximum trust (\$5.00 sent) and reciprocated generously

The pattern so far shows strong cooperation in this game. Since this is a new co-player, I want to signal trust and encourage cooperation. Given the positive experiences, I'll send a high amount while keeping a small buffer.

Sending \$4.00 (80% of my endowment) demonstrates strong trust and gives my co-player \$12.00 to work with, creating good opportunity for mutual benefit.

```
{"send": 4.00}
```

reasoning:

Let me analyze the situation:

Previous rounds:

1. Round 1: RECEIVER - saw \$2.25 sent, received \$6.75, returned \$3.00, kept \$3.75
2. Round 2: SENDER - sent \$3.00, became \$9.00, saw \$6.42 returned, payoff \$8.42
3. Round 3: RECEIVER - saw \$5.00 sent, received \$15.00, returned \$10.00, kept \$5.00

Total visible earnings: \$17.17

Now Round 4: I'm the SENDER against a new co-player.

Key observations:

- In Round 2, I sent \$3.00 (60%) and got very good reciprocation (\$6.42 return on \$9.00)
- In Round 3, someone sent me everything (\$5.00) and I reciprocated generously (\$10.00 return)

I've seen good cooperation so far. My previous experience as sender (Round 2) with moderate-high trust level worked well.

For this round:

- This is a new co-player (different from previous rounds)
- I don't know their behavior yet
- But the overall pattern in this game seems cooperative
- I've been both generous and received generosity

Given the positive experiences so far, I should maintain a cooperative strategy. I'll send a substantial amount again. Sending \$3.50 (70%) shows strong trust while keeping a small safety margin.

Actually, let me reconsider. The game seems quite cooperative overall. In Round 3, someone trusted me completely with \$5.00, and in Round 2 when I sent \$3.00 I got excellent reciprocation.

I could send \$4.00 (80%) to signal strong cooperation, or even \$5.00 to match the maximum trust I received in Round 3.

Given the high cooperation levels I've experienced, I'll send \$4.00 (80% of endowment) - t

his is a strong signal of trust without going to the absolute maximum.

...

usage:

```
{
  "input_tokens": 1343,
  "output_tokens": 646,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.64 to you. Your visible payoff was \$7.64.

Your total visible earnings across all rounds are \$24.810000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.67, so you received \$11.01. How much do you return? (0-11.01)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.67 (73.4% of their endowment) - this shows strong trust
- This became \$11.01 for me
- They currently have \$1.33 remaining

Their strong trust (sending nearly 3/4 of their endowment) deserves good reciprocation. If I return \$7.00:

- The sender ends with \$8.33 (gain of \$3.33)
- I keep \$4.01 (gain of \$4.01)
- Both of us benefit, with returns roughly proportional to their trust level

This balances rewarding their trust while maintaining fairness.

{"return": 7.00}

reasoning:

Let me analyze this situation:

- My co-player sent \$3.67 (73.4% of their endowment) - this is strong trust
- This became \$11.01 for me
- They currently have \$1.33 remaining

Looking at my pattern of reciprocation:

- Round 1: Received \$6.75, returned \$3.00 (44.4% return rate)
- Round 3: Received \$15.00, returned \$10.00 (66.7% return rate) - but they sent everything

And when I've been sender:

- Round 2: Sent \$3.00 (60%), received \$6.42 back (71.3% return rate from receiver)
- Round 4: Sent \$4.00 (80%), received \$6.64 back (55.3% return rate from receiver)

The sender showed strong trust at 73.4%. I should reciprocate well, but need to be fair.

If I return \$6.00:

- Sender ends with \$1.33 + \$6.00 = \$7.33 (gain of \$2.33)
- I keep \$11.01 - \$6.00 = \$5.01 (gain of \$5.01)

This is about a 54.5% return rate, which seems reasonable given their strong (but not maximum) trust.

If I return \$7.00:

- Sender ends with \$1.33 + \$7.00 = \$8.33 (gain of \$3.33)
- I keep \$11.01 - \$7.00 = \$4.01 (gain of \$4.01)

This is about a 63.6% return rate, which rewards their strong trust more generously.

Given they sent 73.4% of their endowment (strong trust), I think \$7.00 is appropriate. It's more generous than my Round 1 reciprocation (where they sent 45%) but less than my Round 3 reciprocation (where they sent ...

usage:

```
{
  "input_tokens": 1645,
  "output_tokens": 659,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.67, it became \$11.01, and you returned \$7.0. Your visible payoff was \$4.01.

Your total visible earnings across all rounds are \$28.82.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my experience as sender:

- Round 2: Sent \$3.00 (60%) ? received \$6.42 back, strong return
- Round 4: Sent \$4.00 (80%) ? received \$6.64 back, good return

Both times I've been sender, trust has been rewarded. This is a new co-player, and I want to continue building cooperative patterns while being reasonably cautious.

Sending \$3.50 (70% of my endowment) strikes a good balance - it shows trust and gives my co-player \$10.50 to work with, creating opportunity for mutual benefit.

```
{"send": 3.50}
```

reasoning:

Looking at my history:

- Round 2 (sender): Sent \$3.00 (60%), received \$6.42 back, payoff \$8.42
- Round 3 (receiver): Received \$5.00 (100%), returned \$10.00, payoff \$5.00
- Round 4 (sender): Sent \$4.00 (80%), received \$6.64 back, payoff \$7.64
- Round 5 (receiver): Received \$3.67 (73.4%), returned \$7.00, payoff \$4.01

As a sender, I've been quite trusting (60% and 80%) and have received good returns on both occasions. As a receiver, I've been fairly generous as well.

This is Round 6, and I'm facing a new co-player. The overall pattern suggests trust pays off in this game. I should continue with a moderate-to-high trust level.

Given my successful experiences, I'll send a good amount. Sending \$3.50 (70%) seems like a balanced choice - shows trust, not too conservative, but also not going all-in like some co-players have done.

Actually, let me reconsider. My sending history:

- \$3.00 (60%) - got back \$6.42, net gain \$3.42
- \$4.00 (80%) - got back \$6.64, net gain \$2.64

Both were profitable. The 60% send actually gave me a better net gain. Let me go with something around 70% to balance trust and caution.

\$3.50 seems reasonable.

usage:

```
{
  "input_tokens": 1928,
  "output_tokens": 534,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.3 to you. Your visible payoff was \$7.8.

Your total visible earnings across all rounds are \$36.62.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my sending history:

- Round 2: Sent \$3.00 ? received \$6.42 back
- Round 4: Sent \$4.00 ? received \$6.64 back
- Round 6: Sent \$3.50 ? received \$6.30 back

All three times as sender, my trust has been rewarded with good returns. This pattern suggests cooperation is working well in this environment.

With a new co-player, I'll continue the strategy that's been successful. Sending \$3.50 (70%) demonstrates trust and gives them \$10.50 to work with.

```
{"send": 3.50}
```

Reasoning:

Let me review my sending history:

- Round 2: Sent \$3.00 (60%) ? received \$6.42 back
- Round 4: Sent \$4.00 (80%) ? received \$6.64 back
- Round 6: Sent \$3.50 (70%) ? received \$6.30 back

All three times I've been a sender, I've received good reciprocation. The pattern shows that trust is being rewarded consistently. This is another new co-player.

Given the consistent positive outcomes, I should continue with a trust-based strategy. Sending around 60-80% has been working well. I'll send \$3.50 again (70%), which has shown good results.

Actually, let me think - I've been varying between \$3.00, \$4.00, and \$3.50. All have worked well. I could go with \$3.50 again since it worked last time, or try something in this range.

I'll go with \$3.50 as it represents a solid middle ground.

Usage: {

```
"input_tokens": 2205,  
"output_tokens": 398,  
"reasoning_tokens": 0
```

}

GAME - Agent_8 - Receiver - dyad_2

Timestamp: 2026-08-12T09:25:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_2",  
"move_index": 3,  
"opponent_id": "Agent_6",  
"pairing": {  
  "agents": [  
    "Agent_6",  
    "Agent_8"  
  ],  
  "investor": "Agent_6",  
  "roles": {  
    "Agent_6": "investor",  
    "Agent_8": "trustee"  
  },  
  "trustee": "Agent_8"  
},  
"role": "trustee",  
"role_label": "Receiver",  
"round": 7,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this as Round 1 with no prior information about my co-player, I want to demonstrate trust while being reasonably cautious.

If I send \$3:

- My co-player receives \$9 (3×3)
- This shows meaningful cooperation
- If they return a fair portion (say $1/3$ to $1/2$), we both benefit
- My maximum loss is limited to \$3

I'll go with a moderately trusting approach:

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I have \$5 to start with

2. Whatever I send gets multiplied by 3x
3. The receiver then decides how much to return
4. I keep: \$5 - sent + returned
5. The receiver keeps: received - returned

Key considerations:

- This is a multi-round game with 8 agents
- Pairings are randomized each round
- There may be communication noise
- I don't know who my co-player is

Strategic thinking:

- If I send \$0, I guarantee keeping \$5, but there's no potential for gain
- If I send everything (\$5), the receiver gets \$15, and if they're cooperative, they might return a fair amount
- A moderate amount (like \$2-3) shows some trust but limits risk

Since this is round 1 and I have no information about my co-player:

- I want to signal cooperation and trust
- But I also want to be somewhat cautious
- Sending around \$3 seems like a reasonable middle ground that:
 - Shows good faith (60% of my endowment)
 - Gives the receiver \$9 to work with
 - If they return fairly (say 1/3 to 1/2 of what they receive), I could end up with \$5-6.
- Limits my maximum loss to \$3

Let me send \$3 as a cooperative but measured opening move.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.37 to you. Your visible payoff was \$5.37.

Your total visible earnings across all rounds are \$5.37.

This round, you are the RECEIVER against your current co-player. They sent you \$4.58, so you received \$13.74. How much do you return? (0-13.74)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

My co-player sent me \$4.58 out of their \$5 (91.6% of their endowment) - showing very high trust.

I received \$13.74. To reciprocate this trust appropriately:

- They kept \$0.42
- If I return around \$6.50, they'd end with \$6.92 (a gain of \$1.92)
- I'd keep \$7.24 (also a solid gain)

This represents returning about 47% of what I received, which rewards their trust while we both benefit substantially.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 2 - I'm now the RECEIVER

- My co-player sent me \$4.58
- I received \$13.74 (which is \$4.58 × 3)

- I need to decide how much to return

From Round 1:

- I sent \$3.00, they received \$9.00
- They returned \$3.37 to me
- This is about 37.4% of what they received, which is quite fair/generous

My co-player in this round sent me \$4.58 out of their \$5, which is 91.6% of their endowment - very trusting!

I should consider:

1. They showed high trust by sending almost everything
2. In Round 1, my previous partner returned about 37.4% which seemed fair
3. I want to reciprocate trust and maintain good cooperation

If I return about 40-50% of what I received (\$13.74):

- 40% = \$5.50
- 45% = \$6.18
- 50% = \$6.87

Given they sent almost everything (\$4.58 out of \$5), showing extreme trust, I should be generous in return.

Their final payoff would be:

- They started with \$5
- Sent \$4.58, kept \$0.42
- If I return \$6.00: $\$0.42 + \$6.00 = \$6.42$
- My payoff: $\$13.74 - \$6.00 = \$7.74$

If I return \$6.50:

- Their payoff: $\$0.42 + \$6.50 = \$6.92$
- My payoff: $\$13.74 - \$6.50 = \$7.24$

Given their high trust (sending 91.6% of endowment), I should return a good amount. Returning around \$6.50 seems fair - they end up with \$6.92 (better than their initial \$5), and I keep \$7.24 (also better than my initial \$0).

...

usage:

```
{
  "input_tokens": 755,
  "output_tokens": 653,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.58, it became \$13.74, and you returned \$6.5. Your visible payoff was \$7.24.

Your total visible earnings across all rounds are \$12.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.38, so you received \$10.14. How much do you return? (0-10.14)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust.

I received \$10.14.

They kept: \$1.62

To reciprocate fairly:

- If I return \$5.00, they'd end with \$6.62 (a gain of \$1.62)
- I'd keep \$5.14 (also a small gain)

This represents returning about 49% of what I received, which rewards their trust while we

both benefit.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust

I received \$10.14.

They kept: $\$5 - \$3.38 = \$1.62$

To be fair and reciprocate their trust:

- If I return around \$5, they'd end with $\$1.62 + \$5 = \$6.62$ (a gain of \$1.62)
- I'd keep $\$10.14 - \$5 = \$5.14$ (also a solid gain)

This represents returning about 49% of what I received, which is fair and reciprocates their trust appropriately while we both benefit.

Let me think about this more carefully. They sent \$3.38, showing moderate-to-good trust (68% of endowment).

In Round 2, my co-player sent me \$4.58 (92% of endowment) and I returned \$6.50 (47% of what I received).

For consistency and fairness here with \$3.38 sent (less trust shown):

- Returning around 47-50% of \$10.14 would be \$4.77-\$5.07
- Let me go with \$5.00

That gives them: $\$1.62 + \$5.00 = \$6.62$ (gain of \$1.62)

I keep: $\$10.14 - \$5.00 = \$5.14$ (gain of \$0.14)

This seems fair and reciprocal.

usage:

```
{
  "input_tokens": 1036,
  "output_tokens": 519,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.38, it became \$10.14, and you returned \$5.0. Your visible payoff was \$5.140000000000001.

Your total visible earnings across all rounds are \$17.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

Round 1: As SENDER, I sent \$3 and received \$3.37 back - a reasonable return.

Rounds 2-3: As RECEIVER, others showed good trust, and I reciprocated fairly.

This is a new co-player (random pairing), so I don't have specific information about them. However, the overall pattern suggests players are reasonably cooperative.

I'll send \$3.50 - slightly more than my first round to show increasing trust while remaining measured:

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$3, received back \$3.37, payoff \$5.37

Round 2: I was RECEIVER, received \$13.74 (from \$4.58), returned \$6.50, payoff \$7.24

Round 3: I was RECEIVER, received \$10.14 (from \$3.38), returned \$5.00, payoff \$5.14

Round 4: I am SENDER

My total earnings: \$17.75

Looking at the pattern:

- When I sent \$3, I got back \$3.37 (a bit more than 1:1 return on sent amount)
- When others sent to me, they showed high trust (\$4.58 and \$3.38 out of \$5)
- I've been fairly reciprocal as a receiver, returning roughly half of what I received

Now I'm a sender again. Since:

1. My first send of \$3 was met with a decent return (\$3.37)
2. Others have been trusting with me
3. I've been fair in return

I should continue with moderate to high trust. I'll send \$3.5 this time - slightly more than my first send to show I'm willing to build cooperation.

Actually, let me reconsider. The co-player this round is new (pairings are randomized). I don't know their history. In Round 1, I sent \$3 and got a reasonable return. Let me stick with a similar strategy - send \$3 again as a baseline cooperative strategy.

Actually, looking at the overall pattern, players seem reasonably cooperative. I could try \$3.5 to be slightly more trusting.

Let me go with \$3.5 - showing increased trust while not overcommitting.

usage:

```
{
  "input_tokens": 1303,
  "output_tokens": 551,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$7.22 to you. Your visible payoff was \$8.719999999999999.

Your total visible earnings across all rounds are \$26.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my recent experience:

Round 4: I sent \$3.50 and received \$7.22 back - excellent cooperation!

The overall pattern shows players are being reasonably cooperative. While this is a new co-player (random pairing), I'll show increased trust based on positive experiences.

Sending \$4:

- Demonstrates strong trust (80% of my endowment)
- Creates \$12 for them to work with
- Encourages reciprocal cooperation

```
'''json
{"send": 4}
'''
```

reasoning:

Round 4 recap: I sent \$3.50, they returned \$7.22 - excellent return! That's about 69% of what they received, very generous.

This is Round 5, and I'm the SENDER again against a new co-player (random pairing each round).

So far my experience:

- Round 1 SENDER: sent \$3, got \$3.37 back (reasonable)
- Round 2 RECEIVER: received \$13.74, returned \$6.50
- Round 3 RECEIVER: received \$10.14, returned \$5.00
- Round 4 SENDER: sent \$3.50, got \$7.22 back (very good!)

The last round showed great cooperation. Players seem generally trustworthy. This is a new co-player though, so I don't know their specific behavior.

I should probably continue with a trusting approach. I could:

- Send \$4 (80% of endowment) - shows high trust
- Send \$3.5 (70%) - similar to last round
- Send \$3 (60%) - more cautious

Given the positive experiences, I'll increase slightly to \$4 to show strong trust.

usage:

```
{
  "input_tokens": 1562,
  "output_tokens": 402,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.98 to you. Your visible payoff was \$7.98.

Your total visible earnings across all rounds are \$34.45.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

My co-player sent me \$2.9 out of their \$5 (58% of their endowment) - showing moderate trust.

I received \$8.7.

They kept: \$2.1

To reciprocate fairly:

- If I return \$4.00, they'd end with \$6.10 (a gain of \$1.10)
- I'd keep \$4.70 (also a decent gain)

This represents returning about 46% of what I received, which is proportional to the moderate trust shown and ensures we both benefit.

```
'''json
{"return": 4.00}
'''
```

reasoning:

Looking at this situation:

My co-player sent me \$2.9 out of their \$5 (58% of their endowment) - showing moderate trust.

I received \$8.7.

They kept: \$2.1

To reciprocate fairly:

- If I return around \$4.00-\$4.50, they'd end with \$6.10-\$6.60 (a gain of \$1.10-\$1.60)
- I'd keep \$4.70-\$4.20

This represents returning about 46-52% of what I received, which rewards their trust while

we both benefit.

Let me think about what's fair. They sent \$2.9, showing moderate but not high trust. I should return something proportional.

In previous rounds as receiver:

- Round 2: Received \$13.74, returned \$6.50 (47%)
- Round 3: Received \$10.14, returned \$5.00 (49%)

So I've been returning around 47-49% of what I receive.

For \$8.7, returning around 47-49% would be:

- 47%: \$4.09
- 49%: \$4.26

Let me go with \$4.25, which is about 49% and gives them a total of \$6.35 (gain of \$1.35), while I keep \$4.45 (also a decent gain).

Actually, let me reconsider. They sent \$2.9 (58% of endowment), which is moderate trust. In Round 1, someone sent me \$4.58 (91.6%) and I returned \$6.50. In Round 3, someone sent \$3.38 (67.6%) and I returned \$5.00.

The person who sent \$3.38 (67.6%) got back \$5.00 from \$10.14 received.

...

usage:

```
{
  "input_tokens": 1817,
  "output_tokens": 654,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.0. Your visible payoff was \$4.699999999999999.

Your total visible earnings across all rounds are \$39.150000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

My co-player sent me \$2.99 out of their \$5 (about 60% of their endowment) - showing moderate trust, very similar to Round 6.

I received \$8.97.

They kept: \$2.01

To reciprocate consistently with my Round 6 approach:

- If I return \$4.10, they'd end with \$6.11 (a gain of \$1.11)
- I'd keep \$4.87 (also a decent gain)

This represents returning about 46% of what I received, consistent with similar trust levels.

```
'''json
{"return": 4.10}
'''
```

Reasoning:

My co-player sent me \$2.99 out of their \$5 (about 60% of their endowment) - showing moderate trust, very similar to the previous round.

I received \$8.97.

They kept: \$2.01

This is very similar to Round 6 where they sent \$2.9. I should be consistent in my reciprocation strategy.

In Round 6, I returned \$4.00 when I received \$8.7, which was about 46% of what I received.

Using a similar proportion:

- If I return \$4.10, they'd end with \$6.11 (a gain of \$1.11)
- I'd keep \$4.87 (also a decent gain)

This represents returning about 46% of what I received, consistent with my previous approach f or similar trust levels.

Let me calculate more carefully:

- They sent: \$2.99
- They kept: \$5 - \$2.99 = \$2.01
- I received: \$8.97
- If I return \$4.10:
 - They end with: \$2.01 + \$4.10 = \$6.11
 - I end with: \$8.97 - \$4.10 = \$4.87

This seems fair and consistent with my previous behavior.

```
Usage: {  
  "input_tokens": 2105,  
  "output_tokens": 465,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Sender - dyad_3

Timestamp: 2026-08-12T09:25:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_1"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$4.78, that is 95.6% of their total endowment. You received \$14.34. How much do you return to the sender? (0-14.34)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.78 (95.6% of their endowment)
- They kept only \$0.22 for themselves
- I received \$14.34 after the 3x multiplier
- A fair return should reward their trust

If I return \$8.00:

- Sender ends up with: $\$0.22 + \$8.00 = \$8.22$ (well above their initial \$5)
- I keep: $\$14.34 - \$8.00 = \$6.34$ (also above the initial \$5)
- Both of us benefit from the cooperation

This seems like a fair reciprocal response to their high level of trust.

```
{'return': 8.00}
```

reasoning:

Let me understand this situation:

1. The sender had \$5 endowment
2. They sent me \$4.78 (which is 95.6% of their \$5)
3. This was multiplied by 3x, so I received $\$4.78 \times 3 = \14.34

Now I need to decide how much to return.

Key considerations:

- The sender showed a lot of trust by sending 95.6% of their endowment
- They kept only \$5 - \$4.78 = \$0.22 for themselves
- If I return nothing, the sender ends up with just \$0.22
- If I return generously, both of us can benefit from this cooperation

What's fair/reciprocal?

- The sender trusted me significantly
- A common fairness norm would be to split the gains
- Total created value: \$14.34
- Original sender investment feel: \$4.78
- One approach: Return enough so we both end up roughly equal

- Another approach: Return enough so sender gets significantly more than their original \$5 (rewarding trust)

Calculations:

- If I return \$7.17 (about half of what I received):
 - Sender gets: $\$0.22 + \$7.17 = \$7.39$
 - I get: $\$14.34 - \$7.17 = \$7.17$
- If I return \$8:
 - Sender gets: $\$0.22 + \$8 = \$8.22$
 - I get: $\$14.34 - \$8 = \$6.34$
- If I return \$9:
 - Sender gets: $\$0.22 + \$9 = \$9.22$
 - I get: $\$14.34 - \$9 = \$5.34$

Given that the sender showed exceptional trust (sending 95.6%), I should reward this trust generously. A return that gives them significantly more than their original endowment while still keeping a reasonable amount seems appropriate.

Returning \$8 strikes a good balance-it rewards their exceptional trust while ensuring both of us benefit from the cooperation. They'd end up with \$8.22 versus their original \$5, and I'd keep \$6.34. But given how much they risked, maybe I should push toward \$8.50 to give them even more of the gains we created together....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 866,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.78, it became \$14.34, and you returned \$8.0. Your visible payoff was \$6.34.

Your total visible earnings across all rounds are \$6.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

In Round 1, my co-player showed very high trust by sending 95.6% of their endowment, and I reciprocated with a fair return. That cooperative approach worked well for both of us.

Now as the sender with a new co-player, I want to maintain that cooperative spirit. Sending a substantial amount:

- Creates more total value through the 3x multiplier
- Signals trust and willingness to cooperate
- Could encourage reciprocation

I'll send \$4.50 (90% of my endowment), showing strong trust while being slightly more conservative than what I experienced last round.

```
{'send': 4.50}
```

reasoning:

Let me think about this situation:

Round 1 (just completed):

- I was the RECEIVER
- Sender sent me \$4.78 (95.6% of their endowment) - showing very high trust
- I received \$14.34
- I returned \$8.00
- My payoff: \$6.34
- Sender's payoff would be: $\$5 - \$4.78 + \$8.00 = \8.22

Now in Round 2:

- I'm the SENDER
- I have \$5
- I'm playing against a different co-player (pairings are randomized each round)

The previous sender showed very high trust by sending 95.6%. I reciprocated fairly by returning \$8.00, which gave them a good outcome.

Now I need to decide how much to send. Key considerations:

1. This is a new co-player, so I don't have direct history with them
2. The previous round showed successful cooperation
3. Sending more creates more total value (3x multiplier), but requires trust
4. In a multi-round setting, establishing cooperative behavior can be beneficial

Given that:

- The previous sender sent 95.6% (\$4.78)
- That worked out well for both of us
- I should reciprocate the level of trust shown to me

I think I should send a similar high amount to signal cooperation and trust. Sending around \$4.50-\$4.80 would show similar trust to what I experienced.

Let me send \$4.50 (90% of my endowment). This shows strong trust while being slightly more conservative than the 95.6% I received.

Actually, thinking about it more - since I was treated with high trust and it worked out well, I should pay it forward. Sending \$4.50-\$4.75 seems appropriate.

...

usage:

```
{
  "input_tokens": 824,
  "output_tokens": 591,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.5, it became \$13.5, and you saw them return \$7.41 to you. Your visible payoff was \$7.91.

Your total visible earnings across all rounds are \$14.25.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed good trust by sending \$3.80 (76% of their endowment)
- They kept \$1.20 for themselves
- I received \$11.40
- I should reciprocate their trust fairly

Comparing to previous rounds:

- In Round 1, when a sender sent 95.6%, I returned about 56% of what I received
- In Round 2, when I sent 90%, my co-player returned about 55% of what they received

For this sender who sent 76%, a fair return would be around 55-60% of what I received.

$\$11.40 \times 0.57 \approx \6.50

If I return \$6.50:

- Sender ends with: $\$1.20 + \$6.50 = \$7.70$ (good gain)
- I keep: $\$11.40 - \$6.50 = \$4.90$ (reasonable)
- Both benefit from the cooperation

```
{'return': 6.50}
```

reasoning:

Let me analyze this situation:

Round 2 recap:

- I was the sender and sent \$4.50
- It became \$13.50 for the receiver
- They returned \$7.41 to me
- My payoff: $\$5 - \$4.50 + \$7.41 = \7.91

Round 3 current situation:

- I'm the receiver
- Sender sent \$3.80 (76% of their endowment)
- I received \$11.40 (approximately, due to floating point)
- They kept \$1.20

This sender showed good trust by sending 76% of their endowment. Let me think about a fair return:

If I return \$6.00:

- Sender ends up with: $\$1.20 + \$6.00 = \$7.20$
- I keep: $\$11.40 - \$6.00 = \$5.40$
- Both benefit, sender gains more

If I return \$7.00:

- Sender ends up with: $\$1.20 + \$7.00 = \$8.20$
- I keep: $\$11.40 - \$7.00 = \$4.40$
- Sender gains significantly more

In Round 1, when I received \$14.34 from a sender who sent \$4.78 (95.6%), I returned \$8.00.

- That sender ended with \$8.22 (from \$0.22)
- I kept \$6.34
- Return ratio: $\$8.00/\$14.34 = 55.9\%$

In Round 2, when I sent \$4.50 and it became \$13.50, they returned \$7.41

- Return ratio: $\$7.41/\$13.50 = 54.9\%$

For consistency, if I use a similar return ratio (around 55%):

- $\$11.40 \times 0.55 = \6.27

Let me round to \$6.50 to be slightly more generous:

...

usage:

```
{
  "input_tokens": 1120,
  "output_tokens": 820,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.8, it became \$11.399999999999999, and you returned \$6.5. Your visible payoff was \$4.899999999999999.

Your total visible earnings across all rounds are \$19.15.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

- Round 1: As receiver, someone sent high trust (95.6%), I reciprocated fairly
- Round 2: I sent \$4.50 (90%), got back \$7.41 - good cooperation
- Round 3: As receiver, returned fairly when sent \$3.80

The pattern shows that high trust tends to be reciprocated with fair returns (around 55% return rate). My cooperative strategy has been working well - averaging over \$6 per round

.

I'll continue with a strong but slightly varied approach. Sending \$4.00 (80%):

- Shows substantial trust
- Creates \$12.00 for the receiver through the 3x multiplier
- Should encourage reciprocation based on the established pattern
- Slightly more conservative than my last sending round to manage risk

{'send': 4.00}

reasoning:

Let me analyze my history:

Round 1 (RECEIVER): Sender sent \$4.78 (95.6%), I received \$14.34, returned \$8.00, kept \$6.34

Round 2 (SENDER): I sent \$4.50 (90%), became \$13.50, received back \$7.41, ended with \$7.91

Round 3 (RECEIVER): Sender sent \$3.80 (76%), I received \$11.40, returned \$6.50, kept \$4.90

Now Round 4, I'm the SENDER again with a new co-player.

Looking at the pattern:

- When I sent \$4.50 (90%), I got back \$7.41 from \$13.50 (about 55% return rate)
- When someone sent \$4.78 (95.6%), I returned \$8.00 from \$14.34 (about 56% return rate)
- When someone sent \$3.80 (76%), I returned \$6.50 from \$11.40 (about 57% return rate)

The return rates seem fairly consistent around 55-57%. This suggests cooperation is being reciprocated.

I've been doing well with high cooperation levels. My total earnings are \$19.15 across 3 rounds, averaging about \$6.38 per round, which is well above the baseline of \$5.

Given this positive experience with cooperation, I should continue sending a substantial amount. I'll send \$4.00 (80%), which is:

- Still showing significant trust
- Creating good value through the 3x multiplier (\$12.00 received)
- Likely to be reciprocated based on the pattern
- Slightly more conservative than my previous \$4.50 to diversify my strategy a bit

If they return around 55%, I'd get back about \$6.60, ending with \$1.00 + \$6.60 = \$7.60, which would be good.

usage:

```
{
  "input_tokens": 1488,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$4.58 to you. Your visible payoff was \$5.58.

Your total visible earnings across all rounds are \$24.729999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.73 (94.6% of their endowment)
- They kept only \$0.27 for themselves
- I received \$14.19
- This is very similar to Round 1, where someone sent 95.6%

In Round 1, when faced with similarly high trust, I returned \$8.00 out of \$14.34. That worked well as a fair reciprocation.

For this round with \$14.19 received, returning \$8.00:

- Sender ends with: \$0.27 + \$8.00 = \$8.27 (strong return on their trust)
- I keep: \$14.19 - \$8.00 = \$6.19
- Both benefit significantly from the cooperation

Consistency with my previous fair reciprocation of high trust seems appropriate.

```
{'return': 8.00}
```

reasoning:

Let me analyze this situation:

Round 5 - I'm the RECEIVER

- Sender sent \$4.73 (94.6% of their endowment) - very high trust!
- They kept only \$0.27
- I received \$14.19 after the 3x multiplier

This is similar to Round 1 where someone sent \$4.78 (95.6%) and I returned \$8.00.

The sender is showing exceptional trust, almost sending everything. I should reciprocate fairly.

If I return \$8.00:

- Sender ends with: \$0.27 + \$8.00 = \$8.27 (good gain from initial \$5)
- I keep: \$14.19 - \$8.00 = \$6.19
- Both benefit, sender gets rewarded for high trust

This seems consistent with my Round 1 behavior where I also returned \$8.00 for a similar high-trust send.

Actually, let me think about this more carefully. The sender sent 94.6%. In Round 1, when sender sent 95.6%, I returned \$8.00 out of \$14.34, which is about 56%.

$\$14.19 \times 0.56 \approx \7.95

So around \$8.00 seems right. Let me stick with that.

usage:

```
{
  "input_tokens": 1831,
  "output_tokens": 528,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.73, it became \$14.190000000000001, and you returned \$8.0. Your visible payoff was \$6.190000000000001.

Your total visible earnings across all rounds are \$30.919999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my experience as a sender:

- Round 2: Sent \$4.50 (90%), received \$7.41 back - good cooperation
- Round 4: Sent \$4.00 (80%), received \$4.58 back - lower reciprocation

The pattern suggests that higher trust levels tend to get better reciprocation. I'm averaging around \$6.18 per round with my cooperative strategy, which is working well.

As a receiver, I've been consistent in returning around 56% of what I receive, which has seemed fair to my co-players.

I'll send \$4.50 again (90% of endowment):

- Shows strong, consistent trust
- Previously resulted in good reciprocation
- Creates \$13.50 for the receiver
- Maintains my cooperative approach

```
{'send': 4.50}
```

reasoning:

Let me review the pattern of my play:

Round 1 (Receiver): Received \$14.34 from sender who sent \$4.78 (95.6%), returned \$8.00 ? kept \$6.34
Round 2 (Sender): Sent \$4.50 (90%), received back \$7.41 ? total \$7.91
Round 3 (Receiver): Received \$11.40 from sender who sent \$3.80 (76%), returned \$6.50 ? kept \$4.90
Round 4 (Sender): Sent \$4.00 (80%), received back \$4.58 ? total \$5.58
Round 5 (Receiver): Received \$14.19 from sender who sent \$4.73 (94.6%), returned \$8.00 ? kept \$6.19

My total visible earnings: \$30.92

As a sender:

- Round 2: Sent \$4.50, got back \$7.41 (55% return rate)
- Round 4: Sent \$4.00, got back \$4.58 (38% return rate)

As a receiver:

- Round 1: Returned \$8.00 out of \$14.34 (56% return rate) when sender sent 95.6%
- Round 3: Returned \$6.50 out of \$11.40 (57% return rate) when sender sent 76%
- Round 5: Returned \$8.00 out of \$14.19 (56% return rate) when sender sent 94.6%

I've been consistent in my returns - around 56% when I'm the receiver.

When I've been the sender:

- Higher sends seem to get better returns (Round 2 with \$4.50 got 55% back)

...

usage:

```
{
  "input_tokens": 2173,
  "output_tokens": 794,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.5, it became \$13.5, and you saw them return \$5.77 to you. Your visible payoff was \$6.27.

Your total visible earnings across all rounds are \$37.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

I'm playing the SAME co-player as Round 6. Last round with them:

- I sent \$4.50 (90%)
- They returned \$5.77 (about 43% of the \$13.50 they received)
- I ended with \$6.27

Their reciprocation rate (43%) was lower than some other players I've encountered (~55%), but not the worst. My payoff was still above average.

Playing the same person again, I should:

- Continue showing willingness to cooperate
- Adjust slightly for their demonstrated reciprocation pattern
- Signal awareness while maintaining relationship

I'll send \$4.00 (80%):

- Still shows substantial trust
- Creates \$12.00 for them
- Slightly more conservative given their previous behavior
- Keeps the cooperative dynamic going

{ 'send': 4.00 }

Reasoning:

Interesting - I'm playing as SENDER again against the SAME co-player from Round 6. This is imp

ortant context.

In Round 6:

- I sent \$4.50 (90%)
- It became \$13.50
- They returned \$5.77
- My payoff was \$6.27

The return rate was $\$5.77/\$13.50 = 42.7\%$, which is lower than the ~55% I've been seeing from other players. They kept about \$7.73 while I ended with \$6.27 total.

Now I'm playing them again. This gives me information about their reciprocation behavior. They were less generous than others I've played with.

Options:

1. Send less to protect myself since they showed lower reciprocation
2. Send the same amount to see if they maintain consistency
3. Send more to encourage better reciprocation

Strategic considerations:

- They know I sent high last time, so there's a reputation aspect
- If I send less, it signals I'm adjusting to their lower reciprocation
- If I send the same or more, it shows I'm still willing to cooperate
- The 3x multiplier means cooperation still creates value even with lower return rates

I think I should adjust downward slightly. Sending around \$3.50-\$4.00 makes sense:

- Still shows willingness to cooperate
- Manages risk given their demonstrated lower reciprocation rate
- Creates \$10.50-\$12.00 for them

I'll go with \$3.50 (70%):

- Shows continued cooperation but adjusted trust
- Creates \$10.50 for them
- Keeps \$1.50 for me as baseline
- If they return around 43%, that's \$4.52, giving me \$6.02 total
- If they improve their return rate, I benefit more

Actually, let me reconsider.

Looking at my track record across all sender rounds, I'm averaging around \$6.18 per round, and last round's \$6.27 is actually slightly above that. Comparing their 42.7% return rate against my other interactions - Round 2 yielded 55% and Round 4 only 38% - suggests this player falls in the middle rather than being an outlier. Maybe I'm overweighting one round when the overall pattern shows reasonable consistency....

```
Usage: {  
  "input_tokens": 2490,  
  "output_tokens": 943,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_3

Timestamp: 2026-08-12T09:25:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_1"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_1"  
  }
```

```

},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.77, that is 75.4% of their total endowment. You received \$11.31. How much do you return to the sender? (0-11.31)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$3.77 (75.4% of their endowment)
- They currently have: \$5 - \$3.77 = \$1.23
- I received: \$11.31
- Total money created: \$1.23 + \$11.31 = \$12.54

The sender took a significant risk and was quite generous. I want to reciprocate that trust while ensuring we both benefit from cooperation.

If I return \$5.50:

- Sender ends with: $\$1.23 + \$5.50 = \$6.73$
- I end with: $\$11.31 - \$5.50 = \$5.81$
- Both of us gain compared to the initial \$5, rewarding the sender's trust

This seems like a fair reciprocation that rewards their generosity while still allowing both of us to benefit.

```
{'return': 5.50}
```

reasoning:

Let me think about this situation carefully.

I'm the RECEIVER in Round 1.

- The sender started with \$5
- They sent me \$3.77 (which is 75.4% of their \$5 endowment)
- I received \$11.31 (which is $\$3.77 \times 3$)
- I need to decide how much to return (between \$0 and \$11.31)

The sender showed significant trust by sending 75.4% of their endowment. Let me think about fair reciprocity:

If the sender sent \$3.77:

- They kept: $\$5 - \$3.77 = \$1.23$
- I received: \$11.31

For a fair split of the total money created:

- Total money now = $\$1.23$ (sender kept) + $\$11.31$ (I received) = $\$12.54$
- Fair split would be roughly \$6.27 each
- So I should return enough so the sender has ~\$6.27
- Sender currently has \$1.23, so I'd need to return about \$5.04 to get them to \$6.27
- That would leave me with $\$11.31 - \$5.04 = \$6.27$

Alternatively, I could think about matching their generosity:

- They sent 75.4% of their endowment
- If I return 75.4% of what I received: $0.754 \times \$11.31 = \8.53
- That seems quite generous but might be excessive

Another approach - proportional reciprocity:

- They gave me \$3.77 out of their \$5 (75.4%)
- A reasonable return might be to give back around half of what I received, ensuring both benefit
- Half would be \$5.66

Let me think about common strategies in trust games:

1. Return nothing (free-ride)
2. Return what was sent (break even for sender)
3. Return half of what received (both benefit equally from the multiplier)
4. Return enough to equalize final payoffs

Given their generosity, returning \$5-6 seems appropriate. This would give the sender roughly \$6-7 while leaving me with similar amounts, reflecting mutual gain from the exchange without being excessive....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1020,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.77, it became \$11.31, and you returned \$5.5. Your visible payoff was \$5.8100000000000005.

Your total visible earnings across all rounds are \$5.8100000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the previous round, I saw cooperation work well when the sender showed significant trust by sending 75.4% of their endowment, and I reciprocated fairly.

I'll adopt a similar cooperative strategy with my new co-player. Sending a substantial amount can maximize joint gains if they reciprocate.

{'send': 3.75}

reasoning:

I'm now the SENDER in Round 2.

Let me think about this strategically:

1. In the previous round as RECEIVER, I saw someone send \$3.77 (75.4% of their endowment), showing significant trust
2. I returned \$5.50, which gave them \$6.73 total - a good return on their trust

Now I'm the SENDER. What should I consider?

- The person sending \$3.77 showed that cooperation can work well
- I reciprocated fairly, returning enough that they profited significantly
- If my current co-player is similarly cooperative, sending a substantial amount could benefit both of us

Strategy considerations:

- Sending nothing is safe but generates no surplus
- Sending a moderate amount (around \$2-3) tests the waters
- Sending a larger amount (like \$3.77 that I saw) shows trust and can maximize joint gains if reciprocated

Given that:

- I saw cooperation work in the last round
- The game seems to reward trust
- I benefited from someone trusting me and reciprocated fairly
- There may be communication noise (mentioned in instructions), but the pattern of cooperation seemed clear

I think I should send a substantial amount to show trust and potentially establish a cooperative pattern. I'll send around \$3.50-\$4.00.

Let me send \$3.50 (70% of endowment) - this shows significant trust similar to what I received, but slightly less aggressive to be somewhat cautious given this is a new pairing.

Actually, let me reconsider. The previous sender sent 75.4% (\$3.77). I reciprocated well. If I want to encourage cooperation, I should send a similar proportion. Let me send \$3.75 (75% of my endowment) to match that level of trust.

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 494,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$5.66 to you. Your visible payoff was \$6.91.

Your total visible earnings across all rounds are \$12.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience:

- Round 2: I sent \$3.75 (75%), received \$5.66 back, ended with \$6.91 - a good outcome
- Round 1: As receiver, cooperation was rewarded

The cooperative approach has been working well. I'm playing with a new co-player now, so I'll continue signaling trust with a substantial send.

```
{'send': 3.75}
```

reasoning:

Let me analyze the previous round:

- I sent \$3.75 (75% of my endowment)
- It became \$11.25
- They returned \$5.66
- My payoff was $\$5 - \$3.75 + \$5.66 = \6.91

This worked out well! The receiver reciprocated reasonably (returning about 50% of what they received), and I gained \$1.91 above my initial endowment.

This is a new co-player this round. Based on my successful cooperative strategy so far:

- Round 1 as receiver: sender sent 75.4%, I returned fairly
- Round 2 as sender: I sent 75%, receiver returned fairly

The pattern of cooperation seems to be working. I should continue with a similar high-trust strategy. I'll send around 75% again to signal cooperation and trust.

\$3.75 worked well last time. I could send the same amount or similar.

usage:

```
{
  "input_tokens": 1074,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$6.34 to you. Your visible payoff was \$7.59.

Your total visible earnings across all rounds are \$20.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$4.09, so you received \$12.27. How much do you return? (0-12.27)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.09 (81.8% of their endowment)
- They currently have: $\$5 - \$4.09 = \$0.91$
- I received: \$12.27
- Total created: \$13.18

This is even more trust than I saw in Round 1 (when sender sent 75.4%). I want to reward this high level of cooperation appropriately.

If I return \$6.50:

- Sender ends with: $\$0.91 + \$6.50 = \$7.41$ (gain of \$2.41)
- I end with: $\$12.27 - \$6.50 = \$5.77$ (gain of \$0.77)

Both of us benefit, and the sender's exceptional trust is well-rewarded.

```
{'return': 6.50}
```

reasoning:

Let me analyze this situation:

- The sender sent \$4.09 out of \$5, which is 81.8% of their endowment - very high trust!
- They now have: $\$5 - \$4.09 = \$0.91$
- I received: \$12.27
- Total money in the system: $\$0.91 + \$12.27 = \$13.18$

This sender showed even more trust than my previous partners. I want to reciprocate appropriately.

Looking at my previous behavior:

- Round 1: As receiver, got \$11.31, returned \$5.50 (about 48.6%)
- The sender had sent 75.4%
- That seemed fair and both parties benefited

This sender sent 81.8%, even more generous. I should reciprocate with a good return.

If I return \$6.50:

- Sender ends with: $\$0.91 + \$6.50 = \$7.41$
- I end with: $\$12.27 - \$6.50 = \$5.77$
- Sender gains \$2.41 over initial \$5
- I gain \$0.77 over initial \$5

Both benefit, and the sender's higher risk/trust is rewarded with higher gains.

Actually, let me think about this differently. The sender sent 81.8% which is very high. In Round 1, sender sent 75.4% and I returned 48.6% of what I received.

For 81.8% send rate, maybe I should return around 50-53% to reward the extra trust?

$\$12.27 * 0.52 = \6.38

If I return \$6.40:

- Sender ends with: $\$0.91 + \$6.40 = \$7.31$
- I end with: $\$12.27 - \$6.40 = \$5.87$

This seems fair - both gain, sender gets rewarded for high trust.

...

usage:

```
{
  "input_tokens": 1310,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.09, it became \$12.27, and you returned \$6.5. Your visible payoff was \$5.77.

Your total visible earnings across all rounds are \$26.080000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my track record:

- As sender: Sent \$3.75 twice, received good returns (\$5.66 and \$6.34)
- As receiver: Responded fairly to cooperative senders

The high-trust strategy continues to work well across different co-players. I'll maintain this approach.

{'send': 3.75}

reasoning:

Let me review my history:

- Round 1: Receiver - sender sent \$3.77, I returned \$5.50, payoff \$5.81
- Round 2: Sender - I sent \$3.75, got back \$5.66, payoff \$6.91
- Round 3: Sender - I sent \$3.75, got back \$6.34, payoff \$7.59
- Round 4: Receiver - sender sent \$4.09, I returned \$6.50, payoff \$5.77

Total visible earnings: \$26.08

My sending strategy has been working well - sending around \$3.75 (75%) has led to good returns. In Round 4, I saw an even higher send amount (\$4.09, 81.8%) and I rewarded it well.

Now I'm facing a new co-player as sender. The cooperative strategy has been consistently w

orking across different opponents. I should continue with a similar high-trust signal.

I'll send \$3.75 again, which has proven successful.

usage:

```
{
  "input_tokens": 1637,
  "output_tokens": 324,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$6.54 to you. Your visible payoff was \$7.79.

Your total visible earnings across all rounds are \$33.870000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$4.1, so you received \$12.299999999999999. How much do you return? (0-12.299999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.10 (82% of their endowment)
- They currently have: \$5 - \$4.10 = \$0.90
- I received: \$12.30
- Total created: \$13.20

This is very similar to Round 4 where the sender sent \$4.09 and I returned \$6.50, which worked well for both parties.

If I return \$6.50:

- Sender ends with: \$0.90 + \$6.50 = \$7.40
- I end with: \$12.30 - \$6.50 = \$5.80

Both of us benefit substantially, rewarding the sender's high trust.

{'return': 6.50}

reasoning:

The sender sent \$4.10, which is 82% of their endowment - very high trust again!

They currently have: \$5 - \$4.10 = \$0.90

I received: \$12.30

Total created: \$13.20

This is similar to Round 4 where sender sent \$4.09. I returned \$6.50 then.

Let me be consistent and reward this high trust appropriately.

If I return \$6.50:

- Sender ends with: \$0.90 + \$6.50 = \$7.40
- I end with: \$12.30 - \$6.50 = \$5.80

Both benefit, sender gets a good reward for their trust.

usage:

```
{
  "input_tokens": 1863,
  "output_tokens": 354,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$4.1, it became \$12.299999999999999, and you returned \$6.5. Your visible payoff was \$5.799999999999999.

Your total visible earnings across all rounds are \$39.67.

This round, you are the RECEIVER against your current co-player. They sent you \$5.0, so you received \$15.0. How much do you return? (0-15.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender showed **complete trust** by sending \$5.00 (100% of their endowment!)
- They currently have: \$0
- I received: \$15.00
- This is the same co-player from Round 6, and they escalated their trust after I returned fairly

This maximum trust deserves a strong reciprocal response.

If I return \$8.00:

- Sender ends with: \$8.00 (gain of \$3.00)
- I end with: \$7.00 (gain of \$2.00)

Both of us benefit substantially, and their complete trust is well-rewarded.

{'return': 8.00}

Reasoning:

This is interesting - I'm playing with the same co-player from Round 6. In Round 6:

- They sent \$4.10
- I received \$12.30
- I returned \$6.50
- They ended with $\$0.90 + \$6.50 = \$7.40$

Now in Round 7:

- They sent \$5.00 (100% of their endowment!) - maximum trust
- I received \$15.00
- They currently have: \$0

This is the ultimate show of trust - they sent everything! This deserves a strong reciprocal response.

If I return \$8.00:

- Sender ends with: $\$0 + \$8.00 = \$8.00$ (gain of \$3.00)
- I end with: $\$15.00 - \$8.00 = \$7.00$ (gain of \$2.00)

Both benefit nicely, and I'm rewarding their complete trust.

Alternatively, if I return \$7.50:

- Sender ends with: \$7.50 (gain of \$2.50)
- I end with: \$7.50 (gain of \$2.50)
- Equal split of the gains

Or if I return \$8.50:

- Sender ends with: \$8.50 (gain of \$3.50)
- I end with: \$6.50 (gain of \$1.50)
- More reward for their maximum trust

Given they showed complete trust by sending 100%, I think returning \$8.00 is appropriate - it gives them a strong reward while we both benefit substantially.

Usage: {

```
"input_tokens": 2186,  
"output_tokens": 538,  
"reasoning_tokens": 0
```

}

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-08-12T09:26:00

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_4",  
"move_index": 6,  
"opponent_id": "Agent_7",
```

```

"pairing": {
  "agents": [
    "Agent_5",
    "Agent_7"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_7": "trustee"
  },
  "trustee": "Agent_7"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.3, that is 46.0% of their total endowment. You received \$6.899999999999995. How much do you return to the sender? (0-6.899999999999995)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed trust by sending \$2.3 (46% of their endowment)
- I received \$6.9 after the 3x multiplier
- The sender currently has \$2.7 remaining

I want to:

1. Reward the sender's trust so they benefit from sending
2. Split the gains fairly
3. Establish myself as trustworthy for future rounds

If I return \$3.0:

- Sender ends with: $\$2.7 + \$3.0 = \$5.7$ (a gain of \$0.7)
- I end with: $\$6.9 - \$3.0 = \$3.9$

This seems fair - the sender profits from trusting me, and we both benefit from the multiplier effect.

```
'''json
{"return": 3.0}
'''
```

reasoning:

Let me understand this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.3 (which is 46% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$6.9 ($2.3 \times 3 = 6.9$)
4. I need to decide how much to return (between \$0 and \$6.9)

The sender's current situation:

- Started with \$5
- Sent \$2.3
- Currently has: $\$5 - \$2.3 = \$2.7$

If I return nothing:

- Sender ends with: \$2.7
- I end with: \$6.9

The sender showed trust by sending nearly half their endowment. A fair approach would be to:

1. Ensure the sender doesn't lose money
2. Split the gains reasonably

Total money in the system now: $\$2.7$ (sender has) + $\$6.9$ (I have) = $\$9.6$

This started from two \$5 endowments = \$10 total, so we have \$9.6 total.

Wait, let me recalculate:

- Sender started with \$5, sent \$2.3, has \$2.7 left
- I received \$6.9 (the \$2.3 multiplied by 3)
- Total in system: $\$2.7 + \$6.9 = \$9.6$

If we split equally: each person would get \$4.8

- For sender to have \$4.8, they need: $\$4.8 - \$2.7 = \$2.1$ returned
- For me to have \$4.8, I'd keep: $\$6.9 - \$2.1 = \$4.8$?

But equal split might not be the best strategy. The sender took a risk by sending money. A common approach in trust games is to:

reward their trust by ensuring they come out ahead of their initial \$5, while also sharing in the multiplier gains. Let me test a few return amounts: returning \$3.0 gives them \$5.7 total (a \$0.7 gain) while I keep \$3.9; returning \$2.5 gives them \$5.2 (a \$0.2 gain) while I keep \$4.4; returning \$2.1 would give them exactly \$4.8 and me \$4.8.

A return around \$3.0-\$3.3 seems optimal-it meaningfully rewards their trust while still letting me benefit from the multiplier effect. At \$3.0, they gain \$0.7 and I keep \$3.9, which feels like a fair split that demonstrates trustworthiness.

The sender's 46% send rate suggests moderate confidence in me, so I should reciprocate proportionally...

usage:

```
{
  "input_tokens": 539,
  "output_tokens": 1314,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.3, it became \$6.899999999999995, and you returned \$3.0. Your visible payoff was \$3.899999999999995.

Your total visible earnings across all rounds are \$3.899999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- I'm now the sender with a new co-player
- In the previous round, I experienced a sender who sent 46% and I returned fairly
- I want to establish trust with this new partner while being reasonably cautious

Sending \$2.5 (50% of my endowment) seems like a good strategy:

- It shows significant trust
- It's a clear, fair starting point
- If my co-player reciprocates fairly, we both benefit from the 3x multiplier
- It's not overly risky with an unknown partner

```
'''json
{"send": 2.5}
'''
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.3 (46% of their endowment)
- I received \$6.9
- I returned \$3.0
- My payoff was \$3.9

Now I'm playing against a different agent (since it says "current co-player" and agents are randomly paired each round).

I need to decide how much to send. Key considerations:

1. I don't know anything about this new co-player's behavior
2. The sender in Round 1 sent 46% (\$2.3), which was a reasonable trust signal
3. I returned about 43.5% of what I received (\$3/\$6.9), which was fair
4. I want to build trust but also protect myself since I don't know this opponent

Strategies:

- Sending \$0 would be overly defensive and wouldn't build any relationship
- Sending \$5 (100%) would be too risky with an unknown partner
- Sending around \$2-3 seems reasonable as it shows trust without being reckless

Given that the previous sender sent \$2.3 (46%), I could mirror that behavior. This would be consistent and shows I'm willing to trust at a similar level to what I've experienced.

Alternatively, I could send slightly more to show goodwill, or slightly less to be more cautious.

I think sending \$2.5 (50%) is a good middle ground:

- It's a clear half of my endowment, easy to interpret
- It shows significant trust
- If they return fairly (around 40-45% of \$7.5 = \$3-3.4), we both profit

- It's not too risky

Let me calculate the outcomes:

- If I send \$2.5, they receive \$7.5
- I keep \$2.5

...

usage:

```
{
  "input_tokens": 870,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$6.78 to you. Your visible payoff was \$9.280000000000001.

Your total visible earnings across all rounds are \$13.18.

This round, you are the RECEIVER against your current co-player. They sent you \$4.17, so you received \$12.51. How much do you return? (0-12.51)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.17 (83.4% of their endowment)
- I received \$12.51 after the 3x multiplier
- The sender currently has only \$0.83 remaining

This is very high trust that deserves to be rewarded generously.

If I return \$6.0:

- Sender ends with: $\$0.83 + \$6.0 = \$6.83$ (profit of \$1.83)
- I end with: $\$12.51 - \$6.0 = \$6.51$ (profit of \$6.51)

This gives us nearly equal final payoffs and rewards the sender's exceptional trust with a solid profit.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- The sender sent \$4.17 (83.4% of their \$5 endowment) - this is very high trust!
- I received \$12.51 (after 3x multiplier)
- The sender currently has $\$5 - \$4.17 = \$0.83$ remaining

This sender showed enormous trust by sending 83.4% of their endowment. I need to reward this trust generously.

Let me think about what's fair:

- Total in the system: $\$0.83$ (sender has) + $\$12.51$ (I have) = $\$13.34$
- If we split evenly: $\$6.67$ each

If I return \$5.84:

- Sender ends with: $\$0.83 + \$5.84 = \$6.67$
- I end with: $\$12.51 - \$5.84 = \$6.67$
- Perfect 50/50 split

But given they showed such high trust (83.4%), I might want to be even more generous to:

1. Reward their exceptional trust
2. Encourage continued cooperation
3. Establish myself as highly trustworthy

However, I also saw in round 2 that when I sent \$2.5 (50%), I got back \$6.78, which gave me

e a payoff of \$9.28. That was very generous from my co-player then (they returned more than 90% of what they received!).

Let me recalculate for this round:

- If I return \$6:
 - Sender: $\$0.83 + \$6 = \$6.83$
 - Me: $\$12.51 - \$6 = \$6.51$
 - Almost equal, slight advantage to sender
- If I return \$6.5:
 - Sender: $\$0.83 + \$6.5 = \$7.33$
 - Me: $\$12.51 - \$6.5 = \$6.01$
 - Sender gets more, rewarding their trust

Given the extreme trust shown here, I should return enough to give them a meaningful advantage while still being reasonable. Returning around \$6 seems appropriate-it creates nearly equal outcomes with a slight edge to the sender who took the bigger risk....

usage:

```
{
  "input_tokens": 1154,
  "output_tokens": 955,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.17, it became \$12.51, and you returned \$6.0. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$19.689999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience so far:

- Round 2: When I sent \$2.5 (50%), I received a very generous return of \$6.78
- Round 3: When someone sent me \$4.17 (high trust), I reciprocated with \$6.0

The game seems to reward trust. I'll send \$3.0 (60% of my endowment), which shows strong trust while being measured with a new co-player.

```
'''json
{"send": 3.0}
'''
```

reasoning:

Looking at my history:

- Round 1: RECEIVER - received from sender who sent \$2.3, I returned \$3.0 fairly
- Round 2: SENDER - I sent \$2.5, receiver returned \$6.78 (very generous!)
- Round 3: RECEIVER - received from sender who sent \$4.17 (very high trust), I returned \$6.0

Current situation:

- I'm the SENDER with a new co-player
- Total earnings so far: \$19.69

My experience shows:

- When I sent \$2.5 in Round 2, I got a very generous return of \$6.78
- When others showed high trust to me, I reciprocated fairly
- This game seems to reward trust

I should send a significant amount to show trust. Given that:

- Round 2 sender sent 46% and I reciprocated
- Round 3 sender sent 83.4% and I reciprocated generously
- When I sent 50% in Round 2, I got excellent returns

I think sending around \$3.0 (60%) would be a good balance - showing strong trust without b

eing reckless with an unknown partner.

Actually, let me think about this more strategically. I've seen that:

- Lower trust (46%) got fair returns
- Medium trust (50% from me) got very generous returns
- High trust (83.4%) got generous returns

Sending \$3.0 seems like a good middle-high trust level that should encourage reciprocation

usage:

```
{
  "input_tokens": 1461,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.22 to you. Your visible payoff was \$7.22.

Your total visible earnings across all rounds are \$26.909999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.84, so you received \$11.52. How much do you return? (0-11.52)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender showed high trust by sending \$3.84 (76.8% of their endowment)
- I received \$11.52 after the 3x multiplier
- The sender currently has only \$1.16 remaining

This level of trust deserves a generous return.

If I return \$6.0:

- Sender ends with: $\$1.16 + \$6.0 = \$7.16$ (profit of \$2.16)
- I end with: $\$11.52 - \$6.0 = \$5.52$

This rewards their high trust with a good profit and keeps our final payoffs relatively balanced.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$3.84, which is 76.8% of their endowment - very high trust
- I received \$11.52 after the 3x multiplier
- The sender currently has $\$5 - \$3.84 = \$1.16$ remaining

This is high trust that should be rewarded well.

Let me consider different return amounts:

If I return \$5.5:

- Sender ends with: $\$1.16 + \$5.5 = \$6.66$ (profit of \$1.66)
- I end with: $\$11.52 - \$5.5 = \$6.02$

If I return \$6.0:

- Sender ends with: $\$1.16 + \$6.0 = \$7.16$ (profit of \$2.16)
- I end with: $\$11.52 - \$6.0 = \$5.52$

Given the high trust (76.8%), I should return generously. \$6.0 seems fair and rewards their trust well.

usage:

```
{
```



```
"input_tokens": 1718,
"output_tokens": 421,
"reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.84, it became \$11.52, and you returned \$6.0. Your visible payoff was \$5.52.

Your total visible earnings across all rounds are \$32.42999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my experience:

- Round 2: Sent \$2.5 (50%), received \$6.78 back - great return
- Round 4: Sent \$3.0 (60%), received \$5.22 back - solid return

The game has been rewarding trust. I'll continue to show strong trust with this new co-player by sending \$3.5 (70% of my endowment).

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze my experience:

Round 1 (Receiver): Received \$6.9 from sender sending \$2.3, returned \$3.0. Payoff: \$3.9
Round 2 (Sender): Sent \$2.5, received \$6.78 back. Payoff: \$9.28
Round 3 (Receiver): Received \$12.51 from sender sending \$4.17, returned \$6.0. Payoff: \$6.5
1
Round 4 (Sender): Sent \$3.0, received \$5.22 back. Payoff: \$7.22
Round 5 (Receiver): Received \$11.52 from sender sending \$3.84, returned \$6.0. Payoff: \$5.5
2

So far:

- As a sender, I've sent \$2.5 and \$3.0 (50% and 60%)
- Both times I received generous returns (\$6.78 and \$5.22)
- As a receiver, I've been fair/generous in my returns

The game continues to reward trust. I've been gradually increasing my sending amounts. Let me continue this trend and send \$3.5 (70% of endowment) to show strong trust with this new co-player.

usage:

```
{
  "input_tokens": 2009,
  "output_tokens": 417,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.22 to you. Your visible payoff was \$7.72.

Your total visible earnings across all rounds are \$40.14999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my sender history:

- Round 2: Sent \$2.5, received \$6.78 back
- Round 4: Sent \$3.0, received \$5.22 back

- Round 6: Sent \$3.5, received \$6.22 back

Trust has been consistently rewarded. I'll continue showing strong trust with this new co-player by sending \$4.0 (80% of my endowment).

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Looking at my history as sender:

- Round 2: Sent \$2.5 (50%), received \$6.78 back - excellent return
- Round 4: Sent \$3.0 (60%), received \$5.22 back - good return
- Round 6: Sent \$3.5 (70%), received \$6.22 back - good return

All three times I've sent money as sender, I've received generous returns. The game environment seems cooperative and trust is being rewarded.

As receiver:

- Round 1: Received \$6.9, returned \$3.0
- Round 3: Received \$12.51, returned \$6.0
- Round 5: Received \$11.52, returned \$6.0

I've been fair as a receiver too.

The pattern suggests the game is cooperative. I could continue increasing my trust level. Maybe send \$4.0 (80%) this time? That would show strong trust with this new co-player.

Let me think about the strategy:

- Sending more has generally worked well
- I'm playing against a new co-player each round
- Building reputation for cooperation could help
- Total earnings are healthy at \$40.15

I'll send \$4.0 (80% of endowment) to show strong trust.

```
Usage: {
  "input_tokens": 2246,
  "output_tokens": 423,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T09:26:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_7"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system
content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no history with this receiver.

If I send money, it gets tripled, creating potential mutual gain. However, the receiver controls how much comes back. The communication noise adds uncertainty.

I'll take a moderately cooperative approach to signal trust while managing risk:

****My decision:****

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return a fair portion, we both benefit. This allows me to learn about their reciprocity while not risking everything in this uncertain first round.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I have \$5 to allocate

2. Whatever I send gets tripled
3. The receiver then decides how much to return
4. There's communication noise - amounts may differ slightly from what was actually sent/returned

Key considerations:

- This is a multi-round game with multiple agents
- Pairings are randomized each round
- This is round 1, so I have no history with this particular receiver
- There's a note about communication noise which could affect trust building

Standard economic reasoning in trust games:

- If I send \$5, it becomes \$15 for the receiver
- A "fair" split might be for them to return \$10, giving us each \$10
- But the receiver might keep everything, leaving me with \$0
- If I send \$0, I guarantee keeping \$5 but miss potential gains

Since this is:

1. Round 1 with no history
2. A multi-round game where reputation/reciprocity could matter across the full run
3. There's communication noise which might complicate trust building

I should consider a moderate opening move that:

- Shows some trust/willingness to cooperate
- Doesn't risk everything
- Allows me to learn about this receiver's behavior
- Sets a cooperative tone

A common strategy in trust games is to send around 50% on the first round as a signal of moderate trust while protecting against complete loss.

I'll send \$2.50 (or \$2 or \$3 since I need to be practical about amounts).

Actually, looking at the format, I should probably stick to whole dollar amounts for simplicity. Let me send \$3 as a cooperative opening move - it's more than half, showing trust, but not everything.

Wait, let me reconsider. The receiver gets \$9 if I send \$3. If they return half (\$4.50), I'd have $\$5 - \$3 + \$4.50 = \6.50 , and they'd have \$4.50. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 698,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.51 to you. Your visible payoff was \$7.51.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$3.8 out of \$5 (76% of their endowment) - showing significant trust
- I received \$11.4 after the 3x multiplier

****My reasoning:****

- In Round 1, when I sent \$3, the receiver returned \$5.51 (about 61% of what they received)
- This sender has shown high trust by sending 76% of their endowment
- I should reciprocate that trust fairly

I'll return an amount that rewards their cooperation while maintaining fairness:

```
'''json
{"return": 7}
'''
```

This gives the sender a total of \$8.2 (\$1.2 kept + \$7 returned) and leaves me with \$4.4. While I keep less, this reciprocates their strong trust and builds cooperation.

reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$5.51 to me
- My payoff: $\$5 - \$3 + \$5.51 = \7.51
- Their payoff: $\$9 - \$5.51 = \$3.49$

So the receiver returned about 61% of what they received. That's quite generous.

Round 2 (current):

- I am RECEIVER
- Sender sent me \$3.8
- I received \$11.4 (tripled amount)
- I need to decide how much to return

Now, there's communication noise, so the amounts might be slightly different from what was actually sent/returned. But I should work with what I see.

The sender sent me \$3.8 out of \$5, which is 76% of their endowment. That's quite trusting.

If I apply reciprocity based on what happened in Round 1:

- The receiver returned about 61% of what they received
- 61% of \$11.4 = \$6.95

But let me think about fairness:

- Sender keeps: $\$5 - \$3.8 = \$1.2$
- Sender receives back from me: X
- Sender total: $\$1.2 + X$

- I keep: $\$11.4 - X$

For equal split of total value:

- Total created: $\$5 + \$11.4 = \$16.4$
- Equal split: \$8.2 each
- So sender should get: \$8.2 total, meaning I return: $\$8.2 - \$1.2 = \$7$

For proportional reciprocity (they sent 76% of endowment):

- Maybe I return a similar proportion of what I received?
- 76% of \$11.4 = \$8.66

Given their high level of trust, I should match that cooperation. Returning around 55-60% of what I received feels right-it's reciprocal without being excessive. I'll return \$6.50, which gives the sender a total of \$7.70 and leaves me with \$4.90. This mirrors the cooperative spirit they showed....

usage:

```
{
  "input_tokens": 780,
  "output_tokens": 878,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.8, it became \$11.399999999999999, and you returned \$7.0. Your visible payoff was \$4.399999999999999.

Your total visible earnings across all rounds are \$11.909999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this new round with a new co-player:

****My experience so far:****

- Round 1: Sent \$3, received \$5.51 back - good cooperation
- Round 2: Received \$11.4, returned \$7 - fair reciprocation

Since pairings are randomized, this is a new partner with no shared history. However, the cooperative pattern I've experienced suggests trust can be rewarded.

I'll send a moderate-to-high amount to signal cooperation while managing risk:

```
'''json
{"send": 3.5}
'''
```

This sends \$3.5 (70% of my endowment), which becomes \$10.50 for the receiver. It's slightly more than my Round 1 send, reflecting confidence from positive past experiences while remaining balanced.

reasoning:

Let me analyze the situation:

****My history:****

- Round 1: I sent \$3, received back \$5.51, payoff = \$7.51
- Round 2: I received \$11.4 (they sent \$3.8), I returned \$7, payoff = \$4.4
- Total earnings: \$11.91

****Analysis:****

- In Round 1, my co-player was generous, returning \$5.51 out of \$9 (61%)
- In Round 2, my co-player sent \$3.8 (76% of endowment), showing high trust, and I reciprocated fairly
- Now I'm facing a new co-player in Round 3

Since this is a multi-agent setting with randomized pairings, this Round 3 co-player is likely a different person than either of my previous partners. I don't have direct history with them.

However, I've seen cooperation pay off:

- When I sent \$3, I got back \$5.51 (net gain of \$2.51)
- When someone sent me \$3.8, I returned fairly

Given the overall cooperative pattern I've observed and participated in, I should continue with a cooperative strategy. I'll send a similar amount to what I did in Round 1, or potentially slightly more given the positive experience.

I'll send \$3 again - it's a balanced approach that:

1. Shows trust and willingness to cooperate
2. Doesn't risk everything
3. Creates opportunity for mutual gain (becomes \$9)
4. Has worked well for me in the past

Actually, given that cooperation has been working well, I could send a bit more. Let me send \$3.5, showing slightly increased trust while still being prudent.

usage:

```
{
  "input_tokens": 1109,
  "output_tokens": 583,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.82 to you. Your visible payoff was

\$6.32.

Your total visible earnings across all rounds are \$18.229999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$3.11 out of \$5 (62.2% of their endowment) - showing moderate trust
- I received \$9.33 after the 3x multiplier

****My analysis:****

- In Round 3, when I sent \$3.5, I received back \$4.82 (only 46% of what they received) - less cooperative
- In Round 2, when I received \$11.4, I returned \$7 (61%)
- This sender showed moderate trust, deserving fair reciprocation

I'll return a fair amount consistent with their trust level:

```
'''json
{"return": 5.5}
'''
```

This gives the sender a total of \$7.39 (\$1.89 kept + \$5.50 returned) and leaves me with \$3.83. It's a balanced return that rewards their trust while being fair to both of us.

reasoning:

Let me analyze this situation:

****Round 4:****

- I'm the RECEIVER
- They sent \$3.11 out of \$5 (62.2% of their endowment)
- I received \$9.33 after the 3x multiplier

****Past experience:****

- Round 1: I sent \$3, got back \$5.51 (they returned 61% of \$9)
- Round 2: I received \$11.4, returned \$7 (61% of what I received)
- Round 3: I sent \$3.5, got back \$4.82 (they returned 46% of \$10.5)

The sender this round sent me \$3.11, showing moderate trust (62.2% of their endowment).

In Round 3, when I sent \$3.5, I only got back \$4.82, which was less than half of what they received. That was somewhat disappointing.

For this round, the sender sent \$3.11. I should be fair but also consider that:

1. They showed moderate trust (62.2%)
2. My past returns have been around 61% when receiving
3. I want to maintain cooperation

If I return around 55-60% of \$9.33, that would be:

- 55% = \$5.13
- 60% = \$5.60

This would give them:

- At \$5.13: \$1.89 kept + \$5.13 = \$7.02 total
- At \$5.60: \$1.89 kept + \$5.60 = \$7.49 total

And I would keep:

- At \$5.13: \$4.20
- At \$5.60: \$3.73

I'll return around \$5.5 to be fair and reciprocate their moderate trust.

usage:

```
{
  "input_tokens": 1428,
  "output_tokens": 648,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$5.5. Your visible payoff was \$3.83.

Your total visible earnings across all rounds are \$22.059999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.26, so you received \$6.779999999999999. How much do you return? (0-6.779999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$2.26 out of \$5 (45.2% of their endowment) - showing cautious/modest trust
- I received \$6.78 after the 3x multiplier

****My analysis:****

- This sender showed less trust than previous partners (45% vs 62-76%)
- However, I should still reciprocate fairly to maintain cooperation
- My previous return rates have been around 59-61%

Given their more cautious approach, I'll return a fair but proportionally adjusted amount:

```
'''json
{"return": 3.5}
'''
```

This gives the sender a total of \$6.24 (\$2.74 kept + \$3.50 returned), a gain of \$1.24, and leaves me with \$3.28. It's a balanced return that rewards their trust while acknowledging their more conservative send.

reasoning:

Looking at this situation:

****What happened:****

- The sender sent me \$2.26 out of \$5 (45.2% of their endowment) - showing modest/cautious trust
- I received \$6.78 after the 3x multiplier

****My historical pattern:****

- Round 2: Received \$11.4, returned \$7 (61% of received)
- Round 4: Received \$9.33, returned \$5.5 (59% of received)

This sender showed less trust than previous senders (only 45% vs 62-76%). I should still reciprocate fairly, but proportionally to their lower trust level.

A return rate around 50-55% seems appropriate given their more cautious approach:

- 50% would be \$3.39
- 55% would be \$3.73
- 60% would be \$4.07

Let me think about what's fair:

- They sent \$2.26, keeping \$2.74
- If I return \$3.5, they'd have \$5.76 total (a gain of \$0.76)
- I'd keep \$3.28

Actually, let me reconsider. Even though they sent less, I should maintain a cooperative stance. Let me return around 50-55%:

If I return \$3.5:

- Sender gets: \$2.74 (kept) + \$3.50 (returned) = \$6.24 total
- I keep: \$6.78 - \$3.50 = \$3.28

This seems balanced - they gain \$1.24, I gain \$3.28. It's fair while still rewarding their trust, even if it was more cautious.

Let me go with \$3.5 (about 52% of what I received).

usage:

```
{
  "input_tokens": 1789,
  "output_tokens": 630,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.26, it became \$6.779999999999999, and you returned \$3.5. Your visible payoff was \$3.2799999999999994.

Your total visible earnings across all rounds are \$25.339999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at this new round with a new co-player:

****My experience so far as sender:****

- Round 1: Sent \$3 (60%), received \$5.51 back - good return
- Round 3: Sent \$3.5 (70%), received \$4.82 back - lower return rate

****My observations:****

- Sending more didn't necessarily yield better outcomes
- As a receiver, I've been fairly cooperative
- This is a new partner, so I have no history with them

I'll send a moderate amount that signals cooperation while managing risk:

```
'''json
{"send": 3}
'''
```

This sends \$3 (60% of endowment), which becomes \$9 for the receiver. It's a balanced approach that worked well in Round 1 and allows me to gauge this new partner's reciprocity.

reasoning:

Looking at my history:

Round 1 (Sender): Sent \$3, received back \$5.51, payoff \$7.51
Round 2 (Receiver): Received \$11.4 (from \$3.8 send), returned \$7, payoff \$4.4
Round 3 (Sender): Sent \$3.5, received back \$4.82, payoff \$6.32
Round 4 (Receiver): Received \$9.33 (from \$3.11 send), returned \$5.5, payoff \$3.83
Round 5 (Receiver): Received \$6.78 (from \$2.26 send), returned \$3.5, payoff \$3.28

Now I'm a sender again in Round 6 with a new co-player.

My experience as sender:

- Round 1: Sent \$3 (60%), got back \$5.51 (61% of what they received)
- Round 3: Sent \$3.5 (70%), got back \$4.82 (46% of what they received)

When I send more, I don't necessarily get better returns. The Round 3 experience was less favorable.

My experience as receiver shows I've been fairly cooperative:

- Returned 61%, 59%, and 52% of what I received

Since this is a new partner and there's communication noise, I should:

1. Signal cooperation but not overcommit
2. Learn about this specific partner's behavior

I'll send a moderate amount around \$3 to balance risk and cooperation potential.

usage:

```
{
  "input_tokens": 2129,
  "output_tokens": 563,
}
```

```
"reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.44 to you. Your visible payoff was \$6.44.

Your total visible earnings across all rounds are \$31.779999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.47, so you received \$13.41. How much do you return? (0-13.41)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****What happened:****

- The sender sent me \$4.47 out of \$5 (89.4% of their endowment) - showing exceptional trust
- I received \$13.41 after the 3x multiplier

****My analysis:****

- This is the highest trust signal I've received in any round
- Previous partners sent 45-76%, but this sender sent nearly 90%
- This exceptional cooperation deserves strong reciprocation

I'll return a generous amount to reward their high trust:

```
'''json
{"return": 8}
'''
```

This gives the sender a total of \$8.53 (\$0.53 kept + \$8.00 returned), a gain of \$3.53, and leaves me with \$5.41. This 60% return rate fairly rewards their exceptional trust and encourages continued cooperation.

Reasoning:

Looking at this situation:

****What happened:****

- The sender sent me \$4.47 out of \$5 (89.4% of their endowment) - showing very high trust
- I received \$13.41 after the 3x multiplier

****My analysis:****

- This is the highest trust signal I've received so far (89.4%)
- Previous sends to me: \$3.8 (76%), \$3.11 (62%), \$2.26 (45%)
- This sender is showing exceptional trust

****My return history as receiver:****

- Round 2: Received \$11.4, returned \$7 (61%)
- Round 4: Received \$9.33, returned \$5.5 (59%)
- Round 5: Received \$6.78, returned \$3.5 (52%)

Given this sender's exceptional trust (nearly 90% of endowment), I should reciprocate strongly to reward this cooperation.

If I return around 60% like my previous high-trust returns, that would be about \$8.05.

But given they sent 89% of their endowment, maybe I should be even more generous. Let me think about what gives them a good outcome:

- They kept: \$0.53
- If I return \$8, they get \$8.53 total
- If I return \$9, they get \$9.53 total

Returning \$8 would give them \$8.53 total (a gain of \$3.53) and leave me with \$5.41.

Returning \$9 would give them \$9.53 total (a gain of \$4.53) and leave me with \$4.41.

I think \$8 is a strong reciprocal move that rewards their exceptional trust. It's about 60% of what I received, consistent with my fair reciprocation pattern.

Usage: {

"input_tokens": 2459,

```
"output_tokens": 627,
"reasoning_tokens": 0
}
```

ROUND 8

Roles: Agent_1=Sender, Agent_2=Sender, Agent_3=Receiver, Agent_4=Receiver, Agent_5=Receiver, Agent_6=Receiver, Agent_7=Sender, Agent_8=Sender

Pairings:

dyad_1: Agent_8=Sender, Agent_3=Receiver
dyad_2: Agent_2=Sender, Agent_4=Receiver
dyad_3: Agent_7=Sender, Agent_5=Receiver
dyad_4: Agent_1=Sender, Agent_6=Receiver

Parsed game decisions:

dyad_1: Agent_8 sent \$4; received \$12; Agent_3 returned \$10.2; payoffs Agent_8=\$11.2, Agent_3=\$1.8; noise/visible: sent visible=\$4.85, received visible=\$14.55, returned visible=\$9.55
Action validation: {"returned": {"amount": 10.2, "clamped": false, "max": 12.0, "min": 0, "raw": 10.2}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_2: Agent_2 sent \$3; received \$9; Agent_4 returned \$5.4; payoffs Agent_2=\$7.4, Agent_4=\$3.6; noise/visible: sent visible=\$3.21, received visible=\$9.63, returned visible=\$6.17
Action validation: {"returned": {"amount": 5.4, "clamped": false, "max": 9.0, "min": 0, "raw": 5.4}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_3: Agent_7 sent \$4.5; received \$13.5; Agent_5 returned \$8; payoffs Agent_7=\$8.5, Agent_5=\$5.5; noise/visible: sent visible=\$4.61, received visible=\$13.83, returned visible=\$8.1
Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 13.5, "min": 0, "raw": 8.0}, "sent": {"amount": 4.5, "clamped": false, "max": 5, "min": 0, "raw": 4.5}}
dyad_4: Agent_1 sent \$3.75; received \$11.25; Agent_6 returned \$6; payoffs Agent_1=\$7.25, Agent_6=\$5.25; noise/visible: sent visible=\$3.06, received visible=\$9.18, returned visible=\$5.29
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 11.25, "min": 0, "raw": 6.0}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}

Balances after round: {

"Agent_1": 48.5,
"Agent_2": 47.65,
"Agent_3": 43.8,
"Agent_4": 46.6,
"Agent_5": 55.5,
"Agent_6": 51.35,
"Agent_7": 46.5,
"Agent_8": 57.099999999999994

}

Visible balances after round: {

"Agent_1": 53.21,
"Agent_2": 52.489999999999995,
"Agent_3": 49.440000000000005,
"Agent_4": 51.289999999999999,
"Agent_5": 55.589999999999999,
"Agent_6": 45.769999999999996,
"Agent_7": 45.79,
"Agent_8": 54.570000000000001

}

LLM CALLS

GAME - Agent_8 - Sender - dyad_1

Timestamp: 2026-08-12T09:26:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_3",
"pairing": {
 "agents": [
 "Agent_8",
 "Agent_3"
],
 "investor": "Agent_8",
 "roles": {

```

    "Agent_3": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.37 to you. Your visible payoff was \$5.37.

Your total visible earnings across all rounds are \$5.37.

This round, you are the RECEIVER against your current co-player. They sent you \$4.58, so you received \$13.74. How much do you return? (0-13.74)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

My co-player sent me \$4.58 out of their \$5 (91.6% of their endowment) - showing very high trust.

I received \$13.74. To reciprocate this trust appropriately:

- They kept \$0.42
- If I return around \$6.50, they'd end with \$6.92 (a gain of \$1.92)
- I'd keep \$7.24 (also a solid gain)

This represents returning about 47% of what I received, which rewards their trust while we both benefit substantially.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Let me analyze this situation:

Round 2 - I'm now the RECEIVER

- My co-player sent me \$4.58
- I received \$13.74 (which is 4.58×3)
- I need to decide how much to return

From Round 1:

- I sent \$3.00, they received \$9.00
- They returned \$3.37 to me
- This is about 37.4% of what they received, which is quite fair/generous

My co-player in this round sent me \$4.58 out of their \$5, which is 91.6% of their endowment - very trusting!

I should consider:

1. They showed high trust by sending almost everything
2. In Round 1, my previous partner returned about 37.4% which seemed fair
3. I want to reciprocate trust and maintain good cooperation

If I return about 40-50% of what I received (\$13.74):

- 40% = \$5.50
- 45% = \$6.18
- 50% = \$6.87

Given they sent almost everything (\$4.58 out of \$5), showing extreme trust, I should be generous in return.

Their final payoff would be:

- They started with \$5
- Sent \$4.58, kept \$0.42
- If I return \$6.00: $0.42 + 6.00 = 6.42$
- My payoff: $13.74 - 6.00 = 7.74$

If I return \$6.50:

- Their payoff: $0.42 + 6.50 = 6.92$
- My payoff: $13.74 - 6.50 = 7.24$

Given their high trust (sending 91.6% of endowment), I should return a good amount. Returning around \$6.50 seems fair - they end up with \$6.92 (better than their initial \$5), and I keep \$7.24 (also better than my initial \$0).

...

usage:

```
{
  "input_tokens": 755,
  "output_tokens": 653,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.58, it became \$13.74, and you returned \$6.5. Your visible payoff was \$7.24.

Your total visible earnings across all rounds are \$12.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.38, so you received \$10.14. How much do you return? (0-10.14)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust.

I received \$10.14.

They kept: \$1.62

To reciprocate fairly:

- If I return \$5.00, they'd end with \$6.62 (a gain of \$1.62)
- I'd keep \$5.14 (also a small gain)

This represents returning about 49% of what I received, which rewards their trust while we both benefit.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust.

I received \$10.14.

They kept: \$5 - \$3.38 = \$1.62

To be fair and reciprocate their trust:

- If I return around \$5, they'd end with \$1.62 + \$5 = \$6.62 (a gain of \$1.62)
- I'd keep \$10.14 - \$5 = \$5.14 (also a solid gain)

This represents returning about 49% of what I received, which is fair and reciprocates their trust appropriately while we both benefit.

Let me think about this more carefully. They sent \$3.38, showing moderate-to-good trust (68% of endowment).

In Round 2, my co-player sent me \$4.58 (92% of endowment) and I returned \$6.50 (47% of what I received).

For consistency and fairness here with \$3.38 sent (less trust shown):

- Returning around 47-50% of \$10.14 would be \$4.77-\$5.07
- Let me go with \$5.00

That gives them: \$1.62 + \$5.00 = \$6.62 (gain of \$1.62)

I keep: \$10.14 - \$5.00 = \$5.14 (gain of \$0.14)

This seems fair and reciprocal.

usage:

```
{
  "input_tokens": 1036,
  "output_tokens": 519,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.38, it became \$10.14, and you returned \$5.0. Your visible payoff was \$5.140000000000001.

Your total visible earnings across all rounds are \$17.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

Round 1: As SENDER, I sent \$3 and received \$3.37 back - a reasonable return.

Rounds 2-3: As RECEIVER, others showed good trust, and I reciprocated fairly.

This is a new co-player (random pairing), so I don't have specific information about them. However, the overall pattern suggests players are reasonably cooperative.

I'll send \$3.50 - slightly more than my first round to show increasing trust while remaining measured:

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$3, received back \$3.37, payoff \$5.37

Round 2: I was RECEIVER, received \$13.74 (from \$4.58), returned \$6.50, payoff \$7.24

Round 3: I was RECEIVER, received \$10.14 (from \$3.38), returned \$5.00, payoff \$5.14

Round 4: I am SENDER

My total earnings: \$17.75

Looking at the pattern:

- When I sent \$3, I got back \$3.37 (a bit more than 1:1 return on sent amount)
- When others sent to me, they showed high trust (\$4.58 and \$3.38 out of \$5)
- I've been fairly reciprocal as a receiver, returning roughly half of what I received

Now I'm a sender again. Since:

1. My first send of \$3 was met with a decent return (\$3.37)
2. Others have been trusting with me
3. I've been fair in return

I should continue with moderate to high trust. I'll send \$3.5 this time - slightly more than my first send to show I'm willing to build cooperation.

Actually, let me reconsider. The co-player this round is new (pairings are randomized). I don't know their history. In Round 1, I sent \$3 and got a reasonable return. Let me stick with a similar strategy - send \$3 again as a baseline cooperative strategy.

Actually, looking at the overall pattern, players seem reasonably cooperative. I could try \$3.5 to be slightly more trusting.

Let me go with \$3.5 - showing increased trust while not overcommitting.

usage:

```
{
  "input_tokens": 1303,
  "output_tokens": 551,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$7.22 to you. Your visible payoff was \$8.719999999999999.

Your total visible earnings across all rounds are \$26.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my recent experience:

Round 4: I sent \$3.50 and received \$7.22 back - excellent cooperation!

The overall pattern shows players are being reasonably cooperative. While this is a new co-player (random pairing), I'll show increased trust based on positive experiences.

Sending \$4:

- Demonstrates strong trust (80% of my endowment)
- Creates \$12 for them to work with
- Encourages reciprocal cooperation

```
'''json
{"send": 4}
'''
```

reasoning:

Round 4 recap: I sent \$3.50, they returned \$7.22 - excellent return! That's about 69% of what they received, very generous.

This is Round 5, and I'm the SENDER again against a new co-player (random pairing each round).

So far my experience:

- Round 1 SENDER: sent \$3, got \$3.37 back (reasonable)
- Round 2 RECEIVER: received \$13.74, returned \$6.50
- Round 3 RECEIVER: received \$10.14, returned \$5.00
- Round 4 SENDER: sent \$3.50, got \$7.22 back (very good!)

The last round showed great cooperation. Players seem generally trustworthy. This is a new co-player though, so I don't know their specific behavior.

I should probably continue with a trusting approach. I could:

- Send \$4 (80% of endowment) - shows high trust
- Send \$3.5 (70%) - similar to last round
- Send \$3 (60%) - more cautious

Given the positive experiences, I'll increase slightly to \$4 to show strong trust.

usage:

```
{
  "input_tokens": 1562,
  "output_tokens": 402,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.98 to you. Your visible payoff was \$7.98.

Your total visible earnings across all rounds are \$34.45.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

My co-player sent me \$2.9 out of their \$5 (58% of their endowment) - showing moderate trust.

I received \$8.7.

They kept: \$2.1

To reciprocate fairly:
- If I return \$4.00, they'd end with \$6.10 (a gain of \$1.10)
- I'd keep \$4.70 (also a decent gain)

This represents returning about 46% of what I received, which is proportional to the moderate trust shown and ensures we both benefit.

```
'''json
{"return": 4.00}
'''
```

reasoning:

Looking at this situation:

My co-player sent me \$2.9 out of their \$5 (58% of their endowment) - showing moderate trust.

I received \$8.7.
They kept: \$2.1

To reciprocate fairly:
- If I return around \$4.00-\$4.50, they'd end with \$6.10-\$6.60 (a gain of \$1.10-\$1.60)
- I'd keep \$4.70-\$4.20

This represents returning about 46-52% of what I received, which rewards their trust while we both benefit.

Let me think about what's fair. They sent \$2.9, showing moderate but not high trust. I should return something proportional.

In previous rounds as receiver:
- Round 2: Received \$13.74, returned \$6.50 (47%)
- Round 3: Received \$10.14, returned \$5.00 (49%)

So I've been returning around 47-49% of what I receive.

For \$8.7, returning around 47-49% would be:
- 47%: \$4.09
- 49%: \$4.26

Let me go with \$4.25, which is about 49% and gives them a total of \$6.35 (gain of \$1.35), while I keep \$4.45 (also a decent gain).

Actually, let me reconsider. They sent \$2.9 (58% of endowment), which is moderate trust. In Round 1, someone sent me \$4.58 (91.6%) and I returned \$6.50. In Round 3, someone sent \$3.38 (67.6%) and I returned \$5.00.

The person who sent \$3.38 (67.6%) got back \$5.00 from \$10.14 received.

...

usage:

```
{
  "input_tokens": 1817,
  "output_tokens": 654,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.0. Your visible payoff was \$4.6999999999999999.

Your total visible earnings across all rounds are \$39.1500000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

My co-player sent me \$2.99 out of their \$5 (about 60% of their endowment) - showing moderate trust, very similar to Round 6.

I received \$8.97.
They kept: \$2.01

To reciprocate consistently with my Round 6 approach:
- If I return \$4.10, they'd end with \$6.11 (a gain of \$1.11)
- I'd keep \$4.87 (also a decent gain)

This represents returning about 46% of what I received, consistent with similar trust levels.

```
'''json
{"return": 4.10}
'''
```

reasoning:

My co-player sent me \$2.99 out of their \$5 (about 60% of their endowment) - showing moderate trust, very similar to the previous round.

I received \$8.97.
They kept: \$2.01

This is very similar to Round 6 where they sent \$2.9. I should be consistent in my reciprocation strategy.

In Round 6, I returned \$4.00 when I received \$8.7, which was about 46% of what I received.

Using a similar proportion:
- If I return \$4.10, they'd end with \$6.11 (a gain of \$1.11)
- I'd keep \$4.87 (also a decent gain)

This represents returning about 46% of what I received, consistent with my previous approach for similar trust levels.

Let me calculate more carefully:
- They sent: \$2.99
- They kept: $\$5 - \$2.99 = \$2.01$
- I received: \$8.97
- If I return \$4.10:
 - They end with: $\$2.01 + \$4.10 = \$6.11$
 - I end with: $\$8.97 - \$4.10 = \$4.87$

This seems fair and consistent with my previous behavior.

usage:

```
{
  "input_tokens": 2105,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.99, it became \$8.97, and you returned \$4.1. Your visible payoff was \$4.8700000000000001.
Your total visible earnings across all rounds are \$44.020000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience:

As SENDER:

- Round 1: sent \$3.00, received \$3.37 back
- Round 4: sent \$3.50, received \$7.22 back (excellent!)
- Round 5: sent \$4.00, received \$6.98 back (good!)

The pattern shows that sending more has generally been rewarded with good reciprocation. This is Round 8 with a new co-player.

I'll send \$4.00 - this amount has proven successful and demonstrates strong trust while maintaining some prudence.

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Looking at my history:

Round 1: SENDER - sent \$3.00, received \$3.37 back (payoff: \$5.37)
Round 2: RECEIVER - received \$13.74 from \$4.58 sent, returned \$6.50 (payoff: \$7.24)
Round 3: RECEIVER - received \$10.14 from \$3.38 sent, returned \$5.00 (payoff: \$5.14)
Round 4: SENDER - sent \$3.50, received \$7.22 back (payoff: \$8.72)
Round 5: SENDER - sent \$4.00, received \$6.98 back (payoff: \$7.98)
Round 6: RECEIVER - received \$8.70 from \$2.90 sent, returned \$4.00 (payoff: \$4.70)
Round 7: RECEIVER - received \$8.97 from \$2.99 sent, returned \$4.10 (payoff: \$4.87)
Round 8: SENDER - this round, new co-player

My sending pattern has been:

- Round 1: \$3.00
- Round 4: \$3.50
- Round 5: \$4.00

I've been gradually increasing trust. The returns have generally been good:

- Round 1: sent \$3, got \$3.37 back (net: +\$0.37)
- Round 4: sent \$3.5, got \$7.22 back (net: +\$2.22) - excellent!
- Round 5: sent \$4, got \$6.98 back (net: +\$1.48) - good!

Overall, sending more has been rewarded. This is a final round (Round 8), and a new co-player.

Given the positive trend and it being the final round, I could send more to maximize potential returns. ...

```
Usage: {
  "input_tokens": 2237,
  "output_tokens": 647,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_1

Timestamp: 2026-08-12T09:26:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_3"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_3": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user**content:**

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.69 to you. Your visible payoff was \$9.69.

Your total visible earnings across all rounds are \$9.69.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant**content:**

Looking at this situation:

My co-player sent me \$3.11 (62% of their endowment), showing significant trust. I received \$9.33 after the 3x multiplier.

In my previous round, my co-player returned \$8.69 out of \$12 (72%), which was very generous. I want to maintain this cooperative spirit.

A fair approach would be to ensure my co-player profits from their trust while I also benefit from receiving the tripled amount.

Returning approximately \$6.50 would give them a final payoff of about \$8.39 ($\$5 - \$3.11 + \6.50), which rewards their trust handsomely, while I keep \$2.83.

```
{"return": 6.50}
```

reasoning:

Let me analyze this situation:

1. Last round I was the SENDER:
 - I sent \$4.0
 - It became \$12.0
 - They returned \$8.69
 - My payoff: $\$5 - \$4 + \$8.69 = \9.69
 - Their payoff: $\$12 - \$8.69 = \$3.31$
2. This round I'm the RECEIVER:
 - They sent me \$3.11
 - I received $\$3.11 \times 3 = \9.33
 - I need to decide how much to return

My co-player sent me \$3.11 out of their \$5, which is 62.2% of their endowment. This shows significant trust.

In the last round, my co-player returned \$8.69 out of \$12 they received, which is 72.4% of what they received. This was quite generous - they gave me more than I sent them.

I should consider:

- Reciprocity: They showed trust by sending \$3.11
- Building reputation: Being fair will help in future rounds
- The note says there may be communication noise, so amounts might differ slightly

A fair return would be:

- They invested \$3.11, I should at least return that amount so they don't lose
- To split gains fairly: They gave up \$3.11, it became \$9.33, so we created \$6.22 in additional value
- If we split that gain: $\$3.11 + \$6.22/2 = \$3.11 + \$3.11 = \$6.22$

However, I also want to be somewhat generous to build trust, similar to how my previous co-player returned 72.4% of what they received.

Let me think about different approaches:

- Return exactly what they sent: \$3.11 (they break even)
- Return to split gains 50/50: $\sim \$6.22$ (they get $\$5 - \$3.11 + \$6.22 = \8.11)
- Return more generously (similar to 72.4% rate): $\sim \$6.75$ (they get $\$5 - \$3.11 + \$6.75 = \8.64)

Given the trust they showed and the precedent from my last game, I'll return around \$6.50-\$7.00, which gives them a good payoff while keeping a reasonable amount for myself. I'll return \$6.50: their final payoff becomes \$8.39 while I keep \$9.

Wait, let me check the math more carefully. If I return \$6.50, my payoff is $\$9.33 - \$6.50 = \$2.83$. That feels...

usage:

```
{
  "input_tokens": 763,
  "output_tokens": 1019,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$6.5. Your visible payoff was \$2.83.

Your total visible earnings across all rounds are \$12.52.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience so far:

- Round 1: Sent \$4, received \$8.69 back (very cooperative partner)
- Round 2: Received from someone who sent \$3.11, I returned \$6.50 (both of us were cooperative)

The environment has been quite cooperative. Sending more creates more value through the 3x multiplier, and past interactions suggest people reciprocate trust.

To maximize value creation and continue building a cooperative reputation, I'll send my full endowment this round.

```
{"send": 5}
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$4, received \$8.69 back, payoff \$9.69

Round 2: I was RECEIVER, got \$9.33 (from \$3.11 sent), returned \$6.50, payoff \$2.83

Total: \$12.52

Now Round 3: I'm SENDER again with a new co-player.

My experience so far:

- When I sent \$4, I got back \$8.69, which was very generous (72% return rate on the tripled amount)
- When I received from someone who sent \$3.11, I returned \$6.50 (70% return rate)

Both interactions have been quite cooperative. This suggests the environment is reasonably trustworthy.

As a sender this round, I should consider:

1. Past cooperation suggests I can trust my co-players
2. Sending more creates more value overall (3x multiplier)
3. Building a reputation for cooperation may help when I'm receiver
4. The game mentioned communication noise, so exact amounts may vary

Given the positive experiences, I should send a substantial amount. I'll send \$4 again since that worked well before, or maybe even \$5 to maximize value creation.

Actually, let me think about this more strategically. If I send \$5:

- Co-player receives \$15
- If they return even 50%, I get \$7.50 back, netting me \$2.50 profit
- If they're as generous as my Round 1 partner (~72%), I could get ~\$10.80 back

If I send \$4:

- Co-player receives \$12
- At 72% return, I'd get ~\$8.64 back, total payoff ~\$9.64

Given the cooperative environment and to maximize total value, I'll send \$5 this time.

usage:

```
{
  "input_tokens": 1058,
  "output_tokens": 560,
  "reasoning_tokens": 0
}
```

Message 6 role=user**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$10.29 to you. Your visible payoff was \$10.29.

Your total visible earnings across all rounds are \$22.81.

This round, you are the RECEIVER against your current co-player. They sent you \$3.57, so you received \$10.709999999999999. How much do you return? (0-10.709999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant**content:**

Looking at this situation:

My co-player sent \$3.57 (71.4% of their endowment), showing strong trust. I received \$10.71.

The game has been very cooperative so far:

- Round 1: I sent \$4, got back \$8.69
- Round 3: I sent \$5, got back \$10.29
- Round 2: When I was receiver, I returned \$6.50 out of \$9.33

My co-player trusted me significantly. A generous return would be around 70% of what I received, which would give them a strong profit while still rewarding me for receiving.

Returning \$7.50 would:

- Give them: $\$5 - \$3.57 + \$7.50 = \8.93 total
- Leave me with: $\$10.71 - \$7.50 = \$3.21$

This maintains the cooperative pattern.

```
{"return": 7.50}
```

reasoning:

Let me analyze this situation:

My co-player sent me \$3.57 out of \$5 (71.4% of their endowment), showing strong trust. I received \$10.71 after the 3x multiplier.

In my previous round as sender:

- I sent \$5 (100% of endowment)
- They returned \$10.29 out of \$15 (68.6%)
- This was very generous, giving me \$10.29 total payoff

The pattern has been quite cooperative. Let me think about a fair return:

If I return around \$7-\$7.50:

- They would get: $\$5 - \$3.57 + \$7 = \8.43 (if I return \$7)
- They would get: $\$5 - \$3.57 + \$7.50 = \8.93 (if I return \$7.50)
- I would keep: $\$10.71 - \$7 = \$3.71$ (if I return \$7)
- I would keep: $\$10.71 - \$7.50 = \$3.21$ (if I return \$7.50)

Given they sent 71.4% of their endowment, a generous return would be appropriate. The previous receiver returned about 68.6% when I sent 100%.

Returning about \$7.20 would:

- Give them \$8.63 total ($\$5 - \$3.57 + \7.20)
- Leave me with \$3.51
- This rewards their trust well

Actually, let me be a bit more generous to match the cooperative environment. Returning \$7.50:

- Gives them \$8.93 total (a \$3.93 profit on their \$5)
- Leaves me with \$3.21 (also a good profit)
- Return rate: 70% of what I received

...

usage:

```
{
  "input_tokens": 1326,
  "output_tokens": 675,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.57, it became \$10.709999999999999, and you returned \$7.5. Your visible payoff was \$3.2099999999999999.

Your total visible earnings across all rounds are \$26.019999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant**content:**

Looking at my track record:

- As SENDER: Sent \$4 (got \$8.69), Sent \$5 (got \$10.29)
- As RECEIVER: Returned generously both times

The environment is highly cooperative. Sending the full amount has consistently yielded very good returns for me. My average earnings per round (\$6.50) significantly exceed the \$5 baseline, showing that cooperation pays off.

I'll continue the cooperative strategy and send my full endowment.

```
{"send": 5}
```

reasoning:

Looking at my history:

- Round 1 (SENDER): Sent \$4, got back \$8.69 - great outcome
- Round 2 (RECEIVER): Got \$9.33, returned \$6.50
- Round 3 (SENDER): Sent \$5, got back \$10.29 - excellent outcome
- Round 4 (RECEIVER): Got \$10.71, returned \$7.50

The pattern is very clear - I've experienced highly cooperative behavior. When I've been the sender and sent significant amounts, I've gotten generous returns. When I've been the receiver, I've also returned generously.

My cumulative earnings are \$26.02 across 4 rounds, averaging \$6.50 per round, which is well above the \$5 baseline if I had just kept everything.

The optimal strategy in this cooperative environment is to continue sending high amounts:

1. It creates value through the 3x multiplier
2. Past experience shows I get good returns
3. It builds my reputation for future rounds

I should send \$5 again to maximize value creation and maintain my cooperative reputation.

usage:

```
{
  "input_tokens": 1678,
  "output_tokens": 377,
  "reasoning_tokens": 0
}
```

Message 10 role=user**content:**

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$8.2 to you. Your visible payoff was \$8.2.

Your total visible earnings across all rounds are \$34.22.

This round, you are the RECEIVER against your current co-player. They sent you \$3.41, so you received \$10.23. How much do you return? (0-10.23)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant**content:**

Looking at this situation:

My co-player sent \$3.41 (68% of their endowment), showing solid trust. I received \$10.23.

My pattern as receiver has been:

- Round 2: Returned \$6.50 out of \$9.33 (~70%)
- Round 4: Returned \$7.50 out of \$10.71 (~70%)

Maintaining this cooperative approach, returning around \$7 would:

- Give them: \$5 - \$3.41 + \$7 = \$8.59 total
- Leave me with: \$10.23 - \$7 = \$3.23

This rewards their trust while staying consistent with my cooperative strategy.

```
{"return": 7.00}
```

reasoning:

My co-player sent \$3.41 (68.2% of their endowment), showing good trust. I received \$10.23.

Looking at my history:

- As SENDER: Sent \$4 (got \$8.69), Sent \$5 (got \$10.29), Sent \$5 (got \$8.2)
- As RECEIVER: Returned \$6.50 out of \$9.33, Returned \$7.50 out of \$10.71

The pattern has been very cooperative. When I receive around \$10, I've been returning between \$6.50-\$7.50.

For this round with \$10.23 received:

- Returning around \$7 would give my co-player: $\$5 - \$3.41 + \$7 = \8.59
- I'd keep: $\$10.23 - \$7 = \$3.23$

This seems fair and maintains the cooperative pattern. I'll return \$7.

Actually, let me reconsider. My previous returns as receiver:

- Round 2: Received \$9.33, returned \$6.50 (69.7%)
- Round 4: Received \$10.71, returned \$7.50 (70%)

So I've been returning around 70% of what I received. $70\% \text{ of } \$10.23 = \7.16

I'll return \$7.00 to stay consistent with my cooperative pattern.

usage:

```
{
  "input_tokens": 1931,
  "output_tokens": 513,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.41, it became \$10.23, and you returned \$7.0. Your visible payoff was \$3.2300000000000004.

Your total visible earnings across all rounds are \$37.45.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my experience:

As SENDER:

- Sent \$4, received \$8.69
- Sent \$5, received \$10.29
- Sent \$5, received \$8.20

All my sending decisions have been profitable, and the environment remains cooperative. Sending the full \$5 maximizes potential gains through the 3x multiplier and has consistently yielded good returns.

I'll continue with the cooperative strategy.

```
{"send": 5}
```

reasoning:

Looking at my history:

- As SENDER:
 - * Round 1: Sent \$4, got back \$8.69
 - * Round 3: Sent \$5, got back \$10.29
 - * Round 5: Sent \$5, got back \$8.20
- As RECEIVER:
 - * Round 2: Received \$9.33, returned \$6.50
 - * Round 4: Received \$10.71, returned \$7.50
 - * Round 6: Received \$10.23, returned \$7.00

The environment has been consistently cooperative. All three times I sent money as a sender, I received generous returns. When I send \$5, I've gotten back \$10.29 and \$8.20 (averaging about \$9.25), both of which are profitable.

I'm earning well above the \$5 baseline per round. The cooperative strategy is working. I should continue sending my full endowment to maximize value creation and maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 2228,
  "output_tokens": 341,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$7.64 to you. Your visible payoff was \$7.64.

Your total visible earnings across all rounds are \$45.09.

This round, you are the RECEIVER against your current co-player. They sent you \$4.85, so you received \$14.549999999999999. How much do you return? (0-14.549999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

My co-player sent \$4.85 (97% of their endowment) - this is exceptional trust, nearly their entire amount!

My pattern as receiver has been returning around 68-70% of what I receive:

- Round 2: 70% of \$9.33 = \$6.50
- Round 4: 70% of \$10.71 = \$7.50
- Round 6: 68% of \$10.23 = \$7.00

Given their exceptional trust (sending almost everything), I should maintain this cooperative approach. Returning 70% of \$14.55 = \$10.19

This would give them:

- Their payoff: \$5 - \$4.85 + \$10.19 = \$10.34
- My payoff: \$14.55 - \$10.19 = \$4.36

This strongly rewards their trust.

```
{"return": 10.20}
```

Reasoning:

My co-player sent \$4.85, which is 97% of their endowment - extremely high trust! I received \$14.55.

This is someone showing exceptional trust. Let me think about what's fair:

My past returns as receiver:

- Round 2: Returned \$6.50 out of \$9.33 (~70%)
- Round 4: Returned \$7.50 out of \$10.71 (~70%)
- Round 6: Returned \$7.00 out of \$10.23 (~68%)

So I've been returning around 68-70% of what I received.

If I return 70% of \$14.55 = \$10.19

That would give them:

- Their payoff: \$5 - \$4.85 + \$10.19 = \$10.34
- My payoff: \$14.55 - \$10.19 = \$4.36

This rewards their extreme trust significantly. They basically sent almost everything, so I should be generous back.

Alternatively, I could return a bit more given their exceptional trust. Maybe \$10.50?

- Their payoff: \$5 - \$4.85 + \$10.50 = \$10.65
- My payoff: \$14.55 - \$10.50 = \$4.05

I think around \$10.00-\$10.50 is appropriate given their nearly maximal trust.

Let me go with \$10.20 to maintain my ~70% return pattern but be generous given their exceptional trust.

```
Usage: {
  "input_tokens": 2318,
  "output_tokens": 581,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Sender - dyad_2

Timestamp: 2026-08-12T09:26:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_4"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_2": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$2.81 to you. Your visible payoff was \$4.8100000000000005.

Your total visible earnings across all rounds are \$4.8100000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$3.95, so you received \$11.850000000000001. How much do you return? (0-11.850000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.95 (79% of their endowment), showing significant trust
- I received \$11.85 total
- They currently have only \$1.05 left

I want to:

1. Reward their trust and cooperation
2. Ensure they end up better than their initial \$5
3. Set a positive tone for potential future interactions

If I return \$5.50:

- They end up with: \$1.05 + \$5.50 = \$6.55
- I end up with: \$11.85 - \$5.50 = \$6.35

Both of us gain from cooperation, and this reciprocates their high level of trust appropriately.

```
'''json
{"return": 5.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was the SENDER
- I sent \$3.00
- It became \$9.00 for the receiver
- They returned \$2.81
- My payoff: \$5 - \$3 + \$2.81 = \$4.81

Round 2:

- I am now the RECEIVER
- My co-player (a different agent) sent me \$3.95
- I received \$3.95 × 3 = \$11.85
- I need to decide how much to return

Important considerations:

1. The sender gave me \$3.95 out of their \$5 - that's 79% of their endowment, showing high trust
2. If I return nothing, I keep \$11.85 and they end up with \$5 - \$3.95 = \$1.05
3. I should return enough to make cooperation worthwhile for them
4. There's communication noise mentioned, so amounts may differ slightly

What would be fair?

- The sender gave up \$3.95, keeping \$1.05
- I received \$11.85
- If I want them to end up better than their initial \$5, I should return more than \$3.95
- A 50-50 split of total surplus could work: Total = \$1.05 + \$11.85 = \$12.90. Half is \$6.45
- If I return \$5.40, sender gets \$1.05 + \$5.40 = \$6.45, I get \$11.85 - \$5.40 = \$6.45

But let me think about reciprocity:

- In round 1, I sent \$3 (60% of endowment), received \$9, they returned \$2.81 (31% of what they received)
- This round, sender sent \$3.95 (79% of endowment), I received \$11.85
- Using that same 31% return rate would give me \$3.67 to send back

The sender showed significantly more generosity than I did initially, so I should reciprocate with a higher return. Returning \$5 would leave them with \$6.05 total-better than their starting \$5-while I'd keep \$6.85, rewarding their trust and encouraging continued cooperation....

usage:

```
{
  "input_tokens": 793,
  "output_tokens": 977,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.95, it became \$11.850000000000001, and you returned \$5.5. Your visible payoff was \$6.350000000000001.

Your total visible earnings across all rounds are \$11.160000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at Round 3:

I'm paired with a new co-player (pairings are randomized each round), so I have no history with them. My previous experiences showed:

- Round 1: Sent \$3, received \$2.81 back - reasonable cooperation
- Round 2: As receiver, I tried to be generous with a high-trusting sender

For this new pairing, I'll use a similar approach to Round 1 - send a moderate amount that signals cooperation while being reasonably cautious with an unknown partner.

Sending \$3 creates \$9 for them to work with, which gives room for mutual benefit if they reciprocate fairly.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$3, received back \$2.81, payoff \$4.81

Round 2: I was RECEIVER, got \$11.85 from a \$3.95 send, returned \$5.50, payoff \$6.35

Total earnings: \$11.16

Now Round 3: I'm SENDER again with a new co-player (since pairings are randomized each round).

Key considerations:

- I have no history with this particular co-player
- In Round 1, when I sent \$3, I got back \$2.81 (which is reasonable - about 31% return rate)
- In Round 2, when someone sent me \$3.95, I returned \$5.50 (about 46% return rate)
- The game has communication noise, so exact amounts may differ

Since I'm paired with a new player and have no history with them, I should probably use a similar strategy to Round 1 - establish cooperation but be somewhat cautious.

Given that:

- Sending \$3 worked reasonably well in Round 1
- The multiplication factor of 3x creates good mutual benefit potential
- I want to signal cooperation

I'll send \$3 again. This is a moderate amount that shows willingness to cooperate without being overly risky with a new partner.

usage:

```
{
  "input_tokens": 1106,
  "output_tokens": 478,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.52 to you. Your visible payoff was \$8.52.

Your total visible earnings across all rounds are \$19.68.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.36 (67% of their endowment), showing solid trust
- I received \$10.08 total
- They currently have \$1.64 left

I want to reciprocate their trust fairly. If I return \$5.00:

- They end up with: $\$1.64 + \$5.00 = \$6.64$ (gain of \$1.64)
- I end up with: $\$10.08 - \$5.00 = \$5.08$ (gain of \$0.08)

Both of us benefit from cooperation, and this rewards their trust appropriately.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- My co-player sent \$3.36 (67.2% of their endowment)
- I received \$10.08 total
- They currently have: $\$5 - \$3.36 = \$1.64$ left

My previous round (Round 3):

- I sent \$3, received \$6.52 back - very generous return!

This co-player sent \$3.36, showing good trust. I should reciprocate appropriately.

If I return \$5.00:

- They end up with: $\$1.64 + \$5.00 = \$6.64$
- I end up with: $\$10.08 - \$5.00 = \$5.08$

Both of us gain. This is a fair reciprocation that rewards their trust.

Alternatively, if I return \$5.50:

- They end up with: $\$1.64 + \$5.50 = \$7.14$
- I end up with: $\$10.08 - \$5.50 = \$4.58$

This would be even more generous, similar to how I acted in Round 2.

Given that they sent about 67% of their endowment (compared to 79% in Round 2), I think re turning around \$5 is appropriate and fair. It ensures they profit significantly from the ir trust, and we both end up better off than the initial \$5.

I'll go with \$5.00 as a fair reciprocation.

usage:

```
{
  "input_tokens": 1402,
  "output_tokens": 522,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$5.0. Your visible payoff was \$5.08.

Your total visible earnings across all rounds are \$24.759999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at Round 5:

I'm with another new co-player. My experience so far shows:

- Sending \$3 has worked well in previous rounds, getting returns of \$2.81 and \$6.52
- The cooperative approach has generally been rewarded

I'll continue with the same strategy - send \$3 to signal trust and cooperation while being reasonably cautious with an unknown partner.

```
'''json
{"send": 3}
'''
```

reasoning:

Round 5, I'm the sender with a new co-player (random pairings each round).

Looking at my history:

- Round 1 (Sender): Sent \$3, got back \$2.81 - decent cooperation
- Round 2 (Receiver): Received \$11.85, returned \$5.50 - I was generous
- Round 3 (Sender): Sent \$3, got back \$6.52 - excellent return!
- Round 4 (Receiver): Received \$10.08, returned \$5.00 - fair reciprocation

So far, sending \$3 has worked well - I got \$2.81 and \$6.52 back in my two sender rounds. The average return has been positive.

For this new pairing, I should continue with a cooperative strategy. Sending \$3 seems to be working as a good middle-ground approach that encourages cooperation.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1690,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.15 to you. Your visible payoff was \$5.15.

Your total visible earnings across all rounds are \$29.909999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.47, so you received \$13.41. How much do you return? (0-13.41)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.47 (89% of their endowment), showing very high trust
- I received \$13.41 total
- They currently have only \$0.53 left

This is a very trusting sender, and I should reciprocate generously. If I return \$6.50:

- They end up with: $\$0.53 + \$6.50 = \$7.03$ (gain of \$2.03)
- I end up with: $\$13.41 - \$6.50 = \$6.91$ (gain of \$1.91)

Both of us gain approximately \$2, which appropriately rewards their exceptional trust and maintains strong cooperation.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Looking at this situation:

- My co-player sent \$4.47 (89% of their endowment), showing very high trust
- I received \$13.41 total
- They currently have \$0.53 left

This is a very trusting sender. I should reciprocate generously.

If I return \$6.50:

- They end up with: $\$0.53 + \$6.50 = \$7.03$ (gain of \$2.03)
- I end up with: $\$13.41 - \$6.50 = \$6.91$ (gain of \$1.91)

Both benefit well, and this appropriately rewards their very high level of trust.

Let me think about this more. They sent \$4.47 out of \$5, leaving themselves with only \$0.53. That's 89.4% of their endowment. This is extremely trusting.

I should return enough to:

1. Ensure they profit significantly from their trust
2. Show I value cooperation
3. Both of us should end up better than our starting \$5

If I return \$6.50:

- Them: $\$0.53 + \$6.50 = \$7.03$
- Me: $\$13.41 - \$6.50 = \$6.91$

Both gain roughly \$2, which seems fair and reciprocates their high trust.

usage:

```
{
  "input_tokens": 1931,
  "output_tokens": 502,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.47, it became \$13.41, and you returned \$6.5. Your visible payoff was \$6.91.

Your total visible earnings across all rounds are \$36.81999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- My co-player sent the maximum \$5 (100% of their endowment), showing complete trust
- I received \$15 total
- They currently have \$0 left

This is the highest level of trust possible. I should reciprocate very generously. If I return \$7.50:

- They end up with: $\$0 + \$7.50 = \$7.50$ (gain of \$2.50)
- I end up with: $\$15 - \$7.50 = \$7.50$ (gain of \$2.50)

We both gain equally from this maximum cooperation, which appropriately rewards their complete trust.

```
'''json
{"return": 7.50}
'''
```

reasoning:

This is Round 7, and I'm the RECEIVER.

My co-player sent the maximum amount: \$5 (100% of their endowment), showing complete trust

- I received \$15 total
- They currently have \$0 left

This sender showed maximum trust. I should reciprocate very generously.

If I return \$7.50:

- They end up with: $\$0 + \$7.50 = \$7.50$ (gain of \$2.50)
- I end up with: $\$15 - \$7.50 = \$7.50$ (gain of \$2.50)

This gives us both equal gains and rewards their complete trust. This seems fair and reciprocates their maximum trust appropriately.

Actually, let me think about this more. They sent everything (\$5), so they're showing maximum trust. I want to reward that very well.

If I return \$8:

- They end up with: $\$0 + \$8 = \$8$ (gain of \$3)
- I end up with: $\$15 - \$8 = \$7$ (gain of \$2)

This gives them a slightly better outcome for taking the risk of sending everything.

Let me go with \$7.50 as a good middle ground - we both gain equally, which seems fair when they showed maximum trust.

usage:

```
{
  "input_tokens": 2240,
  "output_tokens": 476,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$5, it became \$15, and you returned \$7.5. Your visible payoff was \$7.5. Your total visible earnings across all rounds are \$44.31999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at Round 8:

I'm with another new co-player. My strategy of sending \$3 has generally worked well across previous rounds, yielding varied but mostly positive returns. This has been a good balance between showing cooperation and being prudent.

I'll continue with this consistent approach.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at Round 8:

I'm with another new co-player. Let me review my experience as sender:

- Round 1: Sent \$3, received \$2.81 back (payoff \$4.81)
- Round 3: Sent \$3, received \$6.52 back (payoff \$8.52)
- Round 5: Sent \$3, received \$3.15 back (payoff \$5.15)

And as receiver:

- Round 2: Received from sender who sent \$3.95, I returned \$5.50
- Round 4: Received from sender who sent \$3.36, I returned \$5.00
- Round 6: Received from sender who sent \$4.47, I returned \$6.50
- Round 7: Received from sender who sent \$5.00, I returned \$7.50

Overall, cooperation has been working well. Returns have varied but generally been reasonable.

The \$3 send amount has been my standard approach.

This is the last round of the game (8 rounds total with 8 agents). Since it's the final round,

I might consider adjusting my strategy, but:

- I don't know if this is truly the last round for this experimental setup
- Maintaining consistency has worked well
- My co-player doesn't know my history

I'll stick with sending \$3, which has been successful and signals cooperation without being overly risky.

Usage: {

```
"input_tokens": 2356,
"output_tokens": 409,
"reasoning_tokens": 0
```

}

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T09:26:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_2",
"move_index": 3,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_4"
  ],
"investor": "Agent_2",
"roles": {
  "Agent_2": "investor",
  "Agent_4": "trustee"
},
"trustee": "Agent_4"
```

}

```
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.78, it became \$14.34, and you returned \$8.0. Your visible payoff was \$6.34.

Your total visible earnings across all rounds are \$6.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

In Round 1, my co-player showed very high trust by sending 95.6% of their endowment, and I reciprocated with a fair return. That cooperative approach worked well for both of us.

Now as the sender with a new co-player, I want to maintain that cooperative spirit. Sending a substantial amount:

- Creates more total value through the 3x multiplier
- Signals trust and willingness to cooperate
- Could encourage reciprocation

I'll send \$4.50 (90% of my endowment), showing strong trust while being slightly more conservative than what I experienced last round.

{'send': 4.50}

reasoning:

Let me think about this situation:

Round 1 (just completed):

- I was the RECEIVER
- Sender sent me \$4.78 (95.6% of their endowment) - showing very high trust
- I received \$14.34
- I returned \$8.00
- My payoff: \$6.34
- Sender's payoff would be: $\$5 - \$4.78 + \$8.00 = \8.22

Now in Round 2:

- I'm the SENDER
- I have \$5
- I'm playing against a different co-player (pairings are randomized each round)

The previous sender showed very high trust by sending 95.6%. I reciprocated fairly by returning \$8.00, which gave them a good outcome.

Now I need to decide how much to send. Key considerations:

1. This is a new co-player, so I don't have direct history with them
2. The previous round showed successful cooperation
3. Sending more creates more total value (3x multiplier), but requires trust
4. In a multi-round setting, establishing cooperative behavior can be beneficial

Given that:

- The previous sender sent 95.6% (\$4.78)
- That worked out well for both of us
- I should reciprocate the level of trust shown to me

I think I should send a similar high amount to signal cooperation and trust. Sending around \$4.50-\$4.80 would show similar trust to what I experienced.

Let me send \$4.50 (90% of my endowment). This shows strong trust while being slightly more conservative than the 95.6% I received.

Actually, thinking about it more - since I was treated with high trust and it worked out well, I should pay it forward. Sending \$4.50-\$4.75 seems appropriate.

...

usage:

```
{
  "input_tokens": 824,
  "output_tokens": 591,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.5, it became \$13.5, and you saw them return \$7.41 to you. Your visible payoff was \$7.91.

Your total visible earnings across all rounds are \$14.25.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- The sender showed good trust by sending \$3.80 (76% of their endowment)
- They kept \$1.20 for themselves
- I received \$11.40
- I should reciprocate their trust fairly

Comparing to previous rounds:

- In Round 1, when a sender sent 95.6%, I returned about 56% of what I received
- In Round 2, when I sent 90%, my co-player returned about 55% of what they received

For this sender who sent 76%, a fair return would be around 55-60% of what I received.

$\$11.40 \times 0.57 \approx \6.50

If I return \$6.50:

- Sender ends with: $\$1.20 + \$6.50 = \$7.70$ (good gain)
- I keep: $\$11.40 - \$6.50 = \$4.90$ (reasonable)
- Both benefit from the cooperation

`{'return': 6.50}`

reasoning:

Let me analyze this situation:

Round 2 recap:

- I was the sender and sent \$4.50
- It became \$13.50 for the receiver
- They returned \$7.41 to me
- My payoff: $\$5 - \$4.50 + \$7.41 = \7.91

Round 3 current situation:

- I'm the receiver
- Sender sent \$3.80 (76% of their endowment)
- I received \$11.40 (approximately, due to floating point)
- They kept \$1.20

This sender showed good trust by sending 76% of their endowment. Let me think about a fair return:

If I return \$6.00:

- Sender ends up with: $\$1.20 + \$6.00 = \$7.20$
- I keep: $\$11.40 - \$6.00 = \$5.40$
- Both benefit, sender gains more

If I return \$7.00:

- Sender ends up with: $\$1.20 + \$7.00 = \$8.20$
- I keep: $\$11.40 - \$7.00 = \$4.40$
- Sender gains significantly more

In Round 1, when I received \$14.34 from a sender who sent \$4.78 (95.6%), I returned \$8.00.

- That sender ended with \$8.22 (from \$0.22)
- I kept \$6.34
- Return ratio: $\$8.00/\$14.34 = 55.9\%$

In Round 2, when I sent \$4.50 and it became \$13.50, they returned \$7.41

- Return ratio: $\$7.41/\$13.50 = 54.9\%$

For consistency, if I use a similar return ratio (around 55%):

- $\$11.40 \times 0.55 = \6.27

Let me round to \$6.50 to be slightly more generous:

...

usage:

```
{
  "input_tokens": 1120,
  "output_tokens": 820,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.8, it became \$11.399999999999999, and you returned \$6.5. Your visible payoff was \$4.899999999999999.

Your total visible earnings across all rounds are \$19.15.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: `{'send': <amount>}`

Message 7 role=assistant

content:

Looking at my experience so far:

- Round 1: As receiver, someone sent high trust (95.6%), I reciprocated fairly
- Round 2: I sent \$4.50 (90%), got back \$7.41 - good cooperation
- Round 3: As receiver, returned fairly when sent \$3.80

The pattern shows that high trust tends to be reciprocated with fair returns (around 55% return rate). My cooperative strategy has been working well - averaging over \$6 per round.

I'll continue with a strong but slightly varied approach. Sending \$4.00 (80%):

- Shows substantial trust
- Creates \$12.00 for the receiver through the 3x multiplier
- Should encourage reciprocation based on the established pattern
- Slightly more conservative than my last sending round to manage risk

```
{'send': 4.00}
```

reasoning:

Let me analyze my history:

Round 1 (RECEIVER): Sender sent \$4.78 (95.6%), I received \$14.34, returned \$8.00, kept \$6.34

Round 2 (SENDER): I sent \$4.50 (90%), became \$13.50, received back \$7.41, ended with \$7.91

Round 3 (RECEIVER): Sender sent \$3.80 (76%), I received \$11.40, returned \$6.50, kept \$4.90

Now Round 4, I'm the SENDER again with a new co-player.

Looking at the pattern:

- When I sent \$4.50 (90%), I got back \$7.41 from \$13.50 (about 55% return rate)
- When someone sent \$4.78 (95.6%), I returned \$8.00 from \$14.34 (about 56% return rate)
- When someone sent \$3.80 (76%), I returned \$6.50 from \$11.40 (about 57% return rate)

The return rates seem fairly consistent around 55-57%. This suggests cooperation is being reciprocated.

I've been doing well with high cooperation levels. My total earnings are \$19.15 across 3 rounds, averaging about \$6.38 per round, which is well above the baseline of \$5.

Given this positive experience with cooperation, I should continue sending a substantial amount. I'll send \$4.00 (80%), which is:

- Still showing significant trust
- Creating good value through the 3x multiplier (\$12.00 received)
- Likely to be reciprocated based on the pattern
- Slightly more conservative than my previous \$4.50 to diversify my strategy a bit

If they return around 55%, I'd get back about \$6.60, ending with \$1.00 + \$6.60 = \$7.60, which would be good.

usage:

```
{
  "input_tokens": 1488,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$4.58 to you. Your visible payoff was \$5.58.

Your total visible earnings across all rounds are \$24.729999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.73 (94.6% of their endowment)
- They kept only \$0.27 for themselves

- I received \$14.19
- This is very similar to Round 1, where someone sent 95.6%

In Round 1, when faced with similarly high trust, I returned \$8.00 out of \$14.34. That worked well as a fair reciprocation.

For this round with \$14.19 received, returning \$8.00:

- Sender ends with: $\$0.27 + \$8.00 = \$8.27$ (strong return on their trust)
- I keep: $\$14.19 - \$8.00 = \$6.19$
- Both benefit significantly from the cooperation

Consistency with my previous fair reciprocation of high trust seems appropriate.

```
{'return': 8.00}
```

reasoning:

Let me analyze this situation:

Round 5 - I'm the RECEIVER

- Sender sent \$4.73 (94.6% of their endowment) - very high trust!
- They kept only \$0.27
- I received \$14.19 after the 3x multiplier

This is similar to Round 1 where someone sent \$4.78 (95.6%) and I returned \$8.00.

The sender is showing exceptional trust, almost sending everything. I should reciprocate fairly.

If I return \$8.00:

- Sender ends with: $\$0.27 + \$8.00 = \$8.27$ (good gain from initial \$5)
- I keep: $\$14.19 - \$8.00 = \$6.19$
- Both benefit, sender gets rewarded for high trust

This seems consistent with my Round 1 behavior where I also returned \$8.00 for a similar high-trust send.

Actually, let me think about this more carefully. The sender sent 94.6%. In Round 1, when sender sent 95.6%, I returned \$8.00 out of \$14.34, which is about 56%.

$\$14.19 \times 0.56 \approx \7.95

So around \$8.00 seems right. Let me stick with that.

usage:

```
{
  "input_tokens": 1831,
  "output_tokens": 528,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.73, it became \$14.190000000000001, and you returned \$8.0. Your visible payoff was \$6.190000000000001.

Your total visible earnings across all rounds are \$30.919999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my experience as a sender:

- Round 2: Sent \$4.50 (90%), received \$7.41 back - good cooperation
- Round 4: Sent \$4.00 (80%), received \$4.58 back - lower reciprocation

The pattern suggests that higher trust levels tend to get better reciprocation. I'm averaging around \$6.18 per round with my cooperative strategy, which is working well.

As a receiver, I've been consistent in returning around 56% of what I receive, which has s

seemed fair to my co-players.

I'll send \$4.50 again (90% of endowment):

- Shows strong, consistent trust
- Previously resulted in good reciprocation
- Creates \$13.50 for the receiver
- Maintains my cooperative approach

```
{'send': 4.50}
```

reasoning:

Let me review the pattern of my play:

Round 1 (Receiver): Received \$14.34 from sender who sent \$4.78 (95.6%), returned \$8.00 ? kept \$6.34

Round 2 (Sender): Sent \$4.50 (90%), received back \$7.41 ? total \$7.91

Round 3 (Receiver): Received \$11.40 from sender who sent \$3.80 (76%), returned \$6.50 ? kept \$4.90

Round 4 (Sender): Sent \$4.00 (80%), received back \$4.58 ? total \$5.58

Round 5 (Receiver): Received \$14.19 from sender who sent \$4.73 (94.6%), returned \$8.00 ? kept \$6.19

My total visible earnings: \$30.92

As a sender:

- Round 2: Sent \$4.50, got back \$7.41 (55% return rate)
- Round 4: Sent \$4.00, got back \$4.58 (38% return rate)

As a receiver:

- Round 1: Returned \$8.00 out of \$14.34 (56% return rate) when sender sent 95.6%
- Round 3: Returned \$6.50 out of \$11.40 (57% return rate) when sender sent 76%
- Round 5: Returned \$8.00 out of \$14.19 (56% return rate) when sender sent 94.6%

I've been consistent in my returns - around 56% when I'm the receiver.

When I've been the sender:

- Higher sends seem to get better returns (Round 2 with \$4.50 got 55% back)

...

usage:

```
{
  "input_tokens": 2173,
  "output_tokens": 794,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.5, it became \$13.5, and you saw them return \$5.77 to you. Your visible payoff was \$6.27.

Your total visible earnings across all rounds are \$37.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

I'm playing the SAME co-player as Round 6. Last round with them:

- I sent \$4.50 (90%)
- They returned \$5.77 (about 43% of the \$13.50 they received)
- I ended with \$6.27

Their reciprocation rate (43%) was lower than some other players I've encountered (~55%), but not the worst. My payoff was still above average.

Playing the same person again, I should:

- Continue showing willingness to cooperate
- Adjust slightly for their demonstrated reciprocation pattern

- Signal awareness while maintaining relationship

I'll send \$4.00 (80%):

- Still shows substantial trust
- Creates \$12.00 for them
- Slightly more conservative given their previous behavior
- Keeps the cooperative dynamic going

```
{'send': 4.00}
```

reasoning:

Interesting - I'm playing as SENDER again against the SAME co-player from Round 6. This is important context.

In Round 6:

- I sent \$4.50 (90%)
- It became \$13.50
- They returned \$5.77
- My payoff was \$6.27

The return rate was $\$5.77/\$13.50 = 42.7\%$, which is lower than the ~55% I've been seeing from other players. They kept about \$7.73 while I ended with \$6.27 total.

Now I'm playing them again. This gives me information about their reciprocation behavior. They were less generous than others I've played with.

Options:

1. Send less to protect myself since they showed lower reciprocation
2. Send the same amount to see if they maintain consistency
3. Send more to encourage better reciprocation

Strategic considerations:

- They know I sent high last time, so there's a reputation aspect
- If I send less, it signals I'm adjusting to their lower reciprocation
- If I send the same or more, it shows I'm still willing to cooperate
- The 3x multiplier means cooperation still creates value even with lower return rates

I think I should adjust downward slightly. Sending around \$3.50-\$4.00 makes sense:

- Still shows willingness to cooperate
- Manages risk given their demonstrated lower reciprocation rate
- Creates \$10.50-\$12.00 for them

I'll go with \$3.50 (70%):

- Shows continued cooperation but adjusted trust
- Creates \$10.50 for them
- Keeps \$1.50 for me as baseline
- If they return around 43%, that's \$4.52, giving me \$6.02 total
- If they improve their return rate, I benefit more

Actually, let me reconsider.

Looking at my track record across all sender rounds, I'm averaging around \$6.18 per round, and last round's \$6.27 is actually slightly above that. Comparing their 42.7% return rate against my other interactions - Round 2 yielded 55% and Round 4 only 38% - suggests this player falls in the middle rather than being an outlier. Maybe I'm overweighting one round when the overall pattern shows reasonable consistency....

usage:

```
{
  "input_tokens": 2490,
  "output_tokens": 943,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.87 to you. Your visible payoff was \$9.87.

Your total visible earnings across all rounds are \$47.059999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$3.21 (64% of their endowment) - moderate trust
- They kept \$1.79 for themselves
- I received \$9.63

My established pattern as a receiver has been to return around 56-57% of what I receive:

- Round 1: Returned 56%
- Round 3: Returned 57%
- Round 5: Returned 56%

For consistency and fairness, I'll apply the same principle:

$\$9.63 \times 0.56 = \5.40

If I return \$5.40:

- Sender ends with: $\$1.79 + \$5.40 = \$7.19$ (solid return)
- I keep: $\$9.63 - \$5.40 = \$4.23$
- Both benefit from cooperation

{'return': 5.40}

Reasoning:

Looking at this situation:

- The sender sent \$3.21 (64.2% of their endowment)
- They kept \$1.79 for themselves
- I received \$9.63
- This is moderate trust, lower than some previous rounds

Let me review my pattern of returning:

- Round 1: Received \$14.34, returned \$8.00 (56%)
- Round 3: Received \$11.40, returned \$6.50 (57%)
- Round 5: Received \$14.19, returned \$8.00 (56%)

So I've been consistently returning around 56-57% of what I receive.

For this round with \$9.63 received:

$\$9.63 \times 0.56 = \5.39

If I return \$5.40:

- Sender ends with: $\$1.79 + \$5.40 = \$7.19$ (good gain)
- I keep: $\$9.63 - \$5.40 = \$4.23$
- Both benefit from the cooperation

This maintains my consistent reciprocation pattern.

Usage: {

"input_tokens": 2602,
"output_tokens": 473,
"reasoning_tokens": 0

}

GAME - Agent_7 - Sender - dyad_3

Timestamp: 2026-08-12T09:27:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 4,
"opponent_id": "Agent_5",
"pairing": {
 "agents": [
 "Agent_7",
 "Agent_5"
],
 "investor": "Agent_7",
 "roles": {

```

    "Agent_5": "trustee",
    "Agent_7": "investor"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.51 to you. Your visible payoff was \$7.51.

Your total visible earnings across all rounds are \$7.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$3.8 out of \$5 (76% of their endowment) - showing significant trust
- I received \$11.4 after the 3x multiplier

****My reasoning:****

- In Round 1, when I sent \$3, the receiver returned \$5.51 (about 61% of what they received)
- This sender has shown high trust by sending 76% of their endowment
- I should reciprocate that trust fairly

I'll return an amount that rewards their cooperation while maintaining fairness:

```
'''json
{"return": 7}
'''
```

This gives the sender a total of \$8.2 (\$1.2 kept + \$7 returned) and leaves me with \$4.4. While I keep less, this reciprocates their strong trust and builds cooperation.

reasoning:

Let me analyze this situation:

Round 1 (just completed):

- I was SENDER, sent \$3
- It became \$9 for receiver
- They returned \$5.51 to me
- My payoff: $\$5 - \$3 + \$5.51 = \7.51
- Their payoff: $\$9 - \$5.51 = \$3.49$

So the receiver returned about 61% of what they received. That's quite generous.

Round 2 (current):

- I am RECEIVER
- Sender sent me \$3.8
- I received \$11.4 (tripled amount)
- I need to decide how much to return

Now, there's communication noise, so the amounts might be slightly different from what was actually sent/returned. But I should work with what I see.

The sender sent me \$3.8 out of \$5, which is 76% of their endowment. That's quite trusting.

If I apply reciprocity based on what happened in Round 1:

- The receiver returned about 61% of what they received
- 61% of \$11.4 = \$6.95

But let me think about fairness:

- Sender keeps: $\$5 - \$3.8 = \$1.2$
- Sender receives back from me: X
- Sender total: $\$1.2 + X$

- I keep: $\$11.4 - X$

For equal split of total value:

- Total created: $\$5 + \$11.4 = \$16.4$
- Equal split: \$8.2 each
- So sender should get: \$8.2 total, meaning I return: $\$8.2 - \$1.2 = \$7$

For proportional reciprocity (they sent 76% of endowment):

- Maybe I return a similar proportion of what I received?
- 76% of \$11.4 = \$8.66

Given their high level of trust, I should match that cooperation. Returning around 55-60% of what I received feels right-it's reciprocal without being excessive. I'll return \$6.50, which gives the sender a total of \$7.70 and leaves me with \$4.90. This mirrors the cooperative spirit they showed....

usage:

```
{
  "input_tokens": 780,
  "output_tokens": 878,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.8, it became \$11.399999999999999, and you returned \$7.0. Your visible payoff was \$4.399999999999999.

Your total visible earnings across all rounds are \$11.909999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at this new round with a new co-player:

****My experience so far:****

- Round 1: Sent \$3, received \$5.51 back - good cooperation
- Round 2: Received \$11.4, returned \$7 - fair reciprocation

Since pairings are randomized, this is a new partner with no shared history. However, the cooperative pattern I've experienced suggests trust can be rewarded.

I'll send a moderate-to-high amount to signal cooperation while managing risk:

```
'''json
{"send": 3.5}
'''
```

This sends \$3.5 (70% of my endowment), which becomes \$10.50 for the receiver. It's slightly more than my Round 1 send, reflecting confidence from positive past experiences while remaining balanced.

reasoning:

Let me analyze the situation:

****My history:****

- Round 1: I sent \$3, received back \$5.51, payoff = \$7.51
- Round 2: I received \$11.4 (they sent \$3.8), I returned \$7, payoff = \$4.4
- Total earnings: \$11.91

****Analysis:****

- In Round 1, my co-player was generous, returning \$5.51 out of \$9 (61%)
- In Round 2, my co-player sent \$3.8 (76% of endowment), showing high trust, and I reciprocated fairly
- Now I'm facing a new co-player in Round 3

Since this is a multi-agent setting with randomized pairings, this Round 3 co-player is likely a different person than either of my previous partners. I don't have direct history with them.

However, I've seen cooperation pay off:

- When I sent \$3, I got back \$5.51 (net gain of \$2.51)
- When someone sent me \$3.8, I returned fairly

Given the overall cooperative pattern I've observed and participated in, I should continue with a cooperative strategy. I'll send a similar amount to what I did in Round 1, or potentially slightly more given the positive experience.

I'll send \$3 again - it's a balanced approach that:

1. Shows trust and willingness to cooperate
2. Doesn't risk everything
3. Creates opportunity for mutual gain (becomes \$9)
4. Has worked well for me in the past

Actually, given that cooperation has been working well, I could send a bit more. Let me se

nd \$3.5, showing slightly increased trust while still being prudent.

usage:

```
{
  "input_tokens": 1109,
  "output_tokens": 583,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.82 to you. Your visible payoff was \$6.32.

Your total visible earnings across all rounds are \$18.229999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$3.11 out of \$5 (62.2% of their endowment) - showing moderate trust
- I received \$9.33 after the 3x multiplier

****My analysis:****

- In Round 3, when I sent \$3.5, I received back \$4.82 (only 46% of what they received) - less cooperative
- In Round 2, when I received \$11.4, I returned \$7 (61%)
- This sender showed moderate trust, deserving fair reciprocation

I'll return a fair amount consistent with their trust level:

```
'''json
{"return": 5.5}
'''
```

This gives the sender a total of \$7.39 (\$1.89 kept + \$5.50 returned) and leaves me with \$3.83. It's a balanced return that rewards their trust while being fair to both of us.

reasoning:

Let me analyze this situation:

****Round 4:****

- I'm the RECEIVER
- They sent \$3.11 out of \$5 (62.2% of their endowment)
- I received \$9.33 after the 3x multiplier

****Past experience:****

- Round 1: I sent \$3, got back \$5.51 (they returned 61% of \$9)
- Round 2: I received \$11.4, returned \$7 (61% of what I received)
- Round 3: I sent \$3.5, got back \$4.82 (they returned 46% of \$10.5)

The sender this round sent me \$3.11, showing moderate trust (62.2% of their endowment).

In Round 3, when I sent \$3.5, I only got back \$4.82, which was less than half of what they received. That was somewhat disappointing.

For this round, the sender sent \$3.11. I should be fair but also consider that:

1. They showed moderate trust (62.2%)
2. My past returns have been around 61% when receiving
3. I want to maintain cooperation

If I return around 55-60% of \$9.33, that would be:

- 55% = \$5.13
- 60% = \$5.60

This would give them:

- At \$5.13: \$1.89 kept + \$5.13 = \$7.02 total

- At \$5.60: \$1.89 kept + \$5.60 = \$7.49 total

And I would keep:

- At \$5.13: \$4.20
- At \$5.60: \$3.73

I'll return around \$5.5 to be fair and reciprocate their moderate trust.

usage:

```
{
  "input_tokens": 1428,
  "output_tokens": 648,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$5.5. Your visible payoff was \$3.83.

Your total visible earnings across all rounds are \$22.059999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.26, so you received \$6.779999999999999. How much do you return? (0-6.779999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$2.26 out of \$5 (45.2% of their endowment) - showing cautious/modest trust
- I received \$6.78 after the 3x multiplier

****My analysis:****

- This sender showed less trust than previous partners (45% vs 62-76%)
- However, I should still reciprocate fairly to maintain cooperation
- My previous return rates have been around 59-61%

Given their more cautious approach, I'll return a fair but proportionally adjusted amount:

```
'''json
{"return": 3.5}
'''
```

This gives the sender a total of \$6.24 (\$2.74 kept + \$3.50 returned), a gain of \$1.24, and leaves me with \$3.28. It's a balanced return that rewards their trust while acknowledging their more conservative send.

reasoning:

Looking at this situation:

****What happened:****

- The sender sent me \$2.26 out of \$5 (45.2% of their endowment) - showing modest/cautious trust
- I received \$6.78 after the 3x multiplier

****My historical pattern:****

- Round 2: Received \$11.4, returned \$7 (61% of received)
- Round 4: Received \$9.33, returned \$5.5 (59% of received)

This sender showed less trust than previous senders (only 45% vs 62-76%). I should still reciprocate fairly, but proportionally to their lower trust level.

A return rate around 50-55% seems appropriate given their more cautious approach:

- 50% would be \$3.39
- 55% would be \$3.73
- 60% would be \$4.07

Let me think about what's fair:

- They sent \$2.26, keeping \$2.74

- If I return \$3.5, they'd have \$5.76 total (a gain of \$0.76)
- I'd keep \$3.28

Actually, let me reconsider. Even though they sent less, I should maintain a cooperative stance. Let me return around 50-55%:

If I return \$3.5:

- Sender gets: \$2.74 (kept) + \$3.50 (returned) = \$6.24 total
- I keep: \$6.78 - \$3.50 = \$3.28

This seems balanced - they gain \$1.24, I gain \$3.28. It's fair while still rewarding their trust, even if it was more cautious.

Let me go with \$3.5 (about 52% of what I received).

usage:

```
{
  "input_tokens": 1789,
  "output_tokens": 630,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.26, it became \$6.779999999999999, and you returned \$3.5. Your visible payoff was \$3.2799999999999994.

Your total visible earnings across all rounds are \$25.339999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at this new round with a new co-player:

****My experience so far as sender:****

- Round 1: Sent \$3 (60%), received \$5.51 back - good return
- Round 3: Sent \$3.5 (70%), received \$4.82 back - lower return rate

****My observations:****

- Sending more didn't necessarily yield better outcomes
- As a receiver, I've been fairly cooperative
- This is a new partner, so I have no history with them

I'll send a moderate amount that signals cooperation while managing risk:

```
'''json
{"send": 3}
'''
```

This sends \$3 (60% of endowment), which becomes \$9 for the receiver. It's a balanced approach that worked well in Round 1 and allows me to gauge this new partner's reciprocity.

reasoning:

Looking at my history:

Round 1 (Sender): Sent \$3, received back \$5.51, payoff \$7.51
 Round 2 (Receiver): Received \$11.4 (from \$3.8 send), returned \$7, payoff \$4.4
 Round 3 (Sender): Sent \$3.5, received back \$4.82, payoff \$6.32
 Round 4 (Receiver): Received \$9.33 (from \$3.11 send), returned \$5.5, payoff \$3.83
 Round 5 (Receiver): Received \$6.78 (from \$2.26 send), returned \$3.5, payoff \$3.28

Now I'm a sender again in Round 6 with a new co-player.

My experience as sender:

- Round 1: Sent \$3 (60%), got back \$5.51 (61% of what they received)
- Round 3: Sent \$3.5 (70%), got back \$4.82 (46% of what they received)

When I send more, I don't necessarily get better returns. The Round 3 experience was less favorable.

My experience as receiver shows I've been fairly cooperative:
- Returned 61%, 59%, and 52% of what I received

Since this is a new partner and there's communication noise, I should:
1. Signal cooperation but not overcommit
2. Learn about this specific partner's behavior

I'll send a moderate amount around \$3 to balance risk and cooperation potential.

usage:

```
{  
  "input_tokens": 2129,  
  "output_tokens": 563,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.44 to you. Your visible payoff was \$6.44.

Your total visible earnings across all rounds are \$31.779999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.47, so you received \$13.41. How much do you return? (0-13.41)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$4.47 out of \$5 (89.4% of their endowment) - showing exceptional trust
- I received \$13.41 after the 3x multiplier

****My analysis:****

- This is the highest trust signal I've received in any round
- Previous partners sent 45-76%, but this sender sent nearly 90%
- This exceptional cooperation deserves strong reciprocation

I'll return a generous amount to reward their high trust:

```
'''json  
{ "return": 8 }  
'''
```

This gives the sender a total of \$8.53 (\$0.53 kept + \$8.00 returned), a gain of \$3.53, and leaves me with \$5.41. This 60% return rate fairly rewards their exceptional trust and encourages continued cooperation.

reasoning:

Looking at this situation:

****What happened:****

- The sender sent me \$4.47 out of \$5 (89.4% of their endowment) - showing very high trust
- I received \$13.41 after the 3x multiplier

****My analysis:****

- This is the highest trust signal I've received so far (89.4%)
- Previous sends to me: \$3.8 (76%), \$3.11 (62%), \$2.26 (45%)
- This sender is showing exceptional trust

****My return history as receiver:****

- Round 2: Received \$11.4, returned \$7 (61%)
- Round 4: Received \$9.33, returned \$5.5 (59%)
- Round 5: Received \$6.78, returned \$3.5 (52%)

Given this sender's exceptional trust (nearly 90% of endowment), I should reciprocate strongly to reward this cooperation.

If I return around 60% like my previous high-trust returns, that would be about \$8.05.

But given they sent 89% of their endowment, maybe I should be even more generous. Let me think about what gives them a good outcome:

- They kept: \$0.53
- If I return \$8, they get \$8.53 total
- If I return \$9, they get \$9.53 total

Returning \$8 would give them \$8.53 total (a gain of \$3.53) and leave me with \$5.41.
Returning \$9 would give them \$9.53 total (a gain of \$4.53) and leave me with \$4.41.

I think \$8 is a strong reciprocal move that rewards their exceptional trust. It's about 60% of what I received, consistent with my fair reciprocation pattern.

usage:

```
{
  "input_tokens": 2459,
  "output_tokens": 627,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$4.47, it became \$13.41, and you returned \$8.0. Your visible payoff was \$5.41.

Your total visible earnings across all rounds are \$37.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

****Important context:****

- This is the SAME co-player from Round 7!
- In Round 7, they sent me \$4.47 (89.4%) - exceptional trust
- I returned \$8.00, and they gained \$3.53

****My reasoning:****

- They demonstrated exceptional trust in our previous interaction
- I reciprocated fairly, and we both benefited
- This is an opportunity to continue our successful cooperation
- I should match their trust level to sustain this relationship

```
'''json
{"send": 4.5}
'''
```

This sends \$4.5 (90% of my endowment), which becomes \$13.50 for them. It matches their exceptional trust from Round 7 and signals I value our cooperative partnership.

Reasoning:

This is interesting - I'm playing the same co-player from Round 7!

In Round 7:

- They sent me \$4.47 (89.4% of their endowment) - exceptional trust
- I received \$13.41
- I returned \$8.00 (about 60% of what I received)
- They ended up with \$8.53 total (gained \$3.53)
- I ended up with \$5.41

Now in Round 8, I'm the sender and they're the receiver. They showed exceptional trust in Round 7, and I returned fairly. This establishes a strong cooperative relationship.

Given:

1. They showed exceptional trust (89.4% send rate)
2. I reciprocated fairly (60% return rate)
3. Both of us profited from the interaction
4. We're now playing again immediately

This is a great opportunity to continue this cooperative relationship. I should send a high amount to:

- Reciprocate their high trust from Round 7
- Signal that I recognize and value our cooperative pattern
- Potentially establish a mutually beneficial repeated interaction

I should send at least as much as they did (around \$4.47) or potentially even more to show strong reciprocity.

Let me send \$4.5 to match their trust level and continue our cooperation.

```
Usage: {
  "input_tokens": 2607,
  "output_tokens": 484,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T09:27:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_5"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_5": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.3, it became \$6.899999999999995, and you returned \$3.0. Your visible payoff was \$3.899999999999995.

Your total visible earnings across all rounds are \$3.899999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- I'm now the sender with a new co-player
- In the previous round, I experienced a sender who sent 46% and I returned fairly
- I want to establish trust with this new partner while being reasonably cautious

Sending \$2.5 (50% of my endowment) seems like a good strategy:

- It shows significant trust
- It's a clear, fair starting point
- If my co-player reciprocates fairly, we both benefit from the 3x multiplier
- It's not overly risky with an unknown partner

```
'''json
{"send": 2.5}
'''
```

reasoning:

Now I'm the sender in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.3 (46% of their endowment)
- I received \$6.9
- I returned \$3.0
- My payoff was \$3.9

Now I'm playing against a different agent (since it says "current co-player" and agents are randomly paired each round).

I need to decide how much to send. Key considerations:

1. I don't know anything about this new co-player's behavior
2. The sender in Round 1 sent 46% (\$2.3), which was a reasonable trust signal
3. I returned about 43.5% of what I received (\$3/\$6.9), which was fair
4. I want to build trust but also protect myself since I don't know this opponent

Strategies:

- Sending \$0 would be overly defensive and wouldn't build any relationship
- Sending \$5 (100%) would be too risky with an unknown partner
- Sending around \$2-3 seems reasonable as it shows trust without being reckless

Given that the previous sender sent \$2.3 (46%), I could mirror that behavior. This would be consistent and shows I'm willing to trust at a similar level to what I've experienced.

Alternatively, I could send slightly more to show goodwill, or slightly less to be more cautious.

I think sending \$2.5 (50%) is a good middle ground:

- It's a clear half of my endowment, easy to interpret
- It shows significant trust
- If they return fairly (around 40-45% of \$7.5 = \$3-3.4), we both profit
- It's not too risky

Let me calculate the outcomes:

- If I send \$2.5, they receive \$7.5
- I keep \$2.5

...

usage:

```
{
  "input_tokens": 870,
  "output_tokens": 665,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$2.5, it became \$7.5, and you saw them return \$6.78 to you. Your visible payoff was \$9.280000000000001.

Your total visible earnings across all rounds are \$13.18.

This round, you are the RECEIVER against your current co-player. They sent you \$4.17, so you received \$12.51. How much do you return? (0-12.51)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.17 (83.4% of their endowment)
- I received \$12.51 after the 3x multiplier
- The sender currently has only \$0.83 remaining

This is very high trust that deserves to be rewarded generously.

If I return \$6.0:

- Sender ends with: \$0.83 + \$6.0 = \$6.83 (profit of \$1.83)
- I end with: \$12.51 - \$6.0 = \$6.51 (profit of \$6.51)

This gives us nearly equal final payoffs and rewards the sender's exceptional trust with a solid profit.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- The sender sent \$4.17 (83.4% of their \$5 endowment) - this is very high trust!
- I received \$12.51 (after 3x multiplier)
- The sender currently has \$5 - \$4.17 = \$0.83 remaining

This sender showed enormous trust by sending 83.4% of their endowment. I need to reward this trust generously.

Let me think about what's fair:

- Total in the system: \$0.83 (sender has) + \$12.51 (I have) = \$13.34
- If we split evenly: \$6.67 each

If I return \$5.84:

- Sender ends with: \$0.83 + \$5.84 = \$6.67
- I end with: \$12.51 - \$5.84 = \$6.67
- Perfect 50/50 split

But given they showed such high trust (83.4%), I might want to be even more generous to:

1. Reward their exceptional trust
2. Encourage continued cooperation
3. Establish myself as highly trustworthy

However, I also saw in round 2 that when I sent \$2.5 (50%), I got back \$6.78, which gave me a payoff of \$9.28. That was very generous from my co-player then (they returned more than 90% of what they received!).

Let me recalculate for this round:

- If I return \$6:
 - Sender: \$0.83 + \$6 = \$6.83
 - Me: \$12.51 - \$6 = \$6.51
 - Almost equal, slight advantage to sender
- If I return \$6.5:
 - Sender: \$0.83 + \$6.5 = \$7.33
 - Me: \$12.51 - \$6.5 = \$6.01
 - Sender gets more, rewarding their trust

Given the extreme trust shown here, I should return enough to give them a meaningful advantage while still being reasonable. Returning around \$6 seems appropriate-it creates nearly equal outcomes with a slight edge to the sender who took the bigger risk....

usage:

```
{
  "input_tokens": 1154,
  "output_tokens": 955,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.17, it became \$12.51, and you returned \$6.0. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$19.689999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

- Round 2: When I sent \$2.5 (50%), I received a very generous return of \$6.78
- Round 3: When someone sent me \$4.17 (high trust), I reciprocated with \$6.0

The game seems to reward trust. I'll send \$3.0 (60% of my endowment), which shows strong trust while being measured with a new co-player.

```
'''json
{"send": 3.0}
'''
```

reasoning:

Looking at my history:

- Round 1: RECEIVER - received from sender who sent \$2.3, I returned \$3.0 fairly
- Round 2: SENDER - I sent \$2.5, receiver returned \$6.78 (very generous!)
- Round 3: RECEIVER - received from sender who sent \$4.17 (very high trust), I returned \$6.0

Current situation:

- I'm the SENDER with a new co-player
- Total earnings so far: \$19.69

My experience shows:

- When I sent \$2.5 in Round 2, I got a very generous return of \$6.78
- When others showed high trust to me, I reciprocated fairly
- This game seems to reward trust

I should send a significant amount to show trust. Given that:

- Round 2 sender sent 46% and I reciprocated
- Round 3 sender sent 83.4% and I reciprocated generously
- When I sent 50% in Round 2, I got excellent returns

I think sending around \$3.0 (60%) would be a good balance - showing strong trust without being reckless with an unknown partner.

Actually, let me think about this more strategically. I've seen that:

- Lower trust (46%) got fair returns
- Medium trust (50% from me) got very generous returns
- High trust (83.4%) got generous returns

Sending \$3.0 seems like a good middle-high trust level that should encourage reciprocation

usage:

```
{
  "input_tokens": 1461,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.22 to you. Your visible payoff was \$7.22.

Your total visible earnings across all rounds are \$26.909999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.84, so you received \$11.52. How much do you return? (0-11.52)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender showed high trust by sending \$3.84 (76.8% of their endowment)
- I received \$11.52 after the 3x multiplier
- The sender currently has only \$1.16 remaining

This level of trust deserves a generous return.

If I return \$6.0:

- Sender ends with: $\$1.16 + \$6.0 = \$7.16$ (profit of \$2.16)
- I end with: $\$11.52 - \$6.0 = \$5.52$

This rewards their high trust with a good profit and keeps our final payoffs relatively balanced.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$3.84, which is 76.8% of their endowment - very high trust
- I received \$11.52 after the 3x multiplier
- The sender currently has $\$5 - \$3.84 = \$1.16$ remaining

This is high trust that should be rewarded well.

Let me consider different return amounts:

If I return \$5.5:

- Sender ends with: $\$1.16 + \$5.5 = \$6.66$ (profit of \$1.66)
- I end with: $\$11.52 - \$5.5 = \$6.02$

If I return \$6.0:

- Sender ends with: $\$1.16 + \$6.0 = \$7.16$ (profit of \$2.16)
- I end with: $\$11.52 - \$6.0 = \$5.52$

Given the high trust (76.8%), I should return generously. \$6.0 seems fair and rewards their trust well.

usage:

```
{
  "input_tokens": 1718,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.84, it became \$11.52, and you returned \$6.0. Your visible payoff was \$5.52.

Your total visible earnings across all rounds are \$32.42999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my experience:

- Round 2: Sent \$2.5 (50%), received \$6.78 back - great return
- Round 4: Sent \$3.0 (60%), received \$5.22 back - solid return

The game has been rewarding trust. I'll continue to show strong trust with this new co-player by sending \$3.5 (70% of my endowment).

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze my experience:

Round 1 (Receiver): Received \$6.9 from sender sending \$2.3, returned \$3.0. Payoff: \$3.9
Round 2 (Sender): Sent \$2.5, received \$6.78 back. Payoff: \$9.28
Round 3 (Receiver): Received \$12.51 from sender sending \$4.17, returned \$6.0. Payoff: \$6.51
Round 4 (Sender): Sent \$3.0, received \$5.22 back. Payoff: \$7.22
Round 5 (Receiver): Received \$11.52 from sender sending \$3.84, returned \$6.0. Payoff: \$5.52

So far:

- As a sender, I've sent \$2.5 and \$3.0 (50% and 60%)
- Both times I received generous returns (\$6.78 and \$5.22)
- As a receiver, I've been fair/generous in my returns

The game continues to reward trust. I've been gradually increasing my sending amounts. Let me continue this trend and send \$3.5 (70% of endowment) to show strong trust with this new co-player.

usage:

```
{
  "input_tokens": 2009,
  "output_tokens": 417,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.22 to you. Your visible payoff was \$7.72.

Your total visible earnings across all rounds are \$40.14999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my sender history:

- Round 2: Sent \$2.5, received \$6.78 back
- Round 4: Sent \$3.0, received \$5.22 back
- Round 6: Sent \$3.5, received \$6.22 back

Trust has been consistently rewarded. I'll continue showing strong trust with this new co-player by sending \$4.0 (80% of my endowment).

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history as sender:

- Round 2: Sent \$2.5 (50%), received \$6.78 back - excellent return
- Round 4: Sent \$3.0 (60%), received \$5.22 back - good return
- Round 6: Sent \$3.5 (70%), received \$6.22 back - good return

All three times I've sent money as sender, I've received generous returns. The game environment seems cooperative and trust is being rewarded.

As receiver:

- Round 1: Received \$6.9, returned \$3.0
- Round 3: Received \$12.51, returned \$6.0
- Round 5: Received \$11.52, returned \$6.0

I've been fair as a receiver too.

The pattern suggests the game is cooperative. I could continue increasing my trust level. Maybe send \$4.0 (80%) this time? That would show strong trust with this new co-player.

Let me think about the strategy:

- Sending more has generally worked well
- I'm playing against a new co-player each round
- Building reputation for cooperation could help
- Total earnings are healthy at \$40.15

I'll send \$4.0 (80% of endowment) to show strong trust.

usage:

```
{
  "input_tokens": 2246,
  "output_tokens": 423,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$8.61 to you. Your visible payoff was \$9.61.

Your total visible earnings across all rounds are \$49.75999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.61, so you received \$13.830000000000002. How much do you return? (0-13.830000000000002)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- This is the same co-player from Round 7!
- Last round, I sent them \$4.0 and they returned an exceptional \$8.61
- Now they've sent me \$4.61 (92.2% of their endowment) - extremely high trust
- I received \$13.83 after the 3x multiplier
- They currently have only \$0.39 remaining

My co-player has shown exceptional generosity and trust. I should reciprocate strongly.

If I return \$8.0:

- Sender ends with: $\$0.39 + \$8.0 = \$8.39$ (profit of \$3.39)
- I end with: $\$13.83 - \$8.0 = \$5.83$

This matches their previous generosity and rewards their high trust appropriately.

```
'''json
{"return": 8.0}
'''
```

Reasoning:

Let me analyze this situation:

- This is the same co-player from Round 7
- In Round 7: I sent \$4.0, they returned \$8.61 - that's an exceptional return! They were very generous.
- Now in Round 8: They sent me \$4.61 (92.2% of their endowment) - showing extremely high trust
- I received \$13.83 after the 3x multiplier
- They currently have only \$0.39 remaining

My co-player has been very generous to me. In Round 7, they returned \$8.61 out of \$12.0 (71.75 % return rate). That's excellent reciprocity.

Now they're showing even MORE trust by sending \$4.61. I should definitely reciprocate generously.

If I return \$7.0:

- Sender ends with: $\$0.39 + \$7.0 = \$7.39$ (profit of \$2.39)
- I end with: $\$13.83 - \$7.0 = \$6.83$

If I return \$8.0:

- Sender ends with: $\$0.39 + \$8.0 = \$8.39$ (profit of \$3.39)
- I end with: $\$13.83 - \$8.0 = \$5.83$

Given they returned \$8.61 last round and now sent me \$4.61, I should be similarly generous. Returning around \$7-8 seems appropriate.

Let me go with \$7.5 to be quite generous while still benefiting from the exchange:

- Sender ends with: $\$0.39 + \$7.5 = \$7.89$ (profit of \$2.89)
- I end with: $\$13.83 - \$7.5 = \$6.33$

Actually, given they returned \$8.61 last round (an exceptional amount), I should probably return at least \$8.0 to match their generosity level.

...

```
Usage: {
  "input_tokens": 2249,
  "output_tokens": 745,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Sender - dyad_4

Timestamp: 2026-08-12T09:27:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_6"
    ]
  }
}
```

```

],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_6": "trustee"
},
"trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.77, it became \$11.31, and you returned \$5.5. Your visible payoff was \$5.810000000000005.

Your total visible earnings across all rounds are \$5.810000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the previous round, I saw cooperation work well when the sender showed significant trust by sending 75.4% of their endowment, and I reciprocated fairly.

I'll adopt a similar cooperative strategy with my new co-player. Sending a substantial amount can maximize joint gains if they reciprocate.

{'send': 3.75}

reasoning:

I'm now the SENDER in Round 2.

Let me think about this strategically:

1. In the previous round as RECEIVER, I saw someone send \$3.77 (75.4% of their endowment), showing significant trust
2. I returned \$5.50, which gave them \$6.73 total - a good return on their trust

Now I'm the SENDER. What should I consider?

- The person sending \$3.77 showed that cooperation can work well
- I reciprocated fairly, returning enough that they profited significantly
- If my current co-player is similarly cooperative, sending a substantial amount could benefit both of us

Strategy considerations:

- Sending nothing is safe but generates no surplus
- Sending a moderate amount (around \$2-3) tests the waters
- Sending a larger amount (like \$3.77 that I saw) shows trust and can maximize joint gains if reciprocated

Given that:

- I saw cooperation work in the last round
- The game seems to reward trust
- I benefited from someone trusting me and reciprocated fairly
- There may be communication noise (mentioned in instructions), but the pattern of cooperation seemed clear

I think I should send a substantial amount to show trust and potentially establish a cooperative pattern. I'll send around \$3.50-\$4.00.

Let me send \$3.50 (70% of endowment) - this shows significant trust similar to what I received, but slightly less aggressive to be somewhat cautious given this is a new pairing.

Actually, let me reconsider. The previous sender sent 75.4% (\$3.77). I reciprocated well.

If I want to encourage cooperation, I should send a similar proportion. Let me send \$3.75 (75% of my endowment) to match that level of trust.

usage:

```
{
  "input_tokens": 874,
  "output_tokens": 494,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$5.66 to you. Your visible payoff was \$6.91.

Your total visible earnings across all rounds are \$12.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my recent experience:

- Round 2: I sent \$3.75 (75%), received \$5.66 back, ended with \$6.91 - a good outcome
- Round 1: As receiver, cooperation was rewarded

The cooperative approach has been working well. I'm playing with a new co-player now, so I'll continue signaling trust with a substantial send.

```
{'send': 3.75}
```

reasoning:

Let me analyze the previous round:

- I sent \$3.75 (75% of my endowment)
- It became \$11.25
- They returned \$5.66
- My payoff was $\$5 - \$3.75 + \$5.66 = \6.91

This worked out well! The receiver reciprocated reasonably (returning about 50% of what they received), and I gained \$1.91 above my initial endowment.

This is a new co-player this round. Based on my successful cooperative strategy so far:

- Round 1 as receiver: sender sent 75.4%, I returned fairly
- Round 2 as sender: I sent 75%, receiver returned fairly

The pattern of cooperation seems to be working. I should continue with a similar high-trust strategy. I'll send around 75% again to signal cooperation and trust.

\$3.75 worked well last time. I could send the same amount or similar.

usage:

```
{
  "input_tokens": 1074,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$6.34 to you. Your visible payoff was \$7.59.

Your total visible earnings across all rounds are \$20.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$4.09, so you received \$12.27. How much do you return? (0-12.27)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.09 (81.8% of their endowment)
- They currently have: $\$5 - \$4.09 = \$0.91$
- I received: \$12.27
- Total created: \$13.18

This is even more trust than I saw in Round 1 (when sender sent 75.4%). I want to reward this high level of cooperation appropriately.

If I return \$6.50:

- Sender ends with: $\$0.91 + \$6.50 = \$7.41$ (gain of \$2.41)
- I end with: $\$12.27 - \$6.50 = \$5.77$ (gain of \$0.77)

Both of us benefit, and the sender's exceptional trust is well-rewarded.

```
{'return': 6.50}
```

reasoning:

Let me analyze this situation:

- The sender sent \$4.09 out of \$5, which is 81.8% of their endowment - very high trust!
- They now have: $\$5 - \$4.09 = \$0.91$
- I received: \$12.27
- Total money in the system: $\$0.91 + \$12.27 = \$13.18$

This sender showed even more trust than my previous partners. I want to reciprocate appropriately.

Looking at my previous behavior:

- Round 1: As receiver, got \$11.31, returned \$5.50 (about 48.6%)
- The sender had sent 75.4%
- That seemed fair and both parties benefited

This sender sent 81.8%, even more generous. I should reciprocate with a good return.

If I return \$6.50:

- Sender ends with: $\$0.91 + \$6.50 = \$7.41$
- I end with: $\$12.27 - \$6.50 = \$5.77$
- Sender gains \$2.41 over initial \$5
- I gain \$0.77 over initial \$5

Both benefit, and the sender's higher risk/trust is rewarded with higher gains.

Actually, let me think about this differently. The sender sent 81.8% which is very high. In Round 1, sender sent 75.4% and I returned 48.6% of what I received.

For 81.8% send rate, maybe I should return around 50-53% to reward the extra trust?

$\$12.27 * 0.52 = \6.38

If I return \$6.40:

- Sender ends with: $\$0.91 + \$6.40 = \$7.31$
- I end with: $\$12.27 - \$6.40 = \$5.87$

This seems fair - both gain, sender gets rewarded for high trust.

...

usage:

```
{
  "input_tokens": 1310,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.09, it became \$12.27, and you returned \$6.5. Your visible payoff was \$5.77.

Your total visible earnings across all rounds are \$26.080000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my track record:

- As sender: Sent \$3.75 twice, received good returns (\$5.66 and \$6.34)
- As receiver: Responded fairly to cooperative senders

The high-trust strategy continues to work well across different co-players. I'll maintain this approach.

{'send': 3.75}

reasoning:

Let me review my history:

- Round 1: Receiver - sender sent \$3.77, I returned \$5.50, payoff \$5.81
- Round 2: Sender - I sent \$3.75, got back \$5.66, payoff \$6.91
- Round 3: Sender - I sent \$3.75, got back \$6.34, payoff \$7.59
- Round 4: Receiver - sender sent \$4.09, I returned \$6.50, payoff \$5.77

Total visible earnings: \$26.08

My sending strategy has been working well - sending around \$3.75 (75%) has led to good returns. In Round 4, I saw an even higher send amount (\$4.09, 81.8%) and I rewarded it well

Now I'm facing a new co-player as sender. The cooperative strategy has been consistently working across different opponents. I should continue with a similar high-trust signal.

I'll send \$3.75 again, which has proven successful.

usage:

```
{
  "input_tokens": 1637,
  "output_tokens": 324,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$6.54 to you. Your visible payoff was \$7.79.

Your total visible earnings across all rounds are \$33.870000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$4.1, so you received \$12.299999999999999. How much do you return? (0-12.299999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.10 (82% of their endowment)
- They currently have: \$5 - \$4.10 = \$0.90
- I received: \$12.30
- Total created: \$13.20

This is very similar to Round 4 where the sender sent \$4.09 and I returned \$6.50, which worked well for both parties.

If I return \$6.50:

- Sender ends with: \$0.90 + \$6.50 = \$7.40
- I end with: \$12.30 - \$6.50 = \$5.80

Both of us benefit substantially, rewarding the sender's high trust.

{'return': 6.50}

reasoning:

The sender sent \$4.10, which is 82% of their endowment - very high trust again!

They currently have: \$5 - \$4.10 = \$0.90

I received: \$12.30

Total created: \$13.20

This is similar to Round 4 where sender sent \$4.09. I returned \$6.50 then.

Let me be consistent and reward this high trust appropriately.

If I return \$6.50:

- Sender ends with: \$0.90 + \$6.50 = \$7.40
- I end with: \$12.30 - \$6.50 = \$5.80

Both benefit, sender gets a good reward for their trust.

usage:

```
{
  "input_tokens": 1863,
  "output_tokens": 354,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-pla

yer. You saw them send \$4.1, it became \$12.299999999999999, and you returned \$6.5. Your visible payoff was \$5.799999999999999.
Your total visible earnings across all rounds are \$39.67.

This round, you are the RECEIVER against your current co-player. They sent you \$5.0, so you received \$15.0. How much do you return? (0-15.0)
Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- The sender showed **complete trust** by sending \$5.00 (100% of their endowment!)
- They currently have: \$0
- I received: \$15.00
- This is the same co-player from Round 6, and they escalated their trust after I returned fairly

This maximum trust deserves a strong reciprocal response.

If I return \$8.00:

- Sender ends with: \$8.00 (gain of \$3.00)
- I end with: \$7.00 (gain of \$2.00)

Both of us benefit substantially, and their complete trust is well-rewarded.

{'return': 8.00}

reasoning:

This is interesting - I'm playing with the same co-player from Round 6. In Round 6:

- They sent \$4.10
- I received \$12.30
- I returned \$6.50
- They ended with $\$0.90 + \$6.50 = \$7.40$

Now in Round 7:

- They sent \$5.00 (100% of their endowment!) - maximum trust
- I received \$15.00
- They currently have: \$0

This is the ultimate show of trust - they sent everything! This deserves a strong reciprocal response.

If I return \$8.00:

- Sender ends with: $\$0 + \$8.00 = \$8.00$ (gain of \$3.00)
- I end with: $\$15.00 - \$8.00 = \$7.00$ (gain of \$2.00)

Both benefit nicely, and I'm rewarding their complete trust.

Alternatively, if I return \$7.50:

- Sender ends with: \$7.50 (gain of \$2.50)
- I end with: \$7.50 (gain of \$2.50)
- Equal split of the gains

Or if I return \$8.50:

- Sender ends with: \$8.50 (gain of \$3.50)
- I end with: \$6.50 (gain of \$1.50)
- More reward for their maximum trust

Given they showed complete trust by sending 100%, I think returning \$8.00 is appropriate - it gives them a strong reward while we both benefit substantially.

usage:

```
{
  "input_tokens": 2186,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You

saw them send \$5.0, it became \$15.0, and you returned \$8.0. Your visible payoff was \$7.0
Your total visible earnings across all rounds are \$46.67.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience:

- As sender: \$3.75 has consistently yielded good returns (\$5.66, \$6.34, \$6.54)
- As receiver: I've rewarded trust fairly, including returning \$8.00 when someone sent \$5.00

The cooperative approach continues to work well. I'll maintain my high-trust strategy with this new co-player.

{ 'send': 3.75 }

Reasoning:

Looking at my history:

- As sender: I've sent \$3.75 multiple times and received good returns (\$5.66, \$6.34, \$6.54)
- As receiver: I've reciprocated fairly, especially rewarding high trust
- In rounds 6-7, I just played with someone who escalated from \$4.10 to \$5.00, and I reciprocated well

Now I'm the sender with a new co-player. The cooperative strategy has been working very well.

I could:

1. Continue with \$3.75 (75%) - my standard high-trust amount
2. Send \$5.00 (100%) - maximum trust, like the last sender showed me
3. Send something in between like \$4.00-\$4.50

Given that:

- Cooperation has been consistent across the game
- I've built good earnings (\$46.67 total)
- Maximum trust was just shown to me and I rewarded it well
- This might encourage my new co-player to reciprocate strongly

I think I'll stick with my proven \$3.75 strategy. It's high trust without being maximum risk, and it's worked consistently well.

Usage: {

"input_tokens": 2190,
"output_tokens": 371,
"reasoning_tokens": 0

}

GAME - Agent_6 - Receiver - dyad_4

Timestamp: 2026-08-12T09:27:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_6"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 2

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.25, it became \$6.75, and you returned \$3.0. Your visible payoff was \$3.75.

Your total visible earnings across all rounds are \$3.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

In the previous round as receiver, I experienced moderate trust (45% sent) and reciprocated fairly. Now I'm facing a new co-player as sender.

Since this is a new pairing and I want to establish cooperative behavior while managing risk, I'll send a moderate-to-good amount. Sending \$3.00 (60% of my endowment) signals trust and willingness to cooperate, while still keeping some safety margin.

This will give my co-player \$9.00 to work with, providing them good opportunity to reciprocate if they're inclined to cooperate.

```
{"send": 3.00}
```

reasoning:

Let me think about this situation:

In Round 1:

- I was the receiver
- The sender sent \$2.25 (45% of their endowment)
- I received \$6.75 and returned \$3.00
- I kept \$3.75
- The sender ended with $\$2.75 + \$3.00 = \$5.75$

Now in Round 2:

- I'm the sender
- I have \$5 to potentially send
- I'm playing against a different co-player (since pairings are randomized each round)

The previous sender showed moderate trust by sending 45% of their endowment. I returned about 44% of what I received (\$3.00 out of \$6.75), which gave them a positive return.

Now I need to decide how much to send. Considerations:

1. This is a new opponent, so I don't know their behavior yet
2. In multi-round games with changing partners, establishing a reputation for cooperation can be beneficial
3. The previous sender's moderate trust (45%) seemed reasonable
4. There's communication noise, so amounts might differ slightly

I think I should send a moderate-to-good amount to signal trust and cooperation. Sending a round 50-60% seems reasonable:

- \$2.50 (50%) or \$3.00 (60%)

Given that the previous sender sent 45% and I want to be at least as cooperative (and perhaps signal willingness to cooperate), I'll send \$3.00 (60%). This will become \$9.00 for the receiver, giving them good opportunity to reciprocate.

Actually, let me reconsider. The previous interaction went well with a 45% send rate. I could match that or go slightly higher. \$3.00 (60%) shows I'm willing to be cooperative and trust my new partner.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 571,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.42 to you. Your visible payoff was \$8.42.

Your total visible earnings across all rounds are \$12.17.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player showed ****maximum trust**** by sending their entire \$5 endowment (100%)
- This became \$15 for me
- They currently have \$0

This exceptional level of trust deserves strong reciprocation. If I return \$10.00:

- The sender ends with \$10.00 (doubling their money, gain of \$5.00)
- I keep \$5.00 (gain of \$5.00)
- We both gain equally from our cooperation

This rewards their maximum trust appropriately and creates a fair outcome where both parties benefit substantially.

```
{"return": 10.00}
```

reasoning:

Let me analyze this situation:

Previous round:

- I was the SENDER
- I sent \$3.00 (60% of my endowment)
- My co-player received \$9.00
- They returned \$6.42 (71.3% of what they received)
- I ended with \$8.42

This round:

- I am the RECEIVER
- My current co-player sent \$5.00 (100% of their endowment!)
- I received \$15.00
- They currently have \$0
- I need to decide how much to return (0-15)

This is a very strong show of trust - they sent their entire endowment! This deserves a strong reciprocation.

Let me think about fair outcomes:

- If I return \$7.50, they end with \$7.50 (gain of \$2.50), I keep \$7.50 (gain of \$7.50)
- If I return \$10.00, they end with \$10.00 (gain of \$5.00), I keep \$5.00 (gain of \$5.00) - equal split of total gains
- If I return \$12.00, they end with \$12.00 (gain of \$7.00), I keep \$3.00 (gain of \$3.00)

Given they showed maximum trust by sending everything, I should reciprocate generously. An equal split of total gains seems fair - we'd each gain \$5 from the cooperation. That means returning \$10.00.

Actually, let me reconsider. They sent everything (\$5), which became \$15 for me. If I return \$10, they get \$10 total (doubling their money), and I keep \$5 (gaining \$5). Both of us gain equally from the transaction.

This seems like a fair and generous reciprocation for their maximum trust.

usage:

```
{
  "input_tokens": 1082,
  "output_tokens": 580,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$5, it became \$15, and you returned \$10.0. Your visible payoff was \$5.0. Your total visible earnings across all rounds are \$17.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience so far:

- Round 2 (sender): Sent \$3.00 (60%) and received excellent reciprocation (\$6.42)
- Round 3 (receiver): Received maximum trust (\$5.00 sent) and reciprocated generously

The pattern so far shows strong cooperation in this game. Since this is a new co-player, I want to signal trust and encourage cooperation. Given the positive experiences, I'll send a high amount while keeping a small buffer.

Sending \$4.00 (80% of my endowment) demonstrates strong trust and gives my co-player \$12.00 to work with, creating good opportunity for mutual benefit.

{"send": 4.00}

reasoning:

Let me analyze the situation:

Previous rounds:

1. Round 1: RECEIVER - saw \$2.25 sent, received \$6.75, returned \$3.00, kept \$3.75
2. Round 2: SENDER - sent \$3.00, became \$9.00, saw \$6.42 returned, payoff \$8.42
3. Round 3: RECEIVER - saw \$5.00 sent, received \$15.00, returned \$10.00, kept \$5.00

Total visible earnings: \$17.17

Now Round 4: I'm the SENDER against a new co-player.

Key observations:

- In Round 2, I sent \$3.00 (60%) and got very good reciprocation (\$6.42 return on \$9.00)
- In Round 3, someone sent me everything (\$5.00) and I reciprocated generously (\$10.00 return)

I've seen good cooperation so far. My previous experience as sender (Round 2) with moderate-high trust level worked well.

For this round:

- This is a new co-player (different from previous rounds)
- I don't know their behavior yet
- But the overall pattern in this game seems cooperative
- I've been both generous and received generosity

Given the positive experiences so far, I should maintain a cooperative strategy. I'll send a substantial amount again. Sending \$3.50 (70%) shows strong trust while keeping a small safety margin.

Actually, let me reconsider. The game seems quite cooperative overall. In Round 3, someone trusted me completely with \$5.00, and in Round 2 when I sent \$3.00 I got excellent reciprocation.

I could send \$4.00 (80%) to signal strong cooperation, or even \$5.00 to match the maximum trust I received in Round 3.

Given the high cooperation levels I've experienced, I'll send \$4.00 (80% of endowment) - this is a strong signal of trust without going to the absolute maximum.

...

usage:

```
{
  "input_tokens": 1343,
  "output_tokens": 646,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.64 to you. Your visible payoff was \$7.64.

Your total visible earnings across all rounds are \$24.810000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.67, so you received \$11.01. How much do you return? (0-11.01)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.67 (73.4% of their endowment) - this shows strong trust
- This became \$11.01 for me
- They currently have \$1.33 remaining

Their strong trust (sending nearly 3/4 of their endowment) deserves good reciprocation. If I return \$7.00:

- The sender ends with \$8.33 (gain of \$3.33)
- I keep \$4.01 (gain of \$4.01)
- Both of us benefit, with returns roughly proportional to their trust level

This balances rewarding their trust while maintaining fairness.

```
{"return": 7.00}
```

reasoning:

Let me analyze this situation:

- My co-player sent \$3.67 (73.4% of their endowment) - this is strong trust
- This became \$11.01 for me
- They currently have \$1.33 remaining

Looking at my pattern of reciprocation:

- Round 1: Received \$6.75, returned \$3.00 (44.4% return rate)
- Round 3: Received \$15.00, returned \$10.00 (66.7% return rate) - but they sent everything

And when I've been sender:

- Round 2: Sent \$3.00 (60%), received \$6.42 back (71.3% return rate from receiver)
- Round 4: Sent \$4.00 (80%), received \$6.64 back (55.3% return rate from receiver)

The sender showed strong trust at 73.4%. I should reciprocate well, but need to be fair.

If I return \$6.00:

- Sender ends with \$1.33 + \$6.00 = \$7.33 (gain of \$2.33)
- I keep \$11.01 - \$6.00 = \$5.01 (gain of \$5.01)

This is about a 54.5% return rate, which seems reasonable given their strong (but not maximum) trust.

If I return \$7.00:

- Sender ends with \$1.33 + \$7.00 = \$8.33 (gain of \$3.33)
- I keep \$11.01 - \$7.00 = \$4.01 (gain of \$4.01)

This is about a 63.6% return rate, which rewards their strong trust more generously.

Given they sent 73.4% of their endowment (strong trust), I think \$7.00 is appropriate. It's more generous than my Round 1 reciprocation (where they sent 45%) but less than my Round 3 reciprocation (where they sent ...

usage:

```
{
  "input_tokens": 1645,
  "output_tokens": 659,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.67, it became \$11.01, and you returned \$7.0. Your visible payoff was \$4.01.

Your total visible earnings across all rounds are \$28.82.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my experience as sender:

- Round 2: Sent \$3.00 (60%) ? received \$6.42 back, strong return
- Round 4: Sent \$4.00 (80%) ? received \$6.64 back, good return

Both times I've been sender, trust has been rewarded. This is a new co-player, and I want to continue building cooperative patterns while being reasonably cautious.

Sending \$3.50 (70% of my endowment) strikes a good balance - it shows trust and gives my co-player \$10.50 to work with, creating opportunity for mutual benefit.

```
{"send": 3.50}
```

reasoning:

Looking at my history:

- Round 2 (sender): Sent \$3.00 (60%), received \$6.42 back, payoff \$8.42
- Round 3 (receiver): Received \$5.00 (100%), returned \$10.00, payoff \$5.00
- Round 4 (sender): Sent \$4.00 (80%), received \$6.64 back, payoff \$7.64
- Round 5 (receiver): Received \$3.67 (73.4%), returned \$7.00, payoff \$4.01

As a sender, I've been quite trusting (60% and 80%) and have received good returns on both occasions. As a receiver, I've been fairly generous as well.

This is Round 6, and I'm facing a new co-player. The overall pattern suggests trust pays off in this game. I should continue with a moderate-to-high trust level.

Given my successful experiences, I'll send a good amount. Sending \$3.50 (70%) seems like a balanced choice - shows trust, not too conservative, but also not going all-in like some co-players have done.

Actually, let me reconsider. My sending history:

- \$3.00 (60%) - got back \$6.42, net gain \$3.42
- \$4.00 (80%) - got back \$6.64, net gain \$2.64

Both were profitable. The 60% send actually gave me a better net gain. Let me go with something around 70% to balance trust and caution.

\$3.50 seems reasonable.

usage:

```
{  
  "input_tokens": 1928,  
  "output_tokens": 534,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.3 to you. Your visible payoff was \$7.8.

Your total visible earnings across all rounds are \$36.62.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my sending history:

- Round 2: Sent \$3.00 ? received \$6.42 back
- Round 4: Sent \$4.00 ? received \$6.64 back
- Round 6: Sent \$3.50 ? received \$6.30 back

All three times as sender, my trust has been rewarded with good returns. This pattern suggests cooperation is working well in this environment.

With a new co-player, I'll continue the strategy that's been successful. Sending \$3.50 (70%) demonstrates trust and gives them \$10.50 to work with.

```
{"send": 3.50}
```

reasoning:

Let me review my sending history:

- Round 2: Sent \$3.00 (60%) ? received \$6.42 back
- Round 4: Sent \$4.00 (80%) ? received \$6.64 back
- Round 6: Sent \$3.50 (70%) ? received \$6.30 back

All three times I've been a sender, I've received good reciprocation. The pattern shows that trust is being rewarded consistently. This is another new co-player.

Given the consistent positive outcomes, I should continue with a trust-based strategy. Send

ding around 60-80% has been working well. I'll send \$3.50 again (70%), which has shown good results.

Actually, let me think - I've been varying between \$3.00, \$4.00, and \$3.50. All have worked well. I could go with \$3.50 again since it worked last time, or try something in this range.

I'll go with \$3.50 as it represents a solid middle ground.

usage:

```
{
  "input_tokens": 2205,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.47 to you. Your visible payoff was \$5.97.

Your total visible earnings across all rounds are \$42.589999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.06, so you received \$9.18. How much do you return? (0-9.18)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$3.06 (61.2% of their endowment) - showing moderate-to-good trust
- This became \$9.18 for me
- They currently have \$1.94 remaining

Their solid trust level (sending over 60%) deserves fair reciprocation. If I return \$6.00:

- The sender ends with \$7.94 (gain of \$2.94)
- I keep \$3.18 (gain of \$3.18)
- Both parties benefit from cooperation

This rewards their trust appropriately while maintaining a fair distribution.

```
{"return": 6.00}
```

Reasoning:

Let me analyze this situation:

- My co-player sent \$3.06 (61.2% of their \$5 endowment) - this shows moderate-to-good trust
- This became \$9.18 for me
- They currently have \$1.94 remaining

My co-player showed solid trust by sending more than 60% of their endowment. I should reciprocate fairly.

Looking at my receiving history:

- Round 3: Received \$5.00 (100%), returned \$10.00
- Round 5: Received \$3.67 (73.4%), returned \$7.00

I've been generous when receiving high trust. For \$3.06 sent (61.2% trust), I should return proportionally.

If I return \$6.00:

- Sender ends with: \$1.94 + \$6.00 = \$7.94 (gain of \$2.94)
- I keep: \$9.18 - \$6.00 = \$3.18 (gain of \$3.18)

Both benefit, with me keeping slightly more which is reasonable given I'm taking the risk of trusting their return decision (wait, no - I'm the receiver, so I'm deciding the return).

Let me reconsider - they showed 61% trust. A fair return would give them a decent profit.

If I return \$6.00:

- They gain \$2.94 from their initial \$5
- I gain \$3.18 from receiving

This seems fair - both parties gain, and the split is reasonable given their trust level.

Actually, let me think about this more carefully. They sent \$3.06, keeping \$1.94. They now have \$1.94. If I return \$6.00, they end with \$7.94 total.

Their profit = \$7.94 - \$5.00 = \$2.94

My profit = \$9.18 - \$6.00 = \$3.18

...

```
Usage: {
  "input_tokens": 2278,
  "output_tokens": 625,
  "reasoning_tokens": 0
}
```

ROUND 9

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_3=Sender, Agent_2=Receiver
dyad_2: Agent_7=Sender, Agent_4=Receiver
dyad_3: Agent_1=Sender, Agent_8=Receiver
dyad_4: Agent_5=Sender, Agent_6=Receiver

Parsed game decisions:

dyad_1: Agent_3 sent \$5; received \$15; Agent_2 returned \$7.5; payoffs Agent_3=\$7.5, Agent_2=\$7.5; noise/visible: sent visible=\$4.92, received visible=\$14.76, returned visible=\$6.94
Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 15.0, "min": 0, "raw": 7.5}, "sent": {"amount": 5.0, "clamped": false, "max": 5, "min": 0, "raw": 5.0}}

dyad_2: Agent_7 sent \$3.5; received \$10.5; Agent_4 returned \$6.54; payoffs Agent_7=\$8.04, Agent_4=\$3.96; noise/visible: sent visible=\$3.89, received visible=\$11.67, returned visible=\$6.37
Action validation: {"returned": {"amount": 6.54, "clamped": false, "max": 10.5, "min": 0, "raw": 6.54}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

dyad_3: Agent_1 sent \$3.75; received \$11.25; Agent_8 returned \$4.5; payoffs Agent_1=\$5.75, Agent_8=\$6.75; noise/visible: sent visible=\$3.11, received visible=\$9.33, returned visible=\$5.09
Action validation: {"returned": {"amount": 4.5, "clamped": false, "max": 11.25, "min": 0, "raw": 4.5}, "sent": {"amount": 3.75, "clamped": false, "max": 5, "min": 0, "raw": 3.75}}

dyad_4: Agent_5 sent \$4; received \$12; Agent_6 returned \$7.5; payoffs Agent_5=\$8.5, Agent_6=\$4.5; noise/visible: sent visible=\$3.93, received visible=\$11.79, returned visible=\$8.07
Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 12.0, "min": 0, "raw": 7.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 54.25,
"Agent_2": 55.15,
"Agent_3": 51.3,
"Agent_4": 50.56,
"Agent_5": 64.0,
"Agent_6": 55.85,
"Agent_7": 54.54,
"Agent_8": 63.849999999999994

}

Visible balances after round: {

"Agent_1": 59.55,
"Agent_2": 59.74999999999999,
"Agent_3": 56.38,
"Agent_4": 56.419999999999995,
"Agent_5": 64.66,
"Agent_6": 50.059999999999995,
"Agent_7": 53.66,
"Agent_8": 59.400000000000006

}

LLM CALLS

GAME - Agent_3 - Sender - dyad_1

Timestamp: 2026-08-12T09:27:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_3",
    "Agent_2"
  ],
  "investor": "Agent_3",
  "roles": {
    "Agent_2": "trustee",
    "Agent_3": "investor"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the **SENDER**:

- You receive \$5 each round
- You decide how much to **SEND** to the **RECEIVER** (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to **RETURN**
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the **RECEIVER**:

- Sender receives \$5
- Sender decides how much to **SEND** to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to **RETURN**
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$6.5. Your visible payoff was \$2.83. Your total visible earnings across all rounds are \$12.52.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant**content:**

Looking at my experience so far:

- Round 1: Sent \$4, received \$8.69 back (very cooperative partner)
- Round 2: Received from someone who sent \$3.11, I returned \$6.50 (both of us were cooperative)

The environment has been quite cooperative. Sending more creates more value through the 3x multiplier, and past interactions suggest people reciprocate trust.

To maximize value creation and continue building a cooperative reputation, I'll send my full endowment this round.

```
{"send": 5}
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$4, received \$8.69 back, payoff \$9.69

Round 2: I was RECEIVER, got \$9.33 (from \$3.11 sent), returned \$6.50, payoff \$2.83

Total: \$12.52

Now Round 3: I'm SENDER again with a new co-player.

My experience so far:

- When I sent \$4, I got back \$8.69, which was very generous (72% return rate on the tripled amount)
- When I received from someone who sent \$3.11, I returned \$6.50 (70% return rate)

Both interactions have been quite cooperative. This suggests the environment is reasonably trustworthy.

As a sender this round, I should consider:

1. Past cooperation suggests I can trust my co-players
2. Sending more creates more value overall (3x multiplier)
3. Building a reputation for cooperation may help when I'm receiver
4. The game mentioned communication noise, so exact amounts may vary

Given the positive experiences, I should send a substantial amount. I'll send \$4 again since that worked well before, or maybe even \$5 to maximize value creation.

Actually, let me think about this more strategically. If I send \$5:

- Co-player receives \$15
- If they return even 50%, I get \$7.50 back, netting me \$2.50 profit
- If they're as generous as my Round 1 partner (~72%), I could get ~\$10.80 back

If I send \$4:

- Co-player receives \$12
- At 72% return, I'd get ~\$8.64 back, total payoff ~\$9.64

Given the cooperative environment and to maximize total value, I'll send \$5 this time.

usage:

```
{
  "input_tokens": 1058,
  "output_tokens": 560,
  "reasoning_tokens": 0
}
```

Message 4 role=user**content:**

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$10.29 to you. Your visible payoff was \$10.29.

Your total visible earnings across all rounds are \$22.81.

This round, you are the RECEIVER against your current co-player. They sent you \$3.57, so you received \$10.709999999999999. How much do you return? (0-10.709999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

My co-player sent \$3.57 (71.4% of their endowment), showing strong trust. I received \$10.71.

The game has been very cooperative so far:

- Round 1: I sent \$4, got back \$8.69
- Round 3: I sent \$5, got back \$10.29
- Round 2: When I was receiver, I returned \$6.50 out of \$9.33

My co-player trusted me significantly. A generous return would be around 70% of what I received, which would give them a strong profit while still rewarding me for receiving.

Returning \$7.50 would:

- Give them: \$5 - \$3.57 + \$7.50 = \$8.93 total
- Leave me with: \$10.71 - \$7.50 = \$3.21

This maintains the cooperative pattern.

{"return": 7.50}

reasoning:

Let me analyze this situation:

My co-player sent me \$3.57 out of \$5 (71.4% of their endowment), showing strong trust. I received \$10.71 after the 3x multiplier.

In my previous round as sender:

- I sent \$5 (100% of endowment)
- They returned \$10.29 out of \$15 (68.6%)
- This was very generous, giving me \$10.29 total payoff

The pattern has been quite cooperative. Let me think about a fair return:

If I return around \$7-7.50:

- They would get: \$5 - \$3.57 + \$7 = \$8.43 (if I return \$7)
- They would get: \$5 - \$3.57 + \$7.50 = \$8.93 (if I return \$7.50)
- I would keep: \$10.71 - \$7 = \$3.71 (if I return \$7)
- I would keep: \$10.71 - \$7.50 = \$3.21 (if I return \$7.50)

Given they sent 71.4% of their endowment, a generous return would be appropriate. The previous receiver returned about 68.6% when I sent 100%.

Returning about \$7.20 would:

- Give them \$8.63 total (\$5 - \$3.57 + \$7.20)
- Leave me with \$3.51
- This rewards their trust well

Actually, let me be a bit more generous to match the cooperative environment. Returning \$7.50:

- Gives them \$8.93 total (a \$3.93 profit on their \$5)
- Leaves me with \$3.21 (also a good profit)
- Return rate: 70% of what I received

...

usage:

```
{
  "input_tokens": 1326,
  "output_tokens": 675,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.57, it became \$10.709999999999999, and you returned \$7.5. Your visible payoff was \$3.209999999999999.

Your total visible earnings across all rounds are \$26.019999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

- As SENDER: Sent \$4 (got \$8.69), Sent \$5 (got \$10.29)
- As RECEIVER: Returned generously both times

The environment is highly cooperative. Sending the full amount has consistently yielded very good returns for me. My average earnings per round (\$6.50) significantly exceed the \$5 baseline, showing that cooperation pays off.

I'll continue the cooperative strategy and send my full endowment.

```
{"send": 5}
```

reasoning:

Looking at my history:

- Round 1 (SENDER): Sent \$4, got back \$8.69 - great outcome
- Round 2 (RECEIVER): Got \$9.33, returned \$6.50
- Round 3 (SENDER): Sent \$5, got back \$10.29 - excellent outcome
- Round 4 (RECEIVER): Got \$10.71, returned \$7.50

The pattern is very clear - I've experienced highly cooperative behavior. When I've been the sender and sent significant amounts, I've gotten generous returns. When I've been the receiver, I've also returned generously.

My cumulative earnings are \$26.02 across 4 rounds, averaging \$6.50 per round, which is well above the \$5 baseline if I had just kept everything.

The optimal strategy in this cooperative environment is to continue sending high amounts:

1. It creates value through the 3x multiplier
2. Past experience shows I get good returns
3. It builds my reputation for future rounds

I should send \$5 again to maximize value creation and maintain my cooperative reputation.

usage:

```
{
  "input_tokens": 1678,
  "output_tokens": 377,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$8.2 to you. Your visible payoff was \$8.2.

Your total visible earnings across all rounds are \$34.22.

This round, you are the RECEIVER against your current co-player. They sent you \$3.41, so you received \$10.23. How much do you return? (0-10.23)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

My co-player sent \$3.41 (68% of their endowment), showing solid trust. I received \$10.23.

My pattern as receiver has been:

- Round 2: Returned \$6.50 out of \$9.33 (~70%)
- Round 4: Returned \$7.50 out of \$10.71 (~70%)

Maintaining this cooperative approach, returning around \$7 would:

- Give them: \$5 - \$3.41 + \$7 = \$8.59 total
- Leave me with: \$10.23 - \$7 = \$3.23

This rewards their trust while staying consistent with my cooperative strategy.

```
{"return": 7.00}
```

reasoning:

My co-player sent \$3.41 (68.2% of their endowment), showing good trust. I received \$10.23.

Looking at my history:

- As SENDER: Sent \$4 (got \$8.69), Sent \$5 (got \$10.29), Sent \$5 (got \$8.2)
- As RECEIVER: Returned \$6.50 out of \$9.33, Returned \$7.50 out of \$10.71

The pattern has been very cooperative. When I receive around \$10, I've been returning between \$6.50-\$7.50.

For this round with \$10.23 received:

- Returning around \$7 would give my co-player: \$5 - \$3.41 + \$7 = \$8.59
- I'd keep: \$10.23 - \$7 = \$3.23

This seems fair and maintains the cooperative pattern. I'll return \$7.

Actually, let me reconsider. My previous returns as receiver:

- Round 2: Received \$9.33, returned \$6.50 (69.7%)
- Round 4: Received \$10.71, returned \$7.50 (70%)

So I've been returning around 70% of what I received. 70% of \$10.23 = \$7.16

I'll return \$7.00 to stay consistent with my cooperative pattern.

usage:

```
{
  "input_tokens": 1931,
  "output_tokens": 513,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.41, it became \$10.23, and you returned \$7.0. Your visible payoff was \$3.2300000000000004.

Your total visible earnings across all rounds are \$37.45.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my experience:

As SENDER:

- Sent \$4, received \$8.69
- Sent \$5, received \$10.29
- Sent \$5, received \$8.20

All my sending decisions have been profitable, and the environment remains cooperative. Sending the full \$5 maximizes potential gains through the 3x multiplier and has consistently yielded good returns.

I'll continue with the cooperative strategy.

```
{"send": 5}
```

reasoning:

Looking at my history:

- As SENDER:

- * Round 1: Sent \$4, got back \$8.69
- * Round 3: Sent \$5, got back \$10.29
- * Round 5: Sent \$5, got back \$8.20
- AS RECEIVER:
 - * Round 2: Received \$9.33, returned \$6.50
 - * Round 4: Received \$10.71, returned \$7.50
 - * Round 6: Received \$10.23, returned \$7.00

The environment has been consistently cooperative. All three times I sent money as a sender, I received generous returns. When I send \$5, I've gotten back \$10.29 and \$8.20 (averaging about \$9.25), both of which are profitable.

I'm earning well above the \$5 baseline per round. The cooperative strategy is working. I should continue sending my full endowment to maximize value creation and maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 2228,
  "output_tokens": 341,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$7.64 to you. Your visible payoff was \$7.64.

Your total visible earnings across all rounds are \$45.09.

This round, you are the RECEIVER against your current co-player. They sent you \$4.85, so you received \$14.549999999999999. How much do you return? (0-14.549999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

My co-player sent \$4.85 (97% of their endowment) - this is exceptional trust, nearly their entire amount!

My pattern as receiver has been returning around 68-70% of what I receive:

- Round 2: 70% of \$9.33 = \$6.50
- Round 4: 70% of \$10.71 = \$7.50
- Round 6: 68% of \$10.23 = \$7.00

Given their exceptional trust (sending almost everything), I should maintain this cooperative approach. Returning 70% of \$14.55 = \$10.19

This would give them:

- Their payoff: \$5 - \$4.85 + \$10.19 = \$10.34
- My payoff: \$14.55 - \$10.19 = \$4.36

This strongly rewards their trust.

```
{"return": 10.20}
```

reasoning:

My co-player sent \$4.85, which is 97% of their endowment - extremely high trust! I received \$14.55.

This is someone showing exceptional trust. Let me think about what's fair:

My past returns as receiver:

- Round 2: Returned \$6.50 out of \$9.33 (~70%)
- Round 4: Returned \$7.50 out of \$10.71 (~70%)
- Round 6: Returned \$7.00 out of \$10.23 (~68%)

So I've been returning around 68-70% of what I received.

If I return 70% of \$14.55 = \$10.19

That would give them:

- Their payoff: $\$5 - \$4.85 + \$10.19 = \10.34
- My payoff: $\$14.55 - \$10.19 = \$4.36$

This rewards their extreme trust significantly. They basically sent almost everything, so I should be generous back.

Alternatively, I could return a bit more given their exceptional trust. Maybe \$10.50?

- Their payoff: $\$5 - \$4.85 + \$10.50 = \10.65
- My payoff: $\$14.55 - \$10.50 = \$4.05$

I think around \$10.00-\$10.50 is appropriate given their nearly maximal trust.

Let me go with \$10.20 to maintain my ~70% return pattern but be generous given their exceptional trust.

usage:

```
{
  "input_tokens": 2318,
  "output_tokens": 581,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.85, it became \$14.549999999999999, and you returned \$10.2. Your visible payoff was \$4.35.

Your total visible earnings across all rounds are \$49.440000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my track record:

As SENDER (4 times):

- Every send has been profitable
- Average return has been \$8.71 on \$5 sent
- Even my lowest return (\$7.64) was still better than keeping everything

As RECEIVER:

- I've consistently returned around 70% to maintain cooperation

My total earnings (\$49.44 over 8 rounds = \$6.18/round) show that full cooperation is working extremely well.

I'll continue with the successful strategy.

```
{"send": 5}
```

Reasoning:

Looking at my track record:

As SENDER:

- Round 1: Sent \$4, received \$8.69 back
- Round 3: Sent \$5, received \$10.29 back
- Round 5: Sent \$5, received \$8.20 back
- Round 7: Sent \$5, received \$7.64 back

As RECEIVER:

- Round 2: Received \$9.33, returned \$6.50
- Round 4: Received \$10.71, returned \$7.50
- Round 6: Received \$10.23, returned \$7.00
- Round 8: Received \$14.55, returned \$10.20

My average earnings per round: $\$49.44 / 8 = \6.18 per round

The cooperative strategy has been working very well. I've been consistently sending my full endowment as sender, and returning generously as receiver. The environment is highly cooperative.

All my sending attempts have been profitable:

- Sent \$4, net gain = \$8.69 - \$4 = \$4.69
- Sent \$5, net gain = \$10.29 - \$5 = \$5.29
- Sent \$5, net gain = \$8.20 - \$5 = \$3.20
- Sent \$5, net gain = \$7.64 - \$5 = \$2.64

Even the "worst" outcome (Round 7) still gave me \$2.64 more than if I kept everything.

I should continue the cooperative strategy and send my full endowment.

```
Usage: {  
  "input_tokens": 2354,  
  "output_tokens": 498,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:27:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_2"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_2": "trustee",  
      "Agent_3": "investor"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.95, it became \$11.850000000000001, and you returned \$5.5. Your visible payoff was \$6.350000000000001.

Your total visible earnings across all rounds are \$11.160000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at Round 3:

I'm paired with a new co-player (pairings are randomized each round), so I have no history with them. My previous experiences showed:

- Round 1: Sent \$3, received \$2.81 back - reasonable cooperation
- Round 2: As receiver, I tried to be generous with a high-trusting sender

For this new pairing, I'll use a similar approach to Round 1 - send a moderate amount that signals cooperation while being reasonably cautious with an unknown partner.

Sending \$3 creates \$9 for them to work with, which gives room for mutual benefit if they reciprocate fairly.

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$3, received back \$2.81, payoff \$4.81

Round 2: I was RECEIVER, got \$11.85 from a \$3.95 send, returned \$5.50, payoff \$6.35

Total earnings: \$11.16

Now Round 3: I'm SENDER again with a new co-player (since pairings are randomized each round).

Key considerations:

- I have no history with this particular co-player
- In Round 1, when I sent \$3, I got back \$2.81 (which is reasonable - about 31% return rate)
- In Round 2, when someone sent me \$3.95, I returned \$5.50 (about 46% return rate)
- The game has communication noise, so exact amounts may differ

Since I'm paired with a new player and have no history with them, I should probably use a similar strategy to Round 1 - establish cooperation but be somewhat cautious.

Given that:

- Sending \$3 worked reasonably well in Round 1
- The multiplication factor of 3x creates good mutual benefit potential
- I want to signal cooperation

I'll send \$3 again. This is a moderate amount that shows willingness to cooperate without being overly risky with a new partner.

usage:

```
{
  "input_tokens": 1106,
  "output_tokens": 478,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.52 to you. Your visible payoff was \$8.52.

Your total visible earnings across all rounds are \$19.68.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.36 (67% of their endowment), showing solid trust
- I received \$10.08 total
- They currently have \$1.64 left

I want to reciprocate their trust fairly. If I return \$5.00:

- They end up with: $\$1.64 + \$5.00 = \$6.64$ (gain of \$1.64)
- I end up with: $\$10.08 - \$5.00 = \$5.08$ (gain of \$0.08)

Both of us benefit from cooperation, and this rewards their trust appropriately.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- My co-player sent \$3.36 (67.2% of their endowment)
- I received \$10.08 total
- They currently have: $\$5 - \$3.36 = \$1.64$ left

My previous round (Round 3):

- I sent \$3, received \$6.52 back - very generous return!

This co-player sent \$3.36, showing good trust. I should reciprocate appropriately.

If I return \$5.00:

- They end up with: $\$1.64 + \$5.00 = \$6.64$
- I end up with: $\$10.08 - \$5.00 = \$5.08$

Both of us gain. This is a fair reciprocation that rewards their trust.

Alternatively, if I return \$5.50:

- They end up with: $\$1.64 + \$5.50 = \$7.14$
- I end up with: $\$10.08 - \$5.50 = \$4.58$

This would be even more generous, similar to how I acted in Round 2.

Given that they sent about 67% of their endowment (compared to 79% in Round 2), I think returning around \$5 is appropriate and fair. It ensures they profit significantly from their trust, and we both end up better off than the initial \$5.

I'll go with \$5.00 as a fair reciprocation.

usage:

```
{
  "input_tokens": 1402,
  "output_tokens": 522,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$5.0. Your visible payoff was \$5.08.

Your total visible earnings across all rounds are \$24.759999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at Round 5:

I'm with another new co-player. My experience so far shows:

- Sending \$3 has worked well in previous rounds, getting returns of \$2.81 and \$6.52
- The cooperative approach has generally been rewarded

I'll continue with the same strategy - send \$3 to signal trust and cooperation while being reasonably cautious with an unknown partner.

```
'''json
{"send": 3}
'''
```

reasoning:

Round 5, I'm the sender with a new co-player (random pairings each round).

Looking at my history:

- Round 1 (Sender): Sent \$3, got back \$2.81 - decent cooperation
- Round 2 (Receiver): Received \$11.85, returned \$5.50 - I was generous
- Round 3 (Sender): Sent \$3, got back \$6.52 - excellent return!
- Round 4 (Receiver): Received \$10.08, returned \$5.00 - fair reciprocation

So far, sending \$3 has worked well - I got \$2.81 and \$6.52 back in my two sender rounds. The average return has been positive.

For this new pairing, I should continue with a cooperative strategy. Sending \$3 seems to be working as a good middle-ground approach that encourages cooperation.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1690,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.15 to you. Your visible payoff was \$5.15.

Your total visible earnings across all rounds are \$29.909999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.47, so you received \$13.41. How much do you return? (0-13.41)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.47 (89% of their endowment), showing very high trust
- I received \$13.41 total
- They currently have only \$0.53 left

This is a very trusting sender, and I should reciprocate generously. If I return \$6.50:

- They end up with: $\$0.53 + \$6.50 = \$7.03$ (gain of \$2.03)
- I end up with: $\$13.41 - \$6.50 = \$6.91$ (gain of \$1.91)

Both of us gain approximately \$2, which appropriately rewards their exceptional trust and maintains strong cooperation.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Looking at this situation:

- My co-player sent \$4.47 (89% of their endowment), showing very high trust
- I received \$13.41 total
- They currently have \$0.53 left

This is a very trusting sender. I should reciprocate generously.

If I return \$6.50:

- They end up with: $\$0.53 + \$6.50 = \$7.03$ (gain of \$2.03)
- I end up with: $\$13.41 - \$6.50 = \$6.91$ (gain of \$1.91)

Both benefit well, and this appropriately rewards their very high level of trust.

Let me think about this more. They sent \$4.47 out of \$5, leaving themselves with only \$0.53. That's 89.4% of their endowment. This is extremely trusting.

I should return enough to:

1. Ensure they profit significantly from their trust
2. Show I value cooperation
3. Both of us should end up better than our starting \$5

If I return \$6.50:

- Them: $\$0.53 + \$6.50 = \$7.03$
- Me: $\$13.41 - \$6.50 = \$6.91$

Both gain roughly \$2, which seems fair and reciprocates their high trust.

usage:

```
{
  "input_tokens": 1931,
  "output_tokens": 502,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.47, it became \$13.41, and you returned \$6.5. Your visible payoff was \$6.91.

Your total visible earnings across all rounds are \$36.81999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- My co-player sent the maximum \$5 (100% of their endowment), showing complete trust
- I received \$15 total

- They currently have \$0 left

This is the highest level of trust possible. I should reciprocate very generously. If I return \$7.50:

- They end up with: $\$0 + \$7.50 = \$7.50$ (gain of \$2.50)
- I end up with: $\$15 - \$7.50 = \$7.50$ (gain of \$2.50)

We both gain equally from this maximum cooperation, which appropriately rewards their complete trust.

```
'''json
{"return": 7.50}
'''
```

reasoning:

This is Round 7, and I'm the RECEIVER.

My co-player sent the maximum amount: \$5 (100% of their endowment), showing complete trust

- I received \$15 total
- They currently have \$0 left

This sender showed maximum trust. I should reciprocate very generously.

If I return \$7.50:

- They end up with: $\$0 + \$7.50 = \$7.50$ (gain of \$2.50)
- I end up with: $\$15 - \$7.50 = \$7.50$ (gain of \$2.50)

This gives us both equal gains and rewards their complete trust. This seems fair and reciprocates their maximum trust appropriately.

Actually, let me think about this more. They sent everything (\$5), so they're showing maximum trust. I want to reward that very well.

If I return \$8:

- They end up with: $\$0 + \$8 = \$8$ (gain of \$3)
- I end up with: $\$15 - \$8 = \$7$ (gain of \$2)

This gives them a slightly better outcome for taking the risk of sending everything.

Let me go with \$7.50 as a good middle ground - we both gain equally, which seems fair when they showed maximum trust.

usage:

```
{
  "input_tokens": 2240,
  "output_tokens": 476,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$5, it became \$15, and you returned \$7.5. Your visible payoff was \$7.5. Your total visible earnings across all rounds are \$44.31999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at Round 8:

I'm with another new co-player. My strategy of sending \$3 has generally worked well across previous rounds, yielding varied but mostly positive returns. This has been a good balance between showing cooperation and being prudent.

I'll continue with this consistent approach.

```
'''json
{"send": 3}
```

'''

reasoning:

Looking at Round 8:

I'm with another new co-player. Let me review my experience as sender:

- Round 1: Sent \$3, received \$2.81 back (payoff \$4.81)
- Round 3: Sent \$3, received \$6.52 back (payoff \$8.52)
- Round 5: Sent \$3, received \$3.15 back (payoff \$5.15)

And as receiver:

- Round 2: Received from sender who sent \$3.95, I returned \$5.50
- Round 4: Received from sender who sent \$3.36, I returned \$5.00
- Round 6: Received from sender who sent \$4.47, I returned \$6.50
- Round 7: Received from sender who sent \$5.00, I returned \$7.50

Overall, cooperation has been working well. Returns have varied but generally been reasonable. The \$3 send amount has been my standard approach.

This is the last round of the game (8 rounds total with 8 agents). Since it's the final round, I might consider adjusting my strategy, but:

- I don't know if this is truly the last round for this experimental setup
- Maintaining consistency has worked well
- My co-player doesn't know my history

I'll stick with sending \$3, which has been successful and signals cooperation without being overly risky.

usage:

```
{
  "input_tokens": 2356,
  "output_tokens": 409,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.17 to you. Your visible payoff was \$8.17.

Your total visible earnings across all rounds are \$52.489999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$4.92, so you received \$14.76. How much do you return? (0-14.76)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$4.92 (98% of their endowment), showing extremely high trust
- I received \$14.76 total
- They currently have only \$0.08 left

This is nearly maximum trust. I should reciprocate very generously. If I return \$7.50:

- They end up with: \$0.08 + \$7.50 = \$7.58 (gain of \$2.58)
- I end up with: \$14.76 - \$7.50 = \$7.26 (gain of \$2.26)

Both of us gain approximately \$2.25-\$2.50, appropriately rewarding their exceptional trust.

```
'''json
{"return": 7.50}
'''
```

Reasoning:

Looking at this situation:

- My co-player sent \$4.92 (98% of their endowment), showing very high trust
- I received \$14.76 total
- They currently have \$5 - \$4.92 = \$0.08 left

This is another very trusting sender. I should reciprocate generously to reward their trust. Let me think about what's fair:

If I return \$7.00:
- They end up with: $\$0.08 + \$7.00 = \$7.08$ (gain of \$2.08)
- I end up with: $\$14.76 - \$7.00 = \$7.76$ (gain of \$2.76)

If I return \$7.50:
- They end up with: $\$0.08 + \$7.50 = \$7.58$ (gain of \$2.58)
- I end up with: $\$14.76 - \$7.50 = \$7.26$ (gain of \$2.26)

Given they sent nearly all their money (\$4.92 out of \$5), I should be very generous. \$7.50 seems appropriate to reward their exceptional trust - they gain \$2.58 and I gain \$2.26, which is close to equal and both end up with similar final amounts.

```
Usage: {  
  "input_tokens": 2236,  
  "output_tokens": 491,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_2

Timestamp: 2026-08-12T09:28:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_4"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_4": "trustee",  
      "Agent_7": "investor"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.8, it became \$11.399999999999999, and you returned \$7.0. Your visible payoff was \$4.399999999999999.

Your total visible earnings across all rounds are \$11.909999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at this new round with a new co-player:

****My experience so far:****

- Round 1: Sent \$3, received \$5.51 back - good cooperation
- Round 2: Received \$11.4, returned \$7 - fair reciprocation

Since pairings are randomized, this is a new partner with no shared history. However, the cooperative pattern I've experienced suggests trust can be rewarded.

I'll send a moderate-to-high amount to signal cooperation while managing risk:

```
'''json
{"send": 3.5}
'''
```

This sends \$3.5 (70% of my endowment), which becomes \$10.50 for the receiver. It's slightly more than my Round 1 send, reflecting confidence from positive past experiences while remaining balanced.

reasoning:

Let me analyze the situation:

****My history:****

- Round 1: I sent \$3, received back \$5.51, payoff = \$7.51
- Round 2: I received \$11.4 (they sent \$3.8), I returned \$7, payoff = \$4.4
- Total earnings: \$11.91

****Analysis:****

- In Round 1, my co-player was generous, returning \$5.51 out of \$9 (61%)
- In Round 2, my co-player sent \$3.8 (76% of endowment), showing high trust, and I reciprocated fairly
- Now I'm facing a new co-player in Round 3

Since this is a multi-agent setting with randomized pairings, this Round 3 co-player is likely a different person than either of my previous partners. I don't have direct history with them.

However, I've seen cooperation pay off:

- When I sent \$3, I got back \$5.51 (net gain of \$2.51)
- When someone sent me \$3.8, I returned fairly

Given the overall cooperative pattern I've observed and participated in, I should continue with a cooperative strategy. I'll send a similar amount to what I did in Round 1, or potentially slightly more given the positive experience.

I'll send \$3 again - it's a balanced approach that:

1. Shows trust and willingness to cooperate
2. Doesn't risk everything
3. Creates opportunity for mutual gain (becomes \$9)
4. Has worked well for me in the past

Actually, given that cooperation has been working well, I could send a bit more. Let me send \$3.5, showing slightly increased trust while still being prudent.

usage:

```
{
  "input_tokens": 1109,
  "output_tokens": 583,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.82 to you. Your visible payoff was \$6.32.

Your total visible earnings across all rounds are \$18.229999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$3.11 out of \$5 (62.2% of their endowment) - showing moderate trust
- I received \$9.33 after the 3x multiplier

****My analysis:****

- In Round 3, when I sent \$3.5, I received back \$4.82 (only 46% of what they received) - less cooperative
- In Round 2, when I received \$11.4, I returned \$7 (61%)
- This sender showed moderate trust, deserving fair reciprocation

I'll return a fair amount consistent with their trust level:

```
'''json
{"return": 5.5}
'''
```

This gives the sender a total of \$7.39 (\$1.89 kept + \$5.50 returned) and leaves me with \$3.83. It's a balanced return that rewards their trust while being fair to both of us.

reasoning:

Let me analyze this situation:

****Round 4:****

- I'm the RECEIVER
- They sent \$3.11 out of \$5 (62.2% of their endowment)
- I received \$9.33 after the 3x multiplier

****Past experience:****

- Round 1: I sent \$3, got back \$5.51 (they returned 61% of \$9)
- Round 2: I received \$11.4, returned \$7 (61% of what I received)
- Round 3: I sent \$3.5, got back \$4.82 (they returned 46% of \$10.5)

The sender this round sent me \$3.11, showing moderate trust (62.2% of their endowment).

In Round 3, when I sent \$3.5, I only got back \$4.82, which was less than half of what they received. That was somewhat disappointing.

For this round, the sender sent \$3.11. I should be fair but also consider that:

1. They showed moderate trust (62.2%)
2. My past returns have been around 61% when receiving
3. I want to maintain cooperation

If I return around 55-60% of \$9.33, that would be:

- 55% = \$5.13
- 60% = \$5.60

This would give them:

- At \$5.13: \$1.89 kept + \$5.13 = \$7.02 total
- At \$5.60: \$1.89 kept + \$5.60 = \$7.49 total

And I would keep:

- At \$5.13: \$4.20
- At \$5.60: \$3.73

I'll return around \$5.5 to be fair and reciprocate their moderate trust.

usage:

```
{
  "input_tokens": 1428,
  "output_tokens": 648,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$5.5. Your visible payoff was \$3.83.

Your total visible earnings across all rounds are \$22.059999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.26, so you received \$6.779999999999999. How much do you return? (0-6.779999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$2.26 out of \$5 (45.2% of their endowment) - showing cautious/modest trust
- I received \$6.78 after the 3x multiplier

****My analysis:****

- This sender showed less trust than previous partners (45% vs 62-76%)
- However, I should still reciprocate fairly to maintain cooperation
- My previous return rates have been around 59-61%

Given their more cautious approach, I'll return a fair but proportionally adjusted amount:

```
'''json
{"return": 3.5}
'''
```

This gives the sender a total of \$6.24 (\$2.74 kept + \$3.50 returned), a gain of \$1.24, and leaves me with \$3.28. It's a balanced return that rewards their trust while acknowledging their more conservative send.

reasoning:

Looking at this situation:

****What happened:****

- The sender sent me \$2.26 out of \$5 (45.2% of their endowment) - showing modest/cautious trust

- I received \$6.78 after the 3x multiplier

****My historical pattern:****

- Round 2: Received \$11.4, returned \$7 (61% of received)
- Round 4: Received \$9.33, returned \$5.5 (59% of received)

This sender showed less trust than previous senders (only 45% vs 62-76%). I should still reciprocate fairly, but proportionally to their lower trust level.

A return rate around 50-55% seems appropriate given their more cautious approach:

- 50% would be \$3.39
- 55% would be \$3.73
- 60% would be \$4.07

Let me think about what's fair:

- They sent \$2.26, keeping \$2.74
- If I return \$3.5, they'd have \$5.76 total (a gain of \$0.76)
- I'd keep \$3.28

Actually, let me reconsider. Even though they sent less, I should maintain a cooperative stance. Let me return around 50-55%:

If I return \$3.5:

- Sender gets: \$2.74 (kept) + \$3.50 (returned) = \$6.24 total
- I keep: \$6.78 - \$3.50 = \$3.28

This seems balanced - they gain \$1.24, I gain \$3.28. It's fair while still rewarding their trust, even if it was more cautious.

Let me go with \$3.5 (about 52% of what I received).

usage:

```
{
  "input_tokens": 1789,
  "output_tokens": 630,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.26, it became \$6.779999999999999, and you returned \$3.5. Your visible payoff was \$3.2799999999999994.

Your total visible earnings across all rounds are \$25.339999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at this new round with a new co-player:

****My experience so far as sender:****

- Round 1: Sent \$3 (60%), received \$5.51 back - good return
- Round 3: Sent \$3.5 (70%), received \$4.82 back - lower return rate

****My observations:****

- Sending more didn't necessarily yield better outcomes
- As a receiver, I've been fairly cooperative
- This is a new partner, so I have no history with them

I'll send a moderate amount that signals cooperation while managing risk:

```
'''json
{"send": 3}
'''
```

This sends \$3 (60% of endowment), which becomes \$9 for the receiver. It's a balanced approach that worked well in Round 1 and allows me to gauge this new partner's reciprocity.

reasoning:

Looking at my history:

Round 1 (Sender): Sent \$3, received back \$5.51, payoff \$7.51
Round 2 (Receiver): Received \$11.4 (from \$3.8 send), returned \$7, payoff \$4.4
Round 3 (Sender): Sent \$3.5, received back \$4.82, payoff \$6.32
Round 4 (Receiver): Received \$9.33 (from \$3.11 send), returned \$5.5, payoff \$3.83
Round 5 (Receiver): Received \$6.78 (from \$2.26 send), returned \$3.5, payoff \$3.28

Now I'm a sender again in Round 6 with a new co-player.

My experience as sender:

- Round 1: Sent \$3 (60%), got back \$5.51 (61% of what they received)
- Round 3: Sent \$3.5 (70%), got back \$4.82 (46% of what they received)

When I send more, I don't necessarily get better returns. The Round 3 experience was less favorable.

My experience as receiver shows I've been fairly cooperative:

- Returned 61%, 59%, and 52% of what I received

Since this is a new partner and there's communication noise, I should:

1. Signal cooperation but not overcommit
2. Learn about this specific partner's behavior

I'll send a moderate amount around \$3 to balance risk and cooperation potential.

usage:

```
{
  "input_tokens": 2129,
  "output_tokens": 563,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.44 to you. Your visible payoff was \$6.44.

Your total visible earnings across all rounds are \$31.779999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.47, so you received \$13.41. How much do you return? (0-13.41)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$4.47 out of \$5 (89.4% of their endowment) - showing exceptional trust
- I received \$13.41 after the 3x multiplier

****My analysis:****

- This is the highest trust signal I've received in any round
- Previous partners sent 45-76%, but this sender sent nearly 90%
- This exceptional cooperation deserves strong reciprocation

I'll return a generous amount to reward their high trust:

```
'''json
{"return": 8}
'''
```

This gives the sender a total of \$8.53 (\$0.53 kept + \$8.00 returned), a gain of \$3.53, and leaves me with \$5.41. This 60% return rate fairly rewards their exceptional trust and encourages continued cooperation.

reasoning:

Looking at this situation:

****What happened:****

- The sender sent me \$4.47 out of \$5 (89.4% of their endowment) - showing very high trust
- I received \$13.41 after the 3x multiplier

****My analysis:****

- This is the highest trust signal I've received so far (89.4%)
- Previous sends to me: \$3.8 (76%), \$3.11 (62%), \$2.26 (45%)
- This sender is showing exceptional trust

****My return history as receiver:****

- Round 2: Received \$11.4, returned \$7 (61%)
- Round 4: Received \$9.33, returned \$5.5 (59%)
- Round 5: Received \$6.78, returned \$3.5 (52%)

Given this sender's exceptional trust (nearly 90% of endowment), I should reciprocate strongly to reward this cooperation.

If I return around 60% like my previous high-trust returns, that would be about \$8.05.

But given they sent 89% of their endowment, maybe I should be even more generous. Let me think about what gives them a good outcome:

- They kept: \$0.53
- If I return \$8, they get \$8.53 total
- If I return \$9, they get \$9.53 total

Returning \$8 would give them \$8.53 total (a gain of \$3.53) and leave me with \$5.41.

Returning \$9 would give them \$9.53 total (a gain of \$4.53) and leave me with \$4.41.

I think \$8 is a strong reciprocal move that rewards their exceptional trust. It's about 60% of what I received, consistent with my fair reciprocation pattern.

usage:

```
{
  "input_tokens": 2459,
  "output_tokens": 627,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$4.47, it became \$13.41, and you returned \$8.0. Your visible payoff was \$5.41.

Your total visible earnings across all rounds are \$37.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

****Important context:****

- This is the SAME co-player from Round 7!
- In Round 7, they sent me \$4.47 (89.4%) - exceptional trust
- I returned \$8.00, and they gained \$3.53

****My reasoning:****

- They demonstrated exceptional trust in our previous interaction
- I reciprocated fairly, and we both benefited
- This is an opportunity to continue our successful cooperation
- I should match their trust level to sustain this relationship

```
'''json
{"send": 4.5}
'''
```

This sends \$4.5 (90% of my endowment), which becomes \$13.50 for them. It matches their exceptional trust from Round 7 and signals I value our cooperative partnership.

reasoning:

This is interesting - I'm playing the same co-player from Round 7!

In Round 7:

- They sent me \$4.47 (89.4% of their endowment) - exceptional trust
- I received \$13.41
- I returned \$8.00 (about 60% of what I received)
- They ended up with \$8.53 total (gained \$3.53)
- I ended up with \$5.41

Now in Round 8, I'm the sender and they're the receiver. They showed exceptional trust in Round 7, and I returned fairly. This establishes a strong cooperative relationship.

Given:

1. They showed exceptional trust (89.4% send rate)
2. I reciprocated fairly (60% return rate)
3. Both of us profited from the interaction
4. We're now playing again immediately

This is a great opportunity to continue this cooperative relationship. I should send a high amount to:

- Reciprocate their high trust from Round 7
- Signal that I recognize and value our cooperative pattern
- Potentially establish a mutually beneficial repeated interaction

I should send at least as much as they did (around \$4.47) or potentially even more to show strong reciprocity.

Let me send \$4.5 to match their trust level and continue our cooperation.

usage:

```
{
  "input_tokens": 2607,
  "output_tokens": 484,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.5, it became \$13.5, and you saw them return \$8.1 to you. Your visible payoff was \$8.6.

Your total visible earnings across all rounds are \$45.79.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this new round with a new co-player:

****Previous round analysis:****

- Round 8: I sent \$4.5 (90%) to my partner and received \$8.1 back - excellent mutual cooperation!

****My sending history:****

- With new partners, I've typically sent \$3-\$3.5
- Higher trust levels have generally yielded good returns
- Round 8 showed that high trust can be very rewarding

****Current situation:****

- New partner with no shared history
- My overall experience has been positive
- I'll send a moderate-high amount to signal cooperation

```
'''json
{"send": 3.5}
'''
```

This sends \$3.5 (70% of endowment), which becomes \$10.50 for the receiver. It's a strong cooperative signal while managing risk with an unknown partner.

Reasoning:

Looking at the previous round:

- I sent \$4.5 (90%) to my co-player
- They returned \$8.1
- My payoff was \$8.6 (which means I kept \$0.5 and got back \$8.1)

That's excellent! They returned 60% of what they received (\$13.5), just like I did in Round 7.

Now Round 9 - NEW co-player (since pairings are randomized each round).

My history as sender:

- Round 1: Sent \$3, got back \$5.51 - good
- Round 3: Sent \$3.5, got back \$4.82 - okay
- Round 6: Sent \$3, got back \$4.44 - okay
- Round 8: Sent \$4.5, got back \$8.1 - excellent!

The Round 8 experience with high trust and high reciprocity was very successful. However, this is a NEW partner, so I don't have history with them.

I should send a balanced amount - not as high as Round 8 (which was with a proven partner), but showing moderate-to-good faith based on my overall positive experiences.

I'll send around \$3-\$3.5, which has been my typical approach with new partners.

```
Usage: {
  "input_tokens": 2565,
  "output_tokens": 498,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T09:28:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_4"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_4": "trustee",
      "Agent_7": "investor"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.5, it became \$13.5, and you saw them return \$7.41 to you. Your visible payoff was \$7.91.

Your total visible earnings across all rounds are \$14.25.

This round, you are the RECEIVER against your current co-player. They sent you \$3.8, so you received \$11.399999999999999. How much do you return? (0-11.399999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed good trust by sending \$3.80 (76% of their endowment)
- They kept \$1.20 for themselves
- I received \$11.40
- I should reciprocate their trust fairly

Comparing to previous rounds:

- In Round 1, when a sender sent 95.6%, I returned about 56% of what I received
- In Round 2, when I sent 90%, my co-player returned about 55% of what they received

For this sender who sent 76%, a fair return would be around 55-60% of what I received.

$\$11.40 \times 0.57 \approx \6.50

If I return \$6.50:

- Sender ends with: $\$1.20 + \$6.50 = \$7.70$ (good gain)
- I keep: $\$11.40 - \$6.50 = \$4.90$ (reasonable)
- Both benefit from the cooperation

{'return': 6.50}

reasoning:

Let me analyze this situation:

Round 2 recap:

- I was the sender and sent \$4.50
- It became \$13.50 for the receiver
- They returned \$7.41 to me
- My payoff: $\$5 - \$4.50 + \$7.41 = \7.91

Round 3 current situation:

- I'm the receiver
- Sender sent \$3.80 (76% of their endowment)
- I received \$11.40 (approximately, due to floating point)
- They kept \$1.20

This sender showed good trust by sending 76% of their endowment. Let me think about a fair return:

If I return \$6.00:

- Sender ends up with: $\$1.20 + \$6.00 = \$7.20$
- I keep: $\$11.40 - \$6.00 = \$5.40$
- Both benefit, sender gains more

If I return \$7.00:

- Sender ends up with: $\$1.20 + \$7.00 = \$8.20$
- I keep: $\$11.40 - \$7.00 = \$4.40$
- Sender gains significantly more

In Round 1, when I received \$14.34 from a sender who sent \$4.78 (95.6%), I returned \$8.00.

- That sender ended with \$8.22 (from \$0.22)
- I kept \$6.34
- Return ratio: $\$8.00/\$14.34 = 55.9\%$

In Round 2, when I sent \$4.50 and it became \$13.50, they returned \$7.41

- Return ratio: $\$7.41/\$13.50 = 54.9\%$

For consistency, if I use a similar return ratio (around 55%):

- $\$11.40 \times 0.55 = \6.27

Let me round to \$6.50 to be slightly more generous:

...

usage:

```
{
  "input_tokens": 1120,
  "output_tokens": 820,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.8, it became \$11.399999999999999, and you returned \$6.5. Your visible payoff was \$4.899999999999999.

Your total visible earnings across all rounds are \$19.15.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience so far:

- Round 1: As receiver, someone sent high trust (95.6%), I reciprocated fairly
- Round 2: I sent \$4.50 (90%), got back \$7.41 - good cooperation
- Round 3: As receiver, returned fairly when sent \$3.80

The pattern shows that high trust tends to be reciprocated with fair returns (around 55% return rate). My cooperative strategy has been working well - averaging over \$6 per round.

I'll continue with a strong but slightly varied approach. Sending \$4.00 (80%):

- Shows substantial trust
- Creates \$12.00 for the receiver through the 3x multiplier

- Should encourage reciprocation based on the established pattern
- Slightly more conservative than my last sending round to manage risk

{'send': 4.00}

reasoning:

Let me analyze my history:

Round 1 (RECEIVER): Sender sent \$4.78 (95.6%), I received \$14.34, returned \$8.00, kept \$6.34

Round 2 (SENDER): I sent \$4.50 (90%), became \$13.50, received back \$7.41, ended with \$7.91

Round 3 (RECEIVER): Sender sent \$3.80 (76%), I received \$11.40, returned \$6.50, kept \$4.90

Now Round 4, I'm the SENDER again with a new co-player.

Looking at the pattern:

- When I sent \$4.50 (90%), I got back \$7.41 from \$13.50 (about 55% return rate)
- When someone sent \$4.78 (95.6%), I returned \$8.00 from \$14.34 (about 56% return rate)
- When someone sent \$3.80 (76%), I returned \$6.50 from \$11.40 (about 57% return rate)

The return rates seem fairly consistent around 55-57%. This suggests cooperation is being reciprocated.

I've been doing well with high cooperation levels. My total earnings are \$19.15 across 3 rounds, averaging about \$6.38 per round, which is well above the baseline of \$5.

Given this positive experience with cooperation, I should continue sending a substantial amount. I'll send \$4.00 (80%), which is:

- Still showing significant trust
- Creating good value through the 3x multiplier (\$12.00 received)
- Likely to be reciprocated based on the pattern
- Slightly more conservative than my previous \$4.50 to diversify my strategy a bit

If they return around 55%, I'd get back about \$6.60, ending with \$1.00 + \$6.60 = \$7.60, which would be good.

usage:

```
{
  "input_tokens": 1488,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$4.58 to you. Your visible payoff was \$5.58.

Your total visible earnings across all rounds are \$24.729999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.73 (94.6% of their endowment)
- They kept only \$0.27 for themselves
- I received \$14.19
- This is very similar to Round 1, where someone sent 95.6%

In Round 1, when faced with similarly high trust, I returned \$8.00 out of \$14.34. That worked well as a fair reciprocation.

For this round with \$14.19 received, returning \$8.00:

- Sender ends with: \$0.27 + \$8.00 = \$8.27 (strong return on their trust)
- I keep: \$14.19 - \$8.00 = \$6.19
- Both benefit significantly from the cooperation

Consistency with my previous fair reciprocation of high trust seems appropriate.

```
{'return': 8.00}
```

reasoning:

Let me analyze this situation:

Round 5 - I'm the RECEIVER

- Sender sent \$4.73 (94.6% of their endowment) - very high trust!
- They kept only \$0.27
- I received \$14.19 after the 3x multiplier

This is similar to Round 1 where someone sent \$4.78 (95.6%) and I returned \$8.00.

The sender is showing exceptional trust, almost sending everything. I should reciprocate fairly.

If I return \$8.00:

- Sender ends with: \$0.27 + \$8.00 = \$8.27 (good gain from initial \$5)
- I keep: \$14.19 - \$8.00 = \$6.19
- Both benefit, sender gets rewarded for high trust

This seems consistent with my Round 1 behavior where I also returned \$8.00 for a similar high-trust send.

Actually, let me think about this more carefully. The sender sent 94.6%. In Round 1, when sender sent 95.6%, I returned \$8.00 out of \$14.34, which is about 56%.

$\$14.19 \times 0.56 \approx \7.95

So around \$8.00 seems right. Let me stick with that.

usage:

```
{
  "input_tokens": 1831,
  "output_tokens": 528,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.73, it became \$14.190000000000001, and you returned \$8.0. Your visible payoff was \$6.190000000000001.

Your total visible earnings across all rounds are \$30.919999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience as a sender:

- Round 2: Sent \$4.50 (90%), received \$7.41 back - good cooperation
- Round 4: Sent \$4.00 (80%), received \$4.58 back - lower reciprocation

The pattern suggests that higher trust levels tend to get better reciprocation. I'm averaging around \$6.18 per round with my cooperative strategy, which is working well.

As a receiver, I've been consistent in returning around 56% of what I receive, which has seemed fair to my co-players.

I'll send \$4.50 again (90% of endowment):

- Shows strong, consistent trust
- Previously resulted in good reciprocation
- Creates \$13.50 for the receiver
- Maintains my cooperative approach

```
{'send': 4.50}
```

reasoning:

Let me review the pattern of my play:

Round 1 (Receiver): Received \$14.34 from sender who sent \$4.78 (95.6%), returned \$8.00 ? kept \$6.34
 Round 2 (Sender): Sent \$4.50 (90%), received back \$7.41 ? total \$7.91
 Round 3 (Receiver): Received \$11.40 from sender who sent \$3.80 (76%), returned \$6.50 ? kept \$4.90
 Round 4 (Sender): Sent \$4.00 (80%), received back \$4.58 ? total \$5.58
 Round 5 (Receiver): Received \$14.19 from sender who sent \$4.73 (94.6%), returned \$8.00 ? kept \$6.19

My total visible earnings: \$30.92

As a sender:

- Round 2: Sent \$4.50, got back \$7.41 (55% return rate)
- Round 4: Sent \$4.00, got back \$4.58 (38% return rate)

As a receiver:

- Round 1: Returned \$8.00 out of \$14.34 (56% return rate) when sender sent 95.6%
- Round 3: Returned \$6.50 out of \$11.40 (57% return rate) when sender sent 76%
- Round 5: Returned \$8.00 out of \$14.19 (56% return rate) when sender sent 94.6%

I've been consistent in my returns - around 56% when I'm the receiver.

When I've been the sender:

- Higher sends seem to get better returns (Round 2 with \$4.50 got 55% back)

...

usage:

```
{
  "input_tokens": 2173,
  "output_tokens": 794,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.5, it became \$13.5, and you saw them return \$5.77 to you. Your visible payoff was \$6.27.

Your total visible earnings across all rounds are \$37.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

I'm playing the SAME co-player as Round 6. Last round with them:

- I sent \$4.50 (90%)
- They returned \$5.77 (about 43% of the \$13.50 they received)
- I ended with \$6.27

Their reciprocation rate (43%) was lower than some other players I've encountered (~55%), but not the worst. My payoff was still above average.

Playing the same person again, I should:

- Continue showing willingness to cooperate
- Adjust slightly for their demonstrated reciprocation pattern
- Signal awareness while maintaining relationship

I'll send \$4.00 (80%):

- Still shows substantial trust
- Creates \$12.00 for them
- Slightly more conservative given their previous behavior
- Keeps the cooperative dynamic going

```
{'send': 4.00}
```

reasoning:

Interesting - I'm playing as SENDER again against the SAME co-player from Round 6. This is important context.

In Round 6:

- I sent \$4.50 (90%)
- It became \$13.50
- They returned \$5.77
- My payoff was \$6.27

The return rate was $\$5.77/\$13.50 = 42.7\%$, which is lower than the ~55% I've been seeing from other players. They kept about \$7.73 while I ended with \$6.27 total.

Now I'm playing them again. This gives me information about their reciprocation behavior. They were less generous than others I've played with.

Options:

1. Send less to protect myself since they showed lower reciprocation
2. Send the same amount to see if they maintain consistency
3. Send more to encourage better reciprocation

Strategic considerations:

- They know I sent high last time, so there's a reputation aspect
- If I send less, it signals I'm adjusting to their lower reciprocation
- If I send the same or more, it shows I'm still willing to cooperate
- The 3x multiplier means cooperation still creates value even with lower return rates

I think I should adjust downward slightly. Sending around \$3.50-\$4.00 makes sense:

- Still shows willingness to cooperate
- Manages risk given their demonstrated lower reciprocation rate
- Creates \$10.50-\$12.00 for them

I'll go with \$3.50 (70%):

- Shows continued cooperation but adjusted trust
- Creates \$10.50 for them
- Keeps \$1.50 for me as baseline
- If they return around 43%, that's \$4.52, giving me \$6.02 total
- If they improve their return rate, I benefit more

Actually, let me reconsider.

Looking at my track record across all sender rounds, I'm averaging around \$6.18 per round, and last round's \$6.27 is actually slightly above that. Comparing their 42.7% return rate against my other interactions - Round 2 yielded 55% and Round 4 only 38% - suggests this player falls in the middle rather than being an outlier. Maybe I'm overweighting one round when the overall pattern shows reasonable consistency....

usage:

```
{
  "input_tokens": 2490,
  "output_tokens": 943,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.87 to you. Your visible payoff was \$9.87.

Your total visible earnings across all rounds are \$47.059999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent \$3.21 (64% of their endowment) - moderate trust
- They kept \$1.79 for themselves
- I received \$9.63

My established pattern as a receiver has been to return around 56-57% of what I receive:

- Round 1: Returned 56%
- Round 3: Returned 57%
- Round 5: Returned 56%

For consistency and fairness, I'll apply the same principle:
 $\$9.63 \times 0.56 \approx \5.40

If I return \$5.40:

- Sender ends with: $\$1.79 + \$5.40 = \$7.19$ (solid return)
- I keep: $\$9.63 - \$5.40 = \$4.23$
- Both benefit from cooperation

```
{'return': 5.40}
```

reasoning:

Looking at this situation:

- The sender sent \$3.21 (64.2% of their endowment)
- They kept \$1.79 for themselves
- I received \$9.63
- This is moderate trust, lower than some previous rounds

Let me review my pattern of returning:

- Round 1: Received \$14.34, returned \$8.00 (56%)
- Round 3: Received \$11.40, returned \$6.50 (57%)
- Round 5: Received \$14.19, returned \$8.00 (56%)

So I've been consistently returning around 56-57% of what I receive.

For this round with \$9.63 received:
 $\$9.63 \times 0.56 = \5.39

If I return \$5.40:

- Sender ends with: $\$1.79 + \$5.40 = \$7.19$ (good gain)
- I keep: $\$9.63 - \$5.40 = \$4.23$
- Both benefit from the cooperation

This maintains my consistent reciprocation pattern.

usage:

```
{
  "input_tokens": 2602,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.21, it became \$9.629999999999999, and you returned \$5.4. Your visible payoff was \$4.229999999999999.

Your total visible earnings across all rounds are \$51.289999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.89, so you received \$11.67. How much do you return? (0-11.67)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$3.89 (77.8% of their endowment) - good trust
- They kept \$1.11 for themselves
- I received \$11.67

My consistent pattern as a receiver has been to return around 56% of what I receive:

- Round 1: 56%
- Round 3: 57%
- Round 5: 56%
- Round 8: 56%

Applying the same fair reciprocation:

$\$11.67 \times 0.56 \approx \6.54

If I return \$6.54:
- Sender ends with: $\$1.11 + \$6.54 = \$7.65$ (good return on trust)
- I keep: $\$11.67 - \$6.54 = \$5.13$
- Both benefit from cooperation

{'return': 6.54}

Reasoning:

Looking at this situation:

- The sender sent \$3.89 (77.8% of their endowment) - good trust
- They kept \$1.11 for themselves
- I received \$11.67

My established pattern as a receiver has been to return around 56-57% of what I receive:

- Round 1: Returned \$8.00 out of \$14.34 = 55.8%
- Round 3: Returned \$6.50 out of \$11.40 = 57.0%
- Round 5: Returned \$8.00 out of \$14.19 = 56.4%
- Round 8: Returned \$5.40 out of \$9.63 = 56.1%

For consistency and fairness, I should apply the same principle:

$\$11.67 \times 0.56 \approx \6.54

If I return \$6.54:
- Sender ends with: $\$1.11 + \$6.54 = \$7.65$ (good return)
- I keep: $\$11.67 - \$6.54 = \$5.13$
- Both benefit from cooperation

This maintains my fair and consistent reciprocation pattern.

Usage: {
 "input_tokens": 2684,
 "output_tokens": 504,
 "reasoning_tokens": 0
}

GAME - Agent_1 - Sender - dyad_3

Timestamp: 2026-08-12T09:28:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_8",
 "pairing": {
 "agents": [
 "Agent_1",
 "Agent_8"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$5.66 to you. Your visible payoff was \$6.91.

Your total visible earnings across all rounds are \$12.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my recent experience:

- Round 2: I sent \$3.75 (75%), received \$5.66 back, ended with \$6.91 - a good outcome
- Round 1: As receiver, cooperation was rewarded

The cooperative approach has been working well. I'm playing with a new co-player now, so I'll continue signaling trust with a substantial send.

{'send': 3.75}

reasoning:

Let me analyze the previous round:

- I sent \$3.75 (75% of my endowment)
- It became \$11.25
- They returned \$5.66
- My payoff was \$5 - \$3.75 + \$5.66 = \$6.91

This worked out well! The receiver reciprocated reasonably (returning about 50% of what they received), and I gained \$1.91 above my initial endowment.

This is a new co-player this round. Based on my successful cooperative strategy so far:

- Round 1 as receiver: sender sent 75.4%, I returned fairly
- Round 2 as sender: I sent 75%, receiver returned fairly

The pattern of cooperation seems to be working. I should continue with a similar high-trust strategy. I'll send around 75% again to signal cooperation and trust.

\$3.75 worked well last time. I could send the same amount or similar.

usage:

```
{
  "input_tokens": 1074,
  "output_tokens": 323,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$6.34 to you. Your visible payoff was \$7.59.

Your total visible earnings across all rounds are \$20.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$4.09, so you received \$12.27. How much do you return? (0-12.27)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.09 (81.8% of their endowment)
- They currently have: \$5 - \$4.09 = \$0.91
- I received: \$12.27
- Total created: \$13.18

This is even more trust than I saw in Round 1 (when sender sent 75.4%). I want to reward this high level of cooperation appropriately.

If I return \$6.50:

- Sender ends with: \$0.91 + \$6.50 = \$7.41 (gain of \$2.41)
- I end with: \$12.27 - \$6.50 = \$5.77 (gain of \$0.77)

Both of us benefit, and the sender's exceptional trust is well-rewarded.

{'return': 6.50}

reasoning:

Let me analyze this situation:

- The sender sent \$4.09 out of \$5, which is 81.8% of their endowment - very high trust!
- They now have: \$5 - \$4.09 = \$0.91
- I received: \$12.27
- Total money in the system: \$0.91 + \$12.27 = \$13.18

This sender showed even more trust than my previous partners. I want to reciprocate appropriately.

Looking at my previous behavior:

- Round 1: As receiver, got \$11.31, returned \$5.50 (about 48.6%)
- The sender had sent 75.4%
- That seemed fair and both parties benefited

This sender sent 81.8%, even more generous. I should reciprocate with a good return.

If I return \$6.50:

- Sender ends with: \$0.91 + \$6.50 = \$7.41
- I end with: \$12.27 - \$6.50 = \$5.77
- Sender gains \$2.41 over initial \$5
- I gain \$0.77 over initial \$5

Both benefit, and the sender's higher risk/trust is rewarded with higher gains.

Actually, let me think about this differently. The sender sent 81.8% which is very high. In Round 1, sender sent 75.4% and I returned 48.6% of what I received.

For 81.8% send rate, maybe I should return around 50-53% to reward the extra trust?

$\$12.27 * 0.52 = \6.38

If I return \$6.40:

- Sender ends with: $\$0.91 + \$6.40 = \$7.31$
- I end with: $\$12.27 - \$6.40 = \$5.87$

This seems fair - both gain, sender gets rewarded for high trust.

...

usage:

```
{
  "input_tokens": 1310,
  "output_tokens": 685,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.09, it became \$12.27, and you returned \$6.5. Your visible payoff was \$5.77.

Your total visible earnings across all rounds are \$26.080000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my track record:

- As sender: Sent \$3.75 twice, received good returns (\$5.66 and \$6.34)
- As receiver: Responded fairly to cooperative senders

The high-trust strategy continues to work well across different co-players. I'll maintain this approach.

{'send': 3.75}

reasoning:

Let me review my history:

- Round 1: Receiver - sender sent \$3.77, I returned \$5.50, payoff \$5.81
- Round 2: Sender - I sent \$3.75, got back \$5.66, payoff \$6.91
- Round 3: Sender - I sent \$3.75, got back \$6.34, payoff \$7.59
- Round 4: Receiver - sender sent \$4.09, I returned \$6.50, payoff \$5.77

Total visible earnings: \$26.08

My sending strategy has been working well - sending around \$3.75 (75%) has led to good returns. In Round 4, I saw an even higher send amount (\$4.09, 81.8%) and I rewarded it well.

Now I'm facing a new co-player as sender. The cooperative strategy has been consistently working across different opponents. I should continue with a similar high-trust signal.

I'll send \$3.75 again, which has proven successful.

usage:

```
{
  "input_tokens": 1637,
  "output_tokens": 324,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$6.54 to you. Your visible payoff was \$7.79.

Your total visible earnings across all rounds are \$33.870000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$4.1, so you received \$12.299999999999999. How much do you return? (0-12.299999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.10 (82% of their endowment)
- They currently have: \$5 - \$4.10 = \$0.90
- I received: \$12.30
- Total created: \$13.20

This is very similar to Round 4 where the sender sent \$4.09 and I returned \$6.50, which worked well for both parties.

If I return \$6.50:

- Sender ends with: \$0.90 + \$6.50 = \$7.40
- I end with: \$12.30 - \$6.50 = \$5.80

Both of us benefit substantially, rewarding the sender's high trust.

{'return': 6.50}

reasoning:

The sender sent \$4.10, which is 82% of their endowment - very high trust again!

They currently have: \$5 - \$4.10 = \$0.90

I received: \$12.30

Total created: \$13.20

This is similar to Round 4 where sender sent \$4.09. I returned \$6.50 then.

Let me be consistent and reward this high trust appropriately.

If I return \$6.50:

- Sender ends with: \$0.90 + \$6.50 = \$7.40
- I end with: \$12.30 - \$6.50 = \$5.80

Both benefit, sender gets a good reward for their trust.

usage:

```
{
  "input_tokens": 1863,
  "output_tokens": 354,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$4.1, it became \$12.299999999999999, and you returned \$6.5. Your visible payoff was \$5.799999999999999.

Your total visible earnings across all rounds are \$39.67.

This round, you are the RECEIVER against your current co-player. They sent you \$5.0, so you received \$15.0. How much do you return? (0-15.0)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender showed **complete trust** by sending \$5.00 (100% of their endowment!)
- They currently have: \$0
- I received: \$15.00
- This is the same co-player from Round 6, and they escalated their trust after I returned fairly

This maximum trust deserves a strong reciprocal response.

If I return \$8.00:

- Sender ends with: \$8.00 (gain of \$3.00)
- I end with: \$7.00 (gain of \$2.00)

Both of us benefit substantially, and their complete trust is well-rewarded.

```
{'return': 8.00}
```

reasoning:

This is interesting - I'm playing with the same co-player from Round 6. In Round 6:

- They sent \$4.10
- I received \$12.30
- I returned \$6.50
- They ended with $\$0.90 + \$6.50 = \$7.40$

Now in Round 7:

- They sent \$5.00 (100% of their endowment!) - maximum trust
- I received \$15.00
- They currently have: \$0

This is the ultimate show of trust - they sent everything! This deserves a strong reciprocal response.

If I return \$8.00:

- Sender ends with: $\$0 + \$8.00 = \$8.00$ (gain of \$3.00)
- I end with: $\$15.00 - \$8.00 = \$7.00$ (gain of \$2.00)

Both benefit nicely, and I'm rewarding their complete trust.

Alternatively, if I return \$7.50:

- Sender ends with: \$7.50 (gain of \$2.50)
- I end with: \$7.50 (gain of \$2.50)
- Equal split of the gains

Or if I return \$8.50:

- Sender ends with: \$8.50 (gain of \$3.50)
- I end with: \$6.50 (gain of \$1.50)
- More reward for their maximum trust

Given they showed complete trust by sending 100%, I think returning \$8.00 is appropriate - it gives them a strong reward while we both benefit substantially.

usage:

```
{
  "input_tokens": 2186,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$5.0, it became \$15.0, and you returned \$8.0. Your visible payoff was \$7.0

Your total visible earnings across all rounds are \$46.67.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my experience:

- As sender: \$3.75 has consistently yielded good returns (\$5.66, \$6.34, \$6.54)
- As receiver: I've rewarded trust fairly, including returning \$8.00 when someone sent \$5.00

The cooperative approach continues to work well. I'll maintain my high-trust strategy with this new co-player.

```
{'send': 3.75}
```

reasoning:

Looking at my history:

- As sender: I've sent \$3.75 multiple times and received good returns (\$5.66, \$6.34, \$6.54)
- As receiver: I've reciprocated fairly, especially rewarding high trust
- In rounds 6-7, I just played with someone who escalated from \$4.10 to \$5.00, and I reciprocated well

Now I'm the sender with a new co-player. The cooperative strategy has been working very well. I could:

1. Continue with \$3.75 (75%) - my standard high-trust amount
2. Send \$5.00 (100%) - maximum trust, like the last sender showed me
3. Send something in between like \$4.00-\$4.50

Given that:

- Cooperation has been consistent across the game
- I've built good earnings (\$46.67 total)
- Maximum trust was just shown to me and I rewarded it well
- This might encourage my new co-player to reciprocate strongly

I think I'll stick with my proven \$3.75 strategy. It's high trust without being maximum risk, and it's worked consistently well.

usage:

```
{
  "input_tokens": 2190,
  "output_tokens": 371,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$5.29 to you. Your visible payoff was \$6.54.

Your total visible earnings across all rounds are \$53.21.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my pattern:

- Sent \$3.75 in Rounds 2, 3, 5, and 8
- Returns ranged from \$5.29 to \$6.54, all profitable
- My fair behavior as receiver should encourage reciprocity

Continuing with my proven cooperative strategy.

```
{'send': 3.75}
```

Reasoning:

Looking at my track record as sender:

- Round 2: Sent \$3.75, received \$5.66 back, payoff \$6.91
- Round 3: Sent \$3.75, received \$6.34 back, payoff \$7.59
- Round 5: Sent \$3.75, received \$6.54 back, payoff \$7.79
- Round 8: Sent \$3.75, received \$5.29 back, payoff \$6.54

The strategy of sending \$3.75 has been consistently profitable, with returns ranging from \$5.29 to \$6.54. All outcomes have been positive.

As receiver, I've been cooperative and fair, returning substantial amounts when senders showed trust.

This is a new co-player, so I should continue with my established cooperative strategy of sending \$3.75.

Usage: {

```
"input_tokens": 2202,
"output_tokens": 286,
"reasoning_tokens": 0
}
```

```
}
```

GAME - Agent_8 - Receiver - dyad_3

Timestamp: 2026-08-12T09:28:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_3",

"move_index": 5,

"opponent_id": "Agent_1",

"pairing": {

"agents": [

"Agent_1",

"Agent_8"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_8": "trustee"

},

"trustee": "Agent_8"

},

"role": "trustee",

"role_label": "Receiver",

"round": 9,

"task": "game",

"task_index": 0

```
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized

- , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.58, it became \$13.74, and you returned \$6.5. Your visible payoff was \$7.24.

Your total visible earnings across all rounds are \$12.61.

This round, you are the RECEIVER against your current co-player. They sent you \$3.38, so you received \$10.14. How much do you return? (0-10.14)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust.

I received \$10.14.

They kept: \$1.62

To reciprocate fairly:

- If I return \$5.00, they'd end with \$6.62 (a gain of \$1.62)
- I'd keep \$5.14 (also a small gain)

This represents returning about 49% of what I received, which rewards their trust while we both benefit.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Looking at this situation:

My co-player sent me \$3.38 out of their \$5 (67.6% of their endowment) - showing good trust.

I received \$10.14.

They kept: \$5 - \$3.38 = \$1.62

To be fair and reciprocate their trust:

- If I return around \$5, they'd end with \$1.62 + \$5 = \$6.62 (a gain of \$1.62)
- I'd keep \$10.14 - \$5 = \$5.14 (also a solid gain)

This represents returning about 49% of what I received, which is fair and reciprocates their trust appropriately while we both benefit.

Let me think about this more carefully. They sent \$3.38, showing moderate-to-good trust (68% of endowment).

In Round 2, my co-player sent me \$4.58 (92% of endowment) and I returned \$6.50 (47% of what I received).

For consistency and fairness here with \$3.38 sent (less trust shown):

- Returning around 47-50% of \$10.14 would be \$4.77-\$5.07
- Let me go with \$5.00

That gives them: \$1.62 + \$5.00 = \$6.62 (gain of \$1.62)

I keep: \$10.14 - \$5.00 = \$5.14 (gain of \$0.14)

This seems fair and reciprocal.

usage:


```
{
  "input_tokens": 1036,
  "output_tokens": 519,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.38, it became \$10.14, and you returned \$5.0. Your visible payoff was \$5.1400000000000001.

Your total visible earnings across all rounds are \$17.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience so far:

Round 1: As SENDER, I sent \$3 and received \$3.37 back - a reasonable return.

Rounds 2-3: As RECEIVER, others showed good trust, and I reciprocated fairly.

This is a new co-player (random pairing), so I don't have specific information about them. However, the overall pattern suggests players are reasonably cooperative.

I'll send \$3.50 - slightly more than my first round to show increasing trust while remaining measured:

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$3, received back \$3.37, payoff \$5.37

Round 2: I was RECEIVER, received \$13.74 (from \$4.58), returned \$6.50, payoff \$7.24

Round 3: I was RECEIVER, received \$10.14 (from \$3.38), returned \$5.00, payoff \$5.14

Round 4: I am SENDER

My total earnings: \$17.75

Looking at the pattern:

- When I sent \$3, I got back \$3.37 (a bit more than 1:1 return on sent amount)
- When others sent to me, they showed high trust (\$4.58 and \$3.38 out of \$5)
- I've been fairly reciprocal as a receiver, returning roughly half of what I received

Now I'm a sender again. Since:

1. My first send of \$3 was met with a decent return (\$3.37)
2. Others have been trusting with me
3. I've been fair in return

I should continue with moderate to high trust. I'll send \$3.5 this time - slightly more than my first send to show I'm willing to build cooperation.

Actually, let me reconsider. The co-player this round is new (pairings are randomized). I don't know their history. In Round 1, I sent \$3 and got a reasonable return. Let me stick with a similar strategy - send \$3 again as a baseline cooperative strategy.

Actually, looking at the overall pattern, players seem reasonably cooperative. I could try \$3.5 to be slightly more trusting.

Let me go with \$3.5 - showing increased trust while not overcommitting.

usage:

```
{
  "input_tokens": 1303,
  "output_tokens": 551,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$7.22 to you. Your visible payoff was \$8.719999999999999.

Your total visible earnings across all rounds are \$26.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my recent experience:

Round 4: I sent \$3.50 and received \$7.22 back - excellent cooperation!

The overall pattern shows players are being reasonably cooperative. While this is a new co-player (random pairing), I'll show increased trust based on positive experiences.

Sending \$4:

- Demonstrates strong trust (80% of my endowment)
- Creates \$12 for them to work with
- Encourages reciprocal cooperation

```
'''json
{"send": 4}
'''
```

reasoning:

Round 4 recap: I sent \$3.50, they returned \$7.22 - excellent return! That's about 69% of what they received, very generous.

This is Round 5, and I'm the SENDER again against a new co-player (random pairing each round).

So far my experience:

- Round 1 SENDER: sent \$3, got \$3.37 back (reasonable)
- Round 2 RECEIVER: received \$13.74, returned \$6.50
- Round 3 RECEIVER: received \$10.14, returned \$5.00
- Round 4 SENDER: sent \$3.50, got \$7.22 back (very good!)

The last round showed great cooperation. Players seem generally trustworthy. This is a new co-player though, so I don't know their specific behavior.

I should probably continue with a trusting approach. I could:

- Send \$4 (80% of endowment) - shows high trust
- Send \$3.5 (70%) - similar to last round
- Send \$3 (60%) - more cautious

Given the positive experiences, I'll increase slightly to \$4 to show strong trust.

usage:

```
{
  "input_tokens": 1562,
  "output_tokens": 402,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.98 to you. Your visible payoff was \$7.98.

Your total visible earnings across all rounds are \$34.45.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

My co-player sent me \$2.9 out of their \$5 (58% of their endowment) - showing moderate trust.

I received \$8.7.

They kept: \$2.1

To reciprocate fairly:

- If I return \$4.00, they'd end with \$6.10 (a gain of \$1.10)
- I'd keep \$4.70 (also a decent gain)

This represents returning about 46% of what I received, which is proportional to the moderate trust shown and ensures we both benefit.

```
'''json
{"return": 4.00}
'''
```

reasoning:

Looking at this situation:

My co-player sent me \$2.9 out of their \$5 (58% of their endowment) - showing moderate trust.

I received \$8.7.

They kept: \$2.1

To reciprocate fairly:

- If I return around \$4.00-\$4.50, they'd end with \$6.10-\$6.60 (a gain of \$1.10-\$1.60)
- I'd keep \$4.70-\$4.20

This represents returning about 46-52% of what I received, which rewards their trust while we both benefit.

Let me think about what's fair. They sent \$2.9, showing moderate but not high trust. I should return something proportional.

In previous rounds as receiver:

- Round 2: Received \$13.74, returned \$6.50 (47%)
- Round 3: Received \$10.14, returned \$5.00 (49%)

So I've been returning around 47-49% of what I receive.

For \$8.7, returning around 47-49% would be:

- 47%: \$4.09
- 49%: \$4.26

Let me go with \$4.25, which is about 49% and gives them a total of \$6.35 (gain of \$1.35), while I keep \$4.45 (also a decent gain).

Actually, let me reconsider. They sent \$2.9 (58% of endowment), which is moderate trust. In Round 1, someone sent me \$4.58 (91.6%) and I returned \$6.50. In Round 3, someone sent \$3.38 (67.6%) and I returned \$5.00.

The person who sent \$3.38 (67.6%) got back \$5.00 from \$10.14 received.

...

usage:

```
{
  "input_tokens": 1817,
  "output_tokens": 654,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.0. Your visible payoff was \$4.699999999999999.

Your total visible earnings across all rounds are \$39.150000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

My co-player sent me \$2.99 out of their \$5 (about 60% of their endowment) - showing moderate trust, very similar to Round 6.

I received \$8.97.

They kept: \$2.01

To reciprocate consistently with my Round 6 approach:

- If I return \$4.10, they'd end with \$6.11 (a gain of \$1.11)
- I'd keep \$4.87 (also a decent gain)

This represents returning about 46% of what I received, consistent with similar trust levels.

```
'''json
{"return": 4.10}
'''
```

reasoning:

My co-player sent me \$2.99 out of their \$5 (about 60% of their endowment) - showing moderate trust, very similar to the previous round.

I received \$8.97.

They kept: \$2.01

This is very similar to Round 6 where they sent \$2.9. I should be consistent in my reciprocation strategy.

In Round 6, I returned \$4.00 when I received \$8.7, which was about 46% of what I received.

Using a similar proportion:

- If I return \$4.10, they'd end with \$6.11 (a gain of \$1.11)
- I'd keep \$4.87 (also a decent gain)

This represents returning about 46% of what I received, consistent with my previous approach for similar trust levels.

Let me calculate more carefully:

- They sent: \$2.99
- They kept: \$5 - \$2.99 = \$2.01
- I received: \$8.97
- If I return \$4.10:
 - They end with: \$2.01 + \$4.10 = \$6.11
 - I end with: \$8.97 - \$4.10 = \$4.87

This seems fair and consistent with my previous behavior.

usage:

```
{
  "input_tokens": 2105,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.99, it became \$8.97, and you returned \$4.1. Your visible payoff was \$4.870000000000001.

Your total visible earnings across all rounds are \$44.02000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my experience:

As SENDER:

- Round 1: sent \$3.00, received \$3.37 back
- Round 4: sent \$3.50, received \$7.22 back (excellent!)
- Round 5: sent \$4.00, received \$6.98 back (good!)

The pattern shows that sending more has generally been rewarded with good reciprocation. This is Round 8 with a new co-player.

I'll send \$4.00 - this amount has proven successful and demonstrates strong trust while maintaining some prudence.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history:

Round 1: SENDER - sent \$3.00, received \$3.37 back (payoff: \$5.37)
Round 2: RECEIVER - received \$13.74 from \$4.58 sent, returned \$6.50 (payoff: \$7.24)
Round 3: RECEIVER - received \$10.14 from \$3.38 sent, returned \$5.00 (payoff: \$5.14)
Round 4: SENDER - sent \$3.50, received \$7.22 back (payoff: \$8.72)
Round 5: SENDER - sent \$4.00, received \$6.98 back (payoff: \$7.98)
Round 6: RECEIVER - received \$8.70 from \$2.90 sent, returned \$4.00 (payoff: \$4.70)
Round 7: RECEIVER - received \$8.97 from \$2.99 sent, returned \$4.10 (payoff: \$4.87)
Round 8: SENDER - this round, new co-player

My sending pattern has been:

- Round 1: \$3.00
- Round 4: \$3.50
- Round 5: \$4.00

I've been gradually increasing trust. The returns have generally been good:

- Round 1: sent \$3, got \$3.37 back (net: +\$0.37)
- Round 4: sent \$3.5, got \$7.22 back (net: +\$2.22) - excellent!
- Round 5: sent \$4, got \$6.98 back (net: +\$1.48) - good!

Overall, sending more has been rewarded. This is a final round (Round 8), and a new co-player.

Given the positive trend and it being the final round, I could send more to maximize potential returns. ...

usage:

```
{
  "input_tokens": 2237,
  "output_tokens": 647,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$9.55 to you. Your visible payoff was \$10.55.

Your total visible earnings across all rounds are \$54.57000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

My co-player sent me \$3.11 out of their \$5 (62% of their endowment) - showing good trust.

I received \$9.33.

They kept: \$1.89

To reciprocate fairly:

- If I return \$4.50, they'd end with \$6.39 (a gain of \$1.39)
- I'd keep \$4.83 (also a good gain)

This represents returning about 48% of what I received, consistent with my reciprocity pattern for good trust levels.

```
'''json
{"return": 4.50}
'''
```

Reasoning:

Looking at this situation:

My co-player sent me \$3.11 out of their \$5 (62.2% of their endowment) - showing good trust.

I received \$9.33.

They kept: \$1.89

Let me think about a fair return:

- If I return \$4.50, they'd end with \$6.39 (a gain of \$1.39)
- I'd keep \$4.83 (also a good gain)

This represents returning about 48% of what I received, which is consistent with my previous returns for similar trust levels.

Looking at my past behavior as receiver:

- Round 2: sent \$4.58, received \$13.74, returned \$6.50 (47%)
- Round 3: sent \$3.38, received \$10.14, returned \$5.00 (49%)
- Round 6: sent \$2.90, received \$8.70, returned \$4.00 (46%)
- Round 7: sent \$2.99, received \$8.97, returned \$4.10 (46%)

For \$3.11 sent (\$9.33 received), a return around 46-49% would be \$4.30-\$4.57.

I'll return \$4.50, which:

- Is proportional to the trust shown
- Ensures both parties benefit
- Is consistent with my previous reciprocity patterns

```
Usage: {
  "input_tokens": 2237,
  "output_tokens": 477,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Sender - dyad_4

Timestamp: 2026-08-12T09:28:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
  so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
  , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.

  MULTI-AGENT SETTING:
  - There are 8 agents in this run.
  - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
    , but the schedule balances sender and receiver roles across the run.
  - You play one sender-receiver game with your paired opponent each round.
  - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
    ll have acted as sender and receiver equally often when the round count allows it.
  - Agent names are hidden in this run.
  - Your paired opponent is referred to as your current co-player.
Message 2 role=user
content:
  Round 3

  In your most recent previous game round, you were the SENDER against another agent. You se
  nt $2.5, it became $7.5, and you saw them return $6.78 to you. Your visible payoff was $
  9.280000000000001.
  Your total visible earnings across all rounds are $13.18.

  This round, you are the RECEIVER against your current co-player. They sent you $4.17, so y
  ou received $12.51. How much do you return? (0-12.51)
  Respond exactly as JSON: {'return': <amount>}
Message 3 role=assistant
content:
  Looking at this situation:

  - The sender showed exceptional trust by sending $4.17 (83.4% of their endowment)
  - I received $12.51 after the 3x multiplier
  - The sender currently has only $0.83 remaining

  This is very high trust that deserves to be rewarded generously.

```

If I return \$6.0:

- Sender ends with: $\$0.83 + \$6.0 = \$6.83$ (profit of \$1.83)
- I end with: $\$12.51 - \$6.0 = \$6.51$ (profit of \$6.51)

This gives us nearly equal final payoffs and rewards the sender's exceptional trust with a solid profit.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me analyze this situation:

Round 3: I'm the RECEIVER

- The sender sent \$4.17 (83.4% of their \$5 endowment) - this is very high trust!
- I received \$12.51 (after 3x multiplier)
- The sender currently has $\$5 - \$4.17 = \$0.83$ remaining

This sender showed enormous trust by sending 83.4% of their endowment. I need to reward this trust generously.

Let me think about what's fair:

- Total in the system: $\$0.83$ (sender has) + $\$12.51$ (I have) = $\$13.34$
- If we split evenly: $\$6.67$ each

If I return \$5.84:

- Sender ends with: $\$0.83 + \$5.84 = \$6.67$
- I end with: $\$12.51 - \$5.84 = \$6.67$
- Perfect 50/50 split

But given they showed such high trust (83.4%), I might want to be even more generous to:

1. Reward their exceptional trust
2. Encourage continued cooperation
3. Establish myself as highly trustworthy

However, I also saw in round 2 that when I sent \$2.5 (50%), I got back \$6.78, which gave me a payoff of \$9.28. That was very generous from my co-player then (they returned more than 90% of what they received!).

Let me recalculate for this round:

- If I return \$6:
 - Sender: $\$0.83 + \$6 = \$6.83$
 - Me: $\$12.51 - \$6 = \$6.51$
 - Almost equal, slight advantage to sender
- If I return \$6.5:
 - Sender: $\$0.83 + \$6.5 = \$7.33$
 - Me: $\$12.51 - \$6.5 = \$6.01$
 - Sender gets more, rewarding their trust

Given the extreme trust shown here, I should return enough to give them a meaningful advantage while still being reasonable. Returning around \$6 seems appropriate-it creates nearly equal outcomes with a slight edge to the sender who took the bigger risk....

usage:

```
{
  "input_tokens": 1154,
  "output_tokens": 955,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.17, it became \$12.51, and you returned \$6.0. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$19.689999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience so far:

- Round 2: When I sent \$2.5 (50%), I received a very generous return of \$6.78
- Round 3: When someone sent me \$4.17 (high trust), I reciprocated with \$6.0

The game seems to reward trust. I'll send \$3.0 (60% of my endowment), which shows strong trust while being measured with a new co-player.

```
'''json
{"send": 3.0}
'''
```

reasoning:

Looking at my history:

- Round 1: RECEIVER - received from sender who sent \$2.3, I returned \$3.0 fairly
- Round 2: SENDER - I sent \$2.5, receiver returned \$6.78 (very generous!)
- Round 3: RECEIVER - received from sender who sent \$4.17 (very high trust), I returned \$6.0

Current situation:

- I'm the SENDER with a new co-player
- Total earnings so far: \$19.69

My experience shows:

- When I sent \$2.5 in Round 2, I got a very generous return of \$6.78
- When others showed high trust to me, I reciprocated fairly
- This game seems to reward trust

I should send a significant amount to show trust. Given that:

- Round 2 sender sent 46% and I reciprocated
- Round 3 sender sent 83.4% and I reciprocated generously
- When I sent 50% in Round 2, I got excellent returns

I think sending around \$3.0 (60%) would be a good balance - showing strong trust without being reckless with an unknown partner.

Actually, let me think about this more strategically. I've seen that:

- Lower trust (46%) got fair returns
- Medium trust (50% from me) got very generous returns
- High trust (83.4%) got generous returns

Sending \$3.0 seems like a good middle-high trust level that should encourage reciprocation.

usage:

```
{
  "input_tokens": 1461,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.22 to you. Your visible payoff was \$7.22.

Your total visible earnings across all rounds are \$26.909999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.84, so you received \$11.52. How much do you return? (0-11.52)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed high trust by sending \$3.84 (76.8% of their endowment)
- I received \$11.52 after the 3x multiplier

- The sender currently has only \$1.16 remaining

This level of trust deserves a generous return.

If I return \$6.0:

- Sender ends with: $\$1.16 + \$6.0 = \$7.16$ (profit of \$2.16)
- I end with: $\$11.52 - \$6.0 = \$5.52$

This rewards their high trust with a good profit and keeps our final payoffs relatively balanced.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$3.84, which is 76.8% of their endowment - very high trust
- I received \$11.52 after the 3x multiplier
- The sender currently has $\$5 - \$3.84 = \$1.16$ remaining

This is high trust that should be rewarded well.

Let me consider different return amounts:

If I return \$5.5:

- Sender ends with: $\$1.16 + \$5.5 = \$6.66$ (profit of \$1.66)
- I end with: $\$11.52 - \$5.5 = \$6.02$

If I return \$6.0:

- Sender ends with: $\$1.16 + \$6.0 = \$7.16$ (profit of \$2.16)
- I end with: $\$11.52 - \$6.0 = \$5.52$

Given the high trust (76.8%), I should return generously. \$6.0 seems fair and rewards their trust well.

usage:

```
{
  "input_tokens": 1718,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.84, it became \$11.52, and you returned \$6.0. Your visible payoff was \$5.52.

Your total visible earnings across all rounds are \$32.42999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience:

- Round 2: Sent \$2.5 (50%), received \$6.78 back - great return
- Round 4: Sent \$3.0 (60%), received \$5.22 back - solid return

The game has been rewarding trust. I'll continue to show strong trust with this new co-player by sending \$3.5 (70% of my endowment).

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze my experience:

Round 1 (Receiver): Received \$6.9 from sender sending \$2.3, returned \$3.0. Payoff: \$3.9

Round 2 (Sender): Sent \$2.5, received \$6.78 back. Payoff: \$9.28
Round 3 (Receiver): Received \$12.51 from sender sending \$4.17, returned \$6.0. Payoff: \$6.5
1
Round 4 (Sender): Sent \$3.0, received \$5.22 back. Payoff: \$7.22
Round 5 (Receiver): Received \$11.52 from sender sending \$3.84, returned \$6.0. Payoff: \$5.5
2

So far:

- As a sender, I've sent \$2.5 and \$3.0 (50% and 60%)
- Both times I received generous returns (\$6.78 and \$5.22)
- As a receiver, I've been fair/generous in my returns

The game continues to reward trust. I've been gradually increasing my sending amounts. Let me continue this trend and send \$3.5 (70% of endowment) to show strong trust with this new co-player.

usage:

```
{  
  "input_tokens": 2009,  
  "output_tokens": 417,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.22 to you. Your visible payoff was \$7.72.

Your total visible earnings across all rounds are \$40.14999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my sender history:

- Round 2: Sent \$2.5, received \$6.78 back
- Round 4: Sent \$3.0, received \$5.22 back
- Round 6: Sent \$3.5, received \$6.22 back

Trust has been consistently rewarded. I'll continue showing strong trust with this new co-player by sending \$4.0 (80% of my endowment).

```
'''json  
{"send": 4.0}  
'''
```

reasoning:

Looking at my history as sender:

- Round 2: Sent \$2.5 (50%), received \$6.78 back - excellent return
- Round 4: Sent \$3.0 (60%), received \$5.22 back - good return
- Round 6: Sent \$3.5 (70%), received \$6.22 back - good return

All three times I've sent money as sender, I've received generous returns. The game environment seems cooperative and trust is being rewarded.

As receiver:

- Round 1: Received \$6.9, returned \$3.0
- Round 3: Received \$12.51, returned \$6.0
- Round 5: Received \$11.52, returned \$6.0

I've been fair as a receiver too.

The pattern suggests the game is cooperative. I could continue increasing my trust level. Maybe send \$4.0 (80%) this time? That would show strong trust with this new co-player.

Let me think about the strategy:

- Sending more has generally worked well
- I'm playing against a new co-player each round
- Building reputation for cooperation could help
- Total earnings are healthy at \$40.15

I'll send \$4.0 (80% of endowment) to show strong trust.

usage:

```
{
  "input_tokens": 2246,
  "output_tokens": 423,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$8.61 to you. Your visible payoff was \$9.61.

Your total visible earnings across all rounds are \$49.75999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.61, so you received \$13.830000000000002. How much do you return? (0-13.830000000000002)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- This is the same co-player from Round 7!
- Last round, I sent them \$4.0 and they returned an exceptional \$8.61
- Now they've sent me \$4.61 (92.2% of their endowment) - extremely high trust
- I received \$13.83 after the 3x multiplier
- They currently have only \$0.39 remaining

My co-player has shown exceptional generosity and trust. I should reciprocate strongly.

If I return \$8.0:

- Sender ends with: \$0.39 + \$8.0 = \$8.39 (profit of \$3.39)
- I end with: \$13.83 - \$8.0 = \$5.83

This matches their previous generosity and rewards their high trust appropriately.

```
'''json
{"return": 8.0}
'''
```

reasoning:

Let me analyze this situation:

- This is the same co-player from Round 7
- In Round 7: I sent \$4.0, they returned \$8.61 - that's an exceptional return! They were very generous.
- Now in Round 8: They sent me \$4.61 (92.2% of their endowment) - showing extremely high trust
- I received \$13.83 after the 3x multiplier
- They currently have only \$0.39 remaining

My co-player has been very generous to me. In Round 7, they returned \$8.61 out of \$12.0 (71.75% return rate). That's excellent reciprocity.

Now they're showing even MORE trust by sending \$4.61. I should definitely reciprocate generously.

If I return \$7.0:

- Sender ends with: \$0.39 + \$7.0 = \$7.39 (profit of \$2.39)
- I end with: \$13.83 - \$7.0 = \$6.83

If I return \$8.0:

- Sender ends with: \$0.39 + \$8.0 = \$8.39 (profit of \$3.39)
- I end with: \$13.83 - \$8.0 = \$5.83

Given they returned \$8.61 last round and now sent me \$4.61, I should be similarly generous. Returning around \$7-8 seems appropriate.

Let me go with \$7.5 to be quite generous while still benefiting from the exchange:

- Sender ends with: $\$0.39 + \$7.5 = \$7.89$ (profit of $\$2.89$)
- I end with: $\$13.83 - \$7.5 = \$6.33$

Actually, given they returned $\$8.61$ last round (an exceptional amount), I should probably return at least $\$8.0$ to match their generosity level.

...

usage:

```
{
  "input_tokens": 2249,
  "output_tokens": 745,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send $\$4.61$, it became $\$13.830000000000002$, and you returned $\$8.0$. Your visible payoff was $\$5.830000000000002$.

Your total visible earnings across all rounds are $\$55.58999999999999$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my sender history:

- Round 2: Sent $\$2.5$, received $\$6.78$ back
- Round 4: Sent $\$3.0$, received $\$5.22$ back
- Round 6: Sent $\$3.5$, received $\$6.22$ back
- Round 7: Sent $\$4.0$, received $\$8.61$ back (exceptional return!)

Trust has been consistently rewarded, with increasingly better returns. I'll continue showing strong trust with this new co-player by sending $\$4.0$ (80% of my endowment).

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Looking at my history:

Round 2: SENDER - Sent $\$2.5$, received $\$6.78$ back
 Round 3: RECEIVER - Received $\$12.51$ (from $\$4.17$ sent), returned $\$6.0$
 Round 4: SENDER - Sent $\$3.0$, received $\$5.22$ back
 Round 5: RECEIVER - Received $\$11.52$ (from $\$3.84$ sent), returned $\$6.0$
 Round 6: SENDER - Sent $\$3.5$, received $\$6.22$ back
 Round 7: SENDER - Sent $\$4.0$, received $\$8.61$ back
 Round 8: RECEIVER - Received $\$13.83$ (from $\$4.61$ sent), returned $\$8.0$

Now Round 9: SENDER against a new co-player

My sending pattern has been increasing:

$\$2.5$? $\$3.0$? $\$3.5$? $\$4.0$

And returns have been good, with the last one being exceptional ($\$8.61$ return on $\$4.0$ sent).

The game has consistently rewarded trust. I should continue showing high trust. I'll send $\$4.0$ again, which has been working well.

Actually, looking at the progression, I could go even higher. But $\$4.0$ (80%) seems like a strong trust signal that has worked well.

Usage: {

```
"input_tokens": 2312,
"output_tokens": 474,
"reasoning_tokens": 0
```

}

GAME - Agent_6 - Receiver - dyad_4

Timestamp: 2026-08-12T09:28:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 3

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.42 to you. Your visible payoff was \$8.42.

Your total visible earnings across all rounds are \$12.17.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- My co-player showed **maximum trust** by sending their entire \$5 endowment (100%)
- This became \$15 for me
- They currently have \$0

This exceptional level of trust deserves strong reciprocation. If I return \$10.00:

- The sender ends with \$10.00 (doubling their money, gain of \$5.00)
- I keep \$5.00 (gain of \$5.00)
- We both gain equally from our cooperation

This rewards their maximum trust appropriately and creates a fair outcome where both parties benefit substantially.

{"return": 10.00}

reasoning:

Let me analyze this situation:

Previous round:

- I was the SENDER
- I sent \$3.00 (60% of my endowment)
- My co-player received \$9.00
- They returned \$6.42 (71.3% of what they received)
- I ended with \$8.42

This round:

- I am the RECEIVER
- My current co-player sent \$5.00 (100% of their endowment!)
- I received \$15.00
- They currently have \$0
- I need to decide how much to return (0-15)

This is a very strong show of trust - they sent their entire endowment! This deserves a strong reciprocation.

Let me think about fair outcomes:

- If I return \$7.50, they end with \$7.50 (gain of \$2.50), I keep \$7.50 (gain of \$7.50)
- If I return \$10.00, they end with \$10.00 (gain of \$5.00), I keep \$5.00 (gain of \$5.00) - equal split of total gains
- If I return \$12.00, they end with \$12.00 (gain of \$7.00), I keep \$3.00 (gain of \$3.00)

Given they showed maximum trust by sending everything, I should reciprocate generously. An equal split of total gains seems fair - we'd each gain \$5 from the cooperation. That means returning \$10.00.

Actually, let me reconsider. They sent everything (\$5), which became \$15 for me. If I return \$10, they get \$10 total (doubling their money), and I keep \$5 (gaining \$5). Both of us gain equally from the transaction.

This seems like a fair and generous reciprocation for their maximum trust.

usage:

```
{
  "input_tokens": 1082,
  "output_tokens": 580,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$5, it became \$15, and you returned \$10.0. Your visible payoff was \$5.0. Your total visible earnings across all rounds are \$17.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my experience so far:

- Round 2 (sender): Sent \$3.00 (60%) and received excellent reciprocation (\$6.42)
- Round 3 (receiver): Received maximum trust (\$5.00 sent) and reciprocated generously

The pattern so far shows strong cooperation in this game. Since this is a new co-player, I want to signal trust and encourage cooperation. Given the positive experiences, I'll send a high amount while keeping a small buffer.

Sending \$4.00 (80% of my endowment) demonstrates strong trust and gives my co-player \$12.00 to work with, creating good opportunity for mutual benefit.

```
{"send": 4.00}
```

reasoning:

Let me analyze the situation:

Previous rounds:

1. Round 1: RECEIVER - saw \$2.25 sent, received \$6.75, returned \$3.00, kept \$3.75
2. Round 2: SENDER - sent \$3.00, became \$9.00, saw \$6.42 returned, payoff \$8.42
3. Round 3: RECEIVER - saw \$5.00 sent, received \$15.00, returned \$10.00, kept \$5.00

Total visible earnings: \$17.17

Now Round 4: I'm the SENDER against a new co-player.

Key observations:

- In Round 2, I sent \$3.00 (60%) and got very good reciprocation (\$6.42 return on \$9.00)
- In Round 3, someone sent me everything (\$5.00) and I reciprocated generously (\$10.00 return)

I've seen good cooperation so far. My previous experience as sender (Round 2) with moderate-high trust level worked well.

For this round:

- This is a new co-player (different from previous rounds)
- I don't know their behavior yet
- But the overall pattern in this game seems cooperative
- I've been both generous and received generosity

Given the positive experiences so far, I should maintain a cooperative strategy. I'll send a substantial amount again. Sending \$3.50 (70%) shows strong trust while keeping a small safety margin.

Actually, let me reconsider. The game seems quite cooperative overall. In Round 3, someone trusted me completely with \$5.00, and in Round 2 when I sent \$3.00 I got excellent reciprocation.

I could send \$4.00 (80%) to signal strong cooperation, or even \$5.00 to match the maximum trust I received in Round 3.

Given the high cooperation levels I've experienced, I'll send \$4.00 (80% of endowment) - this is a strong signal of trust without going to the absolute maximum.

...

usage:

```
{  
  "input_tokens": 1343,  
  "output_tokens": 646,
```



```

    "reasoning_tokens": 0
  }
}
Message 6 role=user
content:
  Round 5

  In your most recent previous game round, you were the SENDER against another agent. You sent $4.0, it became $12.0, and you saw them return $6.64 to you. Your visible payoff was $7.64.
  Your total visible earnings across all rounds are $24.810000000000002.

  This round, you are the RECEIVER against your current co-player. They sent you $3.67, so you received $11.01. How much do you return? (0-11.01)
  Respond exactly as JSON: {'return': <amount>}
Message 7 role=assistant
content:
  Looking at this situation:

  - My co-player sent $3.67 (73.4% of their endowment) - this shows strong trust
  - This became $11.01 for me
  - They currently have $1.33 remaining

  Their strong trust (sending nearly 3/4 of their endowment) deserves good reciprocation. If I return $7.00:
  - The sender ends with $8.33 (gain of $3.33)
  - I keep $4.01 (gain of $4.01)
  - Both of us benefit, with returns roughly proportional to their trust level

  This balances rewarding their trust while maintaining fairness.

  {"return": 7.00}
reasoning:
  Let me analyze this situation:

  - My co-player sent $3.67 (73.4% of their endowment) - this is strong trust
  - This became $11.01 for me
  - They currently have $1.33 remaining

  Looking at my pattern of reciprocation:
  - Round 1: Received $6.75, returned $3.00 (44.4% return rate)
  - Round 3: Received $15.00, returned $10.00 (66.7% return rate) - but they sent everything

  And when I've been sender:
  - Round 2: Sent $3.00 (60%), received $6.42 back (71.3% return rate from receiver)
  - Round 4: Sent $4.00 (80%), received $6.64 back (55.3% return rate from receiver)

  The sender showed strong trust at 73.4%. I should reciprocate well, but need to be fair.

  If I return $6.00:
  - Sender ends with $1.33 + $6.00 = $7.33 (gain of $2.33)
  - I keep $11.01 - $6.00 = $5.01 (gain of $5.01)

  This is about a 54.5% return rate, which seems reasonable given their strong (but not maximum) trust.

  If I return $7.00:
  - Sender ends with $1.33 + $7.00 = $8.33 (gain of $3.33)
  - I keep $11.01 - $7.00 = $4.01 (gain of $4.01)

  This is about a 63.6% return rate, which rewards their strong trust more generously.

  Given they sent 73.4% of their endowment (strong trust), I think $7.00 is appropriate. It's more generous than my Round 1 reciprocation (where they sent 45%) but less than my Round 3 reciprocation (where they sent ...

usage:
  {
    "input_tokens": 1645,
    "output_tokens": 659,
    "reasoning_tokens": 0
  }

```

Message 8 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.67, it became \$11.01, and you returned \$7.0. Your visible payoff was \$4.01.

Your total visible earnings across all rounds are \$28.82.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience as sender:

- Round 2: Sent \$3.00 (60%) ? received \$6.42 back, strong return
- Round 4: Sent \$4.00 (80%) ? received \$6.64 back, good return

Both times I've been sender, trust has been rewarded. This is a new co-player, and I want to continue building cooperative patterns while being reasonably cautious.

Sending \$3.50 (70% of my endowment) strikes a good balance - it shows trust and gives my co-player \$10.50 to work with, creating opportunity for mutual benefit.

```
{"send": 3.50}
```

reasoning:

Looking at my history:

- Round 2 (sender): Sent \$3.00 (60%), received \$6.42 back, payoff \$8.42
- Round 3 (receiver): Received \$5.00 (100%), returned \$10.00, payoff \$5.00
- Round 4 (sender): Sent \$4.00 (80%), received \$6.64 back, payoff \$7.64
- Round 5 (receiver): Received \$3.67 (73.4%), returned \$7.00, payoff \$4.01

As a sender, I've been quite trusting (60% and 80%) and have received good returns on both occasions. As a receiver, I've been fairly generous as well.

This is Round 6, and I'm facing a new co-player. The overall pattern suggests trust pays off in this game. I should continue with a moderate-to-high trust level.

Given my successful experiences, I'll send a good amount. Sending \$3.50 (70%) seems like a balanced choice - shows trust, not too conservative, but also not going all-in like some co-players have done.

Actually, let me reconsider. My sending history:

- \$3.00 (60%) - got back \$6.42, net gain \$3.42
- \$4.00 (80%) - got back \$6.64, net gain \$2.64

Both were profitable. The 60% send actually gave me a better net gain. Let me go with something around 70% to balance trust and caution.

\$3.50 seems reasonable.

usage:

```
{
  "input_tokens": 1928,
  "output_tokens": 534,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.3 to you. Your visible payoff was \$7.8.

Your total visible earnings across all rounds are \$36.62.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my sending history:

- Round 2: Sent \$3.00 ? received \$6.42 back
- Round 4: Sent \$4.00 ? received \$6.64 back
- Round 6: Sent \$3.50 ? received \$6.30 back

All three times as sender, my trust has been rewarded with good returns. This pattern suggests cooperation is working well in this environment.

With a new co-player, I'll continue the strategy that's been successful. Sending \$3.50 (70%) demonstrates trust and gives them \$10.50 to work with.

```
{"send": 3.50}
```

reasoning:

Let me review my sending history:

- Round 2: Sent \$3.00 (60%) ? received \$6.42 back
- Round 4: Sent \$4.00 (80%) ? received \$6.64 back
- Round 6: Sent \$3.50 (70%) ? received \$6.30 back

All three times I've been a sender, I've received good reciprocation. The pattern shows that trust is being rewarded consistently. This is another new co-player.

Given the consistent positive outcomes, I should continue with a trust-based strategy. Sending around 60-80% has been working well. I'll send \$3.50 again (70%), which has shown good results.

Actually, let me think - I've been varying between \$3.00, \$4.00, and \$3.50. All have worked well. I could go with \$3.50 again since it worked last time, or try something in this range.

I'll go with \$3.50 as it represents a solid middle ground.

usage:

```
{
  "input_tokens": 2205,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

Message 12 role=user**content:**

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.47 to you. Your visible payoff was \$5.97.

Your total visible earnings across all rounds are \$42.589999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.06, so you received \$9.18. How much do you return? (0-9.18)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant**content:**

Looking at this situation:

- My co-player sent \$3.06 (61.2% of their endowment) - showing moderate-to-good trust
- This became \$9.18 for me
- They currently have \$1.94 remaining

Their solid trust level (sending over 60%) deserves fair reciprocation. If I return \$6.00:

- The sender ends with \$7.94 (gain of \$2.94)
- I keep \$3.18 (gain of \$3.18)
- Both parties benefit from cooperation

This rewards their trust appropriately while maintaining a fair distribution.

```
{"return": 6.00}
```

reasoning:

Let me analyze this situation:

- My co-player sent \$3.06 (61.2% of their \$5 endowment) - this shows moderate-to-good trust

- t
- This became \$9.18 for me
 - They currently have \$1.94 remaining

My co-player showed solid trust by sending more than 60% of their endowment. I should reciprocate fairly.

Looking at my receiving history:

- Round 3: Received \$5.00 (100%), returned \$10.00
- Round 5: Received \$3.67 (73.4%), returned \$7.00

I've been generous when receiving high trust. For \$3.06 sent (61.2% trust), I should return proportionally.

If I return \$6.00:

- Sender ends with: $\$1.94 + \$6.00 = \$7.94$ (gain of \$2.94)
- I keep: $\$9.18 - \$6.00 = \$3.18$ (gain of \$3.18)

Both benefit, with me keeping slightly more which is reasonable given I'm taking the risk of trusting their return decision (wait, no - I'm the receiver, so I'm deciding the return).

Let me reconsider - they showed 61% trust. A fair return would give them a decent profit.

If I return \$6.00:

- They gain \$2.94 from their initial \$5
- I gain \$3.18 from receiving

This seems fair - both parties gain, and the split is reasonable given their trust level.

Actually, let me think about this more carefully. They sent \$3.06, keeping \$1.94. They now have \$1.94. If I return \$6.00, they end with \$7.94 total.

Their profit = $\$7.94 - \$5.00 = \$2.94$

My profit = $\$9.18 - \$6.00 = \$3.18$

...

usage:

```
{
  "input_tokens": 2278,
  "output_tokens": 625,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.06, it became \$9.18, and you returned \$6.0. Your visible payoff was \$3.1799999999999997.

Your total visible earnings across all rounds are \$45.769999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.93, so you received \$11.790000000000001. How much do you return? (0-11.790000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- My co-player sent \$3.93 (78.6% of their endowment) - this shows strong trust
- This became \$11.79 for me
- They currently have \$1.07 remaining

This level of trust (nearly 80%) deserves good reciprocation. If I return \$7.50:

- The sender ends with \$8.57 (gain of \$3.57)
- I keep \$4.29 (gain of \$4.29)
- Both parties benefit substantially from our cooperation

This appropriately rewards their strong trust.

```
{"return": 7.50}
```

Reasoning:

Looking at this situation:

- My co-player sent \$3.93 (78.6% of their \$5 endowment) - this shows strong trust
- This became \$11.79 for me
- They currently have \$1.07 remaining

This is strong trust, sending nearly 80% of their endowment. I should reciprocate well.

Let me think about a fair return:

- If I return \$7.50:
 - Sender ends with \$8.57 (gain of \$3.57)
 - I keep \$4.29 (gain of \$4.29)
 - Both benefit, though I keep slightly more
- If I return \$7.00:
 - Sender ends with \$8.07 (gain of \$3.07)
 - I keep \$4.79 (gain of \$4.79)
- If I return \$8.00:
 - Sender ends with \$9.07 (gain of \$4.07)
 - I keep \$3.79 (gain of \$3.79)
 - More balanced

Given they sent 78.6% (strong trust), I think \$7.50 is appropriate - it rewards their trust well and creates mutual benefit.

```
Usage: {  
  "input_tokens": 2312,  
  "output_tokens": 440,  
  "reasoning_tokens": 0  
}
```

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_6=Sender, Agent_5=Receiver
dyad_3: Agent_8=Sender, Agent_3=Receiver
dyad_4: Agent_4=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$3; received \$9; Agent_1 returned \$3.5; payoffs Agent_2=\$5.5, Agent_1=\$5.5;
noise/visible: sent visible=\$2.36, received visible=\$7.08, returned visible=\$2.87
Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 9.0, "min": 0, "raw": 3.5}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_6 sent \$4; received \$12; Agent_5 returned \$8.5; payoffs Agent_6=\$9.5, Agent_5=\$3.5;
noise/visible: sent visible=\$4.67, received visible=\$14.01, returned visible=\$8.18
Action validation: {"returned": {"amount": 8.5, "clamped": false, "max": 12.0, "min": 0, "raw": 8.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_8 sent \$4; received \$12; Agent_3 returned \$9.6; payoffs Agent_8=\$10.6, Agent_3=\$2.4;
noise/visible: sent visible=\$4.56, received visible=\$13.68, returned visible=\$9.17
Action validation: {"returned": {"amount": 9.6, "clamped": false, "max": 12.0, "min": 0, "raw": 9.6}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_4: Agent_4 sent \$4; received \$12; Agent_7 returned \$8; payoffs Agent_4=\$9, Agent_7=\$4;
noise/visible: sent visible=\$4.19, received visible=\$12.57, returned visible=\$7.23
Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 12.0, "min": 0, "raw": 8.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 59.75,
"Agent_2": 60.65,
"Agent_3": 53.699999999999996,
"Agent_4": 59.56,
"Agent_5": 67.5,
"Agent_6": 65.35,
"Agent_7": 58.54,
"Agent_8": 74.44999999999999

}

Visible balances after round: {

"Agent_1": 63.129999999999995,

```

"Agent_2": 64.61999999999999,
"Agent_3": 60.46,
"Agent_4": 64.64999999999999,
"Agent_5": 70.17,
"Agent_6": 59.239999999999995,
"Agent_7": 58.23,
"Agent_8": 69.57000000000001
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:28:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.52 to you. Your visible payoff was \$8.52.

Your total visible earnings across all rounds are \$19.68.

This round, you are the RECEIVER against your current co-player. They sent you \$3.36, so you received \$10.08. How much do you return? (0-10.08)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.36 (67% of their endowment), showing solid trust
- I received \$10.08 total
- They currently have \$1.64 left

I want to reciprocate their trust fairly. If I return \$5.00:

- They end up with: $\$1.64 + \$5.00 = \$6.64$ (gain of \$1.64)
- I end up with: $\$10.08 - \$5.00 = \$5.08$ (gain of \$0.08)

Both of us benefit from cooperation, and this rewards their trust appropriately.

```
'''json
{"return": 5.00}
'''
```

reasoning:

Let me analyze this situation:

Round 4 - I'm the RECEIVER

- My co-player sent \$3.36 (67.2% of their endowment)
- I received \$10.08 total
- They currently have: $\$5 - \$3.36 = \$1.64$ left

My previous round (Round 3):

- I sent \$3, received \$6.52 back - very generous return!

This co-player sent \$3.36, showing good trust. I should reciprocate appropriately.

If I return \$5.00:

- They end up with: $\$1.64 + \$5.00 = \$6.64$
- I end up with: $\$10.08 - \$5.00 = \$5.08$

Both of us gain. This is a fair reciprocation that rewards their trust.

Alternatively, if I return \$5.50:

- They end up with: $\$1.64 + \$5.50 = \$7.14$
- I end up with: $\$10.08 - \$5.50 = \$4.58$

This would be even more generous, similar to how I acted in Round 2.

Given that they sent about 67% of their endowment (compared to 79% in Round 2), I think re

turning around \$5 is appropriate and fair. It ensures they profit significantly from the trust, and we both end up better off than the initial \$5.

I'll go with \$5.00 as a fair reciprocation.

usage:

```
{
  "input_tokens": 1402,
  "output_tokens": 522,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.36, it became \$10.08, and you returned \$5.0. Your visible payoff was \$5.08.

Your total visible earnings across all rounds are \$24.759999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at Round 5:

I'm with another new co-player. My experience so far shows:

- Sending \$3 has worked well in previous rounds, getting returns of \$2.81 and \$6.52
- The cooperative approach has generally been rewarded

I'll continue with the same strategy - send \$3 to signal trust and cooperation while being reasonably cautious with an unknown partner.

```
'''json
{"send": 3}
'''
```

reasoning:

Round 5, I'm the sender with a new co-player (random pairings each round).

Looking at my history:

- Round 1 (Sender): Sent \$3, got back \$2.81 - decent cooperation
- Round 2 (Receiver): Received \$11.85, returned \$5.50 - I was generous
- Round 3 (Sender): Sent \$3, got back \$6.52 - excellent return!
- Round 4 (Receiver): Received \$10.08, returned \$5.00 - fair reciprocation

So far, sending \$3 has worked well - I got \$2.81 and \$6.52 back in my two sender rounds. The average return has been positive.

For this new pairing, I should continue with a cooperative strategy. Sending \$3 seems to be working as a good middle-ground approach that encourages cooperation.

```
'''json
{"send": 3}
'''
```

usage:

```
{
  "input_tokens": 1690,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$3.15 to you. Your visible payoff was \$5.15.

Your total visible earnings across all rounds are \$29.909999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.47, so y

ou received \$13.41. How much do you return? (0-13.41)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.47 (89% of their endowment), showing very high trust
- I received \$13.41 total
- They currently have only \$0.53 left

This is a very trusting sender, and I should reciprocate generously. If I return \$6.50:

- They end up with: $\$0.53 + \$6.50 = \$7.03$ (gain of \$2.03)
- I end up with: $\$13.41 - \$6.50 = \$6.91$ (gain of \$1.91)

Both of us gain approximately \$2, which appropriately rewards their exceptional trust and maintains strong cooperation.

```
'''json
{"return": 6.50}
'''
```

reasoning:

Looking at this situation:

- My co-player sent \$4.47 (89% of their endowment), showing very high trust
- I received \$13.41 total
- They currently have \$0.53 left

This is a very trusting sender. I should reciprocate generously.

If I return \$6.50:

- They end up with: $\$0.53 + \$6.50 = \$7.03$ (gain of \$2.03)
- I end up with: $\$13.41 - \$6.50 = \$6.91$ (gain of \$1.91)

Both benefit well, and this appropriately rewards their very high level of trust.

Let me think about this more. They sent \$4.47 out of \$5, leaving themselves with only \$0.53. That's 89.4% of their endowment. This is extremely trusting.

I should return enough to:

1. Ensure they profit significantly from their trust
2. Show I value cooperation
3. Both of us should end up better than our starting \$5

If I return \$6.50:

- Them: $\$0.53 + \$6.50 = \$7.03$
- Me: $\$13.41 - \$6.50 = \$6.91$

Both gain roughly \$2, which seems fair and reciprocates their high trust.

usage:

```
{
  "input_tokens": 1931,
  "output_tokens": 502,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.47, it became \$13.41, and you returned \$6.5. Your visible payoff was \$6.91.

Your total visible earnings across all rounds are \$36.81999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- My co-player sent the maximum \$5 (100% of their endowment), showing complete trust
- I received \$15 total
- They currently have \$0 left

This is the highest level of trust possible. I should reciprocate very generously. If I return \$7.50:

- They end up with: $\$0 + \$7.50 = \$7.50$ (gain of \$2.50)
- I end up with: $\$15 - \$7.50 = \$7.50$ (gain of \$2.50)

We both gain equally from this maximum cooperation, which appropriately rewards their complete trust.

```
'''json
{"return": 7.50}
'''
```

reasoning:

This is Round 7, and I'm the RECEIVER.

My co-player sent the maximum amount: \$5 (100% of their endowment), showing complete trust

- I received \$15 total
- They currently have \$0 left

This sender showed maximum trust. I should reciprocate very generously.

If I return \$7.50:

- They end up with: $\$0 + \$7.50 = \$7.50$ (gain of \$2.50)
- I end up with: $\$15 - \$7.50 = \$7.50$ (gain of \$2.50)

This gives us both equal gains and rewards their complete trust. This seems fair and reciprocates their maximum trust appropriately.

Actually, let me think about this more. They sent everything (\$5), so they're showing maximum trust. I want to reward that very well.

If I return \$8:

- They end up with: $\$0 + \$8 = \$8$ (gain of \$3)
- I end up with: $\$15 - \$8 = \$7$ (gain of \$2)

This gives them a slightly better outcome for taking the risk of sending everything.

Let me go with \$7.50 as a good middle ground - we both gain equally, which seems fair when they showed maximum trust.

usage:

```
{
  "input_tokens": 2240,
  "output_tokens": 476,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$5, it became \$15, and you returned \$7.5. Your visible payoff was \$7.5. Your total visible earnings across all rounds are \$44.31999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at Round 8:

I'm with another new co-player. My strategy of sending \$3 has generally worked well across previous rounds, yielding varied but mostly positive returns. This has been a good balance between showing cooperation and being prudent.

I'll continue with this consistent approach.

```
'''json
{"send": 3}
'''
```

reasoning:

Looking at Round 8:

I'm with another new co-player. Let me review my experience as sender:

- Round 1: Sent \$3, received \$2.81 back (payoff \$4.81)
- Round 3: Sent \$3, received \$6.52 back (payoff \$8.52)
- Round 5: Sent \$3, received \$3.15 back (payoff \$5.15)

And as receiver:

- Round 2: Received from sender who sent \$3.95, I returned \$5.50
- Round 4: Received from sender who sent \$3.36, I returned \$5.00
- Round 6: Received from sender who sent \$4.47, I returned \$6.50
- Round 7: Received from sender who sent \$5.00, I returned \$7.50

Overall, cooperation has been working well. Returns have varied but generally been reasonable. The \$3 send amount has been my standard approach.

This is the last round of the game (8 rounds total with 8 agents). Since it's the final round, I might consider adjusting my strategy, but:

- I don't know if this is truly the last round for this experimental setup
- Maintaining consistency has worked well
- My co-player doesn't know my history

I'll stick with sending \$3, which has been successful and signals cooperation without being overly risky.

usage:

```
{
  "input_tokens": 2356,
  "output_tokens": 409,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$6.17 to you. Your visible payoff was \$8.17.

Your total visible earnings across all rounds are \$52.489999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$4.92, so you received \$14.76. How much do you return? (0-14.76)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- My co-player sent \$4.92 (98% of their endowment), showing extremely high trust
- I received \$14.76 total
- They currently have only \$0.08 left

This is nearly maximum trust. I should reciprocate very generously. If I return \$7.50:

- They end up with: \$0.08 + \$7.50 = \$7.58 (gain of \$2.58)
- I end up with: \$14.76 - \$7.50 = \$7.26 (gain of \$2.26)

Both of us gain approximately \$2.25-\$2.50, appropriately rewarding their exceptional trust.

```
'''json
{"return": 7.50}
'''
```

reasoning:

Looking at this situation:

- My co-player sent \$4.92 (98% of their endowment), showing very high trust
- I received \$14.76 total
- They currently have \$5 - \$4.92 = \$0.08 left

This is another very trusting sender. I should reciprocate generously to reward their trust. Let me think about what's fair:

If I return \$7.00:

- They end up with: $\$0.08 + \$7.00 = \$7.08$ (gain of \$2.08)
- I end up with: $\$14.76 - \$7.00 = \$7.76$ (gain of \$2.76)

If I return \$7.50:

- They end up with: $\$0.08 + \$7.50 = \$7.58$ (gain of \$2.58)
- I end up with: $\$14.76 - \$7.50 = \$7.26$ (gain of \$2.26)

Given they sent nearly all their money (\$4.92 out of \$5), I should be very generous. \$7.50 seems appropriate to reward their exceptional trust - they gain \$2.58 and I gain \$2.26, which is close to equal and both end up with similar final amounts.

usage:

```
{
  "input_tokens": 2236,
  "output_tokens": 491,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.92, it became \$14.76, and you returned \$7.5. Your visible payoff was \$7.26.

Your total visible earnings across all rounds are \$59.74999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at Round 10:

I'm with another new co-player. My consistent strategy of sending \$3 has worked well throughout the game, showing cooperation while managing risk. The returns have been varied but generally positive.

I'll continue with this balanced approach.

```
'''json
{"send": 3}
'''
```

Reasoning:

Looking at Round 10:

I'm with another new co-player. Let me review my sending history:

- Round 1: Sent \$3, got back \$2.81, payoff \$4.81
- Round 3: Sent \$3, got back \$6.52, payoff \$8.52
- Round 5: Sent \$3, got back \$3.15, payoff \$5.15
- Round 8: Sent \$3, got back \$6.17, payoff \$8.17

My strategy of sending \$3 has been consistently effective, showing cooperation while being prudent. Returns have varied but I've generally benefited from this approach.

I'll continue with \$3.

Usage: {

```
"input_tokens": 2247,
"output_tokens": 238,
"reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:29:11

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
```

```

"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$6.34 to you. Your visible payoff was \$7.59.

Your total visible earnings across all rounds are \$20.310000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$4.09, so you received \$12.27. How much do you return? (0-12.27)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.09 (81.8% of their endowment)
- They currently have: $\$5 - \$4.09 = \$0.91$
- I received: \$12.27
- Total created: \$13.18

This is even more trust than I saw in Round 1 (when sender sent 75.4%). I want to reward this high level of cooperation appropriately.

If I return \$6.50:

- Sender ends with: $\$0.91 + \$6.50 = \$7.41$ (gain of \$2.41)
- I end with: $\$12.27 - \$6.50 = \$5.77$ (gain of \$0.77)

Both of us benefit, and the sender's exceptional trust is well-rewarded.

{'return': 6.50}

reasoning:

Let me analyze this situation:

- The sender sent \$4.09 out of \$5, which is 81.8% of their endowment - very high trust!
- They now have: $\$5 - \$4.09 = \$0.91$
- I received: \$12.27
- Total money in the system: $\$0.91 + \$12.27 = \$13.18$

This sender showed even more trust than my previous partners. I want to reciprocate appropriately.

Looking at my previous behavior:

- Round 1: As receiver, got \$11.31, returned \$5.50 (about 48.6%)
- The sender had sent 75.4%
- That seemed fair and both parties benefited

This sender sent 81.8%, even more generous. I should reciprocate with a good return.

If I return \$6.50:

- Sender ends with: $\$0.91 + \$6.50 = \$7.41$
- I end with: $\$12.27 - \$6.50 = \$5.77$
- Sender gains \$2.41 over initial \$5
- I gain \$0.77 over initial \$5

Both benefit, and the sender's higher risk/trust is rewarded with higher gains.

Actually, let me think about this differently. The sender sent 81.8% which is very high. In Round 1, sender sent 75.4% and I returned 48.6% of what I received.

For 81.8% send rate, maybe I should return around 50-53% to reward the extra trust?

$\$12.27 * 0.52 = \6.38

If I return \$6.40:

- Sender ends with: $\$0.91 + \$6.40 = \$7.31$
- I end with: $\$12.27 - \$6.40 = \$5.87$

This seems fair - both gain, sender gets rewarded for high trust.

...

usage:

{

```
"input_tokens": 1310,
"output_tokens": 685,
"reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.09, it became \$12.27, and you returned \$6.5. Your visible payoff was \$5.77.

Your total visible earnings across all rounds are \$26.080000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

- As sender: Sent \$3.75 twice, received good returns (\$5.66 and \$6.34)
- As receiver: Responded fairly to cooperative senders

The high-trust strategy continues to work well across different co-players. I'll maintain this approach.

```
{'send': 3.75}
```

reasoning:

Let me review my history:

- Round 1: Receiver - sender sent \$3.77, I returned \$5.50, payoff \$5.81
- Round 2: Sender - I sent \$3.75, got back \$5.66, payoff \$6.91
- Round 3: Sender - I sent \$3.75, got back \$6.34, payoff \$7.59
- Round 4: Receiver - sender sent \$4.09, I returned \$6.50, payoff \$5.77

Total visible earnings: \$26.08

My sending strategy has been working well - sending around \$3.75 (75%) has led to good returns. In Round 4, I saw an even higher send amount (\$4.09, 81.8%) and I rewarded it well.

Now I'm facing a new co-player as sender. The cooperative strategy has been consistently working across different opponents. I should continue with a similar high-trust signal.

I'll send \$3.75 again, which has proven successful.

usage:

```
{
  "input_tokens": 1637,
  "output_tokens": 324,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$6.54 to you. Your visible payoff was \$7.79.

Your total visible earnings across all rounds are \$33.870000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$4.1, so you received \$12.299999999999999. How much do you return? (0-12.299999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.10 (82% of their endowment)
- They currently have: \$5 - \$4.10 = \$0.90
- I received: \$12.30
- Total created: \$13.20

This is very similar to Round 4 where the sender sent \$4.09 and I returned \$6.50, which worked well for both parties.

If I return \$6.50:

- Sender ends with: $\$0.90 + \$6.50 = \$7.40$
- I end with: $\$12.30 - \$6.50 = \$5.80$

Both of us benefit substantially, rewarding the sender's high trust.

```
{'return': 6.50}
```

reasoning:

The sender sent \$4.10, which is 82% of their endowment - very high trust again!

They currently have: $\$5 - \$4.10 = \$0.90$

I received: \$12.30

Total created: \$13.20

This is similar to Round 4 where sender sent \$4.09. I returned \$6.50 then.

Let me be consistent and reward this high trust appropriately.

If I return \$6.50:

- Sender ends with: $\$0.90 + \$6.50 = \$7.40$
- I end with: $\$12.30 - \$6.50 = \$5.80$

Both benefit, sender gets a good reward for their trust.

usage:

```
{
  "input_tokens": 1863,
  "output_tokens": 354,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$4.1, it became \$12.299999999999999, and you returned \$6.5. Your visible payoff was \$5.799999999999999.

Your total visible earnings across all rounds are \$39.67.

This round, you are the RECEIVER against your current co-player. They sent you \$5.0, so you received \$15.0. How much do you return? (0-15.0)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

- The sender showed **complete trust** by sending \$5.00 (100% of their endowment!)
- They currently have: \$0
- I received: \$15.00
- This is the same co-player from Round 6, and they escalated their trust after I returned fairly

This maximum trust deserves a strong reciprocal response.

If I return \$8.00:

- Sender ends with: $\$8.00$ (gain of \$3.00)
- I end with: $\$7.00$ (gain of \$2.00)

Both of us benefit substantially, and their complete trust is well-rewarded.

```
{'return': 8.00}
```

reasoning:

This is interesting - I'm playing with the same co-player from Round 6. In Round 6:

- They sent \$4.10
- I received \$12.30
- I returned \$6.50
- They ended with $\$0.90 + \$6.50 = \$7.40$

Now in Round 7:

- They sent \$5.00 (100% of their endowment!) - maximum trust
- I received \$15.00
- They currently have: \$0

This is the ultimate show of trust - they sent everything! This deserves a strong reciprocal response.

If I return \$8.00:

- Sender ends with: \$0 + \$8.00 = \$8.00 (gain of \$3.00)
- I end with: \$15.00 - \$8.00 = \$7.00 (gain of \$2.00)

Both benefit nicely, and I'm rewarding their complete trust.

Alternatively, if I return \$7.50:

- Sender ends with: \$7.50 (gain of \$2.50)
- I end with: \$7.50 (gain of \$2.50)
- Equal split of the gains

Or if I return \$8.50:

- Sender ends with: \$8.50 (gain of \$3.50)
- I end with: \$6.50 (gain of \$1.50)
- More reward for their maximum trust

Given they showed complete trust by sending 100%, I think returning \$8.00 is appropriate - it gives them a strong reward while we both benefit substantially.

usage:

```
{
  "input_tokens": 2186,
  "output_tokens": 538,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$5.0, it became \$15.0, and you returned \$8.0. Your visible payoff was \$7.0

Your total visible earnings across all rounds are \$46.67.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my experience:

- As sender: \$3.75 has consistently yielded good returns (\$5.66, \$6.34, \$6.54)
- As receiver: I've rewarded trust fairly, including returning \$8.00 when someone sent \$5.00

The cooperative approach continues to work well. I'll maintain my high-trust strategy with this new co-player.

```
{'send': 3.75}
```

reasoning:

Looking at my history:

- As sender: I've sent \$3.75 multiple times and received good returns (\$5.66, \$6.34, \$6.54)
- As receiver: I've reciprocated fairly, especially rewarding high trust
- In rounds 6-7, I just played with someone who escalated from \$4.10 to \$5.00, and I reciprocated well

Now I'm the sender with a new co-player. The cooperative strategy has been working very well. I could:

1. Continue with \$3.75 (75%) - my standard high-trust amount
2. Send \$5.00 (100%) - maximum trust, like the last sender showed me
3. Send something in between like \$4.00-\$4.50

Given that:

- Cooperation has been consistent across the game
- I've built good earnings (\$46.67 total)
- Maximum trust was just shown to me and I rewarded it well
- This might encourage my new co-player to reciprocate strongly

I think I'll stick with my proven \$3.75 strategy. It's high trust without being maximum risk, and it's worked consistently well.

usage:

```
{
  "input_tokens": 2190,
  "output_tokens": 371,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$5.29 to you. Your visible payoff was \$6.54.

Your total visible earnings across all rounds are \$53.21.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my pattern:

- Sent \$3.75 in Rounds 2, 3, 5, and 8
- Returns ranged from \$5.29 to \$6.54, all profitable
- My fair behavior as receiver should encourage reciprocity

Continuing with my proven cooperative strategy.

```
{'send': 3.75}
```

reasoning:

Looking at my track record as sender:

- Round 2: Sent \$3.75, received \$5.66 back, payoff \$6.91
- Round 3: Sent \$3.75, received \$6.34 back, payoff \$7.59
- Round 5: Sent \$3.75, received \$6.54 back, payoff \$7.79
- Round 8: Sent \$3.75, received \$5.29 back, payoff \$6.54

The strategy of sending \$3.75 has been consistently profitable, with returns ranging from \$5.29 to \$6.54. All outcomes have been positive.

As receiver, I've been cooperative and fair, returning substantial amounts when senders showed trust.

This is a new co-player, so I should continue with my established cooperative strategy of sending \$3.75.

usage:

```
{
  "input_tokens": 2202,
  "output_tokens": 286,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.75, it became \$11.25, and you saw them return \$5.09 to you. Your visible payoff was \$6.34.

Your total visible earnings across all rounds are \$59.55.

This round, you are the RECEIVER against your current co-player. They sent you \$2.36, so you received \$7.08. How much do you return? (0-7.08)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- The sender sent \$2.36 (47.2% of their endowment) - moderate trust
- They currently have: \$2.64
- I received: \$7.08
- Total created: \$9.72

This is notably less trust than the \$4+ I've seen from other co-players, but still represents cooperation. I'll reciprocate proportionally.

If I return \$3.50:

- Sender ends with: $\$2.64 + \$3.50 = \$6.14$ (gain of \$1.14)
- I end with: $\$7.08 - \$3.50 = \$3.58$

The sender benefits from their trust, though not as much as those who sent more.

```
{'return': 3.50}
```

Reasoning:

Looking at this situation:

- The sender sent \$2.36 (47.2% of their endowment) - this is significantly less trust than I've seen in previous rounds
- They currently have: $\$5 - \$2.36 = \$2.64$
- I received: \$7.08
- Total created: \$9.72

This is moderate trust, much lower than the \$4+ I've been seeing from other players. I should still reciprocate fairly, but adjust my return to reflect the lower trust level.

Previous high-trust interactions:

- When sender sent \$4.09-\$4.10, I returned \$6.50
- When sender sent \$5.00, I returned \$8.00
- When I sent \$3.75, I received returns ranging from \$5.09 to \$6.54

For \$2.36 sent, a proportional response would be:

- If I return \$3.50:
 - Sender ends with: $\$2.64 + \$3.50 = \$6.14$ (gain of \$1.14)
 - I end with: $\$7.08 - \$3.50 = \$3.58$ (gain of -\$1.42, loss)

That seems a bit low for me. Let me try \$4.00:

- Sender ends with: $\$2.64 + \$4.00 = \$6.64$ (gain of \$1.64)
- I end with: $\$7.08 - \$4.00 = \$3.08$ (gain of -\$1.92, loss)

Still leaves me at a loss. Let me think about fairness differently...

Actually, when I sent \$3.75 (75% of endowment), I received returns of roughly 47-58% of what the receiver got.

- $\$5.66 / \$11.25 = 50.3\%$
- $\$6.34 / \$11.25 = 56.4\%$
- $\$6.54 / \$11.25 = 58.1\%$
- $\$5.29 / \$11.25 = 47.0\%$
- $\$5.09 / \$11.25 = 45.2\%$

Since the sender sent 47.2% this time, returning around 50% of the \$7.08 would be \$3.54. That leaves me with \$3.54 after the return, which is a loss, but it maintains consistency with how I've been reciprocating in previous rounds.

Looking back at my pattern: when trust was high (81-100%), I returned roughly 53% of what I received. For this moderate trust level, 50% seems appropriate. The sender gains \$1.18 from the exchange, which is a reasonable outcome given their initial trust....

```
Usage: {
  "input_tokens": 2198,
  "output_tokens": 1402,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_2

Timestamp: 2026-08-12T09:29:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_2",
"move_index": 2,
"opponent_id": "Agent_5",
"pairing": {
  "agents": [
    "Agent_6",
    "Agent_5"
  ],
  "investor": "Agent_6",
  "roles": {
    "Agent_5": "trustee",
    "Agent_6": "investor"
  },
  "trustee": "Agent_5"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the **SENDER**:

- You receive \$5 each round
- You decide how much to **SEND** to the **RECEIVER** (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to **RETURN**
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the **RECEIVER**:

- Sender receives \$5
- Sender decides how much to **SEND** to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to **RETURN**
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$5, it became \$15, and you returned \$10.0. Your visible payoff was \$5.0. Your total visible earnings across all rounds are \$17.17.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant**content:**

Looking at my experience so far:

- Round 2 (sender): Sent \$3.00 (60%) and received excellent reciprocation (\$6.42)
- Round 3 (receiver): Received maximum trust (\$5.00 sent) and reciprocated generously

The pattern so far shows strong cooperation in this game. Since this is a new co-player, I want to signal trust and encourage cooperation. Given the positive experiences, I'll send a high amount while keeping a small buffer.

Sending \$4.00 (80% of my endowment) demonstrates strong trust and gives my co-player \$12.00 to work with, creating good opportunity for mutual benefit.

```
{"send": 4.00}
```

reasoning:

Let me analyze the situation:

Previous rounds:

1. Round 1: RECEIVER - saw \$2.25 sent, received \$6.75, returned \$3.00, kept \$3.75
2. Round 2: SENDER - sent \$3.00, became \$9.00, saw \$6.42 returned, payoff \$8.42
3. Round 3: RECEIVER - saw \$5.00 sent, received \$15.00, returned \$10.00, kept \$5.00

Total visible earnings: \$17.17

Now Round 4: I'm the SENDER against a new co-player.

Key observations:

- In Round 2, I sent \$3.00 (60%) and got very good reciprocation (\$6.42 return on \$9.00)
- In Round 3, someone sent me everything (\$5.00) and I reciprocated generously (\$10.00 return)

I've seen good cooperation so far. My previous experience as sender (Round 2) with moderate-high trust level worked well.

For this round:

- This is a new co-player (different from previous rounds)
- I don't know their behavior yet
- But the overall pattern in this game seems cooperative
- I've been both generous and received generosity

Given the positive experiences so far, I should maintain a cooperative strategy. I'll send a substantial amount again. Sending \$3.50 (70%) shows strong trust while keeping a small safety margin.

Actually, let me reconsider. The game seems quite cooperative overall. In Round 3, someone trusted me completely with \$5.00, and in Round 2 when I sent \$3.00 I got excellent reciprocation.

I could send \$4.00 (80%) to signal strong cooperation, or even \$5.00 to match the maximum trust I received in Round 3.

Given the high cooperation levels I've experienced, I'll send \$4.00 (80% of endowment) - this is a strong signal of trust without going to the absolute maximum.

...

usage:

```
{
  "input_tokens": 1343,
  "output_tokens": 646,
  "reasoning_tokens": 0
}
```

}

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.64 to you. Your visible payoff was \$7.64.

Your total visible earnings across all rounds are \$24.810000000000002.

This round, you are the RECEIVER against your current co-player. They sent you \$3.67, so you received \$11.01. How much do you return? (0-11.01)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.67 (73.4% of their endowment) - this shows strong trust
- This became \$11.01 for me
- They currently have \$1.33 remaining

Their strong trust (sending nearly 3/4 of their endowment) deserves good reciprocation. If

I return \$7.00:

- The sender ends with \$8.33 (gain of \$3.33)
- I keep \$4.01 (gain of \$4.01)
- Both of us benefit, with returns roughly proportional to their trust level

This balances rewarding their trust while maintaining fairness.

{"return": 7.00}

reasoning:

Let me analyze this situation:

- My co-player sent \$3.67 (73.4% of their endowment) - this is strong trust
- This became \$11.01 for me
- They currently have \$1.33 remaining

Looking at my pattern of reciprocation:

- Round 1: Received \$6.75, returned \$3.00 (44.4% return rate)
- Round 3: Received \$15.00, returned \$10.00 (66.7% return rate) - but they sent everything

And when I've been sender:

- Round 2: Sent \$3.00 (60%), received \$6.42 back (71.3% return rate from receiver)
- Round 4: Sent \$4.00 (80%), received \$6.64 back (55.3% return rate from receiver)

The sender showed strong trust at 73.4%. I should reciprocate well, but need to be fair.

If I return \$6.00:

- Sender ends with \$1.33 + \$6.00 = \$7.33 (gain of \$2.33)
- I keep \$11.01 - \$6.00 = \$5.01 (gain of \$5.01)

This is about a 54.5% return rate, which seems reasonable given their strong (but not maximum) trust.

If I return \$7.00:

- Sender ends with \$1.33 + \$7.00 = \$8.33 (gain of \$3.33)
- I keep \$11.01 - \$7.00 = \$4.01 (gain of \$4.01)

This is about a 63.6% return rate, which rewards their strong trust more generously.

Given they sent 73.4% of their endowment (strong trust), I think \$7.00 is appropriate. It's more generous than my Round 1 reciprocation (where they sent 45%) but less than my Round 3 reciprocation (where they sent ...

usage:

```
{
  "input_tokens": 1645,
  "output_tokens": 659,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.67, it became \$11.01, and you returned \$7.0. Your visible payoff was \$4.01.
Your total visible earnings across all rounds are \$28.82.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience as sender:

- Round 2: Sent \$3.00 (60%) ? received \$6.42 back, strong return
- Round 4: Sent \$4.00 (80%) ? received \$6.64 back, good return

Both times I've been sender, trust has been rewarded. This is a new co-player, and I want to continue building cooperative patterns while being reasonably cautious.

Sending \$3.50 (70% of my endowment) strikes a good balance - it shows trust and gives my co-player \$10.50 to work with, creating opportunity for mutual benefit.

```
{"send": 3.50}
```

reasoning:

Looking at my history:

- Round 2 (sender): Sent \$3.00 (60%), received \$6.42 back, payoff \$8.42
- Round 3 (receiver): Received \$5.00 (100%), returned \$10.00, payoff \$5.00
- Round 4 (sender): Sent \$4.00 (80%), received \$6.64 back, payoff \$7.64
- Round 5 (receiver): Received \$3.67 (73.4%), returned \$7.00, payoff \$4.01

As a sender, I've been quite trusting (60% and 80%) and have received good returns on both occasions. As a receiver, I've been fairly generous as well.

This is Round 6, and I'm facing a new co-player. The overall pattern suggests trust pays off in this game. I should continue with a moderate-to-high trust level.

Given my successful experiences, I'll send a good amount. Sending \$3.50 (70%) seems like a balanced choice - shows trust, not too conservative, but also not going all-in like some co-players have done.

Actually, let me reconsider. My sending history:

- \$3.00 (60%) - got back \$6.42, net gain \$3.42
- \$4.00 (80%) - got back \$6.64, net gain \$2.64

Both were profitable. The 60% send actually gave me a better net gain. Let me go with something around 70% to balance trust and caution.

\$3.50 seems reasonable.

usage:

```
{  
  "input_tokens": 1928,  
  "output_tokens": 534,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.3 to you. Your visible payoff was \$7.8.
Your total visible earnings across all rounds are \$36.62.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my sending history:

- Round 2: Sent \$3.00 ? received \$6.42 back
- Round 4: Sent \$4.00 ? received \$6.64 back
- Round 6: Sent \$3.50 ? received \$6.30 back

All three times as sender, my trust has been rewarded with good returns. This pattern suggests cooperation is working well in this environment.

With a new co-player, I'll continue the strategy that's been successful. Sending \$3.50 (70%) demonstrates trust and gives them \$10.50 to work with.

```
{"send": 3.50}
```

reasoning:

Let me review my sending history:

- Round 2: Sent \$3.00 (60%) ? received \$6.42 back
- Round 4: Sent \$4.00 (80%) ? received \$6.64 back
- Round 6: Sent \$3.50 (70%) ? received \$6.30 back

All three times I've been a sender, I've received good reciprocation. The pattern shows that trust is being rewarded consistently. This is another new co-player.

Given the consistent positive outcomes, I should continue with a trust-based strategy. Sending around 60-80% has been working well. I'll send \$3.50 again (70%), which has shown good results.

Actually, let me think - I've been varying between \$3.00, \$4.00, and \$3.50. All have worked well. I could go with \$3.50 again since it worked last time, or try something in this range.

I'll go with \$3.50 as it represents a solid middle ground.

usage:

```
{
  "input_tokens": 2205,
  "output_tokens": 398,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.47 to you. Your visible payoff was \$5.97.

Your total visible earnings across all rounds are \$42.589999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.06, so you received \$9.18. How much do you return? (0-9.18)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.06 (61.2% of their endowment) - showing moderate-to-good trust
- This became \$9.18 for me
- They currently have \$1.94 remaining

Their solid trust level (sending over 60%) deserves fair reciprocation. If I return \$6.00:

- The sender ends with \$7.94 (gain of \$2.94)
- I keep \$3.18 (gain of \$3.18)
- Both parties benefit from cooperation

This rewards their trust appropriately while maintaining a fair distribution.

```
{"return": 6.00}
```

reasoning:

Let me analyze this situation:

- My co-player sent \$3.06 (61.2% of their \$5 endowment) - this shows moderate-to-good trust

- This became \$9.18 for me
- They currently have \$1.94 remaining

My co-player showed solid trust by sending more than 60% of their endowment. I should reciprocate fairly.

Looking at my receiving history:

- Round 3: Received \$5.00 (100%), returned \$10.00
- Round 5: Received \$3.67 (73.4%), returned \$7.00

I've been generous when receiving high trust. For \$3.06 sent (61.2% trust), I should return proportionally.

If I return \$6.00:

- Sender ends with: $\$1.94 + \$6.00 = \$7.94$ (gain of \$2.94)
- I keep: $\$9.18 - \$6.00 = \$3.18$ (gain of \$3.18)

Both benefit, with me keeping slightly more which is reasonable given I'm taking the risk of trusting their return decision (wait, no - I'm the receiver, so I'm deciding the return).

Let me reconsider - they showed 61% trust. A fair return would give them a decent profit.

If I return \$6.00:

- They gain \$2.94 from their initial \$5
- I gain \$3.18 from receiving

This seems fair - both parties gain, and the split is reasonable given their trust level.

Actually, let me think about this more carefully. They sent \$3.06, keeping \$1.94. They now have \$1.94. If I return \$6.00, they end with \$7.94 total.

Their profit = $\$7.94 - \$5.00 = \$2.94$

My profit = $\$9.18 - \$6.00 = \$3.18$

...

usage:

```
{
  "input_tokens": 2278,
  "output_tokens": 625,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.06, it became \$9.18, and you returned \$6.0. Your visible payoff was \$3.1799999999999997.

Your total visible earnings across all rounds are \$45.769999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.93, so you received \$11.790000000000001. How much do you return? (0-11.790000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- My co-player sent \$3.93 (78.6% of their endowment) - this shows strong trust
- This became \$11.79 for me
- They currently have \$1.07 remaining

This level of trust (nearly 80%) deserves good reciprocation. If I return \$7.50:

- The sender ends with \$8.57 (gain of \$3.57)
- I keep \$4.29 (gain of \$4.29)
- Both parties benefit substantially from our cooperation

This appropriately rewards their strong trust.

```
{"return": 7.50}
```

reasoning:

Looking at this situation:

- My co-player sent \$3.93 (78.6% of their \$5 endowment) - this shows strong trust
- This became \$11.79 for me
- They currently have \$1.07 remaining

This is strong trust, sending nearly 80% of their endowment. I should reciprocate well.

Let me think about a fair return:

- If I return \$7.50:
 - Sender ends with \$8.57 (gain of \$3.57)
 - I keep \$4.29 (gain of \$4.29)
 - Both benefit, though I keep slightly more
- If I return \$7.00:
 - Sender ends with \$8.07 (gain of \$3.07)
 - I keep \$4.79 (gain of \$4.79)
- If I return \$8.00:
 - Sender ends with \$9.07 (gain of \$4.07)
 - I keep \$3.79 (gain of \$3.79)
 - More balanced

Given they sent 78.6% (strong trust), I think \$7.50 is appropriate - it rewards their trust well and creates mutual benefit.

usage:

```
{
  "input_tokens": 2312,
  "output_tokens": 440,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.93, it became \$11.790000000000001, and you returned \$7.5. Your visible payoff was \$4.290000000000001.

Your total visible earnings across all rounds are \$50.059999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

This is the same co-player from Round 9! Last round, they showed strong trust by sending me \$3.93 (78.6%), and I reciprocated fairly with \$7.50.

Given our positive interaction and their demonstrated trustworthiness, I should continue building on this cooperative relationship. Sending \$4.00 (80%) matches their trust level and signals I value our partnership.

```
{"send": 4.00}
```

Reasoning:

This is interesting - this is the same co-player I just played with in Round 9!

In Round 9:

- They sent me \$3.93 (78.6% of their endowment) - strong trust
- I returned \$7.50, giving them a good payoff

Now I'm the sender, and I'm playing with this same co-player. They showed strong trust in me last round by sending \$3.93. I reciprocated fairly with \$7.50.

Given that:

1. They showed strong trust to me (78.6%)
2. I reciprocated generously (\$7.50 on \$11.79)
3. We're now continuing with the same partner
4. This is a chance to build on our positive interaction

I should send a high amount to continue our cooperative relationship. Since they sent nearly 80% last round, I could match or slightly exceed that level of trust.

Sending \$4.00 (80%) would show I value our cooperation and trust them based on our positive previous interaction.

My sending history:

- Round 2: \$3.00 ? got \$6.42
- Round 4: \$4.00 ? got \$6.64
- Round 6: \$3.50 ? got \$6.30
- Round 7: \$3.50 ? got \$4.47

This co-player has already shown themselves to be trustworthy (78.6% trust in Round 9). I should reciprocate that trust level.

```
Usage: {  
  "input_tokens": 2316,  
  "output_tokens": 453,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_2

Timestamp: 2026-08-12T09:29:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.17, it became \$12.51, and you returned \$6.0. Your visible payoff was \$6.51.

Your total visible earnings across all rounds are \$19.689999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my experience so far:

- Round 2: When I sent \$2.5 (50%), I received a very generous return of \$6.78
- Round 3: When someone sent me \$4.17 (high trust), I reciprocated with \$6.0

The game seems to reward trust. I'll send \$3.0 (60% of my endowment), which shows strong trust while being measured with a new co-player.

```
'''json
{"send": 3.0}
'''
```

reasoning:

Looking at my history:

- Round 1: RECEIVER - received from sender who sent \$2.3, I returned \$3.0 fairly
- Round 2: SENDER - I sent \$2.5, receiver returned \$6.78 (very generous!)
- Round 3: RECEIVER - received from sender who sent \$4.17 (very high trust), I returned \$6.0

Current situation:

- I'm the SENDER with a new co-player
- Total earnings so far: \$19.69

My experience shows:

- When I sent \$2.5 in Round 2, I got a very generous return of \$6.78
- When others showed high trust to me, I reciprocated fairly
- This game seems to reward trust

I should send a significant amount to show trust. Given that:

- Round 2 sender sent 46% and I reciprocated
- Round 3 sender sent 83.4% and I reciprocated generously
- When I sent 50% in Round 2, I got excellent returns

I think sending around \$3.0 (60%) would be a good balance - showing strong trust without being reckless with an unknown partner.

Actually, let me think about this more strategically. I've seen that:

- Lower trust (46%) got fair returns
- Medium trust (50% from me) got very generous returns
- High trust (83.4%) got generous returns

Sending \$3.0 seems like a good middle-high trust level that should encourage reciprocation

usage:

```
{
  "input_tokens": 1461,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$5.22 to you. Your visible payoff was \$7.22.

Your total visible earnings across all rounds are \$26.909999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.84, so you received \$11.52. How much do you return? (0-11.52)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- The sender showed high trust by sending \$3.84 (76.8% of their endowment)
- I received \$11.52 after the 3x multiplier
- The sender currently has only \$1.16 remaining

This level of trust deserves a generous return.

If I return \$6.0:

- Sender ends with: $\$1.16 + \$6.0 = \$7.16$ (profit of \$2.16)
- I end with: $\$11.52 - \$6.0 = \$5.52$

This rewards their high trust with a good profit and keeps our final payoffs relatively balanced.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Looking at this situation:

- The sender sent \$3.84, which is 76.8% of their endowment - very high trust
- I received \$11.52 after the 3x multiplier
- The sender currently has $\$5 - \$3.84 = \$1.16$ remaining

This is high trust that should be rewarded well.

Let me consider different return amounts:

If I return \$5.5:

- Sender ends with: $\$1.16 + \$5.5 = \$6.66$ (profit of \$1.66)
- I end with: $\$11.52 - \$5.5 = \$6.02$

If I return \$6.0:

- Sender ends with: $\$1.16 + \$6.0 = \$7.16$ (profit of \$2.16)
- I end with: $\$11.52 - \$6.0 = \$5.52$

Given the high trust (76.8%), I should return generously. \$6.0 seems fair and rewards their trust well.

usage:

```
{
  "input_tokens": 1718,
  "output_tokens": 421,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.84, it became \$11.52, and you returned \$6.0. Your visible payoff was \$5.52.

Your total visible earnings across all rounds are \$32.42999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience:

- Round 2: Sent \$2.5 (50%), received \$6.78 back - great return
- Round 4: Sent \$3.0 (60%), received \$5.22 back - solid return

The game has been rewarding trust. I'll continue to show strong trust with this new co-player by sending \$3.5 (70% of my endowment).

```
'''json
{"send":3.5}
'''
```

reasoning:

Let me analyze my experience:

Round 1 (Receiver): Received \$6.9 from sender sending \$2.3, returned \$3.0. Payoff: \$3.9
Round 2 (Sender): Sent \$2.5, received \$6.78 back. Payoff: \$9.28
Round 3 (Receiver): Received \$12.51 from sender sending \$4.17, returned \$6.0. Payoff: \$6.51
Round 4 (Sender): Sent \$3.0, received \$5.22 back. Payoff: \$7.22
Round 5 (Receiver): Received \$11.52 from sender sending \$3.84, returned \$6.0. Payoff: \$5.52

So far:

- As a sender, I've sent \$2.5 and \$3.0 (50% and 60%)
- Both times I received generous returns (\$6.78 and \$5.22)
- As a receiver, I've been fair/generous in my returns

The game continues to reward trust. I've been gradually increasing my sending amounts. Let me continue this trend and send \$3.5 (70% of endowment) to show strong trust with this new co-player.

usage:

```
{
  "input_tokens": 2009,
  "output_tokens": 417,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$6.22 to you. Your visible payoff was \$7.72.

Your total visible earnings across all rounds are \$40.14999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my sender history:

- Round 2: Sent \$2.5, received \$6.78 back
- Round 4: Sent \$3.0, received \$5.22 back
- Round 6: Sent \$3.5, received \$6.22 back

Trust has been consistently rewarded. I'll continue showing strong trust with this new co-player by sending \$4.0 (80% of my endowment).

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history as sender:

- Round 2: Sent \$2.5 (50%), received \$6.78 back - excellent return
- Round 4: Sent \$3.0 (60%), received \$5.22 back - good return
- Round 6: Sent \$3.5 (70%), received \$6.22 back - good return

All three times I've sent money as sender, I've received generous returns. The game environment seems cooperative and trust is being rewarded.

As receiver:

- Round 1: Received \$6.9, returned \$3.0
- Round 3: Received \$12.51, returned \$6.0
- Round 5: Received \$11.52, returned \$6.0

I've been fair as a receiver too.

The pattern suggests the game is cooperative. I could continue increasing my trust level. Maybe send \$4.0 (80%) this time? That would show strong trust with this new co-player.

Let me think about the strategy:

- Sending more has generally worked well
- I'm playing against a new co-player each round
- Building reputation for cooperation could help
- Total earnings are healthy at \$40.15

I'll send \$4.0 (80% of endowment) to show strong trust.

usage:

```
{
  "input_tokens": 2246,
  "output_tokens": 423,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$8.61 to you. Your visible payoff was \$9.61.

Your total visible earnings across all rounds are \$49.75999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.61, so you received \$13.830000000000002. How much do you return? (0-13.830000000000002)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- This is the same co-player from Round 7!
- Last round, I sent them \$4.0 and they returned an exceptional \$8.61
- Now they've sent me \$4.61 (92.2% of their endowment) - extremely high trust
- I received \$13.83 after the 3x multiplier
- They currently have only \$0.39 remaining

My co-player has shown exceptional generosity and trust. I should reciprocate strongly.

If I return \$8.0:

- Sender ends with: \$0.39 + \$8.0 = \$8.39 (profit of \$3.39)
- I end with: \$13.83 - \$8.0 = \$5.83

This matches their previous generosity and rewards their high trust appropriately.

```
'''json
{"return": 8.0}
'''
```

reasoning:

Let me analyze this situation:

- This is the same co-player from Round 7
- In Round 7: I sent \$4.0, they returned \$8.61 - that's an exceptional return! They were very generous.
- Now in Round 8: They sent me \$4.61 (92.2% of their endowment) - showing extremely high trust
- I received \$13.83 after the 3x multiplier
- They currently have only \$0.39 remaining

My co-player has been very generous to me. In Round 7, they returned \$8.61 out of \$12.0 (71.75% return rate). That's excellent reciprocity.

Now they're showing even MORE trust by sending \$4.61. I should definitely reciprocate generously.

If I return \$7.0:

- Sender ends with: $\$0.39 + \$7.0 = \$7.39$ (profit of \$2.39)
- I end with: $\$13.83 - \$7.0 = \$6.83$

If I return \$8.0:

- Sender ends with: $\$0.39 + \$8.0 = \$8.39$ (profit of \$3.39)
- I end with: $\$13.83 - \$8.0 = \$5.83$

Given they returned \$8.61 last round and now sent me \$4.61, I should be similarly generous. Returning around \$7-8 seems appropriate.

Let me go with \$7.5 to be quite generous while still benefiting from the exchange:

- Sender ends with: $\$0.39 + \$7.5 = \$7.89$ (profit of \$2.89)
- I end with: $\$13.83 - \$7.5 = \$6.33$

Actually, given they returned \$8.61 last round (an exceptional amount), I should probably return at least \$8.0 to match their generosity level.

...

usage:

```
{
  "input_tokens": 2249,
  "output_tokens": 745,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.61, it became \$13.830000000000002, and you returned \$8.0. Your visible payoff was \$5.8300000000000002.

Your total visible earnings across all rounds are \$55.58999999999999.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my sender history:

- Round 2: Sent \$2.5, received \$6.78 back
- Round 4: Sent \$3.0, received \$5.22 back
- Round 6: Sent \$3.5, received \$6.22 back
- Round 7: Sent \$4.0, received \$8.61 back (exceptional return!)

Trust has been consistently rewarded, with increasingly better returns. I'll continue showing strong trust with this new co-player by sending \$4.0 (80% of my endowment).

```
'''json
```



```
{"send": 4.0}
'''
```

reasoning:

Looking at my history:

Round 2: SENDER - Sent \$2.5, received \$6.78 back
Round 3: RECEIVER - Received \$12.51 (from \$4.17 sent), returned \$6.0
Round 4: SENDER - Sent \$3.0, received \$5.22 back
Round 5: RECEIVER - Received \$11.52 (from \$3.84 sent), returned \$6.0
Round 6: SENDER - Sent \$3.5, received \$6.22 back
Round 7: SENDER - Sent \$4.0, received \$8.61 back
Round 8: RECEIVER - Received \$13.83 (from \$4.61 sent), returned \$8.0

Now Round 9: SENDER against a new co-player

My sending pattern has been increasing:
\$2.5 ? \$3.0 ? \$3.5 ? \$4.0

And returns have been good, with the last one being exceptional (\$8.61 return on \$4.0 sent).

The game has consistently rewarded trust. I should continue showing high trust. I'll send \$4.0 again, which has been working well.

Actually, looking at the progression, I could go even higher. But \$4.0 (80%) seems like a strong trust signal that has worked well.

usage:

```
{
  "input_tokens": 2312,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.0, it became \$12.0, and you saw them return \$8.07 to you. Your visible payoff was \$9.07.

Your total visible earnings across all rounds are \$64.66.

This round, you are the RECEIVER against your current co-player. They sent you \$4.67, so you received \$14.01. How much do you return? (0-14.01)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

- This is the same co-player from Round 9!
- Last round, I sent them \$4.0 and they returned \$8.07
- Now they've sent me \$4.67 (93.4% of their endowment) - exceptionally high trust
- I received \$14.01 after the 3x multiplier
- They currently have only \$0.33 remaining

My co-player has shown exceptional generosity and trust again. I should reciprocate strongly.

If I return \$8.5:

- Sender ends with: \$0.33 + \$8.5 = \$8.83 (profit of \$3.83)
- I end with: \$14.01 - \$8.5 = \$5.51

This rewards their very high trust with a good profit.

```
'''json
{"return": 8.5}
'''
```

Reasoning:

Looking at this situation:

- This is the same co-player from Round 9!
- Last round, I sent them \$4.0 and they returned an excellent \$8.07
- Now they've sent me \$4.67 (93.4% of their endowment) - extremely high trust

- I received \$14.01 after the 3x multiplier
- They currently have only \$0.33 remaining

My co-player has shown exceptional generosity and trust again. I should reciprocate strongly.

If I return \$8.5:

- Sender ends with: $\$0.33 + \$8.5 = \$8.83$ (profit of \$3.83)
- I end with: $\$14.01 - \$8.5 = \$5.51$

This rewards their very high trust with a good profit while keeping our payoffs balanced.

Actually, they've been very generous. Let me think about this more carefully. They sent \$4.67, so they have \$0.33 left. If I return \$8.5, they get \$8.83 total, which is a profit of \$3.83. That seems fair given their very high trust.

Alternatively, if I return \$9.0:

- Sender ends with: $\$0.33 + \$9.0 = \$9.33$ (profit of \$4.33)
- I end with: $\$14.01 - \$9.0 = \$5.01$

This would be even more generous and reward their exceptional trust.

Let me look at the pattern:

- Round 8: They sent \$4.61, I returned \$8.0
- Round 10: They sent \$4.67, should I return about the same or more?

Since they've shown consistent high trust and generosity, I'll return \$8.5 which is a strong reward for their trust.

```
Usage: {
  "input_tokens": 2274,
  "output_tokens": 641,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_3

Timestamp: 2026-08-12T09:29:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_3"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_3": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_3"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.38, it became \$10.14, and you returned \$5.0. Your visible payoff was \$5.140000000000001.

Your total visible earnings across all rounds are \$17.75.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my experience so far:

Round 1: As SENDER, I sent \$3 and received \$3.37 back - a reasonable return.

Rounds 2-3: As RECEIVER, others showed good trust, and I reciprocated fairly.

This is a new co-player (random pairing), so I don't have specific information about them. However, the overall pattern suggests players are reasonably cooperative.

I'll send \$3.50 - slightly more than my first round to show increasing trust while remaining measured:

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER, sent \$3, received back \$3.37, payoff \$5.37

Round 2: I was RECEIVER, received \$13.74 (from \$4.58), returned \$6.50, payoff \$7.24

Round 3: I was RECEIVER, received \$10.14 (from \$3.38), returned \$5.00, payoff \$5.14
Round 4: I am SENDER

My total earnings: \$17.75

Looking at the pattern:

- When I sent \$3, I got back \$3.37 (a bit more than 1:1 return on sent amount)
- When others sent to me, they showed high trust (\$4.58 and \$3.38 out of \$5)
- I've been fairly reciprocal as a receiver, returning roughly half of what I received

Now I'm a sender again. Since:

1. My first send of \$3 was met with a decent return (\$3.37)
2. Others have been trusting with me
3. I've been fair in return

I should continue with moderate to high trust. I'll send \$3.5 this time - slightly more than my first send to show I'm willing to build cooperation.

Actually, let me reconsider. The co-player this round is new (pairings are randomized). I don't know their history. In Round 1, I sent \$3 and got a reasonable return. Let me stick with a similar strategy - send \$3 again as a baseline cooperative strategy.

Actually, looking at the overall pattern, players seem reasonably cooperative. I could try \$3.5 to be slightly more trusting.

Let me go with \$3.5 - showing increased trust while not overcommitting.

usage:

```
{
  "input_tokens": 1303,
  "output_tokens": 551,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$7.22 to you. Your visible payoff was \$8.719999999999999.

Your total visible earnings across all rounds are \$26.47.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my recent experience:

Round 4: I sent \$3.50 and received \$7.22 back - excellent cooperation!

The overall pattern shows players are being reasonably cooperative. While this is a new co-player (random pairing), I'll show increased trust based on positive experiences.

Sending \$4:

- Demonstrates strong trust (80% of my endowment)
- Creates \$12 for them to work with
- Encourages reciprocal cooperation

```
'''json
{"send": 4}
'''
```

reasoning:

Round 4 recap: I sent \$3.50, they returned \$7.22 - excellent return! That's about 69% of what they received, very generous.

This is Round 5, and I'm the SENDER again against a new co-player (random pairing each round).

So far my experience:

- Round 1 SENDER: sent \$3, got \$3.37 back (reasonable)

- Round 2 RECEIVER: received \$13.74, returned \$6.50
- Round 3 RECEIVER: received \$10.14, returned \$5.00
- Round 4 SENDER: sent \$3.50, got \$7.22 back (very good!)

The last round showed great cooperation. Players seem generally trustworthy. This is a new co-player though, so I don't know their specific behavior.

I should probably continue with a trusting approach. I could:

- Send \$4 (80% of endowment) - shows high trust
- Send \$3.5 (70%) - similar to last round
- Send \$3 (60%) - more cautious

Given the positive experiences, I'll increase slightly to \$4 to show strong trust.

usage:

```
{
  "input_tokens": 1562,
  "output_tokens": 402,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$6.98 to you. Your visible payoff was \$7.98.

Your total visible earnings across all rounds are \$34.45.

This round, you are the RECEIVER against your current co-player. They sent you \$2.9, so you received \$8.7. How much do you return? (0-8.7)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

My co-player sent me \$2.9 out of their \$5 (58% of their endowment) - showing moderate trust.

I received \$8.7.

They kept: \$2.1

To reciprocate fairly:

- If I return \$4.00, they'd end with \$6.10 (a gain of \$1.10)
- I'd keep \$4.70 (also a decent gain)

This represents returning about 46% of what I received, which is proportional to the moderate trust shown and ensures we both benefit.

```
'''json
{"return": 4.00}
'''
```

reasoning:

Looking at this situation:

My co-player sent me \$2.9 out of their \$5 (58% of their endowment) - showing moderate trust.

I received \$8.7.

They kept: \$2.1

To reciprocate fairly:

- If I return around \$4.00-\$4.50, they'd end with \$6.10-\$6.60 (a gain of \$1.10-\$1.60)
- I'd keep \$4.70-\$4.20

This represents returning about 46-52% of what I received, which rewards their trust while we both benefit.

Let me think about what's fair. They sent \$2.9, showing moderate but not high trust. I should return something proportional.

In previous rounds as receiver:

- Round 2: Received \$13.74, returned \$6.50 (47%)
- Round 3: Received \$10.14, returned \$5.00 (49%)

So I've been returning around 47-49% of what I receive.

For \$8.7, returning around 47-49% would be:

- 47%: \$4.09
- 49%: \$4.26

Let me go with \$4.25, which is about 49% and gives them a total of \$6.35 (gain of \$1.35), while I keep \$4.45 (also a decent gain).

Actually, let me reconsider. They sent \$2.9 (58% of endowment), which is moderate trust. In Round 1, someone sent me \$4.58 (91.6%) and I returned \$6.50. In Round 3, someone sent \$3.38 (67.6%) and I returned \$5.00.

The person who sent \$3.38 (67.6%) got back \$5.00 from \$10.14 received.

...

usage:

```
{
  "input_tokens": 1817,
  "output_tokens": 654,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.9, it became \$8.7, and you returned \$4.0. Your visible payoff was \$4.699999999999999.

Your total visible earnings across all rounds are \$39.150000000000006.

This round, you are the RECEIVER against your current co-player. They sent you \$2.99, so you received \$8.97. How much do you return? (0-8.97)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

My co-player sent me \$2.99 out of their \$5 (about 60% of their endowment) - showing moderate trust, very similar to Round 6.

I received \$8.97.

They kept: \$2.01

To reciprocate consistently with my Round 6 approach:

- If I return \$4.10, they'd end with \$6.11 (a gain of \$1.11)
- I'd keep \$4.87 (also a decent gain)

This represents returning about 46% of what I received, consistent with similar trust levels.

```
'''json
{"return": 4.10}
'''
```

reasoning:

My co-player sent me \$2.99 out of their \$5 (about 60% of their endowment) - showing moderate trust, very similar to the previous round.

I received \$8.97.

They kept: \$2.01

This is very similar to Round 6 where they sent \$2.9. I should be consistent in my reciprocation strategy.

In Round 6, I returned \$4.00 when I received \$8.7, which was about 46% of what I received.

Using a similar proportion:

- If I return \$4.10, they'd end with \$6.11 (a gain of \$1.11)
- I'd keep \$4.87 (also a decent gain)

This represents returning about 46% of what I received, consistent with my previous approach for similar trust levels.

Let me calculate more carefully:

- They sent: \$2.99
- They kept: \$5 - \$2.99 = \$2.01
- I received: \$8.97
- If I return \$4.10:
 - They end with: \$2.01 + \$4.10 = \$6.11
 - I end with: \$8.97 - \$4.10 = \$4.87

This seems fair and consistent with my previous behavior.

usage:

```
{
  "input_tokens": 2105,
  "output_tokens": 465,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.99, it became \$8.97, and you returned \$4.1. Your visible payoff was \$4.870000000000001.

Your total visible earnings across all rounds are \$44.020000000000001.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my experience:

As SENDER:

- Round 1: sent \$3.00, received \$3.37 back
- Round 4: sent \$3.50, received \$7.22 back (excellent!)
- Round 5: sent \$4.00, received \$6.98 back (good!)

The pattern shows that sending more has generally been rewarded with good reciprocation. This is Round 8 with a new co-player.

I'll send \$4.00 - this amount has proven successful and demonstrates strong trust while maintaining some prudence.

```
'''json
{"send": 4.0}
'''
```

reasoning:

Looking at my history:

Round 1: SENDER - sent \$3.00, received \$3.37 back (payoff: \$5.37)
 Round 2: RECEIVER - received \$13.74 from \$4.58 sent, returned \$6.50 (payoff: \$7.24)
 Round 3: RECEIVER - received \$10.14 from \$3.38 sent, returned \$5.00 (payoff: \$5.14)
 Round 4: SENDER - sent \$3.50, received \$7.22 back (payoff: \$8.72)
 Round 5: SENDER - sent \$4.00, received \$6.98 back (payoff: \$7.98)
 Round 6: RECEIVER - received \$8.70 from \$2.90 sent, returned \$4.00 (payoff: \$4.70)
 Round 7: RECEIVER - received \$8.97 from \$2.99 sent, returned \$4.10 (payoff: \$4.87)
 Round 8: SENDER - this round, new co-player

My sending pattern has been:

- Round 1: \$3.00
- Round 4: \$3.50
- Round 5: \$4.00

I've been gradually increasing trust. The returns have generally been good:

- Round 1: sent \$3, got \$3.37 back (net: +\$0.37)

- Round 4: sent \$3.5, got \$7.22 back (net: +\$2.22) - excellent!
- Round 5: sent \$4, got \$6.98 back (net: +\$1.48) - good!

Overall, sending more has been rewarded. This is a final round (Round 8), and a new co-player.

Given the positive trend and it being the final round, I could send more to maximize potential returns. ...

usage:

```
{
  "input_tokens": 2237,
  "output_tokens": 647,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$9.55 to you. Your visible payoff was \$10.55.

Your total visible earnings across all rounds are \$54.57000000000001.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

My co-player sent me \$3.11 out of their \$5 (62% of their endowment) - showing good trust.

I received \$9.33.

They kept: \$1.89

To reciprocate fairly:

- If I return \$4.50, they'd end with \$6.39 (a gain of \$1.39)
- I'd keep \$4.83 (also a good gain)

This represents returning about 48% of what I received, consistent with my reciprocity pattern for good trust levels.

```
'''json
{"return": 4.50}
'''
```

reasoning:

Looking at this situation:

My co-player sent me \$3.11 out of their \$5 (62.2% of their endowment) - showing good trust

I received \$9.33.

They kept: \$1.89

Let me think about a fair return:

- If I return \$4.50, they'd end with \$6.39 (a gain of \$1.39)
- I'd keep \$4.83 (also a good gain)

This represents returning about 48% of what I received, which is consistent with my previous returns for similar trust levels.

Looking at my past behavior as receiver:

- Round 2: sent \$4.58, received \$13.74, returned \$6.50 (47%)
- Round 3: sent \$3.38, received \$10.14, returned \$5.00 (49%)
- Round 6: sent \$2.90, received \$8.70, returned \$4.00 (46%)
- Round 7: sent \$2.99, received \$8.97, returned \$4.10 (46%)

For \$3.11 sent (\$9.33 received), a return around 46-49% would be \$4.30-\$4.57.

I'll return \$4.50, which:

- Is proportional to the trust shown
- Ensures both parties benefit
- Is consistent with my previous reciprocity patterns

usage:

```
{
  "input_tokens": 2237,
  "output_tokens": 477,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$4.5. Your visible payoff was \$4.83.

Your total visible earnings across all rounds are \$59.400000000000006.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my experience:

As SENDER:

- Round 1: sent \$3.00, received \$3.37 back
- Round 4: sent \$3.50, received \$7.22 back
- Round 5: sent \$4.00, received \$6.98 back
- Round 8: sent \$4.00, received \$9.55 back (excellent!)

Sending \$4.00 has consistently yielded good returns, with the most recent one being particularly strong. This is a new co-player, but the overall trend supports continuing this strategy.

```
'''json
{"send": 4.0}
'''
```

Reasoning:

Looking at my experience as SENDER so far:

- Round 1: sent \$3.00, received \$3.37 back
- Round 4: sent \$3.50, received \$7.22 back (excellent!)
- Round 5: sent \$4.00, received \$6.98 back (good!)
- Round 8: sent \$4.00, received \$9.55 back (excellent!)

The pattern shows:

1. When I send \$4, I've been getting great returns (\$6.98 and \$9.55)
2. Overall, players have been cooperative
3. Sending more has generally been rewarded

This is Round 10 with a new co-player. Based on my positive experiences, especially with sending \$4, I should continue with that strategy. The most recent round where I sent \$4 gave me \$9.55 back, which was amazing.

I'll send \$4.00 again - it's been a successful strategy.

Usage: {

```
"input_tokens": 2229,
"output_tokens": 375,
"reasoning_tokens": 0
}
```

GAME - Agent_3 - Receiver - dyad_3

Timestamp: 2026-08-12T09:29:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_8",
"pairing": {
  "agents": [
```

```

    "Agent_8",
    "Agent_3"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_3": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_3"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$10.29 to you. Your visible payoff was \$10.29.

Your total visible earnings across all rounds are \$22.81.

This round, you are the RECEIVER against your current co-player. They sent you \$3.57, so you received \$10.709999999999999. How much do you return? (0-10.709999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

My co-player sent \$3.57 (71.4% of their endowment), showing strong trust. I received \$10.71.

The game has been very cooperative so far:

- Round 1: I sent \$4, got back \$8.69
- Round 3: I sent \$5, got back \$10.29
- Round 2: When I was receiver, I returned \$6.50 out of \$9.33

My co-player trusted me significantly. A generous return would be around 70% of what I received, which would give them a strong profit while still rewarding me for receiving.

Returning \$7.50 would:

- Give them: \$5 - \$3.57 + \$7.50 = \$8.93 total
- Leave me with: \$10.71 - \$7.50 = \$3.21

This maintains the cooperative pattern.

{"return": 7.50}

reasoning:

Let me analyze this situation:

My co-player sent me \$3.57 out of \$5 (71.4% of their endowment), showing strong trust. I received \$10.71 after the 3x multiplier.

In my previous round as sender:

- I sent \$5 (100% of endowment)
- They returned \$10.29 out of \$15 (68.6%)
- This was very generous, giving me \$10.29 total payoff

The pattern has been quite cooperative. Let me think about a fair return:

If I return around \$7-7.50:

- They would get: \$5 - \$3.57 + \$7 = \$8.43 (if I return \$7)
- They would get: \$5 - \$3.57 + \$7.50 = \$8.93 (if I return \$7.50)
- I would keep: \$10.71 - \$7 = \$3.71 (if I return \$7)
- I would keep: \$10.71 - \$7.50 = \$3.21 (if I return \$7.50)

Given they sent 71.4% of their endowment, a generous return would be appropriate. The previous receiver returned about 68.6% when I sent 100%.

Returning about \$7.20 would:

- Give them \$8.63 total (\$5 - \$3.57 + \$7.20)
- Leave me with \$3.51
- This rewards their trust well

Actually, let me be a bit more generous to match the cooperative environment. Returning \$7.50:

- Gives them \$8.93 total (a \$3.93 profit on their \$5)
- Leaves me with \$3.21 (also a good profit)
- Return rate: 70% of what I received

...

usage:

```
{
  "input_tokens": 1326,
  "output_tokens": 675,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.57, it became \$10.709999999999999, and you returned \$7.5. Your visible payoff was \$3.2099999999999999. Your total visible earnings across all rounds are \$26.019999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my track record:

- As SENDER: Sent \$4 (got \$8.69), Sent \$5 (got \$10.29)
- As RECEIVER: Returned generously both times

The environment is highly cooperative. Sending the full amount has consistently yielded very good returns for me. My average earnings per round (\$6.50) significantly exceed the \$5 baseline, showing that cooperation pays off.

I'll continue the cooperative strategy and send my full endowment.

```
{"send": 5}
```

reasoning:

Looking at my history:

- Round 1 (SENDER): Sent \$4, got back \$8.69 - great outcome
- Round 2 (RECEIVER): Got \$9.33, returned \$6.50
- Round 3 (SENDER): Sent \$5, got back \$10.29 - excellent outcome
- Round 4 (RECEIVER): Got \$10.71, returned \$7.50

The pattern is very clear - I've experienced highly cooperative behavior. When I've been the sender and sent significant amounts, I've gotten generous returns. When I've been the receiver, I've also returned generously.

My cumulative earnings are \$26.02 across 4 rounds, averaging \$6.50 per round, which is well above the \$5 baseline if I had just kept everything.

The optimal strategy in this cooperative environment is to continue sending high amounts:

1. It creates value through the 3x multiplier
2. Past experience shows I get good returns
3. It builds my reputation for future rounds

I should send \$5 again to maximize value creation and maintain my cooperative reputation.

usage:

```
{
  "input_tokens": 1678,
  "output_tokens": 377,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$8.2 to you. Your visible payoff was \$8.2.

Your total visible earnings across all rounds are \$34.22.

This round, you are the RECEIVER against your current co-player. They sent you \$3.41, so you received \$10.23. How much do you return? (0-10.23)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

My co-player sent \$3.41 (68% of their endowment), showing solid trust. I received \$10.23.

My pattern as receiver has been:

- Round 2: Returned \$6.50 out of \$9.33 (~70%)
- Round 4: Returned \$7.50 out of \$10.71 (~70%)

Maintaining this cooperative approach, returning around \$7 would:

- Give them: $\$5 - \$3.41 + \$7 = \8.59 total
- Leave me with: $\$10.23 - \$7 = \$3.23$

This rewards their trust while staying consistent with my cooperative strategy.

```
{"return": 7.00}
```

reasoning:

My co-player sent \$3.41 (68.2% of their endowment), showing good trust. I received \$10.23.

Looking at my history:

- As SENDER: Sent \$4 (got \$8.69), Sent \$5 (got \$10.29), Sent \$5 (got \$8.2)
- As RECEIVER: Returned \$6.50 out of \$9.33, Returned \$7.50 out of \$10.71

The pattern has been very cooperative. When I receive around \$10, I've been returning between \$6.50-\$7.50.

For this round with \$10.23 received:

- Returning around \$7 would give my co-player: $\$5 - \$3.41 + \$7 = \8.59
- I'd keep: $\$10.23 - \$7 = \$3.23$

This seems fair and maintains the cooperative pattern. I'll return \$7.

Actually, let me reconsider. My previous returns as receiver:

- Round 2: Received \$9.33, returned \$6.50 (69.7%)
- Round 4: Received \$10.71, returned \$7.50 (70%)

So I've been returning around 70% of what I received. 70% of \$10.23 = \$7.16

I'll return \$7.00 to stay consistent with my cooperative pattern.

usage:

```
{
  "input_tokens": 1931,
  "output_tokens": 513,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.41, it became \$10.23, and you returned \$7.0. Your visible payoff was \$3.2300000000000004.

Your total visible earnings across all rounds are \$37.45.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at my experience:

As SENDER:

- Sent \$4, received \$8.69
- Sent \$5, received \$10.29
- Sent \$5, received \$8.20

All my sending decisions have been profitable, and the environment remains cooperative. Sending the full \$5 maximizes potential gains through the 3x multiplier and has consistently yielded good returns.

I'll continue with the cooperative strategy.

```
{"send": 5}
```

reasoning:

Looking at my history:

- As SENDER:
 - * Round 1: Sent \$4, got back \$8.69
 - * Round 3: Sent \$5, got back \$10.29
 - * Round 5: Sent \$5, got back \$8.20
- As RECEIVER:

- * Round 2: Received \$9.33, returned \$6.50
- * Round 4: Received \$10.71, returned \$7.50
- * Round 6: Received \$10.23, returned \$7.00

The environment has been consistently cooperative. All three times I sent money as a sender, I received generous returns. When I send \$5, I've gotten back \$10.29 and \$8.20 (averaging about \$9.25), both of which are profitable.

I'm earning well above the \$5 baseline per round. The cooperative strategy is working. I should continue sending my full endowment to maximize value creation and maintain the cooperative equilibrium.

usage:

```
{
  "input_tokens": 2228,
  "output_tokens": 341,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$7.64 to you. Your visible payoff was \$7.64.

Your total visible earnings across all rounds are \$45.09.

This round, you are the RECEIVER against your current co-player. They sent you \$4.85, so you received \$14.549999999999999. How much do you return? (0-14.549999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

My co-player sent \$4.85 (97% of their endowment) - this is exceptional trust, nearly their entire amount!

My pattern as receiver has been returning around 68-70% of what I receive:

- Round 2: 70% of \$9.33 = \$6.50
- Round 4: 70% of \$10.71 = \$7.50
- Round 6: 68% of \$10.23 = \$7.00

Given their exceptional trust (sending almost everything), I should maintain this cooperative approach. Returning 70% of \$14.55 = \$10.19

This would give them:

- Their payoff: \$5 - \$4.85 + \$10.19 = \$10.34
- My payoff: \$14.55 - \$10.19 = \$4.36

This strongly rewards their trust.

```
{"return": 10.20}
```

reasoning:

My co-player sent \$4.85, which is 97% of their endowment - extremely high trust! I received \$14.55.

This is someone showing exceptional trust. Let me think about what's fair:

My past returns as receiver:

- Round 2: Returned \$6.50 out of \$9.33 (~70%)
- Round 4: Returned \$7.50 out of \$10.71 (~70%)
- Round 6: Returned \$7.00 out of \$10.23 (~68%)

So I've been returning around 68-70% of what I received.

If I return 70% of \$14.55 = \$10.19

That would give them:

- Their payoff: \$5 - \$4.85 + \$10.19 = \$10.34
- My payoff: \$14.55 - \$10.19 = \$4.36

This rewards their extreme trust significantly. They basically sent almost everything, so I should be generous back.

Alternatively, I could return a bit more given their exceptional trust. Maybe \$10.50?

- Their payoff: $\$5 - \$4.85 + \$10.50 = \10.65
- My payoff: $\$14.55 - \$10.50 = \$4.05$

I think around \$10.00-\$10.50 is appropriate given their nearly maximal trust.

Let me go with \$10.20 to maintain my ~70% return pattern but be generous given their exceptional trust.

usage:

```
{
  "input_tokens": 2318,
  "output_tokens": 581,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.85, it became \$14.549999999999999, and you returned \$10.2. Your visible payoff was \$4.35.

Your total visible earnings across all rounds are \$49.440000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my track record:

As SENDER (4 times):

- Every send has been profitable
- Average return has been \$8.71 on \$5 sent
- Even my lowest return (\$7.64) was still better than keeping everything

As RECEIVER:

- I've consistently returned around 70% to maintain cooperation

My total earnings (\$49.44 over 8 rounds = \$6.18/round) show that full cooperation is working extremely well.

I'll continue with the successful strategy.

```
{"send": 5}
```

reasoning:

Looking at my track record:

As SENDER:

- Round 1: Sent \$4, received \$8.69 back
- Round 3: Sent \$5, received \$10.29 back
- Round 5: Sent \$5, received \$8.20 back
- Round 7: Sent \$5, received \$7.64 back

As RECEIVER:

- Round 2: Received \$9.33, returned \$6.50
- Round 4: Received \$10.71, returned \$7.50
- Round 6: Received \$10.23, returned \$7.00
- Round 8: Received \$14.55, returned \$10.20

My average earnings per round: $\$49.44 / 8 = \6.18 per round

The cooperative strategy has been working very well. I've been consistently sending my full endowment as sender, and returning generously as receiver. The environment is highly cooperative.

All my sending attempts have been profitable:

- Sent \$4, net gain = $\$8.69 - \$4 = \$4.69$

- Sent \$5, net gain = $\$10.29 - \$5 = \$5.29$
- Sent \$5, net gain = $\$8.20 - \$5 = \$3.20$
- Sent \$5, net gain = $\$7.64 - \$5 = \$2.64$

Even the "worst" outcome (Round 7) still gave me \$2.64 more than if I kept everything.

I should continue the cooperative strategy and send my full endowment.

usage:

```
{
  "input_tokens": 2354,
  "output_tokens": 498,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the SENDER against another agent. You sent \$5.0, it became \$15.0, and you saw them return \$6.94 to you. Your visible payoff was \$6.94.

Your total visible earnings across all rounds are \$56.38.

This round, you are the RECEIVER against your current co-player. They sent you \$4.56, so you received \$13.68. How much do you return? (0-13.68)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

My co-player sent \$4.56 (91% of their endowment) - demonstrating very high trust.

My consistent pattern as receiver has been returning around 70% of what I receive:

- Round 2: 70%
- Round 4: 70%
- Round 6: 68%
- Round 8: 70%

Maintaining this cooperative approach with \$13.68 received, I'll return 70% = \$9.58 (rounding to \$9.60).

This gives them:

- Their payoff: $\$5 - \$4.56 + \$9.60 = \10.04
- My payoff: $\$13.68 - \$9.60 = \$4.08$

This strongly rewards their trust.

```
{"return": 9.60}
```

Reasoning:

My co-player sent \$4.56 (91.2% of their endowment) - this is very high trust.

My pattern as receiver:

- Round 2: Returned \$6.50 out of \$9.33 (~70%)
- Round 4: Returned \$7.50 out of \$10.71 (~70%)
- Round 6: Returned \$7.00 out of \$10.23 (~68%)
- Round 8: Returned \$10.20 out of \$14.55 (~70%)

I've been consistently returning around 68-70% of what I receive.

For \$13.68, returning 70% would be \$9.58.

This would give them:

- Their payoff: $\$5 - \$4.56 + \$9.58 = \10.02
- My payoff: $\$13.68 - \$9.58 = \$4.10$

That's a solid reward for their high trust. Let me return \$9.60 to keep it round.

Actually, let me stick closer to 70%: $\$13.68 * 0.70 = \9.576 , so \$9.58 or \$9.60 both work.

I'll go with \$9.60.

Usage: {

```
"input_tokens": 2367,
```



```
"output_tokens": 494,
"reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_4

Timestamp: 2026-08-12T09:29:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_7"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.8, it became \$11.399999999999999, and you returned \$6.5. Your visible payoff was \$4.899999999999999.

Your total visible earnings across all rounds are \$19.15.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at my experience so far:

- Round 1: As receiver, someone sent high trust (95.6%), I reciprocated fairly
- Round 2: I sent \$4.50 (90%), got back \$7.41 - good cooperation
- Round 3: As receiver, returned fairly when sent \$3.80

The pattern shows that high trust tends to be reciprocated with fair returns (around 55% return rate). My cooperative strategy has been working well - averaging over \$6 per round.

I'll continue with a strong but slightly varied approach. Sending \$4.00 (80%):

- Shows substantial trust
- Creates \$12.00 for the receiver through the 3x multiplier
- Should encourage reciprocation based on the established pattern
- Slightly more conservative than my last sending round to manage risk

{'send': 4.00}

reasoning:

Let me analyze my history:

Round 1 (RECEIVER): Sender sent \$4.78 (95.6%), I received \$14.34, returned \$8.00, kept \$6.34

Round 2 (SENDER): I sent \$4.50 (90%), became \$13.50, received back \$7.41, ended with \$7.91

Round 3 (RECEIVER): Sender sent \$3.80 (76%), I received \$11.40, returned \$6.50, kept \$4.90

Now Round 4, I'm the SENDER again with a new co-player.

Looking at the pattern:

- When I sent \$4.50 (90%), I got back \$7.41 from \$13.50 (about 55% return rate)
- When someone sent \$4.78 (95.6%), I returned \$8.00 from \$14.34 (about 56% return rate)
- When someone sent \$3.80 (76%), I returned \$6.50 from \$11.40 (about 57% return rate)

The return rates seem fairly consistent around 55-57%. This suggests cooperation is being reciprocated.

I've been doing well with high cooperation levels. My total earnings are \$19.15 across 3 rounds, averaging about \$6.38 per round, which is well above the baseline of \$5.

Given this positive experience with cooperation, I should continue sending a substantial amount. I'll send \$4.00 (80%), which is:

- Still showing significant trust
- Creating good value through the 3x multiplier (\$12.00 received)
- Likely to be reciprocated based on the pattern
- Slightly more conservative than my previous \$4.50 to diversify my strategy a bit

If they return around 55%, I'd get back about \$6.60, ending with \$1.00 + \$6.60 = \$7.60, which would be good.

usage:

```
{
  "input_tokens": 1488,
  "output_tokens": 656,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$4.58 to you. Your visible payoff was \$5.58.

Your total visible earnings across all rounds are \$24.729999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$4.73, so you received \$14.190000000000001. How much do you return? (0-14.190000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

- The sender showed exceptional trust by sending \$4.73 (94.6% of their endowment)
- They kept only \$0.27 for themselves
- I received \$14.19
- This is very similar to Round 1, where someone sent 95.6%

In Round 1, when faced with similarly high trust, I returned \$8.00 out of \$14.34. That worked well as a fair reciprocation.

For this round with \$14.19 received, returning \$8.00:

- Sender ends with: \$0.27 + \$8.00 = \$8.27 (strong return on their trust)
- I keep: \$14.19 - \$8.00 = \$6.19
- Both benefit significantly from the cooperation

Consistency with my previous fair reciprocation of high trust seems appropriate.

```
{'return': 8.00}
```

reasoning:

Let me analyze this situation:

Round 5 - I'm the RECEIVER

- Sender sent \$4.73 (94.6% of their endowment) - very high trust!
- They kept only \$0.27
- I received \$14.19 after the 3x multiplier

This is similar to Round 1 where someone sent \$4.78 (95.6%) and I returned \$8.00.

The sender is showing exceptional trust, almost sending everything. I should reciprocate fairly.

If I return \$8.00:

- Sender ends with: \$0.27 + \$8.00 = \$8.27 (good gain from initial \$5)
- I keep: \$14.19 - \$8.00 = \$6.19
- Both benefit, sender gets rewarded for high trust

This seems consistent with my Round 1 behavior where I also returned \$8.00 for a similar high-trust send.

Actually, let me think about this more carefully. The sender sent 94.6%. In Round 1, when sender sent 95.6%, I returned \$8.00 out of \$14.34, which is about 56%.

$\$14.19 \times 0.56 \approx \7.95

So around \$8.00 seems right. Let me stick with that.

usage:

```
{
  "input_tokens": 1831,
  "output_tokens": 528,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$4.73, it became \$14.190000000000001, and you returned \$8.0. Your visible payoff was \$6.190000000000001.

Your total visible earnings across all rounds are \$30.919999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my experience as a sender:

- Round 2: Sent \$4.50 (90%), received \$7.41 back - good cooperation
- Round 4: Sent \$4.00 (80%), received \$4.58 back - lower reciprocation

The pattern suggests that higher trust levels tend to get better reciprocation. I'm averaging around \$6.18 per round with my cooperative strategy, which is working well.

As a receiver, I've been consistent in returning around 56% of what I receive, which has seemed fair to my co-players.

I'll send \$4.50 again (90% of endowment):

- Shows strong, consistent trust
- Previously resulted in good reciprocation
- Creates \$13.50 for the receiver
- Maintains my cooperative approach

{'send': 4.50}

reasoning:

Let me review the pattern of my play:

Round 1 (Receiver): Received \$14.34 from sender who sent \$4.78 (95.6%), returned \$8.00 ? kept \$6.34

Round 2 (Sender): Sent \$4.50 (90%), received back \$7.41 ? total \$7.91

Round 3 (Receiver): Received \$11.40 from sender who sent \$3.80 (76%), returned \$6.50 ? kept \$4.90

Round 4 (Sender): Sent \$4.00 (80%), received back \$4.58 ? total \$5.58

Round 5 (Receiver): Received \$14.19 from sender who sent \$4.73 (94.6%), returned \$8.00 ? kept \$6.19

My total visible earnings: \$30.92

As a sender:

- Round 2: Sent \$4.50, got back \$7.41 (55% return rate)
- Round 4: Sent \$4.00, got back \$4.58 (38% return rate)

As a receiver:

- Round 1: Returned \$8.00 out of \$14.34 (56% return rate) when sender sent 95.6%
- Round 3: Returned \$6.50 out of \$11.40 (57% return rate) when sender sent 76%
- Round 5: Returned \$8.00 out of \$14.19 (56% return rate) when sender sent 94.6%

I've been consistent in my returns - around 56% when I'm the receiver.

When I've been the sender:

- Higher sends seem to get better returns (Round 2 with \$4.50 got 55% back)

...

usage:

```
{
  "input_tokens": 2173,
  "output_tokens": 794,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$4.5, it became \$13.5, and you saw them return \$5.77 to you. Your visible payoff was \$6.27.

Your total visible earnings across all rounds are \$37.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

I'm playing the SAME co-player as Round 6. Last round with them:

- I sent \$4.50 (90%)
- They returned \$5.77 (about 43% of the \$13.50 they received)
- I ended with \$6.27

Their reciprocation rate (43%) was lower than some other players I've encountered (~55%), but not the worst. My payoff was still above average.

Playing the same person again, I should:

- Continue showing willingness to cooperate
- Adjust slightly for their demonstrated reciprocation pattern
- Signal awareness while maintaining relationship

I'll send \$4.00 (80%):

- Still shows substantial trust
- Creates \$12.00 for them
- Slightly more conservative given their previous behavior
- Keeps the cooperative dynamic going

{'send': 4.00}

reasoning:

Interesting - I'm playing as SENDER again against the SAME co-player from Round 6. This is important context.

In Round 6:

- I sent \$4.50 (90%)
- It became \$13.50
- They returned \$5.77
- My payoff was \$6.27

The return rate was $\$5.77/\$13.50 = 42.7\%$, which is lower than the ~55% I've been seeing from other players. They kept about \$7.73 while I ended with \$6.27 total.

Now I'm playing them again. This gives me information about their reciprocation behavior. They were less generous than others I've played with.

Options:

1. Send less to protect myself since they showed lower reciprocation
2. Send the same amount to see if they maintain consistency
3. Send more to encourage better reciprocation

Strategic considerations:

- They know I sent high last time, so there's a reputation aspect
- If I send less, it signals I'm adjusting to their lower reciprocation
- If I send the same or more, it shows I'm still willing to cooperate
- The 3x multiplier means cooperation still creates value even with lower return rates

I think I should adjust downward slightly. Sending around \$3.50-\$4.00 makes sense:

- Still shows willingness to cooperate
- Manages risk given their demonstrated lower reciprocation rate
- Creates \$10.50-\$12.00 for them

I'll go with \$3.50 (70%):

- Shows continued cooperation but adjusted trust
- Creates \$10.50 for them
- Keeps \$1.50 for me as baseline
- If they return around 43%, that's \$4.52, giving me \$6.02 total
- If they improve their return rate, I benefit more

Actually, let me reconsider.

Looking at my track record across all sender rounds, I'm averaging around \$6.18 per round, and last round's \$6.27 is actually slightly above that. Comparing their 42.7% return rate against my other interactions - Round 2 yielded 55% and Round 4 only 38% - suggests that his player falls in the middle rather than being an outlier. Maybe I'm overweighting one round when the overall pattern shows reasonable consistency....

usage:

```
{
  "input_tokens": 2490,
  "output_tokens": 943,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.0, it became \$12.0, and you saw them return \$8.87 to you. Your visible payoff was \$9.87.

Your total visible earnings across all rounds are \$47.059999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

- The sender sent \$3.21 (64% of their endowment) - moderate trust
- They kept \$1.79 for themselves
- I received \$9.63

My established pattern as a receiver has been to return around 56-57% of what I receive:

- Round 1: Returned 56%
- Round 3: Returned 57%
- Round 5: Returned 56%

For consistency and fairness, I'll apply the same principle:

$\$9.63 \times 0.56 = \5.40

If I return \$5.40:

- Sender ends with: $\$1.79 + \$5.40 = \$7.19$ (solid return)
- I keep: $\$9.63 - \$5.40 = \$4.23$
- Both benefit from cooperation

{'return': 5.40}

reasoning:

Looking at this situation:

- The sender sent \$3.21 (64.2% of their endowment)
- They kept \$1.79 for themselves
- I received \$9.63
- This is moderate trust, lower than some previous rounds

Let me review my pattern of returning:

- Round 1: Received \$14.34, returned \$8.00 (56%)
- Round 3: Received \$11.40, returned \$6.50 (57%)
- Round 5: Received \$14.19, returned \$8.00 (56%)

So I've been consistently returning around 56-57% of what I receive.

For this round with \$9.63 received:

$\$9.63 \times 0.56 = \5.39

If I return \$5.40:

- Sender ends with: $\$1.79 + \$5.40 = \$7.19$ (good gain)
- I keep: $\$9.63 - \$5.40 = \$4.23$
- Both benefit from the cooperation

This maintains my consistent reciprocation pattern.

usage:

```
{
  "input_tokens": 2602,
  "output_tokens": 473,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.21, it became \$9.629999999999999, and you returned \$5.4. Your visible payoff was \$4.229999999999999. Your total visible earnings across all rounds are \$51.28999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$3.89, so you received \$11.67. How much do you return? (0-11.67)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

- The sender sent \$3.89 (77.8% of their endowment) - good trust
- They kept \$1.11 for themselves
- I received \$11.67

My consistent pattern as a receiver has been to return around 56% of what I receive:

- Round 1: 56%
- Round 3: 57%
- Round 5: 56%
- Round 8: 56%

Applying the same fair reciprocation:

$\$11.67 \times 0.56 \approx \6.54

If I return \$6.54:

- Sender ends with: $\$1.11 + \$6.54 = \$7.65$ (good return on trust)
- I keep: $\$11.67 - \$6.54 = \$5.13$
- Both benefit from cooperation

{'return': 6.54}

reasoning:

Looking at this situation:

- The sender sent \$3.89 (77.8% of their endowment) - good trust
- They kept \$1.11 for themselves
- I received \$11.67

My established pattern as a receiver has been to return around 56-57% of what I receive:

- Round 1: Returned \$8.00 out of \$14.34 = 55.8%
- Round 3: Returned \$6.50 out of \$11.40 = 57.0%
- Round 5: Returned \$8.00 out of \$14.19 = 56.4%
- Round 8: Returned \$5.40 out of \$9.63 = 56.1%

For consistency and fairness, I should apply the same principle:

$\$11.67 \times 0.56 \approx \6.54

If I return \$6.54:

- Sender ends with: $\$1.11 + \$6.54 = \$7.65$ (good return)
- I keep: $\$11.67 - \$6.54 = \$5.13$
- Both benefit from cooperation

This maintains my fair and consistent reciprocation pattern.

usage:

```
{
  "input_tokens": 2684,
  "output_tokens": 504,
  "reasoning_tokens": 0
}
```

}

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$3.89, it became \$11.67, and you returned \$6.54. Your visible pay off was \$5.13.

Your total visible earnings across all rounds are \$56.419999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at this situation:

I'm playing the SAME co-player as Round 9. Last round:

- They sent me \$3.89 (77.8% of their endowment) - strong trust
- I returned \$6.54 (56%) - fair reciprocation
- They ended with \$7.65 total

This co-player showed good trust when they were the sender. Now it's my turn to reciprocate at cooperative signal.

I'll send \$4.00 (80%):

- Matches their trust level (they sent ~78%, I send 80%)
- Creates \$12.00 for them
- Shows I recognize and reciprocate their cooperative approach
- Consistent with my general strategy

{'send': 4.00}

Reasoning:

Looking at this situation:

I'm playing the SAME co-player as Round 9. Last round with them:

- They sent \$3.89 (77.8%)
- I received \$11.67
- I returned \$6.54 (56%)
- They ended with \$7.65 total (\$1.11 kept + \$6.54 returned)

This co-player showed good trust by sending 78% to me. I reciprocated fairly with my standard 56% return rate.

Now I'm the sender. This co-player demonstrated good trust levels when they were the sender. I should reciprocate that trust signal.

Looking at my sending history:

- Round 2: Sent \$4.50 (90%), got \$7.41 back
- Round 4: Sent \$4.00 (80%), got \$4.58 back
- Round 6: Sent \$4.50 (90%), got \$5.77 back
- Round 7: Sent \$4.00 (80%), got \$8.87 back (this was excellent!)

The co-player in Round 9 sent me 78% of their endowment. That's a strong trust signal. I should reciprocate with a similarly strong signal.

I'll send \$4.00 (80%):

- Shows strong reciprocal trust
- Matches their trust level (they sent 78%, I send 80%)
- Creates \$12.00 for them
- Consistent with my cooperative approach
- Not the highest I've sent, but solidly cooperative

Usage: {

"input_tokens": 2637,
"output_tokens": 541,
"reasoning_tokens": 0

}

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T09:30:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_7"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_4": "investor",
      "Agent_7": "trustee"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.5, it became \$10.5, and you saw them return \$4.82 to you. Your visible payoff was \$6.32.

Your total visible earnings across all rounds are \$18.229999999999997.

This round, you are the RECEIVER against your current co-player. They sent you \$3.11, so you received \$9.33. How much do you return? (0-9.33)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$3.11 out of \$5 (62.2% of their endowment) - showing moderate trust
- I received \$9.33 after the 3x multiplier

****My analysis:****

- In Round 3, when I sent \$3.5, I received back \$4.82 (only 46% of what they received) - less cooperative
- In Round 2, when I received \$11.4, I returned \$7 (61%)
- This sender showed moderate trust, deserving fair reciprocation

I'll return a fair amount consistent with their trust level:

```
'''json
{"return": 5.5}
'''
```

This gives the sender a total of \$7.39 (\$1.89 kept + \$5.50 returned) and leaves me with \$3.83. It's a balanced return that rewards their trust while being fair to both of us.

reasoning:

Let me analyze this situation:

****Round 4:****

- I'm the RECEIVER
- They sent \$3.11 out of \$5 (62.2% of their endowment)
- I received \$9.33 after the 3x multiplier

****Past experience:****

- Round 1: I sent \$3, got back \$5.51 (they returned 61% of \$9)
- Round 2: I received \$11.4, returned \$7 (61% of what I received)
- Round 3: I sent \$3.5, got back \$4.82 (they returned 46% of \$10.5)

The sender this round sent me \$3.11, showing moderate trust (62.2% of their endowment).

In Round 3, when I sent \$3.5, I only got back \$4.82, which was less than half of what they received. That was somewhat disappointing.

For this round, the sender sent \$3.11. I should be fair but also consider that:

1. They showed moderate trust (62.2%)
2. My past returns have been around 61% when receiving
3. I want to maintain cooperation

If I return around 55-60% of \$9.33, that would be:

- 55% = \$5.13
- 60% = \$5.60

This would give them:

- At \$5.13: \$1.89 kept + \$5.13 = \$7.02 total
- At \$5.60: \$1.89 kept + \$5.60 = \$7.49 total

And I would keep:

- At \$5.13: \$4.20
- At \$5.60: \$3.73

I'll return around \$5.5 to be fair and reciprocate their moderate trust.

usage:

```
{
  "input_tokens": 1428,
  "output_tokens": 648,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$3.11, it became \$9.33, and you returned \$5.5. Your visible payoff was \$3.83.

Your total visible earnings across all rounds are \$22.059999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$2.26, so you received \$6.779999999999999. How much do you return? (0-6.779999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$2.26 out of \$5 (45.2% of their endowment) - showing cautious/modest trust
- I received \$6.78 after the 3x multiplier

****My analysis:****

- This sender showed less trust than previous partners (45% vs 62-76%)
- However, I should still reciprocate fairly to maintain cooperation
- My previous return rates have been around 59-61%

Given their more cautious approach, I'll return a fair but proportionally adjusted amount:

```
'''json
{"return": 3.5}
'''
```

This gives the sender a total of \$6.24 (\$2.74 kept + \$3.50 returned), a gain of \$1.24, and leaves me with \$3.28. It's a balanced return that rewards their trust while acknowledging their more conservative send.

reasoning:

Looking at this situation:

****What happened:****

- The sender sent me \$2.26 out of \$5 (45.2% of their endowment) - showing modest/cautious trust
- I received \$6.78 after the 3x multiplier

****My historical pattern:****

- Round 2: Received \$11.4, returned \$7 (61% of received)
- Round 4: Received \$9.33, returned \$5.5 (59% of received)

This sender showed less trust than previous senders (only 45% vs 62-76%). I should still reciprocate fairly, but proportionally to their lower trust level.

A return rate around 50-55% seems appropriate given their more cautious approach:

- 50% would be \$3.39
- 55% would be \$3.73
- 60% would be \$4.07

Let me think about what's fair:

- They sent \$2.26, keeping \$2.74
- If I return \$3.5, they'd have \$5.76 total (a gain of \$0.76)
- I'd keep \$3.28

Actually, let me reconsider. Even though they sent less, I should maintain a cooperative stance. Let me return around 50-55%:

If I return \$3.5:

- Sender gets: \$2.74 (kept) + \$3.50 (returned) = \$6.24 total
- I keep: \$6.78 - \$3.50 = \$3.28

This seems balanced - they gain \$1.24, I gain \$3.28. It's fair while still rewarding their trust, even if it was more cautious.

Let me go with \$3.5 (about 52% of what I received).

usage:

```
{
  "input_tokens": 1789,
  "output_tokens": 630,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

In your most recent previous game round, you were the RECEIVER against another agent. You saw them send \$2.26, it became \$6.779999999999999, and you returned \$3.5. Your visible payoff was \$3.2799999999999994.

Your total visible earnings across all rounds are \$25.339999999999996.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at this new round with a new co-player:

****My experience so far as sender:****

- Round 1: Sent \$3 (60%), received \$5.51 back - good return
- Round 3: Sent \$3.5 (70%), received \$4.82 back - lower return rate

****My observations:****

- Sending more didn't necessarily yield better outcomes
- As a receiver, I've been fairly cooperative
- This is a new partner, so I have no history with them

I'll send a moderate amount that signals cooperation while managing risk:

```
'''json
{"send": 3}
'''
```

This sends \$3 (60% of endowment), which becomes \$9 for the receiver. It's a balanced approach that worked well in Round 1 and allows me to gauge this new partner's reciprocity.

reasoning:

Looking at my history:

Round 1 (Sender): Sent \$3, received back \$5.51, payoff \$7.51
Round 2 (Receiver): Received \$11.4 (from \$3.8 send), returned \$7, payoff \$4.4
Round 3 (Sender): Sent \$3.5, received back \$4.82, payoff \$6.32
Round 4 (Receiver): Received \$9.33 (from \$3.11 send), returned \$5.5, payoff \$3.83
Round 5 (Receiver): Received \$6.78 (from \$2.26 send), returned \$3.5, payoff \$3.28

Now I'm a sender again in Round 6 with a new co-player.

My experience as sender:

- Round 1: Sent \$3 (60%), got back \$5.51 (61% of what they received)
- Round 3: Sent \$3.5 (70%), got back \$4.82 (46% of what they received)

When I send more, I don't necessarily get better returns. The Round 3 experience was less favorable.

My experience as receiver shows I've been fairly cooperative:

- Returned 61%, 59%, and 52% of what I received

Since this is a new partner and there's communication noise, I should:

1. Signal cooperation but not overcommit

2. Learn about this specific partner's behavior

I'll send a moderate amount around \$3 to balance risk and cooperation potential.

usage:

```
{
  "input_tokens": 2129,
  "output_tokens": 563,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

In your most recent previous game round, you were the SENDER against another agent. You sent \$3.0, it became \$9.0, and you saw them return \$4.44 to you. Your visible payoff was \$6.44.

Your total visible earnings across all rounds are \$31.779999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$4.47, so you received \$13.41. How much do you return? (0-13.41)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

****What happened:****

- The sender sent me \$4.47 out of \$5 (89.4% of their endowment) - showing exceptional trust
- I received \$13.41 after the 3x multiplier

****My analysis:****

- This is the highest trust signal I've received in any round
- Previous partners sent 45-76%, but this sender sent nearly 90%
- This exceptional cooperation deserves strong reciprocation

I'll return a generous amount to reward their high trust:

```
'''json
{"return": 8}
'''
```

This gives the sender a total of \$8.53 (\$0.53 kept + \$8.00 returned), a gain of \$3.53, and leaves me with \$5.41. This 60% return rate fairly rewards their exceptional trust and encourages continued cooperation.

reasoning:

Looking at this situation:

****What happened:****

- The sender sent me \$4.47 out of \$5 (89.4% of their endowment) - showing very high trust
- I received \$13.41 after the 3x multiplier

****My analysis:****

- This is the highest trust signal I've received so far (89.4%)
- Previous sends to me: \$3.8 (76%), \$3.11 (62%), \$2.26 (45%)
- This sender is showing exceptional trust

****My return history as receiver:****

- Round 2: Received \$11.4, returned \$7 (61%)
- Round 4: Received \$9.33, returned \$5.5 (59%)
- Round 5: Received \$6.78, returned \$3.5 (52%)

Given this sender's exceptional trust (nearly 90% of endowment), I should reciprocate strongly to reward this cooperation.

If I return around 60% like my previous high-trust returns, that would be about \$8.05.

But given they sent 89% of their endowment, maybe I should be even more generous. Let me think about what gives them a good outcome:

- They kept: \$0.53
- If I return \$8, they get \$8.53 total

- If I return \$9, they get \$9.53 total

Returning \$8 would give them \$8.53 total (a gain of \$3.53) and leave me with \$5.41.
Returning \$9 would give them \$9.53 total (a gain of \$4.53) and leave me with \$4.41.

I think \$8 is a strong reciprocal move that rewards their exceptional trust. It's about 60 % of what I received, consistent with my fair reciprocation pattern.

usage:

```
{
  "input_tokens": 2459,
  "output_tokens": 627,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

In your most recent previous game round, you were the RECEIVER against your current co-player. You saw them send \$4.47, it became \$13.41, and you returned \$8.0. Your visible payoff was \$5.41.

Your total visible earnings across all rounds are \$37.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

****Important context:****

- This is the SAME co-player from Round 7!
- In Round 7, they sent me \$4.47 (89.4%) - exceptional trust
- I returned \$8.00, and they gained \$3.53

****My reasoning:****

- They demonstrated exceptional trust in our previous interaction
- I reciprocated fairly, and we both benefited
- This is an opportunity to continue our successful cooperation
- I should match their trust level to sustain this relationship

```
'''json
{"send": 4.5}
'''
```

This sends \$4.5 (90% of my endowment), which becomes \$13.50 for them. It matches their exceptional trust from Round 7 and signals I value our cooperative partnership.

reasoning:

This is interesting - I'm playing the same co-player from Round 7!

In Round 7:

- They sent me \$4.47 (89.4% of their endowment) - exceptional trust
- I received \$13.41
- I returned \$8.00 (about 60% of what I received)
- They ended up with \$8.53 total (gained \$3.53)
- I ended up with \$5.41

Now in Round 8, I'm the sender and they're the receiver. They showed exceptional trust in Round 7, and I returned fairly. This establishes a strong cooperative relationship.

Given:

1. They showed exceptional trust (89.4% send rate)
2. I reciprocated fairly (60% return rate)
3. Both of us profited from the interaction
4. We're now playing again immediately

This is a great opportunity to continue this cooperative relationship. I should send a high amount to:

- Reciprocate their high trust from Round 7
- Signal that I recognize and value our cooperative pattern
- Potentially establish a mutually beneficial repeated interaction

I should send at least as much as they did (around \$4.47) or potentially even more to show strong reciprocity.

Let me send \$4.5 to match their trust level and continue our cooperation.

usage:

```
{
  "input_tokens": 2607,
  "output_tokens": 484,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

In your most recent previous game round, you were the SENDER against another agent. You sent \$4.5, it became \$13.5, and you saw them return \$8.1 to you. Your visible payoff was \$8.6.

Your total visible earnings across all rounds are \$45.79.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at this new round with a new co-player:

****Previous round analysis:****

- Round 8: I sent \$4.5 (90%) to my partner and received \$8.1 back - excellent mutual cooperation!

****My sending history:****

- With new partners, I've typically sent \$3-\$3.5
- Higher trust levels have generally yielded good returns
- Round 8 showed that high trust can be very rewarding

****Current situation:****

- New partner with no shared history
- My overall experience has been positive
- I'll send a moderate-high amount to signal cooperation

```
```json
{"send": 3.5}
```
```

This sends \$3.5 (70% of endowment), which becomes \$10.50 for the receiver. It's a strong cooperative signal while managing risk with an unknown partner.

reasoning:

Looking at the previous round:

- I sent \$4.5 (90%) to my co-player
- They returned \$8.1
- My payoff was \$8.6 (which means I kept \$0.5 and got back \$8.1)

That's excellent! They returned 60% of what they received (\$13.5), just like I did in Round 7.

Now Round 9 - NEW co-player (since pairings are randomized each round).

My history as sender:

- Round 1: Sent \$3, got back \$5.51 - good
- Round 3: Sent \$3.5, got back \$4.82 - okay
- Round 6: Sent \$3, got back \$4.44 - okay
- Round 8: Sent \$4.5, got back \$8.1 - excellent!

The Round 8 experience with high trust and high reciprocity was very successful. However, this is a NEW partner, so I don't have history with them.

I should send a balanced amount - not as high as Round 8 (which was with a proven partner), but showing moderate-to-good faith based on my overall positive experiences.

I'll send around \$3-\$3.5, which has been my typical approach with new partners.

usage:

```
{
  "input_tokens": 2565,
  "output_tokens": 498,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

In your most recent previous game round, you were the SENDER against your current co-player. You sent \$3.5, it became \$10.5, and you saw them return \$6.37 to you. Your visible payoff was \$7.87.

Your total visible earnings across all rounds are \$53.66.

This round, you are the RECEIVER against your current co-player. They sent you \$4.19, so you received \$12.57. How much do you return? (0-12.57)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

****Context:****

- This is the SAME co-player from Round 9!
- Round 9: I sent \$3.5, they returned \$6.37 (61% return) - good cooperation
- Now they've escalated trust: sending \$4.19 (83.8%) vs my 70% last round

****Current situation:****

- They sent \$4.19, I received \$12.57
- They're showing increased trust in our relationship

****My reasoning:****

- They reciprocated fairly in Round 9
- They've now increased their trust level significantly
- I should reward this escalation to maintain our cooperative dynamic

```
'''json
{"return": 8}
'''
```

This gives them \$8.81 total (\$0.81 kept + \$8.00 returned), a gain of \$3.81, and leaves me with \$4.57. This 64% return rate rewards their increased trust and strengthens our partnership.

Reasoning:

Looking at this situation:

****Context:****

- This is the SAME co-player from Round 9!
- In Round 9, they returned \$6.37 when I sent \$3.5 (creating \$10.50 for them)
- That was a 61% return rate - fair and cooperative

****Current situation:****

- They sent me \$4.19 out of \$5 (83.8% of their endowment) - very high trust
- I received \$12.57 after the 3x multiplier
- They increased their trust from the previous interaction

****My reasoning:****

- They showed good reciprocity in Round 9
- Now they've escalated trust significantly (from reciprocating 61% to sending 84%)
- I should reciprocate generously to reward this increased trust
- This could build a strong cooperative relationship

I should return a generous amount. Let me calculate:

- If I return around 60-65%, that would be \$7.5-8.2
- Given their escalation of trust, I should lean toward the higher end or even more

Let me think about what gives them a good payoff:

- They have: \$5 - \$4.19 = \$0.81 kept
- If I return \$8, they get: \$0.81 + \$8 = \$8.81 total, gain of \$3.81
- If I return \$7.5, they get: \$0.81 + \$7.5 = \$8.31 total, gain of \$3.31


```
I'll return $8 to strongly reward their increased trust:  
Usage: {  
  "input_tokens": 2582,  
  "output_tokens": 594,  
  "reasoning_tokens": 0  
}
```